

REINFORCEMENT LEARNING WITH LATENT LOW RANK STRUCTURE

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The practicality of reinforcement learning algorithms has been limited due to poor scaling with respect to the problem size; without structural assumptions, algorithms incur a sample complexity bound or regret bound that scales with the product of the sizes of the state space $|S|$ and action space $|A|$ of a finite horizon MDP, e.g., $|S||A|$ or $\sqrt{|S||A|}$. In this work, we consider a class of MDPs that have latent low rank structure.

We first investigate what are the minimal low rank assumptions that admit efficient reinforcement learning. We first show that without imposing further assumptions beyond low rank of Q^* , if one is constrained to estimate the Q function using only observations from a subset of entries, there is a worst case instance in which one must incur a sample complexity exponential in the horizon H to learn a near optimal policy. We subsequently show that under stronger low rank structural assumptions, given access to a generative model, Low Rank Monte Carlo Policy Iteration (LR-MCPI) and Low Rank Empirical Value Iteration (LR-EVI) achieve the desired sample complexity of $\tilde{O}((|S| + |A|)\text{poly}(d, H)/\epsilon^2)$ for a rank d setting, which is minimax optimal with respect to the scaling of $|S|$, $|A|$, and ϵ . In contrast to literature on linear and low-rank MDPs, we do not require a known feature mapping, our algorithm is computationally simple, and our results hold for long time horizons.

We next consider the problem of transferring a latent low rank representation when the source and target MDPs have transition kernels with Tucker

rank $(|S|, d, |A|)$, $(|S|, |S|, d)$, $(d, |S|, |A|)$, or (d, d, d) . In each setting, we introduce the transfer-ability coefficient α that measures the difficulty of representational transfer. Our algorithm learns latent representations in each source MDP and then exploits the linear structure to remove the dependence on $|S|$, $|A|$, or $|S||A|$ in the target MDP regret bound. We complement our positive results with information theoretic lower bounds that show our algorithms (excluding the (d, d, d) setting) are minimax-optimal with respect to α .

BIOGRAPHICAL SKETCH

Tyler grew up in Philadelphia, Fort Meyers, and Tacoma. He completed a Bachelor of Science degree with a joint major in Computer Science and Mathematics from Harvey Mudd College in 2020. Later that year, he began his Ph.D. study in Operations Research and Information Engineering at Cornell University, co-advised by Professor Yudong Chen and Professor Christina Lee Yu.

This dissertation is dedicated to my family.

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CHAPTER 1

INTRODUCTION

Reinforcement learning (RL) methods have been increasingly popular in sequential decision making tasks due to their empirical success, e.g., Atari Games [47], StarCraft II [65], and robotics [41]. RL algorithms can be applied to any sequential decision making problem which can be modeled by a Markov decision process (MDP) defined over a state space S and an action space A . The agent interacts with the environment across a horizon of length H . In each step of the horizon, the agent observes the current state of the environment and takes an action. In response the environment returns an instantaneous reward and transitions to the next state. The key Markov property that the dynamics of an MDP must satisfy is that the distribution of the instantaneous reward and the next state is only a function of the current state and action. As a result it is sufficient for the agent to only consider policies that define a distribution over the actions given the current state of the environment. The goal of the agent is to find an optimal policy which maximizes its cumulative expected reward over the horizon. When the dynamics and reward function of the MDP are known in advance, that can be solved directly using dynamic programming. Reinforcement learning considers the setting in which the MDP dynamics are unknown and thus the algorithm must query from the MDP to both learn the model as well as find an optimal policy.

Despite the empirical success and popularity of RL, its usage in practical applications is limited by high data sampling costs in the training process, which results from poor scaling of RL algorithms with respect to the size of the state and action spaces. Given a finite-horizon homogeneous MDP with state space

S , action space A , and horizon H , one needs $\tilde{\Omega}(|S||A|H^3/\epsilon^2)$ samples from a generative model to learn an optimal policy [56]. The required number of samples is often too large as many real-world problems when modeled as a Markov decision process (MDP) have very large state and action spaces. For example, the n -city Capacitated Vehicle Routing Problem (CVRP), a classical combinatorial problem from operations research, involves a state space $\{0, 1\}^n$ and an action space being all partial permutations of $n - 1$ cities [19].

A key function that is used in the course of solving for an optimal policy is the Q^π function, which is also referred to as the action-value function of policy π . It is defined over steps $h \in [H]$, states $s \in S$, and actions $a \in A$. $Q_h^\pi(s, a)$ represents the expected cumulative reward that an agent would collect if it were at state s at step h , took action a , and subsequently followed the policy π for all future steps until the end of the horizon. When the state and action space are finite, the Q_h^π function can be represented as a $|S| \times |A|$ matrix. The Q function associated to the optimal policy is denoted by Q^* . Given Q_h^* , the optimal policy at step h is trivial to find as it would follow from simply choosing the action that optimizes Q_h^* for each state. Many RL algorithms rely on estimating the Q_h^* functions across all state action pairs in order to find a near optimal policy, resulting in the $|S||A|$ sample complexity dependence. Furthermore, the tight lower bound also suggests one may need to estimate the full Q_h^* function to find the optimal policy in worst case MDPs.

MDPs with Low Rank Structures. A glaring limitation of general purpose RL algorithms is that they do not exploit application dependent structure that may be known in advance. Many real-world systems in fact have additional structure that if exploited should improve computational and statistical efficiency.

The critical question becomes what structure is reasonable to assume, and how to design new algorithms and analyses to efficiently exploit it. In this work, we focus on the subclass of MDPs that exhibit latent low-dimensional structure with respect to the relationship between states and actions, e.g., Q_h^* is low rank when viewed as a $|S|$ -by- $|A|$ matrix. A sufficient but not necessary condition that would result in such a property is that the transition kernel has low Tucker rank when viewed as a $|S|$ -by- $|S|$ -by- $|A|$ tensor of state transition probabilities, and the expected instantaneous reward function can be represented as a $|S|$ -by- $|A|$ low rank matrix.

In this work, we exploit low-rank structure to construct algorithms with improved theoretical guarantees. In Chapter 2, we investigate the benefits of assuming latent low rank structure inherent in standard tabular MDPs. Specifically, we prove novel sample complexity bounds for learning the optimal Q function under varying low rank assumptions when one has access to a simulator. Furthermore, we empirically validate our theory with simulations on common stochastic control problems. In Chapter 3, we study transfer reinforcement learning when both the source and target MDPs have low Tucker rank. Exploiting the low rank assumptions, our algorithms in this setting admit novel regret and sample complexity bounds.

The term “low rank” has been used to describe other types of low dimensional models in the MDP/RL literature, especially in the context of linear function approximation, and we would like to clarify up front that these models are significantly different. In particular, the typical use of “low rank MDPs” refers to an assumption that the transition kernel when viewed as a tensor is low rank with respect to the relationship between the originating state action

pair (s, a) and the destination state s' . This implies that the relationship across time for a given trajectory exhibits latent low-dimensional structure, where the relationship between the future state and the previous state-action pair is mediated through low-dimensional dynamics. However, this assumption does not imply that the Q function is low rank when viewed as a matrix, which would imply a low dimensional relationship between the current state and the action taken at that state. Another assumption which is easily confused with ours is the assumption that Q^* is linearly-realizable. This implies that Q^* can be written as a linear combination of d matrices $\{\phi_\ell\}_{\ell \in [d]}$ (each of size $|S|$ by $|A|$). While this implies that the set of plausible Q^* lives in a low dimensional space parameterized by $\{\phi_\ell\}_{\ell \in [d]}$, this does not imply that Q^* is low rank with respect to the relationship between S and A . The guarantees for RL algorithms under low rank MDP and linearly-realizable Q^* structure either require prior knowledge of the feature representation as given by $\{\phi_\ell\}_{\ell \in [d]}$, or otherwise do not admit polynomial time algorithms. While assuming a priori knowledge of the feature representation is often restrictive and unlikely in real applications, this assumption enables a reduction to supervised learning such that the sample complexity no longer depends on the size of the state and action space, but only on the dimension of the representation. The low rank structure we assume in this work does not require any knowledge of the latent low dimensional representation, but as a result the optimal sample complexity necessarily must still scale linearly with the size of the state and action space.

1.1 Notation

We introduce notation used throughout this work.

Let $a \wedge b := \min(a, b)$, $a \vee b := \max(a, b)$, δ_a denote the distribution over A that puts probability 1 on action a , $\sigma_i(M)$ denote the i -th largest singular value of a matrix M , and M_i denote the i -th row. The n -by- n identity matrix is denoted by $I_{n \times n}$, and $[H] := \{1, \dots, H\}$. We use several vector and matrix norms: Euclidean/ ℓ_2 norm $\|\cdot\|_2$, spectral norm $\|M\|_{op} = \sigma_1(M)$, nuclear norm $\|M\|_*$, entrywise ℓ_∞ norm $\|M\|_\infty$ (largest absolute value of entries), and Frobenius norm $\|M\|_F$. We define the condition number of a rank- d matrix M as $\kappa_M := \frac{\sigma_1(M)}{\sigma_d(M)}$.

CHAPTER 2

**OVERCOMING THE LONG HORIZON BARRIER FOR
SAMPLE-EFFICIENT REINFORCEMENT LEARNING WITH LATENT
LOW-RANK STRUCTURE**

2.1 Introduction

In this chapter, we discuss the benefits of exploiting low-rank structure inherent in an MDP. While low rank structure has been extensively used in the matrix and tensor estimation literature, it has not been widely studied in the RL literature, except for the theoretical results from [55] and empirical results from [76, 51, 50]. However, we will give examples at the end of Section 2.5 to illustrate that this property is in fact quite widespread and common in many real world systems.

Since the sample complexity under the fully general model scales as $|S||A|$, we would expect that the sample complexity under a rank- d model would scale as $d(|S| + |A|)$, as the low rank assumption on a matrix reduces the degrees of freedom of Q_h^* from $|S||A|$ to $d(|S| + |A|)$. Even though this intuition holds true in the classical low rank matrix estimation setting, the additional dynamics of the MDP introduce complex dependencies that may amplify the error for long horizons. The work in [55] proposes an algorithm that learns an ϵ -optimal Q function under a low rank assumption on Q_h^* , resulting in a sample complexity of $\tilde{O}(\text{poly}(d)(|S| + |A|)\exp(H)/\epsilon^2)$ in the general finite-horizon MDP setting. While they do achieve the reduction from $|S||A|$ to $\text{poly}(d)(|S| + |A|)$, they have an exponential dependence on the horizon that arises from an amplification of the estimation error due to the MDP dynamics and nonlinearity of low rank matrix

estimation. A key contribution of this chapter is to characterize conditions under which we are able to achieve both linear sample complexity on $|S|$ and $|A|$ along with polynomial dependence on H .

Our Contributions. In this chapter, we identify sufficient low-rank structural assumptions that allow for computationally and statistically efficient learning, reducing the sample complexity bounds to scale only linearly in $|S|$, $|A|$ and *polynomially* in H (as opposed to exponential in H in [55] or $|S||A|$ in the general tabular MDP setting). First, we show that there are additional complexities that arise from MDPs with *long horizons*; we provide an example where the optimal action-value function Q^* is low rank, yet the learner must observe an exponential (in H) number of samples to learn a near optimal policy when exploiting the low-rank structure of Q^* . This lower bound illustrates that exploiting low rank structure in RL is significantly more involved than classical matrix estimation. We propose a new computationally simple model-free algorithm, referred to as Low Rank Monte Carlo Policy Iteration (LR-MCPI). Under the assumption that Q^* is low rank, by additionally assuming a constant suboptimality gap, we prove that LR-MCPI achieves the desired sample complexity, avoiding the exponential error amplification in the horizon. Additionally we prove that LR-MCPI also achieves the desired sample complexity when all ϵ -optimal policies π have low rank Q^π functions. Under the stronger assumption that the transition kernel and reward function have low rank, we show that the model-free algorithm in [55], which we refer to as Low Rank Empirical Value Iteration (LR-EVI), also achieves the desired sample complexity. Table 2.1 summarizes our sample complexity bounds in their corresponding settings, and compares them with existing results from literature in the tabular finite-horizon MDP setting; here d refers to the rank parameter.¹

MDP Assumptions	Sample Complexity
Low-rank Q_h^* & suboptimality gap $\Delta_{\min} > 0$ (Theorem 5)	$\tilde{O}\left(\frac{d^3(S + A)H^4}{\Delta_{\min}^2}\right)$
ϵ -optimal policies have low-rank Q_h^π (Theorem 6)	$\tilde{O}\left(\frac{d^3(S + A)H^6}{\epsilon^2}\right)$
Transition kernels and rewards are low-rank (Theorem 7)	$\tilde{O}\left(\frac{d^3(S + A)H^5}{\epsilon^2}\right)$
Low-rank Q_h^* & constant horizon [55]	$\tilde{O}\left(\frac{d^5(S + A)}{\epsilon^2}\right)$
Tabular MDP with homogeneous rewards [56]	$\tilde{\Theta}\left(\frac{ S A H^3}{\epsilon^2}\right)$

Table 2.1: Our sample complexity bounds alongside results from the literature, where d denotes the rank.

We extend our results to approximately low-rank MDPs, for which we show that our algorithm learns action-value functions with error $\epsilon + O(H^2\xi)$, where ξ is the rank- d approximation error, with an efficient number of samples. Furthermore, we empirically validate the improved efficiency of our low-rank algorithms. Additionally, we show that our algorithm learns near-optimal action-value functions in a sample-efficient manner in the continuous setting, similar to the results in the table above. Finally, we prove that using existing convex program based matrix estimation methods instead of the one in [55] also achieves the desired reduction in sample complexity.

2.2 Related Work

Tabular Reinforcement Learning. Sample complexity bounds for reinforcement learning algorithms in the tabular MDP setting have been studied extensively, e.g., [5, 73, 17, 40]. Even with a generative model, $\Omega(|S||A|/\epsilon^2(1-\gamma)^3)$ samples are necessary to estimate an ϵ -optimal action-value function [10]. The

¹The sample complexity bounds of Theorems 5, 6, and 7 presented in the table hide terms that are properties of the matrix, which are constant under common regularity assumptions (and will be discussed in later sections) and terms independent of $|S|$ or $|A|$.

work [56] presents an algorithm and associated analysis that achieves a matching upper bound on the sample complexity (up to logarithmic factors), proving that the lower bound is tight. Our work focuses on decreasing the sample complexity’s dependence on $|S|$ and $|A|$ from $|S||A|$ to $|S| + |A|$ under models with a low-rank structure.

Complexity Measures for RL with General Function Approximation. The search for the most general types of structure that allow for sample-efficient reinforcement learning has resulted in many different complexity measures, including Bellman rank [33, 21], witness rank [60], Bellman Eluder dimension [36], and Bilinear Class [22]. For these classes of MDPs with rank d , finding an ϵ -optimal policy requires $\tilde{O}(\text{poly}(d, H)/\epsilon^2)$ samples. Unfortunately, these complexity measures are so broad that the resulting algorithms that achieve sample efficiency are often not polynomial time computable, and they rely on strong optimization oracles in general, e.g., assuming that we can solve a high dimensional non-convex optimization problem. We remark that our settings, including those under our strongest assumptions, cannot be easily incorporated into those frameworks.

Linear Function Approximation - Linear Realizability and Low Rank MDPs. To combat the curse of dimensionality, there is an active literature that combines linear function approximation with RL algorithms. As mentioned in the introduction, although these models are referred to as “low rank”, they are significantly different than the type of low rank structure that we consider in our model. Most notably, the resulting Q^* matrix may not be low rank. As a result we only provide a brief overview of the results in this literature, largely to illustrate the types of properties that one would hope to study for our type of

low rank model. One model class in this literature assumes that Q^* is linearly-realizable with respect to a known low dimensional feature representation, given by a known feature extractor $\phi : S \times A \rightarrow \mathbb{R}^d$ for $d \ll |S|, |A|$. [70, 71] show that an exponential number of samples in the minimum of the dimension d or the time horizon H may still be required under linear realizability, implying that additionally assumptions are required. These results highlight an interesting phenomenon that the dynamics of the MDP introduce additional complexities for linear function approximation in RL settings that are not present in supervised learning.

A more restrictive model class, sometimes referred to as *Linear/Low-rank MDPs*, imposes linearity on the dynamics of the MDP itself, i.e. the transition kernels and reward functions are linear with respect to a known low dimensional feature extractor ϕ [37, 75, 74, 67, 26]. As this does not impose structure on the relationship between s and a , the resulting Q functions may not be low rank. When the feature extractor is known, there are algorithms that achieve sample complexity or regret bounds that are polynomial in d with no dependence on $|S|$ or $|A|$. There have been attempts to extend these results to a setting where the feature mapping is not known [4, 48, 64], however the resulting algorithms are not polynomial time, as they require access to a strong nonconvex optimization oracle. Furthermore they restrict to a finite class of latent representation functions.

Low Rank Structure with respect to States and Actions. There is a limited set of works which consider a model class similar to ours, in which there is low rank structure with respect to the interaction between the states and actions and hence their interaction decomposes. This structure could be imposed on either

the transition kernel, or only on the optimal Q^* function. [76, 51, 50] provide empirical results showing that Q^* and near-optimal Q functions for common stochastic control tasks have low rank. Their numerical experiments demonstrate that the performance of standard RL algorithms, e.g., value iteration and TD learning, can be significantly improved in combination with low-rank matrix/tensor estimation methods. The theoretical work [55] considers the weakest assumption that only imposes low rankness on Q^* . They develop an algorithm that combines a novel matrix estimation method with value iteration to find an ϵ -optimal action-value function with $\tilde{O}\left(d^5(|S| + |A|)/\epsilon^2\right)$ samples for infinite-horizon γ -discounted MDPs assuming that Q^* has rank d . While this is a significant improvement over the tabular lower bound $\tilde{\Omega}\left(|S||A|/((1 - \gamma)^3\epsilon^2)\right)$ [10], their results require strict assumptions. The primary limitation is that they require the discount factor γ to be bounded from above by a small constant, which effectively limits their results to short, *constant* horizons. Lifting this limitation is left as an open question in their paper. In this work, we provide a concrete example that illustrates why long horizons may pose a challenge for using matrix estimation in RL. Subsequently we show that this long horizon barrier can be overcome by imposing additional structural assumptions. The algorithm in [55] also relies on prior knowledge of special anchor states and actions that span the entire space. We will show that under standard regularity conditions, randomly sampling states and actions will suffice.

Matrix Estimation. Low-rank matrix estimation methods focus on recovering the missing entries of a partially observed low-rank matrix with noise. The field has been studied extensively with provable recovery guarantees; see the surveys [13, 18]. However, the majority of recovery guarantees of matrix estimation are in the Frobenius norm instead of an entry-wise/ ℓ_∞ error bound,

whereas a majority of common analyses for reinforcement learning algorithms rely upon constructing entrywise confidence sets for the estimated values. Matrix estimation methods with entry-wise error bounds are given in [14, 20, 3], but all require strict distributional assumptions on the noise, e.g., independent, mean-zero sub-Gaussian/Gaussian error. The matrix estimation method proposed in [55] provides entry-wise error guarantees for arbitrary bounded noise settings and is the method we use in our algorithm in order to aid our analysis.

2.3 Preliminaries

In this chapter, we consider a standard finite-horizon MDP given by (S, A, P, R, H) [61]. Here S and A are the finite state and action spaces, respectively. $H \in \mathbb{Z}_+$ is the time horizon. $P = \{P_h\}_{h \in [H]}$ is the transition kernel, where $P_h(s'|s, a)$ is the probability of transitioning to state s' when taking action a in state s at step h . $R = \{R_h\}_{h \in [H]}$ is the reward function, where $R_h : S \times A \rightarrow \Delta([0, 1])$ is the distribution of the reward for taking action a in state s at step h . We use $r_h(s, a) := \mathbb{E}_{r \sim R_h(s, a)}[r]$ as the mean reward. A stochastic, time-dependent policy of an agent has the form $\pi = \{\pi_h\}_{h \in [H]}$ with $\pi_h : S \rightarrow \Delta(A)$, where the agent selects an action according to the distribution $\pi_h(s)$ at time step h when at state s .

For each policy π , the value function and action-value function of π represent the expected total future reward obtained from following policy π given a starting state or state-action pair at step h ,

$$V_h^\pi(s) := \mathbb{E} \left[\sum_{t=h}^H r_t(s_t, a_t) \mid s_h = s \right], \quad (2.1)$$

$$Q_h^\pi(s, a) := \mathbb{E} \left[\sum_{t=h}^H r_t(s_t, a_t) \mid s_h = s, a_h = a \right], \quad (2.2)$$

where $a_t \sim \pi_t(s_t)$ and $s_{t+1} \sim P_t(\cdot|s_t, a_t)$. The optimal value and action-value functions are given by $V_h^*(s) := \sup_{\pi} V_h^{\pi}(s)$ and $Q_h^*(s, a) := \sup_{\pi} Q_h^{\pi}(s, a)$, respectively, for all $s \in S, h \in [H]$. These functions satisfy the Bellman equations

$$V_h^*(s) = \max_{a \in A} Q_h^*(s, a), \quad Q_h^*(s, a) = r_h(s, a) + \mathbb{E}_{s' \sim P_h(\cdot|s, a)}[V_{h+1}^*(s')], \quad \forall s, a, h \quad (2.3)$$

with $V_{H+1}^*(s) = 0$. For an MDP with finite spaces and horizon, there always exists an optimal policy π^* that satisfies $V_h^{\pi^*}(s) = V_h^*(s)$ for all s, h .

The primary goal in this chapter is to find a near-optimal policy or action-value function. For $\epsilon > 0$, π is an ϵ -optimal policy if $|V_h^*(s) - V_h^{\pi}(s)| \leq \epsilon, \forall (s, h) \in S \times [H]$. Similarly, $Q = \{Q_h\}_{h \in [H]}$ is called an ϵ -optimal action-value function if $|Q_h^*(s, a) - Q_h(s, a)| \leq \epsilon, \forall (s, a, h) \in S \times A \times [H]$. We will view Q_h^*, Q_h^{π} and r_h as $|S|$ -by- $|A|$ matrices and $P_h(\cdot|\cdot, \cdot)$ as an $|S|$ -by- $|S|$ -by- $|A|$ tensor, for which various low-rank assumptions are considered. For a given function $V : S \rightarrow \mathbb{R}$, we sometimes use the shorthand $[P_h V](s, a) := \mathbb{E}_{s' \sim P_h(\cdot|s, a)}[V(s')]$ for the conditional expectation under P_h .

Throughout this chapter, we assume access to a simulator (a.k.a. the generative model framework, introduced by [38]), which takes as input a tuple $(s, a, h) \in S \times A \times [H]$ and outputs independent samples $s' \sim P_h(\cdot|s, a)$ and $r \sim R_h(s, a)$. This assumption is one of the stronger assumptions in reinforcement learning literature, but common in the line of work that studies sample complexity without directly addressing the issue of exploration, e.g., [56, 5].

2.4 Information Theoretic Lower Bound

While one may hope to learn the optimal action-value functions when only assuming that Q_h^* is low rank, we argue that the problem is more nuanced. Specifically, we present two similar MDPs with rank-one Q^* , where the learner has complete knowledge of the MDP except for one state-action pair at each time step. As the learner is restricted from querying that specified state-action pair, in order to distinguish between the two MDPs and learn the optimal policy, the learner must use the low-rank structure to estimate the unknown entry. We then show that doing so requires an exponential number of observations in the horizon H .

Consider MDPs $M^\theta = (S, A, P, R^\theta, H)$ indexed by a real number θ , where $S = A = \{1, 2\}$. At $h = 1$, $r_1^\theta(s_1, a) = 0$, $P_1(\cdot | s_0, a) = \delta_a$ for all $a \in A$, and the starting state s_1 is deterministic. For $h > 1$,

$$r_H^\theta = \begin{pmatrix} \frac{1}{2} \\ 1 + 2\theta \end{pmatrix}, \quad r_h^\theta = \begin{pmatrix} -\frac{1}{4} & 0 \\ -\frac{1}{2} & 2^{H-h}\theta \end{pmatrix}, \quad \text{and } P_h(\cdot | s, a) = \delta_s, \quad \forall s, a, \forall h \in \{2, \dots, H-1\},$$

where δ_s denotes the Dirac delta distribution at s . The rewards are deterministic except for the terminal reward at state 2, where the reward distribution $R_H^\theta(2)$ is such that the reward takes value 2 with probability $\frac{1}{2} + \theta$, and takes value 0 otherwise.

If action $a = 1$ (resp., 2) is taken at the initial step $h = 1$, the MDP will transition to state 1 (resp., 2) and then stay at this state in all subsequent steps. Thus learning the optimal policy only depends on determining the optimal action in step 1. Let θ take one of two possible values: $\theta_1 = -\frac{3}{4 \cdot 2^{H-1}}$ and $\theta_2 = \frac{3}{4 \cdot 2^{H-1}}$. To determine the correct action at the initial step, one must correctly identify θ . We will

show that identifying θ takes an exponential number of samples in the horizon H .

Lemma 1. *The optimal policy for the above MDP (for both values of θ) for steps $h \geq 1$ is $\pi_h^*(1) = \pi_h^*(2) = 2$ for all $h \in \{2, \dots, H-1\}$. Furthermore,*

$$Q_h^{*,\theta} = \begin{pmatrix} \frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} + 2^{H-h}\theta & 1 + 2^{H-h+1}\theta \end{pmatrix}, \quad V_h^{*,\theta} = \begin{pmatrix} \frac{1}{2} \\ 1 + 2^{H-h+1}\theta \end{pmatrix}, \quad \forall h \in \{2, \dots, H-1\}.$$

Lemma 1, proved in Appendix A.3.1, shows that Q_h^* is rank one. We will calculate the optimal Q function and policy at step $h = 1$ in the proof of Theorem 2 after introducing the observation model.

Observation model: The learner has exact knowledge of $r_H^\theta(1), r_h^\theta(s, a), P_h, r_1^\theta, P_1$ for all $(s, a) \in \Omega := \{(1, 1), (1, 2), (2, 1)\}$ and $h \in [H-1]$. Note that these known rewards and transitions are independent of θ . In addition, the learner is given n iid samples from $R_H^\theta(2)$.

One interpretation of this observation model is that the learner has infinitely many samples of the form $(s, a, s'), s' \sim P_h(\cdot|s, a)$ for each (s, a) , so P_h can be estimated with zero error. Similarly, the learner has infinitely many samples from $r_H^\theta(1)$ and $r_h^\theta(s, a)$ for $(s, a) \in \Omega$. However, the learner cannot observe $r_h^\theta(2, 2)$ and hence must estimate $Q_h^\theta(2, 2)$ using the low-rank structure. Finally, n noisy observations of the terminal reward $r_H^\theta(2)$ at state 2 are given.

Theorem 2. *Consider the above class of MDPs and observation model. To learn a $1/8$ -optimal policy with probability at least 0.9, the learner must observe $n = \Omega(4^H)$ samples from $R_H^\theta(2)$.*

Proof. From Lemma 1, we have $V_2^{*,\theta_1} = \begin{pmatrix} 1/2 \\ 1/4 \end{pmatrix}$ and $V_2^{*,\theta_2} = \begin{pmatrix} 1/2 \\ 7/4 \end{pmatrix}$. Hence, at

$h = 1$ the optimal action is $\pi_1^*(s_0) = 1$ for $\theta = \theta_1$ and $\pi_1^*(s_0) = 2$ for $\theta = \theta_2$. If $\theta = \theta_1$ and action 2 is taken instead, $\pi_1(s_0) = 2$, then this action incurs a $1/4$ penalty in value relative to the optimal action, i.e., $Q_1^{*,\theta_1}(s_0, 2) \leq Q_1^{*,\theta_1}(s_0, 1) - \frac{1}{4}$. If $\theta = \theta_2$ and action 1 is taken, $\pi_1(s_0) = 1$, then this action incurs a $5/4$ penalty relative to the optimal action. Therefore, to learn an ϵ -optimal policy for $\epsilon < 1/4$, e.g., $\epsilon = 1/8$ as stated in the theorem, the learner must correctly determine whether $\theta = \theta_1$ or $\theta = \theta_2$. It is well known from existing literature, see e.g., [69, 7, 44], that one needs $\Omega\left(1/(2\theta_1 - 2\theta_2)^2\right)$ samples to distinguish two (scaled) Bernoulli distributions with mean $1/2 + \theta$, $\theta \in \{\theta_1, \theta_2\}$ with probability at least 0.9. Substituting $(2\theta_1 - 2\theta_2)^2 > \frac{4^{H-1}}{9}$ proves the result. $\square$

Consider the following operational interpretation of the above example. The learner can use the rank-one structure to estimate $Q_h^*(2, 2)$ given $Q_h^*(s, a)$, $(s, a) \in \Omega$ as follows: $Q_h^*(2, 2) = Q_h^*(1, 2)Q_h^*(2, 1)/Q_h^*(1, 1)$, coinciding with the matrix estimation algorithm in [55]. Lemma 1 shows that an $\varepsilon = 2^{H-h}\theta$ error in $Q_h^*(2, 1)$ leads to a $2 \cdot \varepsilon = 2^{H-h+1}\theta$ error in $Q_h^*(2, 2)$ and $Q_{h-1}^*(2, 1)$. As such, the error is amplified exponentially when propagating backwards through the horizon, showing that this low-rank based procedure is inherently unstable.

This example illustrates that reinforcement learning with low-rank structure is more nuanced than low-rank estimation without dynamics, and that the constant horizon assumption in [55] is not merely an artifact of their analysis. Furthermore, as the entries of Q_h^* are similar in magnitude, the blow up in error is not due to the missing entry containing most of the signal. This motivates us to consider additional assumptions beyond Q_h^* being low rank. In the above example, the optimal state-action pair $(2, 2)$ is not observed, and the reward r_h and transition kernel P_h are not low-rank. To achieve stable and sample-efficient

learning with long horizons, we will consider when additional structures in the MDP dynamics can be exploited to identify and sample from the optimal action.

2.5 Assumptions

In this section, we present three low rank settings that enable sample-efficient reinforcement learning, with each setting increasing in the strength of the low rank structural assumption.

Assumption 1 (Low-rank Q_h^*). *For all $h \in [H]$, the rank of the matrix Q_h^* is d . Consequently, Q_h^* can be represented via its singular value decomposition $Q_h^* = U^{(h)}\Sigma^{(h)}(V^{(h)})^\top$, for a $|S| \times d$ orthonormal matrix $U^{(h)}$, a $|A| \times d$ orthonormal matrix $V^{(h)}$, and a $d \times d$ diagonal matrix $\Sigma^{(h)}$.*

Assumption 1 imposes that the action-value function of the optimal policy is low rank. This assumption can be contrasted with another common structural assumption in the literature, namely linearly-realizable Q^* , meaning that $Q_h^*(s, a) = w_h^\top \phi(s, a)$ for some weight vector $w_h \in \mathbb{R}^d$ and a *known* feature mapping $\phi : S \times A \rightarrow \mathbb{R}^d$ [70, 71]. In comparison, Assumption 1 decomposes ϕ into the product of separate feature mappings on the state space $U^{(h)}$ and the action space $V^{(h)}$. Hence, linearly-realizable Q^* does not imply low-rank Q_h^* . Furthermore, we assume the latent factors $U^{(h)}$ and $V^{(h)}$ are completely unknown, whereas the linear function approximation literature typically assumes ϕ is known or approximately known.

Assumption 1 only imposes low-rankness on Q_h^* , allowing for the Q^π function associated to non-optimal policies π to be full rank. Assumption 1 is likely

too weak, as Theorem 2 illustrates a doubly exponential growth in policy evaluation error under only this assumption. Below we present three additional assumptions. *Each* of these assumptions enable our algorithms to achieve the desired sample complexity when coupled with Assumption 1.

Assumption 2 (Suboptimality Gap). *For each $(s, a) \in S \times A$, the suboptimality gap is defined as $\Delta_h(s, a) := V_h^*(s) - Q_h^*(s, a)$. Assume that there exists an $\Delta_{\min} > 0$ such that*

$$\min_{h \in [H], s \in S, a \in A} \{\Delta_h(s, a) : \Delta_h(s, a) > 0\} \geq \Delta_{\min}.$$

Assumption 2 stipulates the existence of a suboptimality gap bounded away from zero. In the finite setting with $|S|, |A|, H < \infty$, there always exists a $\Delta_{\min} > 0$ for any non-trivial MDP in which there is at least one suboptimal action. This is an assumption commonly used in bandit and reinforcement learning literature.

Assumption 3 (ϵ -optimal Policies have Low-rank Q Functions). *For all ϵ -optimal policies π , the associated Q_h^π matrices are rank- d for all $h \in [H]$, i.e., Q_h^π can be represented via $Q_h^\pi = U^{(h)} \Sigma^{(h)} (V^{(h)})^\top$ for some $|S| \times d$ matrix $U^{(h)}$, $|A| \times d$ matrix $V^{(h)}$, and $d \times d$ diagonal matrix $\Sigma^{(h)}$.*

Assumption 3 imposes that all ϵ -optimal policies π have low-rank Q_h^π . We have not seen this assumption in existing literature. It is implied by the stronger assumption that *all policies* have low-rank Q_h^π ; see Appendix 2.5.1 for an MDP that satisfies Assumption 3 but fails the stronger assumption. The stronger assumption is analogous to the property that Q^π is linear in the feature map ϕ for *all* policies, which is commonly used in work on linear function approximation and linear MDPs.

To state our strongest low-rank assumption, we first recall the definition of tensor Tucker rank.

Definition 1 (Tucker Rank [43]). A tensor $M \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ has Tucker rank (d_1, d_2, d_3) if (d_1, d_2, d_3) is the smallest such that there exists a core tensor $X \in \mathbb{R}^{d_1 \times d_2 \times d_3}$ and matrices $G_i \in \mathbb{R}^{n_i \times d_i}$ for $i \in [3]$ with orthonormal columns such that

$$M(a, b, c) = \sum_{i \in [d_1], j \in [d_2], k \in [d_3]} X(i, j, k) G_1(a, i) G_2(b, j) G_3(c, k).$$

Our strongest low-rank assumption imposes that the expected reward functions are low rank, and the transition kernels have low Tucker rank *along one dimension*.

Assumption 4 (Low-rank Transition Kernels and Reward Functions). The expected reward function has rank d , and the transition kernel P_h has Tucker rank $(|S|, |S|, d)$ or $(|S|, d, |A|)$, with shared latent factors. For the Tucker rank $(|S|, |S|, d)$ case, this means that for each $h \in [H]$, there exists a $|S| \times |S| \times d$ tensor $U^{(h)}$, an $|A| \times d$ matrix $V^{(h)}$, and an $|S| \times d$ matrix $W^{(h)}$ such that

$$P_h(s'|s, a) = \sum_{i=1}^d U^{(h)}(s', s, i) V^{(h)}(a, i) \quad \text{and} \quad r_h(s, a) = \sum_{i=1}^d W^{(h)}(s, i) V^{(h)}(a, i).$$

For the Tucker rank $(|S|, d, |A|)$ case, this means that for each $h \in [H]$, there exists a $|S| \times |A| \times d$ tensor $V^{(h)}$, an $|S| \times d$ matrix $U^{(h)}$, and an $|A| \times d$ matrix $W^{(h)}$ such that

$$P_h(s'|s, a) = \sum_{i=1}^d U^{(h)}(s, i) V^{(h)}(s', a, i) \quad \text{and} \quad r_h(s, a) = \sum_{i=1}^d U^{(h)}(s, i) W^{(h)}(a, i).$$

Assumption 4 is our strongest low-rank structural assumption as it implies that the Q_h^π functions associated with *any* policy π are low rank, which subsequently implies both Assumptions 3 and 1. In fact, Assumption 4 implies that for any value function estimate $\hat{V}_h$, the matrix $r_h + [P_h \hat{V}_{h+1}]$ is low rank, as stated in the following proposition.

Proposition 3. If the transition kernel has Tucker rank $(|S|, |S|, d)$ or $(|S|, d, |A|)$ and the expected reward function has rank d with shared latent factors, i.e., Assumption 4 holds, then the matrix $r_h + [P_h \hat{V}_{h+1}]$ has rank at most d for any $\hat{V}_{h+1} \in \mathbb{R}^{|S|}$.

See Appendix A.3.2 for the proof. Proposition 3 results from the fact that for any fixed h , the matrices corresponding to r_h and $P_h(s'|\cdot, \cdot)$ for all s' share either the same column or row space, which is critically used in the analysis of our Low Rank Empirical Value Iteration algorithm.

Next we present several definitions used to characterize the error guarantees of the matrix estimation algorithm. It is commonly understood in the matrix estimation literature that other properties of the matrix beyond low rank, such as its incoherence or condition number, govern how efficiently a matrix can be estimated. Consider a trivial rank-1 MDP where $H = 1$ and the reward is a sparse matrix with only d nonzero entries taking value 1. Since the locations of the nonzero entries are unknown, we will likely observe only zeros upon sampling any small subset of entries. Estimation using a small number of samples would be possible, however, if an expert were to provide knowledge of a special set of rows and columns, which have been referred to as *anchor* states and actions in [55]. For some sets $S_h^\# \subseteq S$ and $A_h^\# \subseteq A$, we use $Q_h(S_h^\#, A_h^\#)$ to denote the submatrix obtained by restricting Q_h to state-action pairs from $S_h^\# \times A_h^\#$.

Definition 2 ($((k, \alpha)$ -Anchor States and Actions). *A set of states $S_h^\# \subset S$ and a set of actions $A_h^\# \subset A$ are $((k, \alpha)$ -anchor states and actions for a rank- d matrix Q_h if $|S_h^\#|, |A_h^\#| \leq k$, the submatrix $Q_h(S_h^\#, A_h^\#)$ has rank d , and $\|Q_h\|_\infty / \sigma_d(Q_h(S_h^\#, A_h^\#)) \leq \alpha$.*

Any set of valid anchor states and anchor actions must have at least size d in order for the associated anchor submatrix to be rank d . As the full matrix Q_h has rank d , this also implies that all rows (resp., columns) of Q_h can be written as a linear combination of the rows associated to states $S_h^\#$ (resp., columns associated to actions $A_h^\#$). The parameter α depends on the quality of the anchor sets; sub-matrices that are close to being singular result in large α . We remark that

assuming knowledge of a minimal set of anchor states and actions is common in literature, i.e., anchor-based topic modelling [9, 12] and linear feature-based RL [67, 75]. Furthermore, Shah et al. [55] posit that it suffices empirically to choose states and actions that are far from each other as anchor states and actions. However, in the worst case, finding valid anchor states and actions may require significant a priori knowledge about the unknown matrix.

Alternately, anchor states and actions can be randomly constructed for matrices that satisfy standard regularity conditions such as incoherence, commonly used in matrix estimation [11].

Definition 3 (Incoherence). *Let $Q_h \in \mathbb{R}^{|S| \times |A|}$ be a rank- d matrix with singular value decomposition $Q_h = U\Sigma V^\top$ with $U \in \mathbb{R}^{|S| \times d}$ and $V \in \mathbb{R}^{|A| \times d}$. Q_h is μ -incoherent if $\max_{i \in [|S|]} \|U_i\|_2 \leq \sqrt{\mu d / |S|}$ and $\max_{j \in [|A|]} \|V_j\|_2 \leq \sqrt{\mu d / |A|}$, where U_i denotes the i -th row of a matrix U .*

A small incoherence parameter μ ensures that the mass of U and V are not too concentrated in a couple of rows or columns. Consequently, a randomly sampled subset of rows (resp., columns) will span the row (resp., column) space, so these subsets of rows and columns contain sufficient information to reconstruct the entire matrix. Both μ and κ , the condition number of Q_h , will be used in the analysis to show that the entrywise error amplification from the matrix estimation method scales with μ, d, κ instead of the size of the state or action space, k , or α .

Discussion of Assumptions. While low rank structure with incoherence is widely accepted in the matrix and tensor estimation literature, we provide a few examples to illustrate how these properties could also naturally arise in

MDPs. Consider a continuous MDP which is converted to a tabular MDP via discretization, which is a common approach for tackling continuous MDPs. As the size of the discretization is artificial, the true complexity of the MDP is governed by the structure of the continuous MDP, which is independent of the discretization size. As long as the reward function and dynamics are sufficiently smooth with respect to the continuous MDP, the resulting tabular MDP will have approximate low-rank structure as d would be at most logarithmic with respect to $|S|, |A|$, due to a universal low rank property of smooth functions [63]. Additionally, the incoherence condition intuitively states that there cannot be a disproportionately small set of rows or columns that represent a disproportionately large amount of the signal. For MDPs that are derived from uniform discretizations of continuous MDPs with smoothness properties, incoherence also arises naturally as there will be a constant fraction of the rows or columns representing any fixed length interval of the continuous state space. Even in inherently discrete settings such as a recommendation system with users and movies, when the population is sufficiently large, one could view the discrete population of states/actions as representing a sample from an underlying continuous population with appropriate smoothness conditions. Finally, in many physical systems as relevant to most stochastic control tasks, there exist low dimensional feature representations that capture the “sufficient statistics” of the state, which fully govern the dynamics of the system.

2.5.1 Example Illustrating Assumption 3 (Low Rank Q functions for Near Optimal Policies)

Assumption 3 states that the ϵ -optimal policies π have associated Q functions that are low rank. At first glance, it might be unclear if this Assumption can be satisfied without requiring the stronger conditions in Assumption 4 of low rank rewards and low Tucker rank transition kernels. In this section, we present an MDP (S, A, P, R, H) in the reward maximization setting with all ϵ -optimal policies π (ϵ -optimal π for all $s \in S$ and $h \in [H]$) having low-rank Q^π without the transition kernel having low Tucker rank. Specifically, we upperbound the rank of Q^π with a function of ϵ and the size of the state/action space. where Π_ϵ is the policy class containing all ϵ -optimal deterministic policies. With the following example, we show that there exists an MDP with a non-trivial relationship between d_ϵ and $\epsilon, |S|$, and $|A|$. We now present the H -step MDP that exhibits this property. Let $S = A = 0 \cup [m]$, and the reward function be $r_h(s, a) = 0$ for all $(s, a, h) \in S \times A \times [H - 1]$ and $r_H(s, a) = 1 - \left(\frac{sa}{(m+1)^2}\right)^{1/2}$ for all $(s, a) \in S \times A$.

For all $h \in [H - 1]$, the transition kernel is

$$P_h(0|s, a) = \begin{cases} 0, & \text{if } s = a, \\ 1, & \text{otherwise,} \end{cases} \quad P_h(s'|s, a) = \begin{cases} 1, & \text{if } s' = s = a, \\ 0, & \text{otherwise,} \end{cases}$$

for $s' \in \{1, \dots, m\}$. We note that

$$P_h(0|\cdot, \cdot) = \begin{bmatrix} \mathbf{0} & 1 & 1 & \dots & 1 \\ 1 & \mathbf{0} & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & \dots & \mathbf{0} & 1 \\ 1 & 1 & \dots & 1 & \mathbf{0} \end{bmatrix},$$

and for $s' \in [m]$, $P_h(s'|s, a) = E^{s'}$ is an all-zero matrix with the s' -th diagonal entry equal to one, so the transition kernels do not have low Tucker rank. Next, we present the main result of this subsection, which upper bounds the rank of the Q_h functions of ϵ -optimal policies. We remark that at time step 2, selecting action 0 is always optimal, regardless of the state.

Lemma 4. *Let π be an ϵ -optimal policy, that is $V_h^*(s) - V_h^\pi(s) \leq \epsilon$ for all $(s, h) \in S \times [H]$. Then, $\text{rank}(Q_H^\pi) = 2$, and $\text{rank}(Q_h^\pi) \leq 1 + \lfloor \epsilon^2(m+1)^2 \rfloor$ for all $h \in [H-1]$.*

2.6 Algorithm

Our algorithm follows from a natural synthesis of matrix estimation with empirical value iteration and Monte Carlo policy iteration. We first describe the vanilla approximate dynamic programming algorithms for the general tabular MDP settings. Empirical value iteration simply replaces the expectation in the Bellman update in Equation (2.3) with empirical samples [27]. Specifically, to estimate $Q_h^*(s, a)$, one collects N samples of one step transitions, which entails sampling a reward and next state from $R_h(s, a)$ and $P_h(\cdot|s, a)$. Let $\hat{r}_h(s, a)$ denote the empirical average reward of the N samples from $R_h(s, a)$. Let $\hat{P}_h(\cdot|s, a)$ denote the empirical distribution over N next states sampled from $P_h(\cdot|s, a)$. Given an estimate $\hat{V}_{h+1}$ for the optimal value function at step $h+1$, the empirical Bellman update equation is

$$\hat{Q}_h(s, a) = \hat{r}_h(s, a) + \mathbb{E}_{s' \sim \hat{P}_h(\cdot|s, a)}[\hat{V}_{h+1}(s')], \text{ and } \hat{V}_h(s) = \max_{a \in A} \hat{Q}_h(s, a). \quad (2.4)$$

Evaluating $\hat{Q}_h$ and $\hat{V}_h$ requires collecting N samples for each of the $|S||A|$ state action pairs (s, a) .

Monte Carlo policy iteration for tabular MDPs approximates $Q_h^\pi(s, a)$ for a policy π by replacing the expectation in the definition (2.2) of Q^π with empirical trajectory samples, which is similar to first-visit Monte Carlo policy evaluation except we use the generative model to start at a specified state-action pair and time step [61]. This involves sampling N independent trajectories starting from state-action pair (s, a) at step h and following a given policy π until the end of the horizon H . For a fixed policy π and state action pair (s, a) , let the sequence of rewards along the i -th sampled trajectory be denoted $(r_h^i, r_{h+1}^i, \dots, r_H^i)$. We will use $\hat{r}_h^{\text{cum}}(s, a)$ to denote the empirical average cumulative reward across the N trajectories, given by

$$\hat{r}_h^{\text{cum}}(s, a) := \frac{1}{N} \sum_{i=1}^N \sum_{t=h}^H r_t^i. \quad (2.5)$$

Given an estimate of the optimal policy for steps greater than h , denoted by $(\hat{\pi}_{h+1}, \hat{\pi}_{h+2}, \dots, \hat{\pi}_H)$, the Monte Carlo estimate for the optimal action-value function and optimal policy at step h are

$$\hat{Q}_h(s, a) = \hat{r}_h^{\text{cum}}(s, a), \text{ and } \hat{\pi}_h(s) = \delta_a \text{ for } a = \operatorname{argmax}_{a' \in A} \hat{Q}_h(s, a'), \quad (2.6)$$

where the trajectories used to compute $\hat{r}_h^{\text{cum}}(s, a)$ are sampled by following the policy $(\hat{\pi}_{h+1}, \hat{\pi}_{h+2}, \dots, \hat{\pi}_H)$, and recall δ_a denotes the distribution that puts probability 1 on action a . Computing $\hat{Q}_h$ and $\hat{\pi}_h$ involves sampling $|S||A|N$ trajectories, which are each of length $H - h$, which results in a sample complexity of $|S||A|N(H - h)$ individual transitions from the MDP.

The dependence on $|S||A|$ in the sample complexity for both of the classical algorithms described above is due to using empirical samples to evaluate $\hat{Q}_h$ for every state-action pair $(s, a) \in S \times A$. The assumption that Q_h^* is at most rank d imposes constraints on the relationship between $Q_h^*(s, a)$ at different state-action pairs, such that by approximating Q_h^* using empirical samples at only $O(d|S| +$

$d|A|)$ locations, we should intuitively be able to use the low rank constraint to predict the remaining entries. Let $\Omega_h \subset S \times A$ denote the subset of entries (s, a) for which we use empirical samples to approximate $\hat{Q}_h(s, a)$, computed via either (2.4) or (2.6). Given estimates of $\hat{Q}_h(s, a)$ at $(s, a) \in \Omega_h$, we can then use a low-rank matrix estimation subroutine to estimate the Q function for $(s, a) \notin \Omega$. This is the core concept of our algorithm, which we then combine with the two classical approaches of empirical value iteration and Monte Carlo policy iteration.

2.6.1 Formal Algorithm Statement

We present two Low Rank RL algorithms, which take as input any matrix estimation algorithm, $\text{ME}(\cdot)$, that takes in a subset of entries of the matrix and returns an estimate of the whole matrix, the sets $\{\Omega_h\}_{h \in [H]}$ that indicate the state action pairs for which data should be collected by querying the MDP generative model, and $\{N_{s,a,h}\}_{(s,a,h) \in S \times A \times H}$, which denotes how many samples to query at state-action pair (s, a) at timestep h . We use “Low Rank Empirical Value Iteration” (LR-EVI) to refer to the algorithm which uses option (a) for Step 1 below, and we use “Low Rank Monte Carlo Policy Iteration” (LR-MCPI) to refer to the algorithm which uses option (b) for Step 1.

Hyperparameters: $\{\Omega_h\}_{h \in [H]}$, $\{N_{s,a,h}\}_{(s,a,h) \in S \times A \times H}$, and $\text{ME}(\cdot)$

Initialize: Set $\hat{V}_{H+1}(s) = 0$ for all s , and let $\hat{\pi}^{H+1}$ be any arbitrary policy.

For each $h \in \{H, H-1, H-2, \dots, 1\}$ in descending order,

- **Step 1:** For each $(s, a) \in \Omega_h$, compute $\hat{Q}_h(s, a)$ using empirical estimates

according to either (a) empirical value iteration or (b) Monte Carlo policy evaluation.

- (a) **Empirical Value Iteration:** Collect $N_{s,a,h}$ samples of a single transition starting from state s and action a at step h . Use the samples to estimate $\hat{Q}_h(s, a)$ according to

$$\hat{Q}_h(s, a) = \hat{r}_h(s, a) + \mathbb{E}_{s' \sim \hat{P}_h(\cdot|s,a)}[\hat{V}_{h+1}(s')],$$

where $\hat{r}_h(s, a)$ denotes the empirical average reward of the $N_{s,a,h}$ samples from $R_h(s, a)$, and $\hat{P}_h(\cdot|s, a)$ denotes the empirical distribution over the $N_{s,a,h}$ states sampled from $P_h(\cdot|s, a)$.

- (b) **Monte Carlo Policy Evaluation:** Collect $N_{s,a,h}$ independent full trajectories starting from state s and action a at step h until the end of the horizon H , where actions are chosen according to the estimated policy $(\hat{\pi}_{h+1}, \hat{\pi}_{h+2}, \dots, \hat{\pi}_H)$. Let $\hat{Q}_h(s, a) = \hat{r}_h^{\text{cum}}(s, a)$, where $\hat{r}_h^{\text{cum}}(s, a)$ denotes the empirical average cumulative reward across the $N_{s,a,h}$ trajectories starting from (s, a) at step h . If $(r_h^i, r_{h+1}^i, \dots, r_H^i)$ denotes the sequence of rewards along the i -th sampled trajectory from (s, a) at step h , then

$$\hat{Q}_h(s, a) = \hat{r}_h^{\text{cum}}(s, a) := \frac{1}{N_{s,a,h}} \sum_{i=1}^{N_{s,a,h}} \sum_{t=h}^H r_t^i.$$

- **Step 2:** Predict the action-value function for all $(s, a) \in S \times A$ according to $\text{ME}(\cdot)$:

$$\bar{Q}_h = \text{ME}\left(\{\hat{Q}_h(s, a)\}_{(s,a) \in \Omega_h}\right).$$

- **Step 3:** Compute the estimates of the value function and the optimal policy according to

$$\hat{V}_h(s) = \max_{a \in A} \bar{Q}_h(s, a) \text{ and } \hat{\pi}_h(s) = \delta_{\arg\max \bar{Q}_h(s, a)}.$$

The tabular MDP variant of the algorithm proposed in [55] is equivalent to LR-EVI where anchor states $S_h^\#$ and actions $A_h^\#$ are given and $\Omega_h = (S_h^\# \times A) \cup (S \times A_h^\#)$. Furthermore, LR-EVI is equivalent to a modification of the algorithm in [76] with a different choice for the matrix estimation algorithm used in Step 2 and the corresponding sample set Ω_h constructed in Step 1.

2.6.2 Matrix Estimation Subroutine

A critical piece to specify for the algorithm above is how to choose the subset Ω_h , and what matrix estimation subroutine $\text{ME}(\cdot)$ to use to predict the full Q_h function, where Q_h is Q_h^* , Q_h^π , or $Q'_h = r_h + P_h \hat{V}_{h+1}$ depending on the low-rank setting, given $\hat{Q}_h(s, a)$ for $(s, a) \in \Omega_h$. The performance of any matrix estimation algorithm will depend both on the selected subset Ω_h , as well as the entrywise noise distribution on $\hat{Q}_h(s, a)$ relative to the “ground truth” low-rank matrix. As a result, the subset Ω_h should be determined jointly with the choice of matrix estimation algorithm.

A limitation of a majority of results in the classical matrix estimation literature is that they do not admit entrywise bounds on the estimation error, and the analyses may be sensitive to the distribution of the observation error, i.e., require mean-zero sub-Gaussian noise. When estimating Q_h^* , the observations are biased unless one has learned the optimal policy at time steps $h + 1$ to H . Since Q_h is low-rank under our assumptions, our estimates of the observations for the matrix estimation method are unbiased with bounded noise, therefore enabling us to relax the small discount factor requirement.

Many standard analyses of RL algorithms rely upon the construction of en-

trywise confidence sets for the estimates of the Q function. Our results and analyses rely on entrywise error bounds for the matrix estimation step that balance the worst case entrywise error amplification with the size of the observation set. As such, similar theoretical guarantees can be obtained for our algorithm under any matrix estimation method that admits suitable entrywise error bounds.

The majority of our theoretical results will be shown for the variant of the algorithm that uses a matrix estimation algorithm from [55], which is incidentally equivalent to exploiting a skeleton decomposition of a low rank matrix [25]. Their algorithm uses a specific sampling pattern, in which Ω_h is constructed according to $\Omega_h = (S^\# \times A) \cup (S \times A^\#)$, where $S_h^\#$ and $A_h^\#$ are assumed to be valid anchor states and actions for the matrix Q_h (cf. Definition 2). Given estimates $\hat{Q}_h(s, a)$ for all $(s, a) \in \Omega_h$, their algorithm estimates the Q function at all state action pairs according to

$$\bar{Q}_h(s, a) = \hat{Q}_h(s, A^\#) \left[\hat{Q}_h(S^\#, A^\#) \right]^\dagger \hat{Q}_h(S^\#, a), \quad (2.7)$$

where $M^\dagger$ denotes the pseudoinverse of M , and $\bar{Q}$ is the output of the matrix estimation algorithm. The simple explicit formula for the estimates enables direct entrywise error bounds. Instead of ensuring a uniform error bound over each state-action pair in $(S^\# \times A) \cup (S \times A^\#)$, we show that additional sampling of the anchor submatrix $\Omega_h^\# = S^\# \times A^\#$ yields a smaller error amplification compared to the method proposed in [55]. In addition, we show that if Q_h is μ -incoherent, introduced in Definition 3, $\tilde{O}(\mu d, \kappa)$ -anchor states and actions can be constructed randomly by including each state in $S^\#$ independently with probability $p_1 = \Theta(\mu d \log(|S|)/|S|)$ and including each action in $A^\#$ independently with probability $p_2 = \Theta(\mu d \log(|S|)/|S|)$. As a result, a priori knowledge of the anchor states and actions is not required under these regularity conditions.

In Section 2.11, we show that our theoretical results also extend to the variation of our algorithm that uses soft nuclear norm minimization for matrix estimation alongside uniform Bernoulli sampling, utilizing entrywise guarantees shown in [14]. One matrix estimation algorithm that solves the soft nuclear norm minimization problem is `Soft-Impute` [45]. `Soft-Impute` proceeds by iteratively filling in the missing values by using a soft-thresholded singular value decomposition on the matrix of observed entries. In contrast to the sampling pattern used in the matrix estimation method given in Equation A.4, the sampling pattern needed to ensure the entrywise guarantees from [14] assumes that each state-action pair is observed with probability p_{SI} .

We use LR-EVI (resp., LR-MCPI) + `SI` to refer to the algorithm that uses option (a) (resp., option (b)) for Step 1 and `Soft-Impute` as the matrix estimation method. In Section 2.12, we empirically compare LR-EVI and LR-MCPI for both variations of matrix estimation algorithms.

2.7 Results

In this section, we present the sample complexity, i.e., an upper bound on the number of observed samples of the reward and next state, guarantees for LR-MCPI and LR-EVI with the matrix estimation method presented in Section 2.6.2 under different low-rank assumptions, from the weakest to the strongest. For $(s, a) \notin \Omega_h^\# = S_h^\# \times A_h^\#$, we denote $N_{s,a,h} = N_h$. For $(s, a) \in \Omega_h^\# = S_h^\# \times A_h^\#$, we denote $N_{s,a,h} = N_h^\# = \alpha^2 k^2 N_h$, such that entries in the anchor submatrix get a factor of $\alpha^2 k^2$ more samples.

Theorem 5. *Assume that Q_h^* is rank d and has suboptimality gap $\Delta_{\min}$ (Assump-*

tions 1 and 2), and $S_h^\#, A_h^\#$ are (k, α) -anchor states and actions for Q_h^* for all $h \in [H]$. Let $N_h = \tilde{O}\left((H - h + 1)^2 \alpha^2 k^2 / \Delta_{\min}^2\right)$ and $N_h^\# = \alpha^2 k^2 N_h$. LR-MCPI returns an optimal policy with probability at least $1 - \delta$ with a sample complexity of $\tilde{O}\left((|S| + |A|) \alpha^2 k^3 H^4 / \Delta_{\min}^2 + \alpha^4 k^6 H^4 / \Delta_{\min}^2\right)$.

The dependence on the rank d is not explicitly shown in the sample complexities stated in these theorems as it is captured by k , which we bound with Lemma 8 (presented later in this section). In the tabular setting, there always exists a $\Delta_{\min} > 0$. This sample complexity improves upon $|S||A|$ when $\Delta_{\min}$ is greater than $|S|^{-1/2} \wedge |A|^{-1/2}$. When $\Delta_{\min}$ is small, if stronger low-rank assumptions also hold, then the results in Theorems 6 and 7 below may provide stronger bounds.

Under the assumption that the Q_h^π function is low rank for all ϵ -optimal policies, Theorem 6 states that LR-MCPI learns an ϵ -optimal policy with a sample complexity independent of $\Delta_{\min}$.

Theorem 6. Assume that for all ϵ -optimal policies π , Q_h^π is rank d (Assumption 3), and $S_h^\#, A_h^\#$ are (k, α) -anchor states and actions for $Q_h^{\hat{\pi}}$, where $\hat{\pi}$ is the learned policy from LR-MCPI for all $h \in [H]$. Let $N_h = \tilde{O}\left((H - h + 1)^2 \alpha^2 k^2 H^2 / \epsilon^2\right)$ and $N_h^\# = \alpha^2 k^2 N_h$. Then, LR-MCPI returns an ϵ -optimal policy and action-value function with probability at least $1 - \delta$ with a sample complexity of $\tilde{O}\left((|S| + |A|) \alpha^2 k^3 H^6 / \epsilon^2 + \alpha^4 k^6 H^6 / \epsilon^2\right)$.

The strongest assumption that the transition kernel has low Tucker rank and the reward function is low rank, implies that Q_h^π for all policies π is low rank. As such, the result in Theorem 6 also implies an efficient sample complexity guarantee for LR-MCPI under Assumption 4. We can further remove a factor of H by using LR-EVI instead. Empirical value iteration (see Step 1(a)) reduces the sample complexity by a factor of H since it does not require sampling a full

rollout of the policy to the end of the horizon, as required for the Monte Carlo estimates (see Step 1(b)).

Theorem 7. *Assume that for any ϵ -optimal value function V_{h+1} , the matrix corresponding to $Q'_h = [r_h + [P_h V_{h+1}]]$ is rank d (a consequence of Assumption 4), and $S_h^\#, A_h^\#$ are (k, α) -anchor states and actions for $\hat{Q}'_h = [r_h + [P_h \hat{V}_{h+1}]]$, where $\hat{V}_{h+1}$ is the learned value function from LR-EVI for all $h \in [H]$. Let $N_h = \tilde{O}\left((H - h + 1)^2 \alpha^2 k^2 H^2 / \epsilon^2\right)$ and $N_h^\# = \alpha^2 k^2 N_h$. Then, LR-EVI returns an ϵ -optimal Q function and policy with probability at least $1 - \delta$ with a sample complexity of $\tilde{O}\left((|S| + |A|) \alpha^2 k^3 H^5 / \epsilon^2 + \alpha^4 k^6 H^5 / \epsilon^2\right)$.*

From Proposition 3, under Assumption 4 (low-rank transition kernel and expected rewards), the matrix corresponding to $[r_h + [P_h \hat{V}_{h+1}]]$ has rank d for any value function estimate $\hat{V}_{h+1}$. This is critical to the analysis of LR-EVI as it guarantees that the expectation of the matrix $\bar{Q}_h$ constructed from Empirical Value Iteration in Step 1(a) is low rank. This property is not satisfied by Assumptions 3 and 1, and as such the analysis for Theorem 7 does not extend to these weaker settings. Additionally, this property eliminates the need for constructing estimates with rollouts, which removes a factor of H in the sample complexity compared to LR-MCPI under Assumption 3.

Our sample complexity bounds depend on k, α , presuming that the algorithm uses some given set of (k, α) -anchor states and actions. When there may not be a domain expert to suggest anchor states and actions, we show in the next lemma that one can construct (k, α) -anchor states and actions with high probability by random sampling, where k and α scale with the incoherence and the bounded condition number of the target matrix.

Lemma 8. *Let Q_h be a rank d , μ -incoherent matrix with condition number κ . Let $S^\#$ and $A^\#$ be constructed randomly such that each state s is included in $S^\#$ with prob-*

ability $p_1 = \Theta(d\mu/\log(|S|))$, and each action a is included in $A^\#$ with probability $p_2 = \Theta(d\mu/\log(|A|))$. With probability $1 - O\left(H(|S| \wedge |A|)^{-10}\right)$, $S^\#$ and $A^\#$ are (k, α) anchor states and actions for Q_h for $k = \tilde{O}(\mu d)$ and $\alpha = O(\kappa)$.

Lemma 8 asserts that without a priori knowledge, one can find a set of $\tilde{O}(\mu d, \kappa)$ -anchor states and actions using the sampling subroutine defined in Section 2.6.2, given that the corresponding matrix is μ -incoherent with condition number κ .

Comparison to Impossibility Result in Theorem 2. Recall that Theorem 2 establishes an exponential 4^H lower bound for learning a near-optimal policy in MDPs with low-rank Q^* . While the constructed MDP has a constant suboptimality gap, the lower bound does not contradict Theorem 5 which achieves a $\text{poly}(H)$ sample complexity for LR-MCPI under a stronger generative model, i.e. after estimating the optimal action at step h , LR-MCPI can subsequently sample full trajectories from the estimate of the optimal policy, which would then include entry $(2, 2)$, which was prohibited in the setup of Theorem 2. In contrast, LR-EVI does not admit an efficient sample complexity for the MDP constructed in Section 2.4, and one can show that it exhibits exponential blowup in the estimation error due to an amplification of the estimation error in the terminal step when propagating the estimates backwards via value iteration. The MDP does not have a low rank transition kernel violating Assumption 4, as needed for Theorem 7.

2.7.1 Discussion of Optimality

Theorems 5, 6 and 7 show that under our various low rank assumptions, LR-MCPI and LR-EVI learn near-optimal policies in a sample efficient manner, decreasing the dependence of sample complexity on S and A from $|S||A|$ to $|S| + |A|$. Furthermore in Lemma 9 we establish a $d(|S| + |A|)H^3/\epsilon^2$ sample complexity lower bound for MDPs with low rank reward and transition kernel in the sense of Assumption 4 via minor modifications of existing lower bounds for tabular MDPs. Since Assumption 4 implies the optimal Q^* function is low rank, the same lower bound holds for the latter setting. Comparing our results to the lower bound, it follows that the dependence on $|S|$, $|A|$, and ϵ in our sample complexity upper bound is minimax optimal.

Lemma 9. *For any algorithm, there exists an MDP $M = (S, A, P, R, H)$ with rank d reward R_h and transition kernel P_h for all $h \in [H]$ such that $\Omega(d(|S| + |A|)H^3/\epsilon^2)$ samples are needed to learn an ϵ -optimal policy with high probability.*

Proof. Existing lower bounds from [56] prove the necessity of $\Omega(d|S|H^3/\epsilon^2)$ samples to learn an ϵ -optimal policy with high probability for a time-homogeneous MDP with $|S|$ states and d actions. Replicating each action $|A|/d$ times results in an MDP $|A|$ actions and rank d reward functions and transition kernels, and this MDP is at least as hard as the original MDP. Repeating this construction with an MDP with d states and $|A|$ actions proves an $\Omega(d|A|H^3/\epsilon^2)$ sample complexity lower bound. Combining these two lower bounds proves the lemma. $\square$

As an aside we also point out that previously shown lower bounds for linearly-realizable MDPs [70, 71] are not directly applicable to our setting, as

the constructed instances therein need not have low-rank Q^* or transition kernels, and the size of their state space scales exponentially in d .

Our sample complexity bounds depend on k and α , the size and quality of the (k, α) -anchor sets. As stated in Lemma 8, we can construct a set of $\tilde{O}(\mu d, \kappa)$ -anchor states and actions for any μ -incoherent matrix with condition number κ simply by randomly sampling a subset of state and action. The results presented in the table in Section 2.1 are obtained by substituting $k = \tilde{O}(d\mu)$ and $\alpha = O(\kappa)$ into the sample complexity bounds in Theorems 5, 6 and 7 and treating μ and κ as constants, as is standard in the matrix estimation literature, e.g., [3].

In the event that there is a domain expert who provides a set of (k, α) -anchor states and actions, then the sample complexity bound may be better by using the given set rather than randomly sampling if μ and κ are large. Note that k must be minimally at least d , and the quality of a given set of anchor states and actions depends on the smallest singular value associated to the anchor submatrix as reflected in α , which for poorly chosen anchor state and actions could scale with H .

In Theorems 5, 6 and 7, the cubic dependence on d is likely suboptimal, but this results from the suboptimal dependence on d in the corresponding entry-wise error bounds in the matrix estimation literature [55, 3, 14]. Without knowledge of good anchor states/actions from a domain expert, the dependence on μ that arises from randomly sampling anchor states and actions is not surprising, as it also commonly arises in the classical matrix estimation literature under uniform sampling models. Any improvements in the matrix estimation literature on the dependence on d, μ would directly translate into improved bounds via our results.

Our dependence on the horizon H is fairly standard as it matches the dependence on H for vanilla Q -value iteration. There is a gap between the dependence on H in our upper bounds and the H^3 lower bound in Lemma 9, which is given for homogeneous MDPs. Our upper bound results allow for nonhomogenous rewards and transition kernels, which would likely increase the lower bound to H^4 . Reducing the upper bounds to H^4 would likely require using the total variance technique from [56], which requires estimates of the variance of the policy at a given state-action pair. One can show that the variance of the Bellman operator is low rank under the strongest assumption of a low Tucker rank transition kernel, but the corresponding rank of the matrix of variances is $O(d^2)$. Hence, while it may be possible to adapt this variance technique to achieve the optimal dependence on H in our low-rank settings, doing so may incur a significantly worse dependence on d , i.e., d^6 .

2.7.2 Proof Sketch

The analysis of LR-MCPI and LR-EVI are fairly similar, and involves first showing that upon each application of the matrix estimation subroutine stated in (2.7), the amplification of the entrywise error is bounded, as stated below in Lemma 10.

Lemma 10. *Let $S^\#$ and $A^\#$ be (k, α) -anchor states and actions for matrix Q_h . For all $(s, a) \in \Omega_h = (S^\# \times A) \cup (S \times A^\#) \setminus (S^\# \times A^\#)$, assume that $\hat{Q}_h(s, a)$ satisfies $|\hat{Q}_h(s, a) - Q_h(s, a)| \leq \eta$, and for all $(s, a) \in S^\# \times A^\#$, assume that $\hat{Q}_h(s, a)$ satisfies $|\hat{Q}_h(s, a) - Q_h(s, a)| \leq \eta^\#$. Then, for all $(s, a) \in S \times A$, the estimates $\bar{Q}_h(s, a)$ computed via (2.7) satisfy*

$$|\bar{Q}_h(s, a) - Q_h(s, a)| = O(\alpha k \eta + \alpha^2 k^2 \eta^\#).$$

Proof Sketch for Lemma 10 As our algorithm constructs $\hat{Q}_h(s, a)$ for $(s, a) \in \Omega_h$ via averaging over samples from the MDP, the condition $|\hat{Q}_h(s, a) - Q_h(s, a)| \leq \eta$ is satisfied with high probability for $\eta = O((H - h)/\sqrt{N_{s,a,h}})$, shown via a simple application of Hoeffding's inequality. To prove Lemma 10, we show that the error is bounded by

$$\begin{aligned}
|\bar{Q}_h(s, a) - Q_h(s, a)| &\lesssim \|[\hat{Q}_h(S^\#, A^\#)]^\dagger\|_{op} \cdot \|\hat{Q}_h(S^\#, a)\hat{Q}_h(s, A^\#) - Q_h(S^\#, a)Q_h(s, A^\#)\|_F \\
&\quad + \left\| [\hat{Q}_h(S^\#, A^\#)]^\dagger - [Q_h(S^\#, A^\#)]^\dagger \right\|_{op} \cdot \|Q_h(S^\#, a)Q_h(s, A^\#)\|_F \\
&\lesssim \left(\frac{1}{\sigma_d(Q_h(S^\#, A^\#))} \right) \cdot k\|Q_h\|_\infty(2\eta + \eta^2) \\
&\quad + \left(\frac{\eta^\# k}{(\sigma_d(Q_h(S^\#, A^\#)))^2} \right) \cdot \|Q_h\|_\infty^2 k = O(\alpha k \eta + \alpha^2 k^2 \eta^\#).
\end{aligned}$$

The first inequality comes from an application of the triangle inequality and the definition of the operator norm since for any rank d matrix Q with (k, α) -anchor states and actions, for all $(s, a) \in S \times A$, $Q(s, a) = Q(s, A^\#)[Q(S^\#, A^\#)]^\dagger Q(S^\#, a)$. The operator norm terms are bounded using Weyl's inequality and a classic result from the perturbation of pseudoinverses. The other two terms are bounded by our assumption on $\hat{Q}_h$ and that the reward functions are bounded by one.

As η is the dominant error term as $\eta^\#$ is the error on the small anchor sub-matrix, $\{\bar{Q}_h(s, a)\}_{(s,a) \in S^\# \times A^\#}$ with size $\tilde{O}(k) \times \tilde{O}(k)$, the critical insight from Lemma 10 is that the amplification of the error due to matrix estimation is only a factor of αk , which is constant for a good choice of anchor states and actions. We set $N_{s,a,h}$ for each $(s, a) \in \Omega_h$ to guarantee $\alpha k \eta$ and $\alpha^2 k^2 \eta^\#$ are sufficiently small for a subsequent induction argument that shows the algorithm maintains near optimal estimates of Q^* and π^* . For each of the Theorems 5, 6, and 7, we will apply Lemma 10 to different choices of Q_h , chosen to guarantee that $\hat{Q}_h(s, a)$ is an unbiased estimate of Q_h . For Theorem 5, we choose $Q_h = Q_h^*$. For Theorem 6, we choose $Q_h = Q_h^{\hat{\pi}}$, where $\hat{\pi}$ is an ϵ -optimal policy. For Theorem 7, we choose

$Q_h = r_h + [P_h \hat{V}_{h+1}]$, where $\hat{V}_{h+1}$ is the value function estimate for step $h + 1$.

Choosing Q_h to be potentially distinct from Q_h^* is a simple yet critical distinction between our analysis and [55]. The analysis in [55] applies a bound similar to Lemma 10 with a choice of $Q_h = Q_h^*$. However, as $\hat{Q}_h$ will not be unbiased estimates of Q_h^* , the initial error η will contain a bias term that is then amplified exponentially in H when combined with an inductive argument for LR-EVI.

Proof Sketch for Lemma 8 To prove that the random sampling method presented in Section 2.6.2 finds $\tilde{O}(\mu d, \kappa)$ -anchor states and actions with high probability, let us denote the singular value decomposition of matrix Q_h with $U\Sigma V^T$. For a randomly sampled set of anchor states and actions $S^\#$ and $A^\#$, let $\tilde{U}$ and $\tilde{V}$ denote the submatrices of U and V limited to $S^\#$ and $A^\#$, such that the anchor submatrix $Q_h(S^\#, A^\#)$ is given by $\tilde{U}\Sigma\tilde{V}^T$. By the matrix Bernstein inequality [62], when rows and columns are sampled uniformly with probability $p_1 = \Theta(d\mu/\log(|S|))$, $p_2 = \Theta(d\mu/\log(|A|))$, the columns of $\tilde{U}$ and $\tilde{V}$ are nearly orthogonal. In particular, with high probability

$$\|p_1^{-1}\tilde{U}^T\tilde{U} - I_{d \times d}\|_{op} \leq \frac{1}{2} \quad \text{and} \quad \|p_2^{-1}\tilde{V}^T\tilde{V} - I_{d \times d}\|_{op} \leq \frac{1}{2},$$

implying that the anchor submatrix is rank d . By an application of the singular value version of the Courant-Fischer minimax theorem [29], we can relate $\sigma_d(Q_h(S^\#, A^\#))$ to $\sigma_d(Q_h)$ to show that

$$\alpha = \max_h \|Q_h\|_\infty / \sigma_d(Q_h(S_h^\#, A_h^\#)) = O(\kappa).$$

Inductive Argument for Main Theorems. The final step is to use the error analysis of each iteration in an inductive argument that argues the estimated policy at each step is near optimal. As the induction argument is similar across

all three theorems, we present the inductive argument for Theorem 6, and refer readers to the Appendix for the full proofs of all the theorems. For Theorem 6, the induction step is that if $\hat{\pi}_{H-t+1}$ is $t\epsilon/H$ -optimal, then for time step $H - t$, the policy found with LR-MCPI, $\hat{\pi}_{H-t}$, is $(t + 1)\epsilon/H$ -optimal. We then induct backwards across horizon, i.e. $t \in \{1, \dots, H\}$.

To show the induction step, first we argue that by Hoeffding's inequality, for $(s, a) \in \Omega_{H-t}$, with high probability $|\hat{Q}_{H-t} - Q_{H-t}^{\hat{\pi}}| = O(\epsilon/\alpha^2 k^2 H)$ for $N_{H-t} = \tilde{O}\left((t+1)^2 \alpha^2 k^2 H^2 / \epsilon^2\right)$, $N_{H-t}^{\#} = \alpha^2 k^2 N_{H-t}$. It is critical that $\hat{Q}_{H-t}$ are indeed unbiased estimates of $Q_{H-t}^{\hat{\pi}}$ as the estimate is constructed via Monte Carlo rollouts. By Assumption 3 and the inductive hypothesis, $Q_{H-t}^{\hat{\pi}}$ is low rank, such that by an application of Lemma 10, it follows that for all $(s, a) \in S \times A$, $|\bar{Q}_{H-t}(s, a) - Q_{H-t}^{\hat{\pi}}(s, a)| \leq \epsilon/2H$ for the appropriate choice of N_{H-t} . Finally we argue that assuming the inductive hypothesis, choosing greedily according to $\bar{Q}_{H-t}$ results in a $(t+1)\epsilon/H$ -optimal policy. For some state s , we denote $a^* = \pi_{H-t}^*(s)$ and $\hat{a} = \hat{\pi}_{H-t}(s) = \max_a \bar{Q}_{H-t}(s, a)$. The final induction step is shown via

$$\begin{aligned} V_{H-t}^*(s) - V_{H-t}^{\hat{\pi}}(s) &= Q_{H-t}^*(s, a^*) - \bar{Q}_{H-t}(s, \hat{a}) + \bar{Q}_{H-t}(s, \hat{a}) - Q_{H-t}^{\hat{\pi}}(s, \hat{a}) \\ &\leq |Q_{H-t}^*(s, a^*) - Q_{H-t}^{\hat{\pi}}(s, a^*)| + |Q_{H-t}^{\hat{\pi}}(s, a^*) - \bar{Q}_{H-t}(s, a^*)| + |\bar{Q}_{H-t}(s, \hat{a}) - Q_{H-t}^{\hat{\pi}}(s, \hat{a})| \\ &\leq \max_{s'} (V_{H-t+1}^*(s') - V_{H-t+1}^{\hat{\pi}}(s')) + \frac{\epsilon}{2H} + \frac{\epsilon}{2H}, \end{aligned}$$

where $\max_{s'} (V_{H-t+1}^*(s') - V_{H-t+1}^{\hat{\pi}}(s')) \leq t\epsilon/H$ from the induction hypothesis.

The proof of Theorem 5 involves a similar inductive argument except that given the stronger suboptimality gap assumption, we guarantee that $\hat{\pi}_h$ is an exactly optimal policy with high probability. This removes the linear growth in the error across the horizon that arises in Theorem 6, enabling us to reduce N_h by H^2 . The proof of Theorem 7 also involves a similar inductive argument, but under Assumption 4, we additionally show that at each time step $Q'_{H-t} =$

$r_{H-t} + [P_{H-t}\hat{V}_{H-t+1}]$, the expected value of $\hat{Q}_{H-t}$ for LR-EVI, is close to not only Q_{H-t}^* but also Q_{H-t}^π , which ensures that LR-EVI not only recovers an ϵ -optimal Q function, but also an ϵ -optimal policy.

Sample Complexity Calculation. The sample complexity of LR-MCPI is given by $\sum_h (H-h)(N_h|\Omega_h| + N_h^\#k^2)$, and the sample complexity of LR-EVI is given by $\sum_h (N_h|\Omega_h| + N_h^\#k^2)$. The set $|\Omega_h|$ scales as $O(k|S| + k|A|)$, where $k = \tilde{O}(\mu d)$ when the anchor states and actions are sampled randomly. The final sample complexity bounds result from substituting the choices of N_h and $N_h^\#$ as specified in the statements of Theorems 5, 6, and 7 into the summation.

2.8 Approximately Low Rank Setting

Our sample complexity results rely on either Q_h^* , Q_h^π , or $[r_h + P_h\hat{V}_{h+1}]$ having rank d , which may only be approximately satisfied. Furthermore, our algorithms require knowledge of the rank of those matrices, which may not be feasible to assume in practice. Hence, we extend our results under the low-rank reward and low Tucker rank transition kernel setting (Assumption 4) to a (d, ξ_R, ξ_P) -approximate low-rank MDP.

Assumption 5 ((d, ξ_R, ξ_P) -Approximate Low-rank MDP). *An MDP specified by (S, A, P, R, H) is a (d, ξ_R, ξ_P) -approximate low-rank MDP if for all $h \in [H]$, there exists a rank d matrix $r_{h,d}$ and a low Tucker rank transition kernel $P_{h,d}$ with Tucker rank either $(|S|, d, |A|)$ or $(|S|, |S|, d)$, such that $\forall h$,*

$$\max_{(s,a) \in S \times A} |r_h(s, a) - r_{h,d}(s, a)| \leq \xi_R \quad \text{and} \quad \sup_{(s,a) \in S \times A} 2d_{\text{TV}}(P_h(\cdot|s, a), P_{h,d}(\cdot|s, a)) \leq \xi_P, \quad (2.8)$$

where d_{TV} is the total variation distance.

Assumption 2.8 extends the exact low-rank Assumption 4, where ξ_R is the entrywise low-rank approximation error of the reward function, and ξ_P is the low-rank approximation error of the transition kernel in total variation distance. For small values of ξ_R and ξ_P , the MDP can be approximated well by a rank d MDP, and subsequently, it follows that for any estimate of the future value function, $r_h + [P_h \hat{V}_{h+1}]$ is close to a corresponding rank d approximation.

Proposition 11. *Consider a (d, ξ_R, ξ_P) -approximate low-rank MDP with $\xi_R, \xi_P, r_{h,d}$, and $P_{h,d}$ as defined in Assumption 2.8, with respect to the low rank approximation. For all $h \in [H]$ and any $\hat{V}_{h+1}$,*

$$\left| [r_{h,d} + P_{h,d} \hat{V}_{h+1}] - [r_h + P_h \hat{V}_{h+1}] \right|_{\infty} \leq \xi_R + (H - h)\xi_P.$$

Theorem 12 shows that LR-EVI with the matrix estimation routine defined in Section 2.6.2 is robust with respect to the low rank approximation error.

Theorem 12. *Assume we have a (d, ξ_R, ξ_P) -approximate low-rank MDP (Assumption 2.8) where $r_{h,d}$ and $P_{h,d}$ refer to the corresponding low rank approximations for the reward function and transition kernel. For all $h \in [H]$, let $S_h^{\#}, A_h^{\#}$ be (k, α) -anchor states and actions for $Q'_{h,d} = [r_{h,d} + P_{h,d} \hat{V}_{h+1}]$, where $\hat{V}_{h+1}$ is the learned value function from Low Rank Empirical Value iteration. Let $N_{H-t} = \tilde{O}(k^2 \alpha^2 H^4 / \epsilon^2)$, $N_{H-t}^{\#} = \alpha^2 k^2 N_{H-t}$ for all $t \in \{0, \dots, H-1\}$. Then LR-EVI returns an $(\epsilon + \tilde{O}(k^2 \alpha^2 (\xi_R H + \xi_P H^2)))$ -optimal policy with probability at least $1 - \delta$ with a sample complexity of $\tilde{O}(k^3 \alpha^2 (|S| + |A|) H^5 / \epsilon^2 + k^6 \alpha^4 H^5 / \epsilon^2)$.*

The proof of this theorem (see Appendix A.3.7) follows the same steps as the proof of Theorem 7 but additionally accounts for the low rank approximation

error using applications of Proposition 11. Proposition 11 is first used to bound the error between $\hat{Q}_h(s, a)$ and $[r_{h,d} + P_{h,d}\hat{V}_{h+1}]Q'_{h,d}(s, a)$ for $(s, a) \in \Omega_h$. Second, the proposition is used to bound the second term in the below inequality which controls the error of our estimate relative to Q_h^* and $Q_h^{\hat{\pi}}$:

$$|\hat{Q}_h(s, a) - Q_h^*(s, a)| \leq |\hat{Q}_h(s, a) - Q'_{h,d}(s, a)| + |Q'_{h,d}(s, a) - Q'_h(s, a)| + |Q'_h(s, a) - Q_h^*(s, a)|$$

for all $(s, a) \in S \times A$ where $Q'_h = r_h + P_h\hat{V}_{h+1}$ and $Q'_{h,d} = r_{h,d} + P_{h,d}\hat{V}_{h+1}$.

Theorem 12 shows that in the approximate rank setting, the error of the policy our algorithm finds is additive with respect to the approximation error while remaining sample efficient. If $Q'_{h,d}$ is μ -incoherent with condition number κ , one can use the result of Lemma 8 to find $\tilde{O}(\mu d, \kappa)$ -anchor states and actions without a priori/domain knowledge.

2.9 Continuous State and Action Spaces

Our results in Theorems 6 and 7 can be extended to the continuous MDP setting where S and A are both continuous spaces. In particular, the action-value function obtained when running LR-EVI and LR-MCPI on a discretized version of the continuous MDP can be used to construct an ϵ -optimal action-value function for the continuous MDP, similar to the reduction used in [55]. We assume the same regularity conditions on the continuous MDP as used in [55].

Assumption 6 (MDP Regularity for Continuous MDPs [55]). *Assume the MDP satisfies*

- (Compact Domain): $S = [0, 1]^n, A = [0, 1]^n$,

- (Lipschitz): Q_h^* is L -Lipschitz with respect to the one-product metric:

$$|Q_h^*(s, a) - Q_h^*(s', a')| \leq L(\|s - s'\|_2 + \|a - a'\|_2) \quad \forall h \in [H].$$

We follow the same steps as in [55] to discretize the state and action spaces into β -nets (S^β and A^β , respectively), i.e. S^β is a set such that for all $s \in S$, there exists an $s' \in S^\beta$ where $\|s' - s\|_2 \leq \beta$. We next define the discretized MDP to be $M^\beta = (S^\beta, A^\beta, P^\beta, r, H)$ where P_h^β is defined as follows:

$$P_h^\beta(s'|s, a) = \int_{\{s'' \in S : \|s'' - s'\|_2 \leq \beta\}} P_h(s''|s, a) ds''.$$

After discretizing the state and action spaces, LR-MCPI or LR-EVI is run on the discretized MDP. Our approach differs from the one from [55] because we only discretize the continuous sets once and then run the tabular algorithm while their algorithm changes the discretization error β at each iteration. To run LR-MCPI or LR-EVI on the discretized MDP, one needs to be able to sample transitions/rollouts from P_h^β instead of P_h . See Appendix A.3.8 for details on how we exploit the generative model to obtain transitions/rollouts on M^β . The following lemma shows how the optimal Q function on M^β can be used to approximate Q^* of the original MDP with small enough β .

Lemma 13. *Let MDP $M^\beta = (S^\beta, A^\beta, P^\beta, R, H)$ be the discretized approximation to MDP $M = (S, A, P, R, H)$ where S^β and A^β are β -nets of S and A , respectively. Let Q^* and Q^β be the optimal Q functions of M and M^β , respectively. For any $s \in S, a \in A$ and $s' \in S^\beta, a' \in A^\beta$ such that $\|s - s'\|_2 \leq \beta, \|a - a'\|_2 \leq \beta$ and for all $h \in [H]$,*

$$|Q_h^*(s, a) - Q_h^\beta(s', a')| \leq 2L(H - h + 1)\beta, \quad |V_h^*(s, a) - V_h^\beta(s', a')| \leq 2L(H - h + 1)\beta.$$

If the transition kernels and reward functions of M^β are low rank, satisfying Assumption 4, then LR-EVI finds an ϵ -optimal Q_h function with an efficient

number of samples. If M^β only satisfies Assumption 3, then LR-MCPI finds an ϵ -optimal Q_h function with an efficient number of samples. For sake of brevity, we present only the sample complexity bound of LR-EVI under Assumption 4. See Appendix A.3.8 for the analogous result with LR-MCPI.

Theorem 14. *Let $Q_h^\beta = [r_h + [P_h \hat{V}_{h+1}]]_{(s,a) \in S^\beta \times A^\beta}$ where $\hat{V}_{h+1}$ is the value function learned when running LR-EVI on the discretized MDP M^β . Let Assumption 4 hold on M^β , and $S_h^\#, A_h^\#$ be (k, α) -anchor states and actions for Q_h^β for all $h \in [H]$. Then, the learned $\bar{Q}_h$ from LR-EVI can be used to construct an ϵ -optimal Q function with probability at least $1 - \delta$ when $\beta = \epsilon/4LH$ and $N_{H-t} = \tilde{O}\left((t+1)^2 k^2 \alpha^2 H^2 / \epsilon^2\right)$, $N_{H-t}^\# = \tilde{O}\left((t+1)^2 k^4 \alpha^4 H^2 / \epsilon^2\right)$ for all $t \in \{0, \dots, H-1\}$ with a sample complexity of $\tilde{O}\left(k^3 \alpha^2 H^{n+5} / \epsilon^{n+2} \text{Vol}(B)\right)$, where B is the unit norm ball in $\mathbb{R}^n$.*

Theorem 14 shows that if the low-rank and matrix estimation assumptions hold on the discretized MDP, then one can use the learned Q_h estimate from LR-EVI to construct an ϵ -optimal estimate of Q function. Both algorithms are sample efficient (with respect to the dimension of the state and action spaces) as the bounds have a $1/\epsilon^{n+2}$ dependence instead of $1/\epsilon^{2n+2}$, which is minimax optimal without the low-rank assumption. Furthermore, if Q_h^β is μ -incoherent with condition number κ , one can use the result of Lemma 8 to find $\tilde{O}(\mu d, \kappa)$ -anchor states and actions without a priori/domain knowledge. Using the finite-horizon version of Corollary 2 from [58], we can construct an $O(\epsilon H)$ -optimal policy by defining a policy greedily with respect to $\bar{Q}$.

The proof of Theorem 14 follows from combining Theorem 7 with a covering number lemma to upper bound the size of the β -nets. β is chosen carefully to account for the error amplification with respect to H from Lemma 13 while ensuring that the algorithms use an efficient number of samples.

2.10 Infinite-Horizon Discounted MDPs

We also show that our results hold for infinite-horizon MDPs. We consider the standard setup for infinite-horizon tabular MDPs, (S, A, P, R, γ) , where S and A denote the finite state and action spaces. $R : S \times A \rightarrow \Delta([0, 1])$ denotes the reward distribution, and use $r_h(s, a) = \mathbb{E}_{r \sim R(s, a)}[r]$ to denote the expected reward. P denotes the transition kernel, and $0 < \gamma < 1$ denotes the discount factor. The value and action-value function of following the policy π are defined as:

$$V^\pi(s) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R_t \middle| s_0 = s \right], \quad Q^\pi(s, a) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R_t \middle| s_0 = s, a_0 = a \right],$$

for $R_t \sim R(s_t, a_t)$, $a_t \sim \pi(s_t)$, and $s_t \sim P(\cdot | s_{t-1}, a_{t-1})$. We define the optimal value function as $V^*(s) = \sup_\pi V^\pi(s)$ for all $s \in S$ and the optimal action-value function as $Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)}[V^*(s')]$. Since the reward function is bounded, for any policy π , $Q^\pi(s, a), V^\pi(s) \leq \frac{1}{1-\gamma}$ for all $(s, a) \in S \times A$. To use matrix estimation methods, we require the transition kernel to have low Tucker rank and the reward function to have shared latent factors, which is our strongest low-rank assumption (Assumption 4).

Assumption 7 (Low-rank Transition Kernels and Reward Functions (Infinite-horizon)). *The expected reward function has rank d , and the transition kernel P has Tucker rank $(|S|, |S|, d)$ or $(|S|, d, |A|)$, with shared latent factors. For the Tucker rank $(|S|, |S|, d)$ case, this means that there exists a $|S| \times |S| \times d$ tensor U , an $|A| \times d$ matrix V , and an $|S| \times d$ matrix W such that*

$$P(s' | s, a) = \sum_{i=1}^d U(s', s, i) V(a, i) \quad \text{and} \quad r(s, a) = \sum_{i=1}^d W(s, i) V(a, i).$$

For the Tucker rank $(|S|, d, |A|)$ case, this means that there exists a $|S| \times |A| \times d$ tensor V , an $|S| \times d$ matrix U , and an $|A| \times d$ matrix W such that

$$P(s' | s, a) = \sum_{i=1}^d U(s, i) V(s', a, i) \quad \text{and} \quad r(s, a) = \sum_{i=1}^d U(s, i) W(a, i).$$

Similar to the finite-horizon setting, this assumption implies that $r + \gamma[P\bar{V}]$ has low rank for any value function estimate.

Proposition 15. *For any MDP that satisfies Assumption 7, for any estimate of the value function, the rank of $r + \gamma[P\bar{V}]$ is upperbounded by d .*

The algorithm we consider that admits an efficient sample complexity is LR-EVI with the same matrix estimation method adapted for the infinite-horizon discounted setting, i.e., including the discount factor in the estimates and running Step 1, Step 2, and Step 3 for T iterations instead of recursing backwards through the horizon. We overload notation and let $\hat{Q}_i$ refer to the Q function estimated in the i -th iteration of the algorithm. The correctness result and sample complexity bound in this setting is as follows.

Theorem 16 (Correctness and Sample Complexity of LR-EVI under Assumption 7). *Assume that for any ϵ -optimal value function $\bar{V}$, the matrix corresponding to $Q'_t = [r + [P\hat{V}_t]]$ has rank d (a consequence of Assumption 7) for all $t \in [T]$, and $S_t^\#, A_t^\#$ are (k, α) -anchor states and actions for $\hat{Q}'_t = [r + [P\hat{V}_t]]$, where $\hat{V}_t$ is the learned value function from LR-EVI at iteration t for all $t \in [T]$. Let $N_t = \tilde{O}(\alpha^2 k^2 / \epsilon^2 (1 - \gamma)^4)$ and $N_t^\# = O(\alpha^2 k^2 N_t)$. Then, LR-EVI returns an ϵ -optimal Q function with probability at least $1 - \delta$ with a sample complexity of $\tilde{O}\left(\frac{(|S|+|A|)\alpha^2 k^3}{\epsilon^2 (1-\gamma)^4} + \frac{\alpha^4 k^6}{\epsilon^2 (1-\gamma)^4}\right)$.*

Theorem 7 states that if the transition kernel has low Tucker rank, one can learn an ϵ -optimal Q function with sample complexity that scales with the sum of the sizes of the state and action space instead of the product. Furthermore, if Q'_t is μ -incoherent, then one can use Lemma 8 to find $\tilde{O}(\mu d, \kappa)$ -anchor states and actions without domain knowledge, where κ is the condition number of Q' . To prove the correctness result in Theorem 7, we show that at each iteration the error of the Q function decreases with the following lemma.

Lemma 17. *Let $S_t^\#, A_t^\#$ and N_t be as defined as in Theorem 16, and let the estimate of the value function at step t satisfy $|\bar{V}_t - V^*|_\infty \leq B_t$. After one iteration of the algorithm, the resulting estimates of the value function and action-value function satisfy*

$$|\bar{Q}_{t+1} - Q^*|_\infty \leq \frac{(1 + \gamma)B_t}{2}, \quad |\bar{V}_{t+1} - V^*|_\infty \leq \frac{(1 + \gamma)B_t}{2}$$

with probability at least $1 - \frac{\delta}{T}$ for each $t \in [T]$.

Running the algorithm for a logarithmic number of times returns an ϵ -optimal Q function, which gives the sample complexity shown in Theorem 7.

2.11 Matrix Completion via Nuclear Norm Regularization

While all of the above results are stated for the variants of LR-MCPI and LR-EVI that use the matrix estimation algorithm as stated in Section 2.6.2, our results are not limited only to this specific choice of the matrix estimation algorithm. As briefly mentioned in Section 2.6.2, the analysis relies on using entrywise error bounds for the outputs of the matrix estimation algorithm. While the algorithm stated in Section 2.6.2 lends itself to explicit entrywise error bounds given its explicit form, it requires the non-standard sampling pattern associated to a set of anchor states and actions.

We show next that similar results can be derived for a different variation of LR-MCPI or LR-EVI that performs matrix estimation by solving the convex relaxation of the low-rank matrix completion problem. We utilize Theorem 1 from [14] to obtain entry-wise bounds on the matrix estimator. Chen et al. [14] state that it is straightforward to extend their results to the rectangular matrix setting, but for ease of notation, they only consider square matrices. To directly

apply the theorem, we let $|S| = |A| = n$ but note that it is easy to extend our results to $|S| \neq |A|$. Their analysis assumes data is gathered via a Bernoulli sampling model; i.e. each state-action pair is added to Ω_h with probability p , i.e., $\Omega_h = \{(s, a) | X_{(s,a)} = 1\}$ where $X_{(s,a)} \sim \text{Bernoulli}(p)$ for $(s, a) \in S \times A$.

The matrix estimator is the minimizer of a least-squares loss function with a nuclear norm regularizer, which is the convex relaxation of the low rank constraint. For the observed matrix $M_{ij} = M_{ij}^* + E_{ij}$, with M^* being the matrix we wish to recover and error matrix E , the formulation is

$$\min_{Z \in \mathbb{R}^{n \times n}} g(Z) \triangleq \frac{1}{2} \sum_{(i,j) \in \Omega_h} (Z_{ij} - M_{ij})^2 + \lambda \|Z\|_*, \quad (2.9)$$

with Ω_h constructed via Bernoulli sampling as mentioned above [14].

We next present the primary result that is needed from [14] for the readers' convenience. Assume that Ω is constructed with the Bernoulli sampling model and the error matrix $E = [E_{i,j}]$ is composed of i.i.d. zero-mean sub-Gaussian random variables with norm at most η .

Theorem 18 (Theorem 1 in [14]). *Let M^* have rank- d and be μ -incoherent with condition number κ , where $d, \kappa \in O(1)$. Let $\lambda = C_\lambda n \sigma \sqrt{np}$ in Equation 2.9 for a large enough positive constant C_λ . Assume that $n^2 p \geq C \mu^2 n \log^3 n$ and $\sigma \leq c \sqrt{\frac{np}{\mu^3 \log n}} \|M^*\|_\infty$ for some sufficiently large constant $C > 0$ and small constant $c > 0$. Then with probability $1 - O(n^{-3})$, any minimizer Z_{cvx} of Equation 2.9 satisfies*

$$\|Z_{cvx} - M^*\|_\infty \leq C_{cvx} \frac{\sigma}{\sigma_d(M^*)} \sqrt{\frac{\mu n \log n}{p}} \|M^*\|_\infty$$

for some constant $C_{cvx} > 0$.

Applying Theorem 18 into our analyses for LR-MCPI and LR-EVI gives us the necessary error bounds to prove the desired linear $|S| + |A|$ sample com-

plexities for LR-MCPI and LR-EVI with Ω_h generated according to the Bernoulli sampling model and

$$\text{ME} \left(\{\hat{Q}_h(s, a)\}_{(s,a) \in \Omega_h} \right) \leftarrow \text{CvxSolver} \left(\min_{Q \in \mathbb{R}^{|S| \times |A|}} g(Q) \triangleq \frac{1}{2} \sum_{(s,a) \in \Omega_h} (Q(s, a) - (\hat{Q}_h(s, a)))^2 + \lambda \|Q\|_* \right).$$

We state only the result for LR-EVI under Assumption 4 (low-rank reward function and low Tucker rank transition kernel). The modifications to the theorems and proofs to show the analogous result for LR-MCPI under Assumption 3 are essentially the same.

Theorem 19. *Let $p_h = \mu^3 d^2 \kappa^2 H^4 C_{\text{cvx}}^2 \log(n)/\epsilon^2 n$. Assume that for any ϵ -optimal value function $\hat{V}_{h+1}$, $Q'_h = [r_h + [P_h \hat{V}_{h+1}]]$ has rank d (Assumption 4), is μ -incoherent, and has condition number κ for all $h \in [H]$. Then, the learned policy from the algorithm specified above is ϵ -optimal with probability at least $1 - O(Hn^{-3} + \exp(-\mu^3 d^2 \kappa^2 H^4 n \log(n)/\epsilon^2))$. Furthermore, the number of samples used is upper bounded by $\tilde{O}(\mu^3 H^5 n/\epsilon^2)$ with the same probability.*

The proof of Theorem 19 follows the same argument as the proof of Theorem 7 but uses Theorem 18 to control the error amplification from the matrix estimation method. Similar to our main results, using this matrix estimation method as a subroutine reduces the sample complexity's dependence on $|S|$ and $|A|$ from $|S||A|$ to $|S| + |A|$. This theorem provides a potential explanation for the successful experimental results in [76], and answers an open question posed in [55]; it guarantees that using existing matrix estimation methods based on convex problems as a subroutine in traditional value iteration has significantly better sample complexity compared to vanilla value iteration when finding ϵ -optimal action-value functions.

2.12 Experiments

We empirically compare the performance of combining low-rank matrix estimation with empirical value iteration and Monte Carlo policy iteration on a tabular version of the Oil Discovery problem [57]. Our results also join other empirical works that show the benefit of using low-rank variants of RL algorithms on stochastic control problems [76, 51, 50].

Experimental Setup: We formulate this problem as a finite-horizon MDP, where the state and action spaces are both $\{0, 1, \dots, D\}$ for $D = 399$, and the horizon length $H = 10$. The learner’s goal is to locate the oil deposits over a 1 dimensional space, where the target location $l_h = \text{round}(400(1 - \frac{1}{h}))$ changes with h to make the learning task more difficult. At step h the learner receives a reward $f_h(s)$ that depends on how close the learner is to the oil deposit at l_h , perturbed by a zero-mean Gaussian noise with variance $\sigma_h^2(s, a) = (0.5 + a/400)^2/10$. The action a chosen indicates what state the learner attempts to move to next, and the learner additionally pays a transportation cost proportional to the distance between s and a , denoted by $c(s, a)$. As a result the reward function is

$$r_h(s, a) = f_h(s) - c(s, a) + \mathcal{N}(0, \sigma_h^2(s, a)),$$

where we choose $f_h(s)$ and $c(s, a)$ according to

$$f_h(s) = 1 - \frac{1}{4} \left\lceil \frac{4}{D} \max(0, |s - l_h| - 20) \right\rceil \quad \text{and} \quad c(s, a) = 0.01 \times \text{round}\left(\frac{|s - a|}{100}\right),$$

where $\text{round}(s)$ rounds s to the nearest integer. $c(s, a)$ is discretized to take on only 5 distinct values, but the level sets of $c(s, a)$ are diagonal bands, such that $c(s, a)$ is in fact full rank. However, the stable rank of $c(s, a)$, as defined by $\|c(s, a)\|_F^2 / \|c(s, a)\|_*^2$ is only 1.46, which implies that $c(s, a)$ is close to a low-rank matrix [53]. See Appendix A.1.1 for further discussion about $c(s, a)$.

The learner’s intended movements are perturbed, resulting in the following transition kernel:

$$\mathbb{P}_h(s'|s, a) = \max\{0, \min\{D, \delta_a + \text{Unif}(-C_h, C_h)\}\}$$

where $\text{Unif}(-C_h, C_h)$ denotes the discrete uniform distribution over $\{-C_h, -C_h + 1, \dots, C_h\}$ and $C_h = 4(H - h + 1)$ determines the amount of noise in the transitions. Since $\mathbb{E}[\mathbb{P}_h(s'|s, a)]$ only depends on the time step h and action a , it follows that the rows of $\mathbb{E}[\mathbb{P}_h(s'| :, :)]$ are the same and the rank of $\mathbb{E}[\mathbb{P}_h(s'| :, :)]$ is one. Hence, the transition kernel has Tucker rank $(|S|, 1, |A|)$.

Because the reward function is approximately low-rank and the transition kernel has low Tucker rank, it follows that this MDP is approximately low rank (satisfying Assumption 5). See Appendix A.1.2 for a visualization of Q_1^* and more discussion on the rank of Q_h^* .

Algorithms: We compare LR-EVI, LR-EVI + SI, LR-MCPI, and LR-MCPI + SI with empirical value iteration (EVI) and Monte Carlo policy iteration (MCPI). Recall that LR-EVI + SI is essentially the same as LR-EVI but uses `Soft-Impute` from the `fancyimpute` package [52] for the matrix estimation method, whereas LR-EVI uses the matrix estimation algorithm presented in section 2.6.2.

The observation set, i.e. Ω_h , that is used for `Soft-Impute` is a Bernoulli sampled subset of entries where the probability of including each entry is denoted p_{SI} . Equivalently LR-MCPI + SI is the same as LR-MCPI except that it uses `Soft-Impute` with Bernoulli sampled $|\Omega_h|$. The vanilla EVI (resp., MCPI) refers to our algorithm using option (a) (resp., option (b)) for Step 1 without the matrix estimation step, setting $\Omega_h = S \times A$ for all $h \in [H]$ and change Step 2 to be

$$\bar{Q}_h = \hat{Q}_h.$$

To empirically validate the performance of the algorithms, for a fixed sample budget $\bar{N}$, we compare the max entrywise error of $\bar{Q}_1$ of all the algorithms. We test five different allocation schemes on how to distribute the $\bar{N}$ samples across the time steps to determine $N_{s,a,h}$ and use the best one for each algorithm. We ensure that an equal number of samples are allocated to each state-action pair. As $\bar{N}$ may not be divisible by D^2 , the true samples used is within D^2 of $\bar{N}$ due to rounding. We show that LR-EVI and LR-MCPI are robust to $p = p_S = p_A$ as both algorithms perform well for a range of values of p , and it suffices to choose p to be small. We perform a grid search to determine p_{SI} for each value of $\bar{N}$, choosing the best performing parameter for each. See Appendix A.1.3 for the details on how we chose and set the hyperparameters of the algorithms.

Results: For each value of $\bar{N} \in [10^6, 10^7, 10^8, 10^9]$, we run each of the above algorithms 10 times. Figures 2.1 and 2.2 show the average ℓ_∞ error of $\bar{Q}_1$ across the 10 simulations, along with error bars whose height indicates one standard deviation above and below the mean. Note that for vanilla EVI to produce an estimate, it requires one sample per (s, a, h) , which already requires 1.6×10^6 samples. For vanilla MCPI to produce an estimate, it requires one trajectory per (s, a, h) of length $H - h + 1$, which requires 8.8×10^6 one-step samples. As a result, there is no bar depicted for either EVI or MCPI for $\bar{N} = 10^6$, as both algorithms require more than 10^6 samples to even produce any estimate.

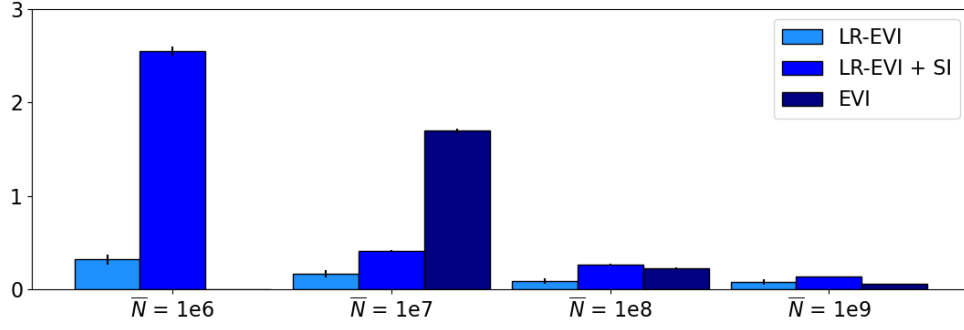


Figure 2.1: Max entrywise error of $\bar{Q}_1$ vs. sample budget for LR-EVI, LR-EVI + `Soft Impute`, empirical value iteration. Note that the optimal Q_1^* function ranges in value from roughly 8.3 to 9.6, such that 0.8 error would be roughly 10% error.

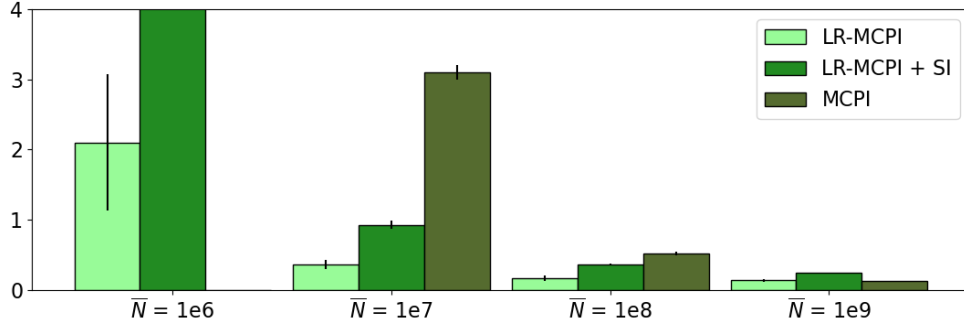


Figure 2.2: Max entrywise error of $\bar{Q}_1$ vs. sample budget for LR-MCPI, LR-MCPI + `Soft-Impute`, Monte Carlo policy iteration at $h = 1$. Note that the optimal Q_1^* function ranges in value from roughly 8.3 to 9.6, such that 0.8 error would be roughly 10% error.

For $\bar{N} = 10^6$, the error bar for LR-MCPI + `SI` has a height of 11.5 but is trimmed to align the y-axis in both graphs.

Figures 2.1 and 2.2 shows that when the sample budget is small ($\bar{N} = 10^6$), the low rank RL algorithms can still produce reasonable estimates even when there are not sufficient samples to even run the vanilla RL algorithms, i.e., less than 8.8×10^6 one-step samples. Our chosen MDP is also not strictly low rank, but only approximately low rank, thus our results validate that our algorithms

are not sensitive to the exact rank, as they perform very well on this approximately low rank MDP as well. The Monte Carlo Policy Iteration variants seem to require more samples to achieve the same performance relative to Empirical Value Iteration variants. This is expected as the sample complexity of MCPI is multiplied by H due to sampling entire trajectories rather than one step samples. The MDP in this illustration is well-behaved for LR-EVI as it has a low rank transition kernel, but the practical benefit of LR-MCPI is that it is more robust to MDPs that may not have low rank structure in the transition kernel, as exhibited by the MDP constructed in Section 2.4.

We also compare the performance of EVI and MCPI and their low-rank variants on the Double Integrator, a stochastic control problem, see Appendix A.2 for full details. The results from the Double Integrator simulations also show the benefit of the low-rank methods when the sample budget is small; LR-MCPI produces a reasonable estimate of Q_1^* even when there are not sufficient samples to run MCPI. However, LR-EVI and LR-MCPI are sensitive to the choice of matrix estimation method, so in practice, one should carefully tune the matrix estimation methods' hyperparameters given computational limits on storage and runtime. When the sample budget is large, the low-rank methods lose their advantage and may even perform worse than tabular variants.

CHAPTER 3

THE LIMITS OF TRANSFER REINFORCEMENT LEARNING WITH LATENT LOW-RANK STRUCTURE

3.1 Introduction

In an online setting where rewards are incurred during the process of learning, we may want to minimize the regret, or the difference between the reward of an optimal policy and the reward collected by the learner. In the finite-horizon online episodic MDP setting with K episodes and state space cardinality S , action space cardinality A , and horizon H , any algorithm must incur regret of at least $\tilde{\Omega}(\sqrt{SAH^3K})$ [34]. This regret bound is often too large in practice as many problems modeled as MDPs have large state and action spaces. For example, in the Atari games [46], the state space is the set of all images as a vector of pixel values.

One way to remove the dependence on the state or action space in RL algorithms is to leverage existing data or models from similar problems through transfer learning. In the transfer learning setting for MDPs, the learner can interact with or query from multiple source MDPs, which the learner can utilize at low-cost to improve their performance in the target MDP [49]. The efficacy of the transfer learning approach depends both on the similarity between the source and the target MDPs, and how information is utilized when transferring knowledge between the source and target.

As feature learning in MDPs is difficult and expensive, reusing learned low rank representations can greatly improve the performance on new problems.

Thus, in this work, we consider the setting when the source and target MDPs exhibit latent low rank structure, and the transfer RL task is to learn latent low rank representation of the state, action, or state-action pair from the source MDPs, and subsequently use it to improve learning on the target MDP. If the learned representations were perfect, then one could remove all dependence on the state space or action space in the learning task on the target MDP. This approach relies on a critical assumption that the transition kernels of the source and target MDPs have low Tucker rank when viewed as an S -by- S -by- A tensor.

The work [6] studied this problem when the transition kernel is low rank along the first mode, i.e., with Tucker rank (d, S, A) where $d < \min\{S, A\}$; this is also referred to as the Low Rank MDP model. We also consider this setting, as well as the complementary settings in which the transition kernel has low Tucker rank along the second or third mode, i.e., Tucker rank (S, d, A) or (S, S, d) . In addition, we study the (d, d, d) Tucker rank setting, where we can further exploit the low rank structure to improve performance in the target problem. (As we elaborate later, up to a factor of d , the (d, d, d) setting subsumes the remaining settings with Tucker ranks (d, d, A) , (S, d, d) and (d, S, d) .)

The different modes are not interchangeable algorithmically as there is a directionality due to the time of the transition between the current and future state. Additionally, the algorithm in [6] is not computationally tractable as it utilizes an optimization oracle which may not be practical. They assume that the latent representations are within a known finite function class. In this chapter, we propose computationally efficient algorithms that can handle latent low rank structure on the transition kernel along any of the three different modes.

As the success of a transfer learning approach depends on the similarity be-

tween the source and target problems, we introduce the *transfer-ability coefficient* α , which quantifies the similarity between the source and target problems under our low rank assumptions. This coefficient, generalizing the task-relatedness in [6], captures the unique challenge in the transfer reinforcement learning setting.

Our Contributions: We propose new computationally efficient transfer RL algorithms that admit the desired efficient regret bounds under all low Tucker rank settings (S, d, A) , (S, S, d) , (d, S, A) , and (d, d, d) . With enough samples from the source MDPs, we remove the dependence on S , A , or SA in the regret bound on the target MDP. In addition, we introduce the transfer-ability coefficient α that measures the ease of transferring a latent representation from the source MDPs to target MDP. We establish information theoretic lower bounds showing that our algorithms are optimal with respect to α (excluding the (d, d, d) Tucker rank setting). In particular, in the (d, S, A) Tucker rank setting, we achieve the optimal dependence of α^2 while [6] is sub-optimal with a dependence on α^5 .

Table 3.1 summarizes our main theoretical results, suppressing the dependence on common matrix estimation terms, and compares them with the results from [6].¹ Furthermore, we consider the setting where one has access to generative models in both the source and target problems, and show that the benefits of transfer learning go beyond alleviating exploration burden.

¹Since $\bar{\alpha} \geq \alpha$, moving $\bar{\alpha}$ into the source sample complexity yields a sample complexity that scales with α^5 . Φ and Γ refer to the function classes that contain the true ϕ and μ , respectively. As we make no assumptions on the function class of ϕ , we replace $\log(|\Phi||\Gamma|)$ with $|S||A|$.

	Tucker Rank	Source Sample Complexity	Target Regret Bound
Theorem 2	(S, S, d)	$\tilde{O}\left(d^4(1 + A/S)M^2H^4\alpha^2T\right)$	$\tilde{O}\left(\sqrt{(dMH)^3ST}\right)$
Theorem 8	(S, d, A)	$\tilde{O}\left(d^4(S/A + 1)M^2H^4\alpha^2T\right)$	$\tilde{O}\left(\sqrt{(dMH)^3AT}\right)$
Theorem 12	(d, S, A)	$\tilde{O}\left(\alpha^2dM^2TSA\right)$	$\tilde{O}\left(\sqrt{(dMH)^3T}\right)$
Theorem 13	(d, d, d)	$\tilde{O}\left(d^{10}(S + A)M^2H^4\alpha^4T\right)$	$\tilde{O}\left(\sqrt{d^6M^6H^3T}\right)$
Theorem 3.1 [6]	(d, S, A)	$\tilde{O}(\alpha^5SA^5H^7d^5M^2T)$	$\tilde{O}(\sqrt{d^3H^4T})$

Table 3.1: Theoretical guarantees of our algorithms alongside results from literature in the different Tucker rank settings. See Table B.2 for the regret bounds of algorithms in each Tucker rank setting with known latent representations.

3.2 Related Work

RL in the (d, S, A) Tucker rank Setting: There exists an extensive literature on Linear MDPs and Low-rank MDPs, which correspond to the (d, S, A) Tucker rank setting. Linear MDPs assume that the dynamics are linear with respect to a known feature mapping ϕ . This allows one to remove all dependence on the size of the state and action spaces and replace it with the rank d of the latent space [37, 75, 74, 67]. The work [28] proposes an algorithm that admits a regret bound of $\tilde{O}(\sqrt{d^2H^2T})$, matching the known lower bounds. Our algorithm in the target phase modifies existing linear MDP algorithms to take advantage of the knowledge learned in the source phase. Low Rank MDPs impose the same low-rank structure as linear MDPs but assume that ϕ is unknown. To avoid dependence on the size of the state space, many works in this setting construct algorithms that utilize a computationally inefficient oracle to admit sample complexity or regret bounds that scale with the size of the action space and the log of the size of the function class the true low rank representation [4, 64, 48, 78]. [59] incurs sample complexity on SA instead of the size of the function class to learn near-optimal policies in reward free low rank MDPs.

Transfer RL: As feature learning in low rank MDPs is difficult, [6, 15, 42, 31] study representational transfer in low rank MDPs. The work most closely related to ours is [6], which performs reward-free exploration in the source problem to learn a sufficient representation to use in the target problem in the (d, S, A) Tucker rank setting. They propose a task relatedness coefficient that captures the existing representational transfer RL settings [15]. This coefficient is similar to our transfer-ability coefficient but differs as our low rank assumption is imposed on different modes of the probability transition tensor. Their algorithm admits a source sample complexity bound of $\tilde{O}(A^4\alpha^3d^5H^7K^2T\log(|\Phi||\Gamma|))$ with a target regret bound of $\tilde{O}(\sqrt{d^3\alpha^2H^4T})$. Other transfer RL works [15, 42] require reach-ability assumptions to learn the latent representations in the source phase. Similar to transfer RL, reward free RL studies the problem of giving the learner access to only the transition kernel in the first phase, and in the second phase the learner must output a good policy when given multiple reward functions [35, 77, 16, 59]. While reward free RL is similar to our setting, enabling the dynamics to differ between the source and target phase greatly increases the difficulty and changes the objective in the source problem, i.e., learning a good latent representation instead of collecting trajectories that sufficiently explore the state space.

RL in the (S, d, A) or (S, S, d) Tucker Rank Setting: The work [54] introduces the RL setting in which the transition kernel has Tucker rank (S, d, A) or (S, S, d) . Assuming access to a generative model, their algorithms learn near-optimal policies with sample complexities that scale with $S + A$, thereby circumventing the SA lower bound for tabular reinforcement learning [10]. The work [72] consider the same low Tucker rank assumption on the dynamics but in the offline RL setting. The works [76, 51, 50] empirically show the benefits of combining ma-

trix/tensor estimation with RL algorithms on common stochastic control tasks. [32] consider the low-rank bandits setting and use a similar algorithm to ours by estimating the singular subspaces of the reward matrix to improve the regret bound's dependence on the dimension of the problem.

3.3 Preliminaries

We consider the transfer reinforcement learning setting introduced by [6], in which the learner interacts with M finite-horizon source MDPs and one finite horizon episodic target MDP. All MDPs share the same discrete state space $\mathcal{S}$ with cardinality S , discrete action space $\mathcal{A}$ with cardinality A , and finite horizon $H \in \mathbb{Z}_+$.² Let $r = \{r_h\}_{h \in [H]}$ be the deterministic unknown reward function of the target MDP, where $r_h : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$. For $m \in [M]$, let $r_m = \{r_{m,h}\}_{h \in [H]}$ be the deterministic unknown reward function for the m th source MDP, where $r_{m,h} : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$.³ Let $\mathcal{P} = \{P_h\}$ be the transition kernel for the target MDP, where $P_h(s'|s, a)$ is the probability of transitioning to state s' from state-action pair (s, a) at time step h . Similarly, for $m \in [M]$, $\mathcal{P}_m$ is the transition kernel for m th source MDP and is defined in the same way as $\mathcal{P}$. An agent's policy has the form $\pi = \{\pi_h\}_{h \in [H]}$, $\pi_h : \mathcal{S} \rightarrow \Delta(\mathcal{A})$, where $\pi_h(a|s)$ is the probability that the agent chooses action a at state s and time step h .

In the transfer RL problem, the learner first interacts with the M source MDPs without access to the target MDP. In this phase, called the source phase, the learner is given access to a generative model in each of the source MDPs. The

²It is possible to relax the assumption of sharing the same state/action spaces. In particular, depending on the Tucker rank of the transition kernel, we only require the source and target MDPs share only one of the spaces.

³Our results can easily be extended to stochastic reward functions.

generative model takes in a state-action pair (s, a) and time step h and outputs $r_h(s, a)$ and an independent sample from $P_h(\cdot|s, a)$ [38]. Access to a generative model in the source phase is necessary; in particular, it has been shown in [6, Theorem 4.1] that without access to a generative model in the source phase, one cannot learn a near-optimal policy in the target phase using the learned representation. Then, in the next phase, called the target phase, the learner loses access to the source MDPs and interacts only with the target MDP in a standard online model, without access to a generative model. The performance of the learner is evaluated by two metrics: i) the regret in the target MDP, i.e., $\text{Regret}(T) = \sum_{k \in [K]} V_1^*(s) - V_1^{\pi^k}(s)$, where π^k is the learner's policy in episode k , and ii) the number of samples taken in the source phase.

In standard MDPs, the transition kernel can have Tucker rank (S, S, A) as there is no low rank structure imposed on the tensor. In this work, we investigate the benefits of assuming that the transition kernel is low rank along each mode/dimension of the transition kernel, which corresponds to the (S, d, A) , (S, S, d) , and (d, S, A) Tucker rank settings. We note in passing that all our sample complexity and regret bounds in Table 3.1 remain valid when d is an upper bound on the exact rank. To illustrate the difference in these three settings, we consider $d = 1$ and present the corresponding structure of the transition kernel in Equations (3.1), (3.2), and (3.3), respectively:

$$\text{Tucker rank } (S, 1, A): \quad P(s'|s, a) = \mu(s)\phi(s', a), \quad (3.1)$$

$$\text{Tucker rank } (S, S, 1): \quad P(s'|s, a) = \mu(a)\phi(s', s), \quad (3.2)$$

$$\text{Tucker rank } (1, S, A): \quad P(s'|s, a) = \mu(s')\phi(s, a), \quad (3.3)$$

for some functions μ, ϕ defined on the appropriate domains. Figure 3.1 pictorially displays the (S, S, d) Tucker rank assumption on the transition kernel.

While the three settings may seem similar, the (S, d, A) and (S, S, d) Tucker rank settings differ significantly from the (d, S, A) Tucker rank setting (i.e., Low Rank MDP). In particular, the (d, S, A) setting assumes low rank structure across time, i.e., the current state and action vs. the next state, while the other Tucker rank assumptions impose low rank structure between the current state and action. These differences are significant as each setting requires different algorithms in the target problem and construction for the lower bounds.

Additionally, we analyze the (d, d, d) Tucker rank setting, where the transition kernel fully factorizes, in which case the next state, current state, and action all have separate low rank representations. This assumption allows us to further exploit the low rank structure to improve the regret bound in the target phase when compared to the (S, d, A) and (S, S, d) Tucker rank settings. Note that this (d, d, d) Tucker rank assumption is equivalent to imposing low rank along any two modes of the transition kernel, up to factors of d in the sample complexity and regret bounds. For example, for an (S, d, d) Tucker rank transition kernel P , one can factor the $S \times d \times d$ core tensor into a $S \times d$ orthonormal matrix and $d^2 \times d \times d$ core tensor, which implies P has Tucker rank (d^2, d, d) .

While the sample complexity and regret bound of each algorithm differ in each Tucker rank setting, one should choose which algorithm to use based on the low rank structure inherent in their problem. For example, in the (S, S, d) Tucker rank setting, one assumes that each action has a low rank representation. One potential application of this setting is a personal movie recommender system. Specifically, the states are user's different states of mind while each action chooses a movie from a large collection. As movies in the same genre evoke similar responses, each action can be mapped to a lower dimensional

space with the axis being each genre. In contrast, each state of mind is distinct. In the (S, d, A) Tucker rank setting, one assumes that the state one transitions from has a low rank representation. The (d, d, d) Tucker rank setting combines the two previous assumptions and asserts that current states and actions have separate low dimensional representations while low rank MDPs with (d, S, A) Tucker rank assume that each state-action pair has a joint low rank representation. For sake of brevity, we will only present our assumption and algorithm for the (S, S, d) Tucker rank setting in the main body of this work and defer the other settings to the appendix.

3.3.1 Low Tucker Rank MDPs

Introduced in [54], we assume that reward functions are low rank and the transition kernels have low Tucker rank in each of the source and target MDPs. The main benefit of Assumption 8 is that for any value function V , the Bellman update of any value function, $Q_h = r_h + P_h V$, is a low rank matrix due to Proposition 4 [54].

Assumption 8. *In each of the M source MDPs, the reward functions have rank d , and the transition kernels have Tucker rank (S, S, d) . The target MDP has reward function with rank d' and transition kernels with Tucker rank (S, S, d') where $d' \leq dM$. Thus, there exists $S \times S \times d$ tensors $U_{m,h}$, $S \times S \times d'$ tensors U_h , $A \times d$ μ -incoherent matrices $G_{m,h}$ with orthonormal columns, $A \times d'$ μ -incoherent matrices G_h with orthonormal columns, and $S \times d$ matrices $W_{m,h}$, $S \times d'$ matrices W_h such that*

$$P_{m,h}(s'|s, a) = \sum_{i \in [d]} G_{m,h}(a, i) U_{m,h}(s', s, i), \quad r_{m,h}(s, a) = \sum_{i \in [d]} G_{m,h}(a, i) W_{m,h}(s, i),$$

$$P_h(s'|s, a) = \sum_{i \in [d']} G_h(a, i) U_h(s', s, i), \quad r_h(s, a) = \sum_{i \in [d']} G_h(a, i) W_h(s, i)$$

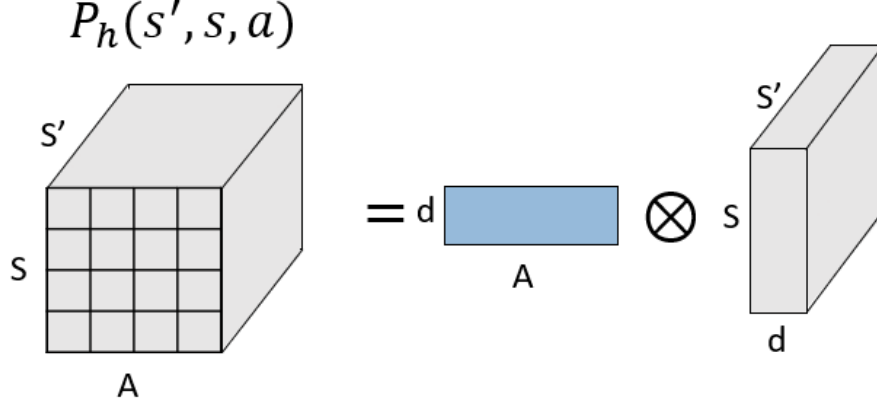


Figure 3.1: Transition kernel with Tucker rank (S, S, d)

where $\|\sum_{s' \in \mathcal{S}} g(s') U_{m,h}(s', s, :)\|_2, \|\sum_{s' \in \mathcal{S}} g(s') U_h(s', s, :)\|_2, \|W_h(s)\|_2, \|W_{m,h}(s)\|_2 \leq \sqrt{A/(d\mu)}$ for all $s \in \mathcal{S}, a \in \mathcal{A}, h \in [H]$, and $m \in [M]$ for any function $g : \mathcal{S} \rightarrow [0, 1]$.

The latent representations of the reward function and transition kernel are not unique under our assumption. Figure 3.1 visualizes the low rank structure Assumption 8 imposes on the transition kernel.

While Assumption 8 states that the source MDPs and target MDPs have latent low rank structure, the assumption does not guarantee that low rank representations are similar. Additionally, we allow target MDP to be more complex than a single source MDP (at most Tucker rank (S, S, dM) compared to Tucker rank (S, S, d) , respectively). Thus, to allow for transfer learning, we assume that for each step $h \in [H]$, the space spanned by the target latent factors is a subset of the space spanned by the latent factors from all the source MDPs.

Assumption 9. Suppose Assumption 8 holds. The target MDP latent factors G_h and source MDP latent factors $G_{m,h}$ satisfy for all $h \in [H]$, $\text{Span}(\{G_h(:, i)\}_{i \in [d']}) \subseteq \text{Span}(\{G_{m,h}(:, i)\}_{i \in [d], m \in [M]})$.

Under Assumption 9, it is possible that no single source MDP captures the

target MDP, but the union of all source MDP does. It is also possible that the source MDPs span a strictly larger subspace than the target MDP.

Thanks to the above assumptions, by estimating the latent features $\{G_{m,h}\}$ of the source MDPs, we can construct an approximation of the target features $\{G_h\}$ **without interacting** with the target MDP, which is essential to transfer learning. Assumption 8 only upper bounds the rank of $Q_{m,h}^*$ by d . However, to learn each source latent factor, one needs to obtain a good estimate of a rank d matrix with at least one large entry. Therefore, we assume that the optimal Q -functions in each source MDP are full rank with the largest entry lower bounded by a constant.

Assumption 10. *For each $m \in [M], h \in [H]$, the optimal Q -function of the m -th source MDP at step h satisfies $\text{rank}(Q_{m,h}^*) = d$ and $\|Q_{m,h}^*\|_\infty \geq C$ for some $C \in \mathbb{R}^+$.*

This assumption allows one to learn the near-optimal Q -functions from noisy samples of the source MDPs. Assumption 10 and its variants are common in the literature of noisy low-rank estimation [2, 14]. Without the above assumption (the rank of Q^* is strictly less than d), then one cannot learn every latent factor $G_{m,h}(:, i)$ since it may have a zero singular value, and thus using the approximate feature mapping would result in regret linear in T . Similarly, $\|Q^*\|_\infty \geq C$ ensures that some entries of Q^* are sufficiently large and prevents the noise from dominating the low rank signal that one learns in the source phase.

3.4 Transfer-ability

Our transfer-ability coefficient is motivated by the following observation. Under Assumption 9, we can construct each target latent factor G_h with a linear combination of the source latent factors $G_{m,h}$. The magnitudes of the coefficients correspond to the difficulty in estimating the target latent factor because the estimation error of $G_{m,h}$ is amplified by its coefficient. Thus, when these coefficients are too large, transfer learning is essentially impossible; one example is when the source latent factors are almost identical or parallel to each other while the target latent factor is orthogonal to the source latent factors (see Figure 3.2).

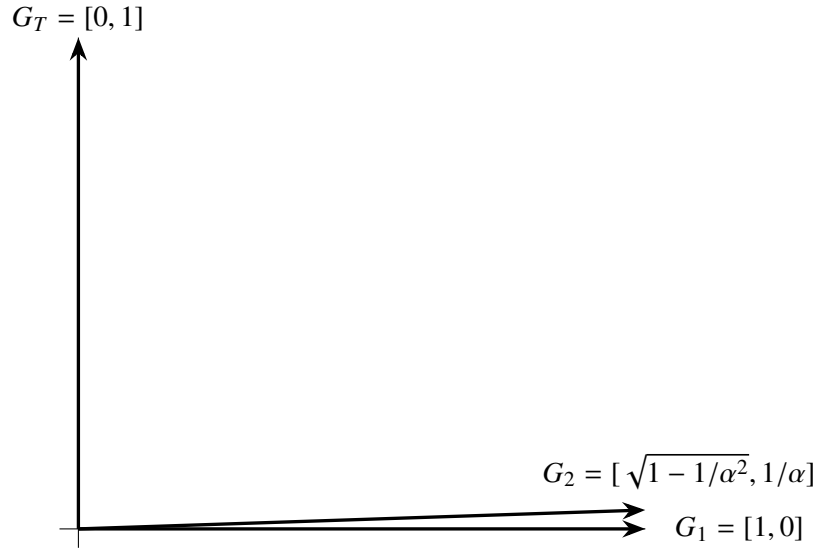


Figure 3.2: A hard transfer learning example where one estimates G_T with G_1 and G_2

However, these coefficients are not unique and are a function of the source and target latent factors. Furthermore, as the source MDPs latent factors of $Q_{m,h}^*$ and target latent factors $\{G_h\}$ are not unique, each choice of latent factor results in a different set of coefficients. Thus, we introduce $\mathcal{B}_h$ as the set containing all such coefficients given the subspaces from the target and source MDPs. Let

$\mathcal{G}_{S,h}$ be the set of all $\mathbb{R}^{S \times d}$ orthonormal latent factor matrices with columns that span the column space of $Q_{m,h}^*$ for all $m \in [M]$, and let $\mathcal{G}_{T,h}$ be the set of all $\mathbb{R}^{S \times d'}$ orthonormal latent factor matrices with columns that span the column space of $P_h(s')$ for all $s' \in \mathcal{S}$. Then, we define the set $\mathcal{B}_h(\mathcal{G}_{T,h}, \mathcal{G}_{S,h})$ to contain all such coefficients, i.e., $B_h \in \mathbb{R}^{d' \times d \times M}$ and $B_h \in \mathcal{B}_h(\mathcal{G}_{T,h}, \mathcal{G}_{S,h})$ if there exists $G_h \in \mathcal{G}_{T,h}$, $G_{m,h} \in \mathcal{G}_{S,h}$ such that for all $i \in [d']$, $h \in [H]$, $G_h(\cdot, i) = \sum_{j \in [d], m \in [M]} B_h(i, j, m) G_{m,h}(\cdot, j)$. Now, we define α , which precisely measures the challenge involved in transferring the latent representation.

Definition 4 (Transfer-ability Coefficient). *Given a transfer RL problem that satisfies Assumptions 8, 9, and 10, with the definition of $\mathcal{B}_h$ above, we define α as*

$$\alpha := \max_{h \in [H]} \min_{B \in \mathcal{B}_h(\mathcal{G}_{T,h}, \mathcal{G}_{S,h})} \max_{i \in [d'], j \in [d], m \in [M]} |B(i, j, m)|.$$

Note that the minimum is taken over all latent factor matrices. When there is only one source MDP, α is small, i.e., less than or equal to one, because the columns of G form an orthonormal basis of the space containing the target latent factor. In the best case, α is $1/(dM)$ (see Appendix B.4). On the other hand, when there are multiple source MDPs, α can be arbitrarily large as in Figure 3.2; in this case, the estimation error of G_1 and G_2 are amplified by α when estimating G_T using G_1 and G_2 . We remark that adding more source MDPs or increasing the rank of $Q_{m,h}^*$ will not increase α as one can use the same set of coefficients without the additional source MDP or larger subspace of $Q_{m,h}^*$. See Appendix B.4 for more discussion on α .

Our transfer-ability coefficient is similar to the quantity $\alpha_{\max}$ defined in [6]. While these terms both quantify the similarity between the source and target MDPs, in our setting, we allow for the source MDPs' state latent factors, $\{G_{m,h}\}$, to differ from target MDP's state latent factors while [6] assumes that all MDPs

share the same latent representation ϕ . Our setting is more general than the one in [6] even when assuming the same Tucker rank assumption.

3.4.1 Information Theoretic Lower Bound

To formalize the importance of α , we prove a lower bound that shows that a dependence on α in the sample complexity of the source phase is necessary to benefit from transfer learning. Before proving our main result, we first present an intermediate lower bound to provide intuition on how α measures the difficulty in transferring a learned representation.

Theorem 1. *There exist two transfer RL instances such that (i) they satisfy Assumptions 8 and 9, (ii) they cannot be distinguished without observing $\Omega(\alpha^2)$ samples in the source phase, and (3) they have target action latent features that are orthogonal to each other.*

Learning a latent representation G that is orthogonal to the true representation is no better than randomly guessing a representation G' in the source phase; using either G or G' in the target phase incurs regret linear in T . Thus, Theorem 1 states that without observing $\Omega(\alpha^2)$ samples in the source MDPs, one cannot learn a good policy in the target phase with G or G' , and one should disregard the information from the source MDPs and use a non-transfer learning RL algorithm to achieve regret bound that scales with $\sqrt{T}$. Therefore, $\Omega(\alpha^2)$ samples are required to benefit from transfer learning.

To prove Theorem 1, we construct two $(S, S, 1)$ transfer RL problems that satisfies Assumption 9 with similar source action latent representations $G_{m,h}^i \in \mathcal{G}$

but orthogonal target action latent representations G_T^i for $i \in \{1, 2\}$. To avoid learning the incorrect representation, one must identify the transfer RL problem by differentiating the Q^* s. By construction, the Q^* s for the two transfer RL problems differ by at most $1/\alpha$ entrywise, so one must observe $\Omega(\alpha^2)$ samples in the source problem from standard hypothesis testing lower bounds [8]. Observing less samples in the source problem results in an estimate orthogonal to the true target representation with constant probability.

See Appendix B.9.1 for the formal proof. One can easily prove a similar result in the (S, d, A) and (d, S, A) Tucker rank setting with similar constructions by switching the states and actions (see Appendices B.5 and B.6).

3.5 Algorithm

We now present our algorithm that achieves the optimal dependence on α . In the source phase, our algorithm can use any algorithm that learns ϵ -optimal Q functions on each source MDP. As the error bound on the estimated feature representation depends on the incoherence of $Q_{m,h}^*$, we use LR-EVI, which combines empirical value iteration with low rank matrix estimation [54]. After obtaining estimates $\bar{Q}_{m,h}$ of $Q_{m,h}^*$, our algorithm computes the singular value decompositions (SVDs) of each $\bar{Q}_{m,h}$ to construct the latent feature representation $\tilde{G}$ by scaling and concatenating the singular vectors. Then, in the target phase, our algorithm deploys a modification of LSVI-UCB [37] adapted to our Tucker rank setting called LSVI-UCB-(S, S, d) using the $\tilde{G}$. For ease of notation, we define $T_{k,h}^s = \{k' \in [k] | s_h^{k'} = s\}$, which is the set of episodes before k , in which one was at s at time step h .

Algorithm 1 Source Phase

Require: $\{N_h\}_{h \in [H]}$

- 1: **for** $m \in [M]$ **do**
 - 2: Run LR-EVI($\{N_h\}_{h \in [H]}$) on source MDP m to obtain $\bar{Q}_{m,h}$.
 - 3: Compute the singular value decomposition of $\bar{Q}_{m,h} = \hat{F}_{m,h} \hat{\Sigma}_{m,h} \hat{G}_{m,h}^\top$.
 - 4: **end for**
 - 5: Compute feature mapping $\hat{G} = \{\hat{G}_h\}_{h \in [H]}$ with $\hat{G}_h = \sqrt{\frac{A}{d\mu}} [\hat{G}_{1,h} \ \dots \ \hat{G}_{M,h}]$.
-

Algorithm 2 Target Phase: LSVI-UCB-(S, S, d)

Require: $\lambda, \beta_{k,h}^s, \hat{G}$

- 1: **for** $k \in [K]$ **do**
 - 2: Receive initial state s_1^k .
 - 3: **for** $h = H, \dots, 1$ **do**
 - 4: **for** $s \in S$ **do**
 - 5: /* Compute Gram matrix Λ_h^s for each state s */
 - 6: Set $\Lambda_h^s \leftarrow \sum_{t \in T_{k-1,h}^s} \hat{G}_h(a_h^t) \hat{G}_h(a_h^t)^\top + \lambda \mathbf{I}$.
 - 7: /* Estimate w_h^s via regularized least squares */
 - 8: Set $w_h^s \leftarrow (\Lambda_h^s)^{-1} \sum_{t \in T_{k-1,h}^s} \hat{G}_h(a_h^t) [r_h(s, a_h^t) + \max_{a' \in A} Q_{h+1}(s_{h+1}^t, a')]$.
 - 9: **end for**
 - 10: /* Estimate Q^* via the low rank structure with optimism */
 - 11: Set $Q_h(\cdot, \cdot) \leftarrow \min \left(H, \langle w_h^s, \hat{G}_h(\cdot) \rangle + \beta_{k,h}^s \sqrt{\hat{G}_h(\cdot)^\top (\Lambda_h^s)^{-1} \hat{G}_h(\cdot)} \right)$.
 - 12: **end for**
 - 13: **for** $h \in [H]$ **do**
 - 14: Take action $a_h^k \leftarrow \max_{a \in A} Q_h(s_h^k, a)$, and observe s_{h+1}^k .
 - 15: **end for**
 - 16: **end for**
-

LSVI-UCB-(S, S, d) uses the same mechanisms as LSVI-UCB, but we compute coefficients and Gram matrices for each action. Specifically, we update the weights w_h^s with the solution to the regularized least squares problem,

$$w_h^s \leftarrow \arg \min_{w \in \mathbb{R}^d} \sum_{t \in T_{k,h}^s} (r_h(s, a_h^t) + \max_{a' \in \mathcal{A}} Q_{h+1}(s_{h+1}^t, a') - w^\top \tilde{G}_h(a_h^t))^2 + \lambda \|w\|_2^2,$$

using the reward when s was observed and add an exploration bonus $\beta_{k,h}^s \sqrt{\hat{G}_h(a)^\top (\Lambda_h^s)^{-1} \hat{G}_h(a)}$ to our estimate of Q^* , which is common in linear MDP/bandits algorithms [37, 1]. Multiplying the latent factors by $\sqrt{A/(d\mu)}$ to

construct $\hat{G}$ ensures that the feature mapping and coefficients are of similar magnitude, which is crucial in adapting the mechanisms of LSVI-UCB to our setting.

While one may wonder why our modification of LSVI-UCB is necessary, Algorithm 2 improves the regret bound by a factor $\sqrt{S}$ compared to using any linear MDP algorithm off the shelf; reducing our setting to a linear MDP results in using a feature mapping with dimension dS . Thus, using any linear MDP algorithm results in a target regret bound of $\tilde{O}(\sqrt{d^2 S^2 H^2 T})$ [79]. However, our algorithm shaves off a factor of $\sqrt{S}$ as our algorithm computes S copies of $d \times d$ dimensional gram matrices in contrast to the $d^2 S^2$ dimensional gram matrices used in linear MDP algorithms.

3.6 Theoretical Results

In this section, we prove our main theoretical results that show with enough samples in the source problem, our algorithms admit regret bounds that are independent of S, A , or both S and A in the target problem. We first state our main result in the (S, S, d) Tucker rank setting.

Theorem 2. *Suppose Assumptions 8, 9, and 10 hold, and set $\delta \in (0, 1)$. Furthermore, assume that, for any $\epsilon \in (0, \sqrt{SH/T})$, $Q_{m,h} = r_{m,h} + P_{m,h}V_{m,h+1}$ has rank d and is μ -incoherent with condition number κ for all ϵ -optimal value functions V_{h+1} . Let $\lambda = 1$ and $\beta_{k,h}^s$ be a function of $d, H, |T_{k,h}^s|, |S|, M, T$. Then, for $T \geq \frac{S}{\alpha^2}$, using at most $\tilde{O}(d^4 \mu^5 \kappa^4 (1 + A/S) M^2 H^4 \alpha^2 T)$ samples in the source problems, our algorithm has regret $\tilde{O}(\sqrt{(dMH)^3 ST})$ with probability at least $1 - \delta$.*

Theorem 2 states that using transfer learning improves the performance on

the target problem by removing the regret bound's dependence on the action space. Thus, with enough samples from the source MDPs one can recover a regret bound in the target problem that matches the best regret bound for any algorithm in the (S, S, d) Tucker rank setting with **known** latent feature representation G concerning S and T . See Appendix B.9.1 for the proof of Theorem 2.

For sake of brevity, we present our results in our other Tucker rank settings in Table 3.1 (see Appendices B.5, B.6, and B.7 for the formal statements and proofs). In each of our Tucker rank settings, our algorithms use transfer learning to remove the dependence on S , A , or SA in the target regret bound, which matches the dependence on S and A of the best regret bounds of algorithms with **known** latent representation. While the source sample complexities of Theorems 2 and 8 may seem to not scale linearly in A and S , respectively, we require the horizon in the target phase to be large enough, i.e., $T\alpha^2(S/A + 1) \geq S + A$ and $T\alpha^2(1 + A/S) \geq S + A$, respectively, so that one observes at least $\Omega(S + A)$ samples in the source phase.

While our dependence on M in the target regret bound may seem sub optimal, our algorithm cannot distinguish which of the d dimensions (out of dM possible dimensions) match the ones in the target MDP without interacting with it. Thus, our algorithm must include all of them, which results in using at worst a dM -dimensional feature representation. However, in the case when the feature representations from source MDPs lie in subspaces that intersect, we can apply a singular value thresholding procedure to remove the unneeded dimensions used in the feature representation at the cost of an increased sample complexity (by a factor of A) in the source phase (see Appendix B.3).

Comparison with [6]: In comparison to Theorem 3.1 from [6], our result, Theorem 12, has similar dependence on A, d, H , and T while our dependence on α is optimal compared to their α^5 dependence⁴. While our dependence on M in the regret bound is worse, this is due to our transfer learning setting being more general; we only require that the space spanned by the target features be a subset of the space spanned by the source features instead of having the target features being a linear combination of the source features. Furthermore, our source sample complexity scales with the size of the state space instead of scaling logarithmically in the size of the given function class.

The proofs of Theorem 2, 8, 13, and 12 synthesize the theoretical guarantees of LR-EVI, singular vector perturbation bounds, and LSVI-UCB or our modification of LSVI-UCB. Given N samples from the source MDPs, we first bound the error between the singular subspaces up to a rotation by $\sqrt{d^3 M(S + A)/N}$ (suppressing the dependence on common matrix estimation terms) to construct our feature mapping. Thus, with enough samples in the source MDPs, LR-EVI returns Q functions with singular vectors that can be used to construct a sufficient feature representation; the additional regret incurred from using the approximate feature representation in LSVI-UCB or Algorithm 2 is dominated by the original regret of using the true representation.

3.6.1 Discussion of Optimality

In the (S, d, A) , (S, S, d) , (d, S, A) , and (d, d, d) Tucker rank settings, the source sample complexity's scalings concerning S, A, H , and T are reasonable when learn-

⁴Since $\bar{\alpha} \geq \alpha$, moving $\bar{\alpha}$ into the source sample complexity yields a sample complexity that scales with α^5 .

ing $\sqrt{A/T}$, $\sqrt{S/T}$, $\sqrt{1/T}$, and $\sqrt{1/T}$ -optimal Q functions, respectively, given the Q learning lower bounds in [54]. Similarly, the dependence on μ and κ are standard for RL algorithms incorporating matrix estimation.

In the (S, d, A) , (S, S, d) , and (d, S, A) Tucker rank settings, our α^2 dependence in the source sample complexity is optimal due to our lower bounds. In the (d, d, d) Tucker rank setting, our α^4 dependence is likely optimal when eliminating the dependence on S and A in the target problem; one must combine both sets of singular vectors together, $F : \mathcal{S} \rightarrow \mathbb{R}^{dM}$ and $G : \mathcal{A} \rightarrow \mathbb{R}^{dM}$, to create a feature mapping $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^{d^2 M^2}$, which causes the α^4 dependence in our upper bound. However, if α^4 is too large, one can use the algorithms from the (S, d, A) or (S, S, d) Tucker rank settings to benefit from transfer learning. Proving an α^4 lower bound is an interesting open question. Additionally, the dependence on d and M in this setting is worse as our low rank latent representation of each state-action pair lies in a $d^2 M^2$ dimensional space in contrast to a space with dimension dM .

The regret bounds of our algorithms used in the target phase, i.e., Algorithm 2, Algorithm 4, and LSVI-UCB [37], in each Tucker rank setting are optimal with respect to S , A , and T . Since each Tucker rank setting captures tabular MDPs, we can construct lower bounds with a reduction from the bound in [34], e.g., in the (S, S, d) setting, the regret is at least $\tilde{\Omega}(\sqrt{dSH^2T})$. However, our dependence on d and H are sub-optimal. The extra factor of $\sqrt{H}$ is likely due to our use of Hoeffding's inequality instead of tracking the variance of the estimates to use Bernstein's inequality. Our dependence on d is sub-optimal as LSVI-UCB is sub-optimal with respect to d ; for linear MDPs, LSVI-UCB++ [28] admits a regret bound of $\tilde{O}(\sqrt{d^2 H^2 T})$, which matches the lower bound from [79]. Thus,

to obtain the optimal dependence on d and H , we would need to adapt LSVI-UCB++ to our Tucker rank setting.

3.7 Empirical Results

We empirically verify our theory by comparing our algorithm with a standard regret minimization tabular RL algorithm, e.g., Q-Learning (UCB-H) [34], on a synthetic example in the (S, S, d) Tucker rank setting.

Experimental Setup: We construct two finite-horizon MDPs with the same state space, action space, and horizon. We consider $S = [50]$, $A = [500]$, and $H = 5$. To satisfy the (S, S, d) Tucker rank assumption, we construct MDPs with d or d' latent actions for the source or target MDP, respectively; both the reward functions and transition kernels are constructed as functions of the original state, next state, and latent action. For the reward functions, we sample an $S \times d$ or $S \times d'$ uniform matrix. For the transition kernel, we sample an $S \times d \times S$ or $S \times d' \times S$ uniform tensor and normalize the tensor along the last dimension with respect to the ℓ_1 norm to ensure that it is a valid transition kernel.

To construct the source latent representation ϕ , we sample a $A \times d$ uniform matrix and normalize the matrix along the second dimension to ensure $\|\phi_h(a)\|_1 = 1, \phi_h(a, i) \geq 0$ for all $a \in A, i \in [d]$. We set $d = 5$ in this example. To show the benefits of transfer RL, we construct the target latent representation ϕ'

with

$$\phi'_h(:, 1) = \phi_h(:, 1)/2 + \phi_h(:, 2)/2$$

$$\phi'_h(:, 2) = \phi_h(:, 3)/2 + \phi_h(:, 4)/2$$

$$\phi'_h(:, 3) = \phi_h(:, 5)$$

for all $h \in [H]$. With this construction, it follows that $\alpha \leq 1$. To illustrate the benefits of our algorithm, we first run LR-EVI on the source MDP with varying sample budgets, $N_{cap} \in \{1 \times 10^4, 5 \times 10^4, 2 \times 10^5, 5 \times 10^5, 1 \times 10^6\}$. We next construct approximate latent representations via SVD of $\hat{Q}_h$ and use them in LSVI-UCB-(S, S, d). We compare our algorithm with Q-Learning (UCB-H) [34]. Figure 3.3 shows the average regret of the last 10,000 episodes over $K = 625,000$ total episodes.

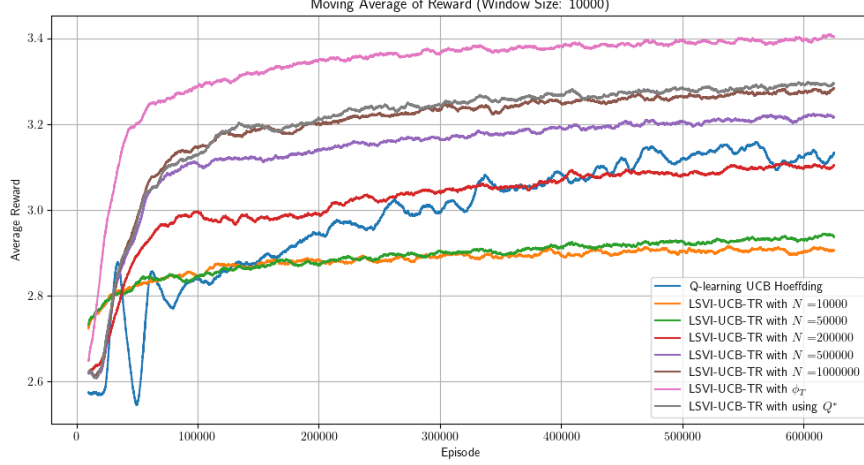


Figure 3.3: Average regret over 10,000 episdoes of our algorithm and Q-Learning (UCB-H)

Figure 3.3 shows that when the latent representation is sufficient ($N \in \{5 \times 10^5, 1 \times 10^6\}$), the transfer learning approach outperforms the standard RL algorithm when $K \leq 6.25 \times 10^5$. However, as K increases, Q-Learning (UCB-H)

outperforms our algorithm due to the error from the learned latent representation. Furthermore, when the sample budget is small ($N \in \{1 \times 10^4, 5 \times 10^4\}$), our algorithm performs poorly due to the poorly learned latent representations. Thus, one should only use the transfer learning approach if samples in the source phase are cheap enough.

APPENDIX A
APPENDICES FOR CHAPTER 2

A.1 Experimental Details for Oil Discovery Problem

In this section we discuss the rank of the cost function, the rank of the Q^* function, and details of how we tuned the hyperparameters of our algorithms for the experiments presented in Section 2.12.

A.1.1 Rank of $c(s, a)$

Recall that $c(s, a) = 0.01 \times \text{round}\left(\frac{|s-a|}{100}\right)$. Figure A.1 displays a heat map of the cost function while figure A.2 displays the singular values of the cost function.

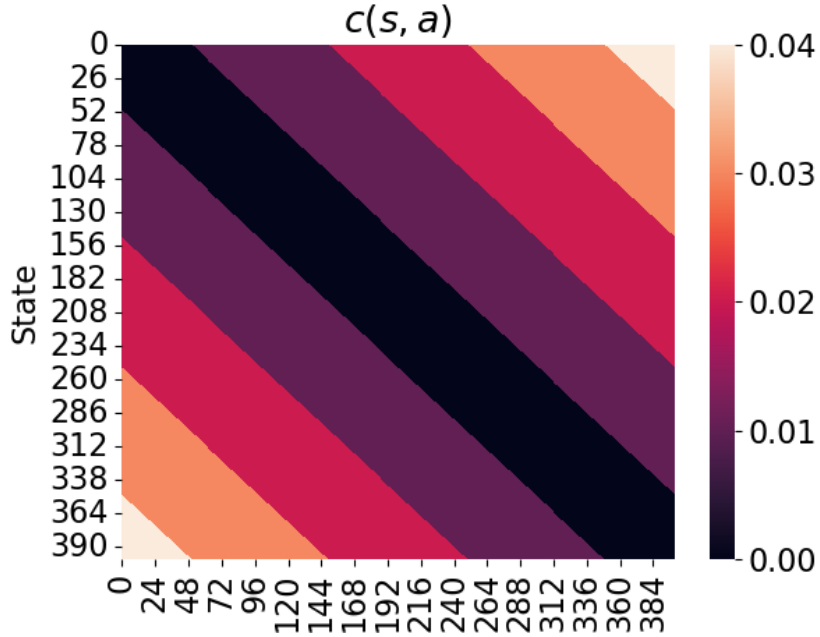


Figure A.1: Heat map of $c(s, a)$

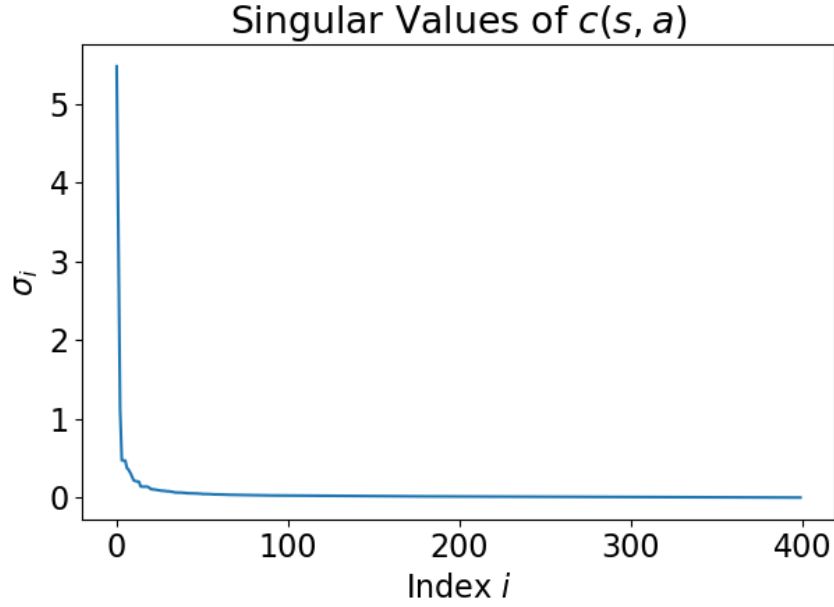


Figure A.2: Singular values (right) of $c(s, a)$

Even though each entry of $c(s, a)$ can only be one of five values, the rank of $c(s, a)$ is 400 as all of the singular values are greater than zero. However, Figure A.2 shows that the magnitude of the singular values decrease quickly. Furthermore, the stable rank of $c(s, a)$ is small $\|c(s, a)\|_F^2 / \|c(s, a)\|_*^2 = 1.46$. It follows that $c(s, a)$ is “approximately” low-rank.

A.1.2 Rank of Q^*

Figure A.3 displays a heat map of Q_1^* while figure A.4 displays a plot of Q_1^* singular values from largest to smallest.

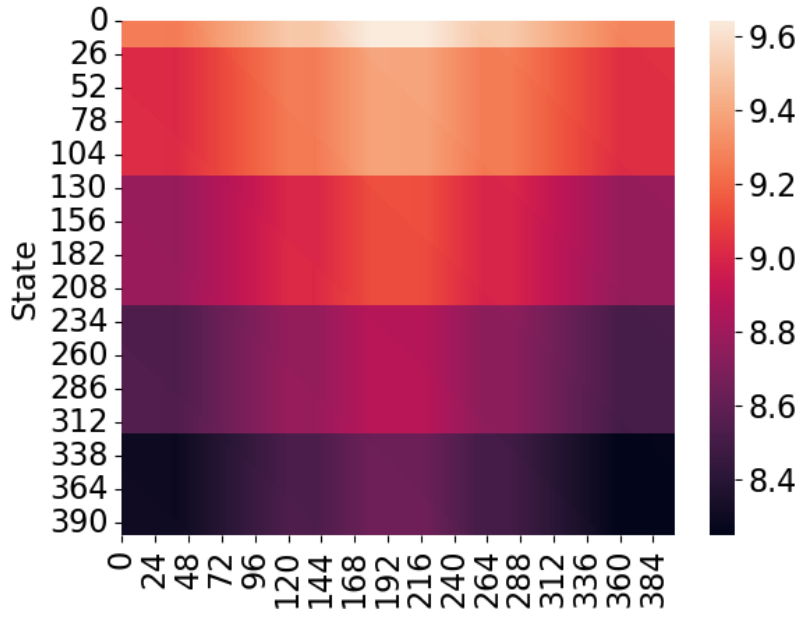


Figure A.3: Heat map of Q_1^* .

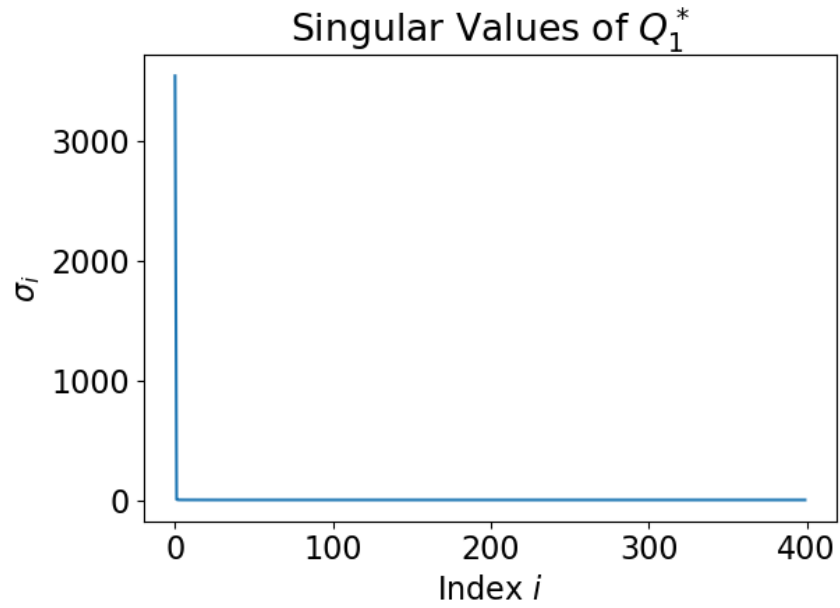


Figure A.4: Singular values (right) of Q_1^* .

While all of the singular values of Q_1^* are greater than zero ($\text{rank}(Q_1^*) = 400$),

the magnitude of the first singular value is significantly larger than all the other singular values; $\sigma_1 / \sum_{i=1}^{400} \sigma_i = 0.995$. Table A.1 displays the rank and stable rank of Q_h^* for $h \in [H]$.

	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$	$h = 7$	$h = 8$	$h = 9$	$h = 10$
Rank	400	400	400	400	400	400	400	400	400	400
Stable Rank	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.004	1.000

Table A.1: Rank and stable rank of Q_h^* for $h \in [H]$.

From the results in Table A.1, it's clear that despite $\text{rank}(Q_h^*) = 400$ for all $h \in [H]$, Q_h^* is approximately low rank for all $h \in [H]$ because all the stable ranks are close to one.

A.1.3 Hyperparameter Tuning

In this section, we discuss the hyperparameters of our algorithms and our methodology on how to choose their value.

Allocation Scheme $N_{s,a,h}$: In the proof of our theorems for LR-EVI and LR-MCPI, the sample allocations $N_{s,a,h}$ are chosen to ensure that at each time step h , the algorithm takes enough samples so that that $\|\bar{Q}_h - Q_h^*\|_\infty \leq \epsilon(H - h + 1)/H$. In practice, the algorithm does not have access to the optimal Q function and cannot use this condition as a criteria to choose $N_{s,a,h}$, so we instead empirically test a few different allocation schemes and choose the best. We choose $N_{s,a,h}$ to be uniform for all s, a , not distinguishing between (s, a) in the anchor submatrix or not.

To determine how to allocate samples across the ten time steps for each algo-

rithm, we run a set of experiments benchmarking the algorithm's performance on a set of different allocation schedules. For an allocation scheme $\tau = \{\tau^i\}_{h \in H}$, where $\tau^i \geq 0$, $\sum_{h \in [H]} \tau^i = 1$, and total sample budget of $\bar{N}$, we will allocate roughly $\tau^i \bar{N}$ samples to the estimation of $\bar{Q}_h$. Essentially τ specifies the proportion of samples that are allocated to the estimates at each step, where there is some rounding involved as the number of samples must be integral. For some sequence of nonnegative numbers $\{a_1^i, a_2^i, \dots, a_H^i\}$, the corresponding allocation scheme τ^i follows by simply normalizing according to $\tau_h^i = a_h^i / \sum_{h' \in [H]} a_{h'}^i$. The five different allocation schemes we consider are, $\tau^i = \{\tau_h^i\}_{h \in [H]}$, corresponding to

$$\begin{aligned} a_h^1 &= H - h + 1, & a_h^4 &= \lfloor (h + 1)/2 \rfloor, \\ a_h^2 &= (H - h + 1)^2, & a_h^5 &= 1. \\ a_h^3 &= (H - h + 1)^3 \end{aligned}$$

τ^5 is simply a constant allocation schedule, which evenly allocates samples across the steps. When there is an insufficient sample budget to implement other allocation schemes, we let the Empirical Value Iteration algorithms default to τ^5 , evenly allocating one-step samples across steps h .

τ^1 is a linearly decreasing allocation schedule. Note that as Monte Carlo Policy Iteration requires samples of length $(H - h + 1)$ trajectories to estimate Q_h , the allocation that would uniformly allocate trajectories across h for MCPI corresponds to τ^1 . When there is an insufficient sample budget to implement other allocation schemes, we let the Monte Carlo Policy Iteration algorithms default to τ^1 , evenly allocating trajectories across steps h .

τ^4 is also a linearly decreasing allocation schedule, but simply at a slower rate. τ^2 is a quadratically decreasing allocation schedule, which matches the allocation schedule chosen in our Theorem 7 for LR-EVI, as indicated by the

number of one-step samples N_h scaling as $(H - h + 1)^2$ in its dependence on h . τ^3 is a cubically decreasing allocation schedule, which matches the allocation schedule chosen in our Theorems 5 and 6 for LR-MCPI. In particular, the number of trajectories N_h scales as $(H - h + 1)^2$ in its dependence on h , but the samples used need to be multiplied by the trajectory length $(H - h + 1)$, resulting in a cubic relationship.

Table A.2 displays the average entrywise error of $\bar{Q}_1$ of all the algorithms over ten trials for each of these allocation schemes, where we set the sample budget $\bar{N} = 10^8$, $p_S, p_A = 0.025$, and $p_{SI} = 0.2$. “—” refers to the algorithm not having enough samples to take even one sample for each state-action pair at one time step according to the specified allocation scheme.

	τ^1	τ^2	τ^3	τ^4	τ^5
LR-EVI	0.0803	0.0736	0.0948	0.0828	0.0825
LR-MCPI	0.1742	0.1419	0.1532	0.1795	0.2031
LR-EVI + SI	0.5638	0.4512	1.0065	0.5541	0.5358
LR-MCPI + SI	0.6126	0.6535	0.6868	0.6285	0.6406
EVI	0.2927	1.5558	—	0.255	0.2264
MCPI	0.52	0.5437	—	0.5498	0.6127

Table A.2: Mean ℓ_∞ error of $\bar{Q}_1$ of LR-EVI, LR-MCPI, LR-EVI + SI, LR-MCPI + SI, EVI, and MCPI.

To calibrate these results, recall that entries in Q_1^* take values from roughly 8.3 to 9.6. While the performances are generally pretty similar, from the results in Table A.2 the best allocation schemes for the algorithms are τ^2 for LR-EVI, τ^2 for LR-MCPI, τ^2 for LR-EVI + SI, τ^1 for LR-MCPI + SI, τ^5 for EVI, and τ^1 for MCPI. We use these allocation schemes for the experiments in Section 2.12.

Note that for our algorithms to run, they require minimally one sample for the value iteration algorithms or one trajectory for the policy iteration algo-

rithms for each $(s, a) \in \Omega_h$. Hence, for smaller values of $\bar{N}$, e.g., $\bar{N} = 10^6$, there may be state-action pairs in Ω_h that do not receive even one sample/trajectory following the best allocation scheme chosen from the data in Table A.2. Hence, if that problem occurs, we default to allocation scheme τ^5 for the value iteration algorithms and we default to τ^1 for the policy iteration algorithm. These allocations uniformly spread the samples/trajectories to ensure that the algorithm still produces an estimate when $\bar{N}$ may be small.

Finally we describe the details of the rounding that we implement to ensure that $N_{s,a,h}$ are integral, yet are distributed as close as possible to the desired allocation schedule $\tau = \{\tau^h\}$ with the sample budget of $\bar{N}$. For Empirical Value Iteration algorithms, $N_{s,a,h}$ denotes one-step samples at (s, a) used to construct the estimate for Q_h . Thus we compute initial values by $N_{s,a,h} = \lfloor \tau^h \bar{N} / |\Omega_h| \rfloor$, where the floor function is applied as the number of samples must be integral. Subsequently, we compute the number of excess samples, given by $N_\Delta = \bar{N} - \sum_{h \in [H]} \sum_{(s,a) \in \Omega_h} N_{s,a,h}$. Then, recursing forwards through the horizon, we add one sample to each state action pair in Ω_h , i.e., $N_{s,a,h} = N_{s,a,h} + 1$, provided that there is sufficient samples $N_\Delta \geq |\Omega_h|$. Then we recompute the number of extra samples, i.e., $N_\Delta = N_\Delta - |\Omega_h|$ and repeat continuing at $h + 1$. With this rounding scheme, the final number of samples used by our algorithm will be within $[\bar{N} - D^2, \bar{N}]$, where $D^2 = 1.6 \times 10^5$.

For Monte Carlo Policy Iteration algorithms, $N_{s,a,h}$ denotes number of trajectories sampled starting from (s, a) , that are used to construct the estimate for Q_h . Thus we compute initial values by $N_{s,a,h} = \lfloor \tau^h \bar{N} / |\Omega_h| (H - h + 1) \rfloor$, where the floor function is applied as the number of trajectories must be integral. Subsequently, we compute the number of excess samples, given by

$N_\Delta = \bar{N} - \sum_{h \in [H]} \sum_{(s,a) \in \Omega_h} N_{s,a,h}(H - h + 1)$. Then, recursing forwards through the horizon, we add one trajectory to each state action pair in Ω_h , i.e., $N_{s,a,h} = N_{s,a,h} + 1$, provided that there is sufficient samples $N_\Delta \geq |\Omega_h|(H - h + 1)$. Then we recompute the number of extra samples, i.e., $N_\Delta = N_\Delta - |\Omega_h|(H - h + 1)$ and repeat continuing at $h + 1$. With this rounding scheme, the final number of samples used by our algorithm will be again within $[\bar{N} - D^2, \bar{N}]$.

Choosing $p = p_S = p_A$ for LR-EVI and LR-MCPI: While our theorems use knowledge of the rank and incoherence to choose p_S , p_A , and N_h , one cannot assume knowledge of these quantities in practice. However, many other matrix estimation methods face similar challenges. For example, in [76], to compute the Q function of the optimal policy, they need to choose p_{SI} , which depends on the rank of Q^* , for their algorithm, which combines value iteration and `Soft-Impute`. They show their algorithm is robust to the choice of p_{SI} by showing their algorithm performed similarly for multiple p_{SI} values. Similarly, we show LR-EVI and LR-MCPI are robust to the choice of $p = p_S = p_A$ (the only parameters in LR-EVI and LR-MCPI that depend on the rank and incoherence for a fixed allocation scheme and $\bar{N}$) in a similar manner. To show that LR-EVI and LR-MCPI are robust to $p = p_S = p_A$, we ran both LR-EVI and LR-MCPI with allocation scheme τ^2 and $\bar{N} = 10^8$ for each $p \in [0.025, 0.05, 0.075, 0.1]$, repeating each experiment 10 times. Since Q_h^* effectively has a rank of one, p should be minimally greater than or equal to $1/400 = 0.0025$; ideally even larger to ensure that with high probability there are a sufficient number of rows and columns sample. We set the smallest value of p to be 0.025, which results in an expected number of sampled rows/columns of 10 out of 400, which is already a fairly small number. Table A.3 shows the average ℓ_∞ error of $\bar{Q}_1$ at time step $h = 1$ for different values of p .

	$p = 0.025$	$p = 0.05$	$p = 0.075$	$p = 0.1$
LR-EVI	0.077	0.084	0.09	0.129
LR-MCPI	0.152	0.179	0.196	0.216

Table A.3: The mean $\|\bar{Q}_1\|_\infty$ error of LR-MCPI and LR-EVI for different values of p .

To calibrate these results, recall that entries in Q_1^* take values from roughly 8.3 to 9.6. The results show that for the different values of p , LR-EVI performs well and the errors are on the same order. Furthermore, the errors are less for smaller values of p . The same results hold for LR-MCPI for the different values of p . As a result, for the experiments in Section 2.12, we set $p = p_S = p_A = 0.025$. As our table suggests, the algorithm has decent performance for different values of p , so empirically one could choose p based on given computational and memory constraints. The tradeoff is that small values of p could reduce computation and memory usage, but it does assume the MDP satisfies the desired low rank conditions. By choosing p to be as larger, one may increase some robustness to the low rank conditions, as the guarantees would be able to tolerate MDPs with larger ranks.

Choosing p_{SI} : In contrast to LR-EVI and LR-MCPI, for larger values of $\bar{N}$, p_{SI} needs to be increased as the gain from decreasing the noise is not as beneficial as observing more samples. For LR-EVI + SI and LR-MCPI + SI, we determine the best value of p_{SI} for the four different values of $\bar{N} \in [10^6, 10^7, 10^8, 10^9]$ used in our experiments in Section 2.12. We test eight values of $p_{SI} \in [0.2, 0.3, \dots, 0.9]$ for the different $\bar{N}$.

For LR-EVI + SI to run, it minimally requires one sample for each $(s, a) \in \Omega_h$, which would mean at least $p_{SI} * 1.6 * 10^6$ total samples in expectation. Thus for

the lowest sample budget of $\bar{N} = 10^6$, we set $p_{SI} = 0.2$ to ensure that LR-EVI + SI has sufficient samples to run successfully for all ten trials.

Similarly, LR-MCPI + SI requires at least one trajectory for each $(s, a) \in \Omega_h$ to run, which would mean a total of $p_{SI} * 8.8 * 10^6$ one-step samples in expectation. Thus for the lowest sample budget of $\bar{N} = 10^6$, we set $p_{SI} = 0.075$ to ensure that LR-MCPI + SI has sufficient samples to run successfully for all ten trials.

For the larger values of $\bar{N} \in [10^7, 10^8, 10^9]$, we test eight values of $p_{SI} \in [0.2, 0.3, \dots, 0.9]$. Figures A.5 and A.6 shows the average ℓ_∞ error of $\bar{Q}_1$ for LR-EVI + SI and LR-MCPI + SI for the different values of p_{SI} and $\bar{N}$ for LR-EVI + SI using allocation scheme τ^2 and LR-MCPI + SI using allocation scheme τ^1 , where each experiment is repeated ten times.

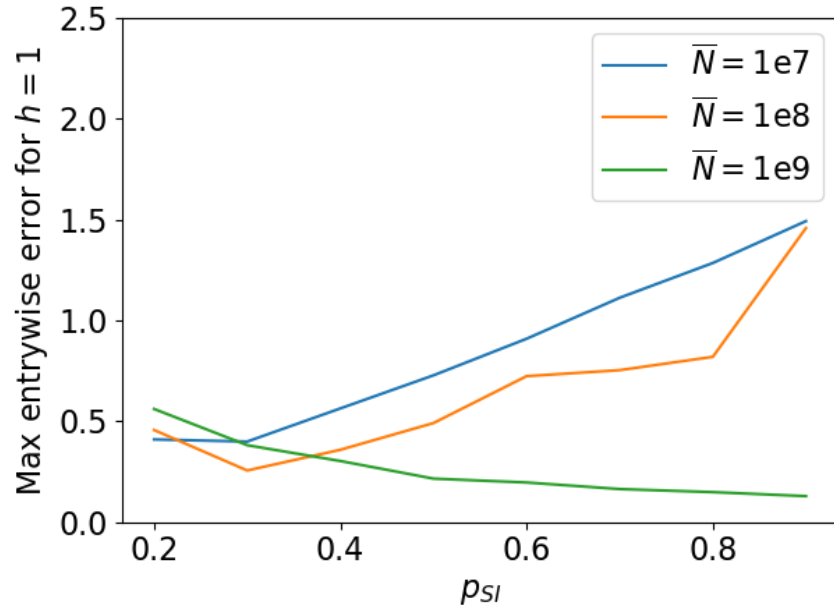


Figure A.5: Max entrywise error of $\bar{Q}_1$ vs. p_{SI} for four different values of $\bar{N}$ for LR-EVI + SI.

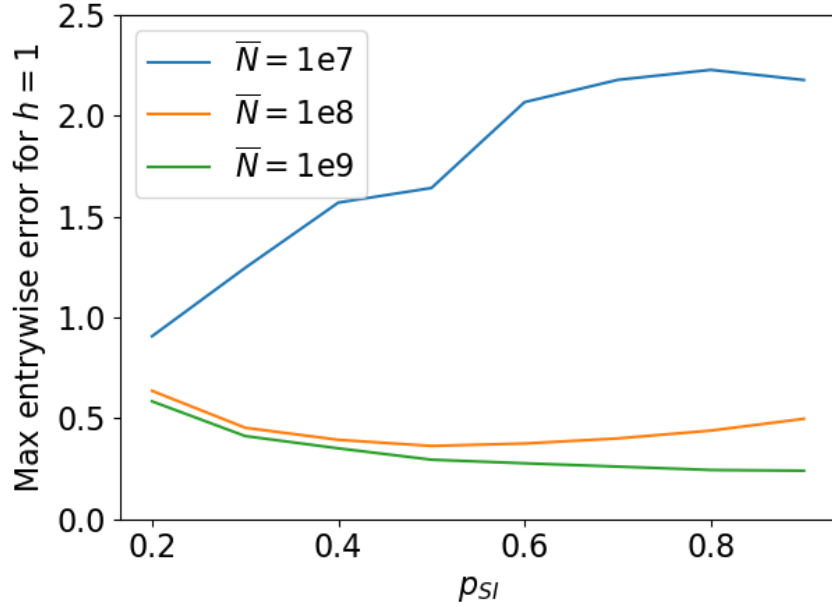


Figure A.6: Max entrywise error of $\bar{Q}_1$ vs. p_{SI} for four different values of $\bar{N}$ for LR-MCPI + SI.

Figures A.5 and A.6 show that when the sample budget is smaller, i.e. $\bar{N} = 10^7$, smaller values of p perform better; this is expected as there is insufficient sample budget so that increasing p means the number of samples or trajectories allocated to each $(s, a) \in \Omega_h$ will be small, resulting in large noise. For large sample budget, i.e. $\bar{N} = 10^9$, the performance is not very sensitive to the choice of p_{SI} , though the larger values of p_{SI} do perform better. This is also expected as there is sufficient samples to both sample more entries while still having $N_{s,a,h}$ large enough that the noise is well controlled. For $\bar{N} = 10^8$ the performance with respect to p_{SI} is quite different in these two plots, and it may be due in part to the different allocation schemes. Allocation scheme τ^2 significantly skews the proportion of samples to the earlier time steps compared to τ^1 . Hence for $\bar{N} = 10^8$, with scheme τ^2 , increasing p_{SI} results in the error LR-EVI + SI growing perhaps due to high noise in the later time steps. In contrast, with scheme τ^1 ,

for $\bar{N} = 10^8$, the error of LR-MCPI + SI does not increase in p_{SI} .

Table A.4 displays the value of p_{SI} we use for our experiments in Section 2.12. As discussed before, the value of p_{SI} is chosen for $\bar{N} = 10^6$ simply to ensure that the observation set is small enough such that the algorithms can produce some estimate for the given sample budget. For $\bar{N} \in [10^7, 10^8, 10^9]$, p_{SI} is chosen according to the value that minimized the error in the results displayed in Figures A.5 and A.6.

	$\bar{N} = 10^6$	$\bar{N} = 10^7$	$\bar{N} = 10^8$	$\bar{N} = 10^9$
LR-EVI + SI	0.2	0.3	0.3	0.9
LR-MCPI + SI	0.075	0.2	0.5	0.9

Table A.4: Values of p_{SI} for each $\bar{N}$ in the experiments in Section 2.12.

A.2 Additional Experiments for Double Integrator Problem

We empirically evaluate the benefit of including a low-rank subroutine in tabular RL algorithms on the discretized finite-horizon version of the Double Integrator problem, a stochastic control problem seen in [76, 55].

Experimental Setup: We formulate the Double Integrator problem as finite-horizon tabular MDP with state space $S = \{(x, \dot{x})\}$ for $x \in \{-2, -1.9, \dots, 1.9\}$, $\dot{x} \in \{-1, -0.9, \dots, 0.9\}$, action space $A = \{-0.5, -0.499, \dots, 0.5\}$, and $H = 5$. With this setup, the size of the state space is $|S| = 40 \times 20 = 800$, and the size of the action space is $|A| = 1000$. The learner’s goal is to control a unit brick on a frictionless surface and guide it to the origin, state $(0, 0)$. x refers to the brick’s position, and $\dot{x}$ denotes the brick’s velocity. At each step, the learner is given a noisy reward

that penalizes them for the brick's current position,

$$r_h((x, \dot{x}), a) = -\frac{x^2 + \dot{x}^2}{2} + \mathcal{N}(0, 1),$$

for all $h \in [H]$, and $\mathcal{N}(0, 1)$ is a standard normal random variable. The learner chooses an action a to change the velocity of the brick. The dynamics of the system for a given state-action pair $((x, \dot{x}), a)$ for all $h \in [H]$ are

$$x' := \min(\max(x + \dot{x}, -2), 1.9), \quad \dot{x}' := \min(\max(\lfloor \dot{x} + a \rfloor, -1), 0.9),$$

where $\lfloor \dot{x} \rfloor$ rounds $\dot{x}$ down to the nearest tenth. Since the reward function does not depend on the action, the rank of the reward function is one. Due to the deterministic dynamics, for a given next state $(x', \dot{x}')$, the current state $(x, \dot{x})$ must minimally satisfy $x' = x + \dot{x}$. Thus, there are at most twenty $(x, \dot{x})$ pairs that satisfy $x' = x + \dot{x}$ (the velocity can only take on twenty different values), so there are at most twenty non-zero entries in $P((x', \dot{x}') | \cdot, \cdot)$. Therefore, the Tucker rank of $P((x', \dot{x}') | (x, \dot{x}), a)$ is upperbounded by $(|S|, 20, |A|)$. Hence, this MDP satisfies Assumption 4. Figure A.7 displays a heat map of Q_1^* , and figure A.8 displays a plot of Q_1^* singular values from largest to smallest.

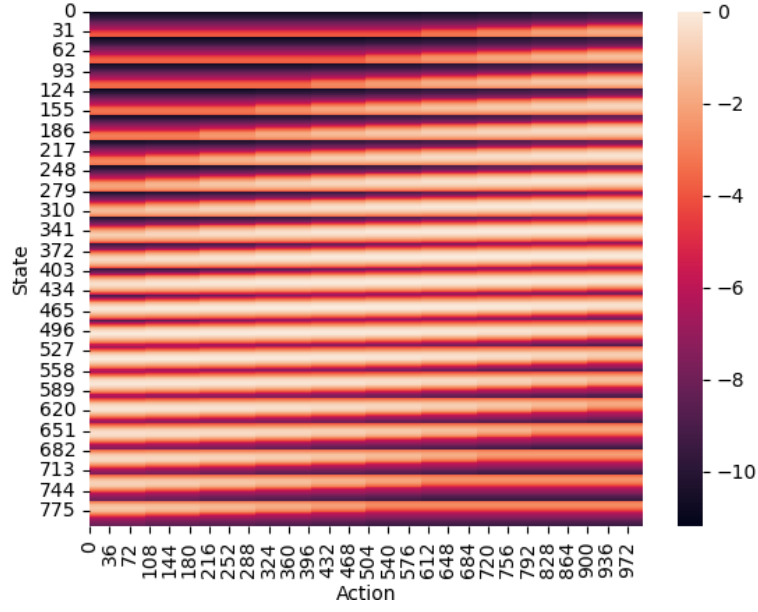


Figure A.7: Heat map of Q_1^*

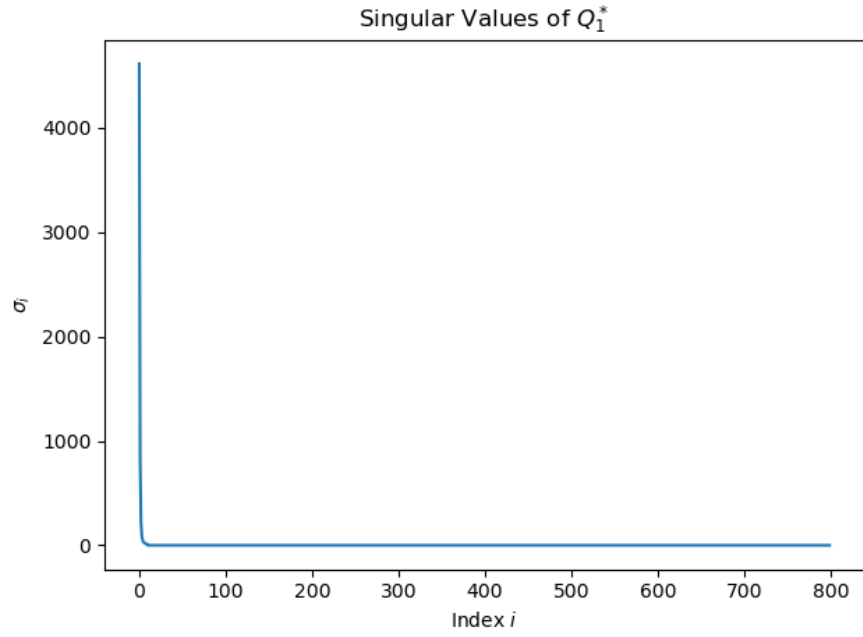


Figure A.8: Singular values of Q_1^*

While the transition kernel has Tucker rank upper bounded by $(|S|, 20, |A|)$,

the rank of Q_h^* is ten for $h \in [4]$ while the rank of $Q_H^* = r_H$ is one. Table A.5 displays the rank and stable rank of Q_h^* for $h \in [H]$.

	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$
Rank	10	10	10	10	1
Stable Rank	1.04	1.02	1.01	1.00	1.00

Table A.5: Rank and stable rank of Q_h^* for $h \in [H]$.

From the results in Table A.5, it's clear that the rank of Q_h^* is much smaller than $|S|$ or $|A|$.

Algorithms: We compare the same algorithms used in the oil discovery experiments. Since $|S| \neq |A|$, we allow for p_S and p_A to be different. Instead of using `Soft-Impute` from the `fancyimpute` package, we use the implementation from [23] because it yielded better results and shorter runtimes.

A.2.1 Hyperparameter Tuning

In this section, we discuss the results of tuning the allocation schemes, p_S, p_A , and p_{SI} for our different algorithms.

Allocation Schemes: To determine how to divide samples across the five time steps, we test our algorithms on the five different allocation schemes introduced in Appendix A.1.3. Recall that the allocation scheme τ^i is $\tau_h^i = a_h^i / \sum_{h' \in [H]} a_{h'}^i$ for $[a_1, \dots, a_H]$. The five different allocation schemes we consider are, $\tau^i = \{\tau_h^i\}_{h \in [H]}$,

corresponding to

$$\begin{aligned} a_h^1 &= H - h + 1, & a_h^4 &= \lfloor (h + 1)/2 \rfloor, \\ a_h^2 &= (H - h + 1)^2, & a_h^5 &= 1. \\ a_h^3 &= (H - h + 1)^3 \end{aligned}$$

With our implementation, roughly $\tau_h^i \bar{N}$ samples are allocated to estimating Q_h^* ($\bar{N}$ is the sample budget). Table A.6 displays the mean entrywise error of $\bar{Q}_1$ of all the algorithms over five trials for each allocation schemes. We set the sample budget $\bar{N} = 10^8$, $p_S = 0.1$, $p_A = 0.08$, and $p_{SI} = 0.4$.

	τ^1	τ^2	τ^3	τ^4	τ^5
LR-EVI	0.761	1.20	4.30	2.15	0.507
LR-MCPI	0.550	0.416	0.633	2.07	1.16
LR-EVI+SI	0.459	0.469	1.343	0.512	0.456
LR-MCPI+SI	0.394	0.405	0.388	0.436	0.441
EVI	1.044	2.921	1.343	3.074	1.469
MCPI	0.642	0.615	1.09	2.225	2.526

Table A.6: Mean ℓ_∞ error of $\bar{Q}_1$ of LR-EVI, LR-MCPI, LR-EVI+SI, LR-MCPI+SI, EVI, and MCPI.

While the errors are roughly similar for many of the allocation schemes for each algorithm, we choose the allocation scheme that corresponds to the lowest error. Hence, we use allocation scheme τ^5 for LR-EVI, τ^2 for LR-MCPI, τ^5 for LR-EVI+SI, τ^3 for LR-MCPI+SI, τ^1 for EVI, and τ^2 for MCPI.

Choosing p_S and p_A : Similar to `Soft-Impute`, varying p_s and p_a as a function of the total number of samples improves the performance of LR-EVI and LR-MPCI. When the sample budget is small ($\bar{N} = 10^7$), one should set p_s and p_a to be smaller, which increases the bias from the matrix estimation method but decreases the noise on the empirical estimates. However, when the sample budget is increased ($\bar{N} = 10^9$), one should increase p_s and p_a to reduce

the bias of the matrix estimation method as the estimation error on $\hat{Q}$ is already very small. Thus, for $\bar{N} \in [10^7, 10^8, 10^9]$, we try the following $(p_s, p_a) \in [(0.025, 0.02), (0.05, 0.04), (0.1, 0.08), (0.2, 0.16)]$ over five trials. Table A.7 displays the entrywise error of $\bar{Q}_1$ obtained from running LR-EVI with allocation scheme τ^5 . Table A.8 displays the entrywise error of $\bar{Q}_1$ obtained from running LR-MCPI with allocation scheme τ^2 .

	$\bar{N} = 10^7$	$\bar{N} = 10^8$	$\bar{N} = 10^9$
$(p_s, p_a) = (0.025, 0.02)$	1.72	1.09	0.650
$(p_s, p_a) = (0.05, 0.04)$	2.74	0.236	0.190
$(p_s, p_a) = (0.1, 0.08)$	5.43	0.488	0.0175
$(p_s, p_a) = (0.2, 0.16)$	10.9	1.05	0.0365

Table A.7: Mean ℓ_∞ error of $\bar{Q}_1$ of LR-EVI for varying values of (p_s, p_a) .

	$\bar{N} = 10^7$	$\bar{N} = 10^8$	$\bar{N} = 10^9$
$(p_s, p_a) = (0.025, 0.02)$	2.26	1.50	1.34
$(p_s, p_a) = (0.05, 0.04)$	4.70	0.293	0.143
$(p_s, p_a) = (0.1, 0.08)$	4.54	0.288	0.0452
$(p_s, p_a) = (0.2, 0.16)$	10.84	0.747	0.108

Table A.8: Mean ℓ_∞ error of $\bar{Q}_1$ of LR-MCPI for varying values of (p_s, p_a) .

Hence, for LR-EVI, we use $(p_s, p_a) = (0.025, 0.02)$ for $\bar{N} = 10^7$, $(p_s, p_a) = (0.05, 0.04)$ for $\bar{N} = 10^8$, and $(p_s, p_a) = (0.1, 0.08)$ for $\bar{N} = 10^9$. For LR-MCPI, we use $(p_s, p_a) = (0.025, 0.02)$ for $\bar{N} = 10^7$, $(p_s, p_a) = (0.1, 0.08)$ for $\bar{N} = 10^8$, and $(p_s, p_a) = (0.1, 0.08)$ for $\bar{N} = 10^9$.

Choosing p_{SI} : For LR-EVI + SI and LR-MCPI + SI, we test different values of p_{SI} for $\bar{N} \in [10^7, 10^8, 10^9]$ to determine what to set p_{SI} to in our final experiments in Section 2.12. We test five values of $p_{SI} \in [0.1, 0.5, \dots, 0.9]$ for the different $\bar{N}$. Figures A.9 and A.10 show the average ℓ_∞ error of $\bar{Q}_1$ for LR-EVI + SI and LR-MCPI + SI for the different values of p_{SI} and $\bar{N}$ for LR-EVI + SI using al-

location scheme τ^5 and LR-MCPI + SI using allocation scheme τ^3 , where each experiment is repeated five times.

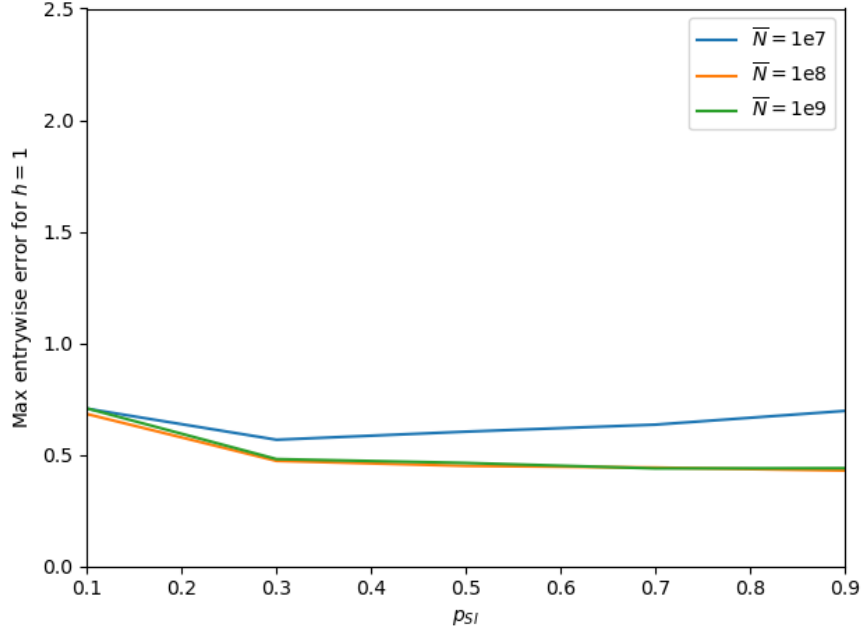


Figure A.9: Mean ℓ_∞ error of $\bar{Q}_1$ of LR-EVI+SI for various values of p_{SI} and $\bar{N}$

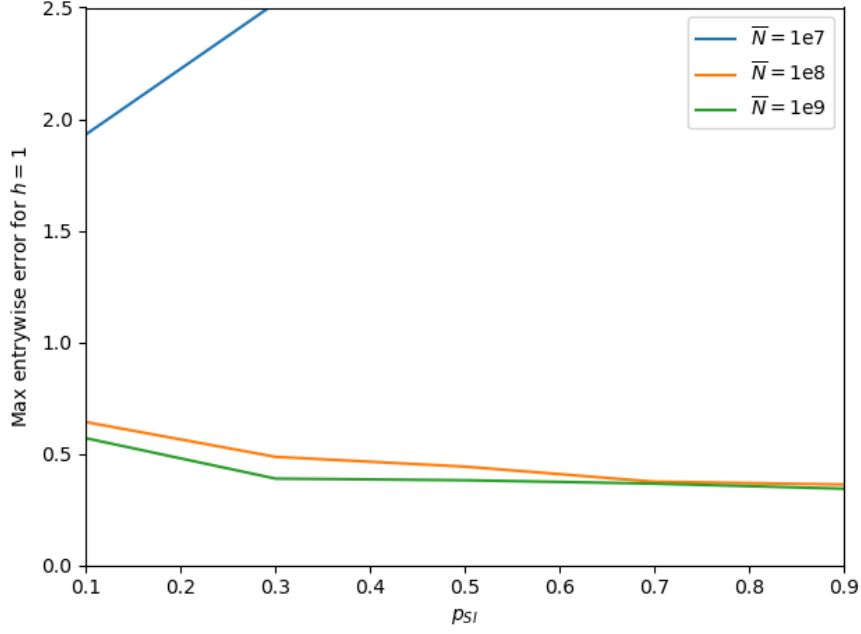


Figure A.10: Mean ℓ_∞ error of $\bar{Q}_1$ of LR-MCPI+SI for various values of p_{SI} and $\bar{N}$

From these results, we choose the p_{SI} value that corresponds to the lowest error for our final experiments. Note that for $\bar{N} = 10^7$, the error is strictly increasing as p_{SI} increases for LR-MCPI+SI. Hence, for LR-EVI+SI, we use $p_{SI} = 0.3$ for $\bar{N} = 10^7$, $p_{SI} = 0.9$ for $\bar{N} = 10^8$, and $p_{SI} = 0.9$ for $\bar{N} = 10^9$. For LR-MPCI+SI, we use $p_{SI} = 0.1$ for $\bar{N} = 10^7$, $p_{SI} = 0.9$ for $\bar{N} = 10^8$, and $p_{SI} = 0.9$ for $\bar{N} = 10^9$.

Results: For each value of the sample budget $\bar{N} \in [10^7, 10^8, 10^9]$, we run each of the six algorithms ten times, with the hyperparameters specified above, and compute the average ℓ_∞ error of $\bar{Q}_1$. Figures A.11 and A.12 display the mean entrywise error of $\bar{Q}_1$ over ten simulations with the error bars corresponding to the standard deviation. For vanilla MCPI to produce an estimate of Q_1^* , it requires at least $\sum_{h=1}^H SAh = 1.2 \times 10^7$ one-step samples. Hence, there is no error bar for MCPI with $\bar{N} = 10^7$.

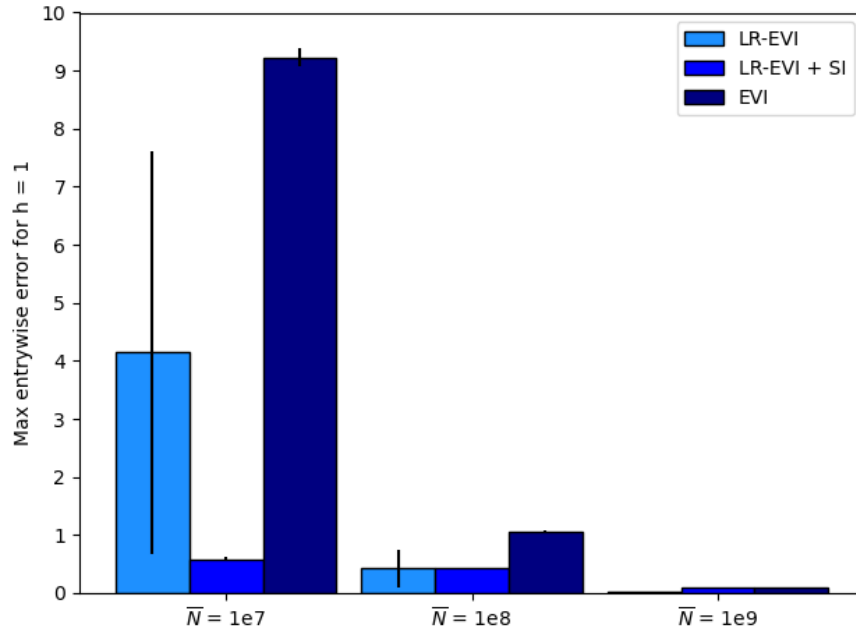


Figure A.11: Max entrywise error of $\bar{Q}_1$ vs. sample budget for LR-EVI, LR-EVI + Soft Impute, empirical value iteration at $h = 1$. Note that the optimal Q_1^* function ranges in value from roughly -11.2 to 0 , such that 1.12 error would be roughly 10% error.

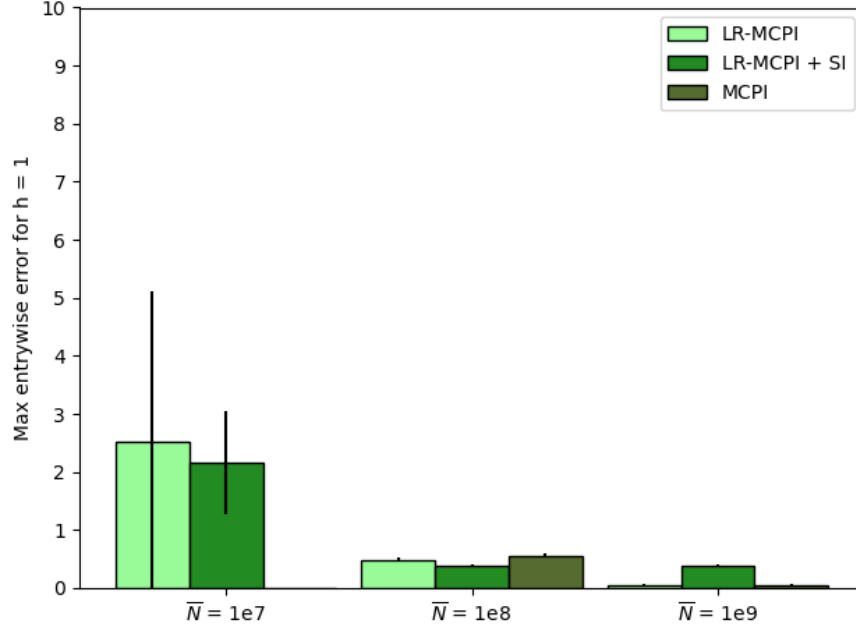


Figure A.12: Max entrywise error of $\bar{Q}_1$ vs. sample budget for LR-MCPI, LR-MCPI + `Soft-Impute`, Monte Carlo policy iteration at $h = 1$. Note that the optimal Q_1^* function ranges in value from roughly -11.2 to 0 , such that 1.12 error would be roughly 10% error.

Similarly to the results from the oil discovery problem, Figure A.11 shows that even when there are not enough samples to MCPI, LR-MCPI produces a reasonable estimate. Furthermore, the low-rank algorithms perform better than the tabular versions, EVI and MCPI, when the sample budget is small, i.e., $\bar{N} = 10^7$ or $\bar{N} = 10^8$. When the sample budget is large, i.e., $\bar{N} = 10^9$, the low-rank methods perform similarly to EVI and MCPI, except for LR-MCPI+SI. The relatively large error from LR-MCPI+SI for $\bar{N} = 10^9$ suggests that the matrix estimation methods are sensitive to the choice of hyperparameters, so in practice, one should carefully tune these given their computational limits, e.g., storage and runtime constraints. In contrast to the oil discovery simulations, the policy iteration algorithms achieve a similar error to the value iteration algorithms with the same sample budget.

A.3 Proofs for Chapter 2

A.3.1 Proof of Lemma 1

Proof of Lemma 1. Consider the MDP defined in Section 2.4. Let $\pi_h(1) = \pi_h(2) = 2$ for all $h \in \{2, \dots, H-1\}$. We prove that

$$Q_h^{\pi, \theta} = \begin{pmatrix} \frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} + 2^{H-h}\theta & 1 + 2^{H-h+1}\theta \end{pmatrix}, \quad V_h^{\pi, \theta} = \begin{pmatrix} \frac{1}{2} \\ 1 + 2^{H-h+1}\theta \end{pmatrix}$$

with backwards induction on h . Since $V_H^{\pi, \theta} = \begin{pmatrix} \frac{1}{2} \\ 1 + 2\theta \end{pmatrix}$, the base case occurs at step $H-1$. Applying the exact Bellman operator, it follows that

$$Q_{H-1}^{\pi, \theta} = \begin{pmatrix} \frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} + 2\theta & 1 + 2^2\theta \end{pmatrix}$$

and

$$V_{H-1}^{\pi, \theta} = \begin{pmatrix} \frac{1}{2} \\ 1 + 2^2\theta \end{pmatrix}$$

because for both values of θ , $1 + 4\theta > 0$. Next, assume that the induction hypothesis holds, that is for some $t \in \{2, \dots, H-1\}$,

$$Q_t^{\pi, \theta} = \begin{pmatrix} \frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} + 2^{H-t}\theta & 1 + 2^{H-t+1}\theta \end{pmatrix}, \quad V_t^{\pi, \theta} = \begin{pmatrix} \frac{1}{2} \\ 1 + 2^{H-t+1}\theta \end{pmatrix}.$$

Applying the exact Bellman operator, it follows that

$$\begin{aligned} Q_{t-1}^{\pi, \theta} &= r_{t-1} + P_{t-1} V_t^{\pi, \theta} \\ &= \begin{pmatrix} -\frac{1}{4} & 0 \\ -\frac{1}{2} & 2^{H-t+1}\theta \end{pmatrix} + \begin{pmatrix} V_t^{\pi, \theta}(1) & V_t^{\pi, \theta}(1) \\ V_t^{\pi, \theta}(2) & (2) \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} + 2^{H-t+1}\theta & 1 + 2^{H-t+2}\theta \end{pmatrix}. \end{aligned}$$

Because $2^H|\theta| = 3/4$, $1 + 2^H\theta > 0$, which implies that $Q_{t-1}^{\pi,\theta}(2, 2) > Q_{t-1}^{\pi,\theta}(2, 1)$. Therefore,

$$V_{t-1}^{\pi,\theta} = \begin{pmatrix} \frac{1}{2} \\ 1 + 2^{H-t+2}\theta \end{pmatrix}$$

and the induction hypothesis holds. Finally, since one stays in the same state at all steps after $h = 1$ by construction, $\pi_h(1) = \pi_h(2) = 2$ for all $h \in \{2, \dots, H-1\}$ is the unique optimal policy because $2^H|\theta| = 3/4$, which implies that $Q_h^{\pi,\theta}(2, 2) = 2Q_h^{\pi,\theta}(2, 1) > 0$. $\square$

A.3.2 Proof of Proposition 3

Proposition 3 states that if the reward function and transition kernel are low rank, then for any value function estimate $\hat{V}_{h+1}$, $r_h + [P_h \hat{V}_{h+1}]$ has rank upper bounded by d .

Proof of Proposition 3. Let MDP $M = (S, A, P, r, H)$ satisfy Assumption 4 (specifically, P_h has Tucker rank $(|S|, |S|, d)$). Hence, for any value function estimate $\hat{V}_{h+1}$,

$$\begin{aligned} r_h(s, a) + P_h \hat{V}_{h+1} &= \sum_{i=1}^d W^{(h)}(s, i) V^{(h)}(a, i) + \sum_{s' \in S} \hat{V}_{h+1}(s') P_h(s' | s, a) \\ &= \sum_{i=1}^d W^{(h)}(s, i) V^{(h)}(a, i) + \sum_{s' \in S} \hat{V}_{h+1}(s') \sum_{i=1}^d U^{(h)}(s', s, i) V^{(h)}(a, i) \\ &= \sum_{i=1}^d V^{(h)}(a, i) \left(W^{(h)}(s, i) + \sum_{s' \in S} \hat{V}_{h+1}(s') U^{(h)}(s', s, i) \right). \end{aligned}$$

Since $W^{(h)}(:, :) + \sum_{s' \in S} \hat{V}_{h+1}(s') U^{(h)}(s', :, :)$ is an $|S| \times d$ matrix, $r_h(s, a) + P_h \hat{V}_{h+1}$ has rank at most d . The same result holds when P_h has Tucker rank $(|S|, d, |A|)$ from a similar argument. $\square$

A.3.3 Proof of Lemma 8 (Random Sampling of Anchor States and Actions)

As stated in Lemma 8, our sampling method is as follows: we sample states and actions using the Bernoulli model. Let $\tilde{U} \in \mathbb{R}^{|S| \times d}$, $\tilde{V} \in \mathbb{R}^{|A| \times d}$ such that

$$\tilde{U}_i = \begin{cases} U_i & \text{with probability } p_1, \\ 0 & \text{otherwise} \end{cases}, \quad \tilde{V}_i = \begin{cases} V_i & \text{with probability } p_2, \\ 0 & \text{otherwise} \end{cases}$$

Let $\tilde{Q}_h := \tilde{U}\Sigma\tilde{V}^\top \in \mathbb{R}^{|S| \times |A|}$. The sampled anchor states and actions are the states corresponding to the non-zero rows and columns, respectively. We remark that the Bernoulli model is chosen for convenience and similar results hold if we sample with replacement. To prove Lemma 8, we present two intermediate lemmas, the first shows $p_1^{-1/2}\tilde{U}$ and $p_2^{-1/2}\tilde{V}$ have near orthonormal columns, which implies that $\tilde{U}$ and $\tilde{V}$ have full column rank, with high probability.

Lemma 20. *Let $Q_h, U, \tilde{U}, \Sigma, V$, and $\tilde{V}$ be defined as above. Let Q_h be μ -incoherent. Then, with probability at least $1 - 4(|S| \wedge |A|)^{-10}$, we have*

$$\begin{aligned} \|p_1^{-1}\tilde{U}^\top\tilde{U} - I_{d \times d}\|_{op} &\leq \sqrt{\frac{40\mu d \log(|S|)}{p_1|S|}} + \frac{40\mu d \log(|S|)}{p_1|S|} \\ \|p_2^{-1}\tilde{V}^\top\tilde{V} - I_{d \times d}\|_{op} &\leq \sqrt{\frac{40\mu d \log(|A|)}{p_2|A|}} + \frac{40\mu d \log(|A|)}{p_2|A|}. \end{aligned}$$

Proof of Lemma 20. For each $i \in [|S|]$, let $Z^{(i)} \in \mathbb{R}^{|S| \times d}$ be the matrix obtained from U by zeroing out all but the i -th row. Let $\delta_1, \dots, \delta_{|S|}$ be i.i.d. Bernoulli(p_1) random variables. We can express

$$U = \sum_{i \in [|S|]} Z^{(i)} \text{ and } \tilde{U} = \sum_{i \in [|S|]} \delta_i Z^{(i)}$$

Note that

$$\tilde{U}^\top \tilde{U} = \sum_{i \in [|S|]} \sum_{j \in [|S|]} \delta_i \delta_j Z^{(i)\top} Z^{(j)} \quad (\text{A.1})$$

$$= \sum_{i \in [|S|]} \delta_i^2 Z^{(i)\top} Z^{(i)} \quad (\text{A.2})$$

by construction of $Z^{(i)}$ and $Z^{(j)}$. Hence,

$$\begin{aligned} \mathbb{E}[\tilde{U}^\top \tilde{U}] &= p_1 \sum_{i \in [|S|]} Z^{(i)\top} Z^{(i)} \\ &= p_1 \sum_{i \in [|S|]} \sum_{j \in [|S|]} Z^{(i)\top} Z^{(j)} \\ &= p_1 U^\top U \\ &= p_1 I_{d \times d} \end{aligned} \quad (\text{A.3})$$

where the last equality is due to U having orthonormal columns. For each $i \in [|S|]$, we define the following the mean-zero matrices

$$X^{(i)} := (\delta_i^2 - \mathbb{E}[\delta_i^2]) Z^{(i)\top} Z^{(i)} = (\delta_i - p_1) Z^{(i)\top} Z^{(i)}.$$

Since Q_h^* is μ -incoherent,

$$\|X^{(i)}\|_{op} \leq |\delta_i - p_1| \|Z^{(i)\top} Z^{(i)}\|_{op} \leq \|Z^{(i)\top} Z^{(i)}\|_{op} = \|U_{i-}\|_2^2 \leq \frac{\mu r d}{|S|} \quad \text{surely.}$$

Furthermore,

$$\begin{aligned} \sum_{i \in [|S|]} \mathbb{E}[X^{(i)\top} X^{(i)}] &= \sum_{i \in [|S|]} \mathbb{E}[X^{(i)} X^{(i)\top}] = \sum_{i \in [|S|]} \mathbb{E}[(\delta_i - p_1)^2 Z^{(i)\top} Z^{(i)} Z^{(i)\top} Z^{(i)}] \\ &= p_1(1 - p_1) \sum_{i \in [|S|]} \|U_{i-}\|_2^2 Z^{(i)\top} Z^{(i)} \\ &\leq p_1 \cdot \frac{d\mu}{|S|} \sum_{i \in [|S|]} Z^{(i)\top} Z^{(i)} \\ &= \frac{d\mu p_1}{|S|} U^\top U \\ &= \frac{d\mu p_1}{|S|} I_{d \times d}. \end{aligned}$$

Thus,

$$\left\| \sum_{i \in [|S|]} \mathbb{E}[X^{(i)\top} X^{(i)}] \right\|_{op} = \left\| \sum_{i \in [|S|]} \mathbb{E}[X^{(i)} X^{(i)\top}] \right\|_{op} \leq \frac{d\mu p_1}{|S|}$$

From the matrix Bernstein inequality (Theorem 29), we have

$$\begin{aligned} \mathbb{P} \left(\left\| \tilde{U}^\top \tilde{U} - p_1 I_{d \times d} \right\|_{op} \geq t \right) &= \mathbb{P} \left(\left\| \sum_{i \in [|S|]} ((\delta_i^2 - p_1) Z^{(i)\top} Z^{(i)}) \right\|_{op} \geq t \right) \\ &= \mathbb{P} \left(\left\| \sum_{i \in [|S|]} X^{(i)} \right\|_{op} \geq t \right) \\ &\leq 2|S| \exp \left(- \frac{t^2/2}{\frac{p_1 \mu d}{|S|} + \frac{\mu d}{3|S|} t} \right) \\ &\leq 2|S| \exp \left(- \frac{t^2}{\frac{2p_1 \mu d}{|S|} + \frac{2\mu d}{|S|} t} \right) \end{aligned}$$

where the first equality follows from equations A.2 and A.3. For $t = \sqrt{\frac{40p_1 \mu d \log(|S|)}{|S|}} + \frac{40\mu d \log(|S|)}{|S|}$, we have

$$\left\| \tilde{U}^\top \tilde{U} - p_1 I_{d \times d} \right\|_{op} \leq \sqrt{\frac{40p_1 \mu d \log(|S|)}{|S|}} + \frac{40\mu d \log(|S|)}{|S|}$$

with probability at least $1 - 2|S|^{-10}$. Dividing both sides by p_1 yields the first inequality in the lemma. The corresponding bound for $\tilde{V}$ holds from a similar argument. Taking a union bound over the two events proves the lemma. $\square$

Now, we present our second lemma that shows that the uniformly sample submatrix ($\tilde{O}(d)$ by $\tilde{O}(d)$ in expectation) has rank- d with its smallest non-zero singular value bounded away from zero.

Lemma 21. *Let $p_1 = \frac{\mu d \log(|S|)}{320|S|}$ and $p_2 = \frac{\mu d \log(|A|)}{320|A|}$. Under the event in Lemma 20, we have*

$$\sigma_d((p_1 \vee p_2)^{-1} \tilde{Q}) \geq \frac{1}{2} \sigma_d(Q_h).$$

Proof Of Lemma 21. Under the assumption that $p_1 = \frac{\mu d \log(|S|)}{320|S|}$ and $p_2 = \frac{\mu d \log(|A|)}{320|A|}$ and the event in Lemma 20, we have $\|p_1^{-1} \tilde{U}^\top \tilde{U} - I_{d \times d}\|_{op} \leq \frac{1}{2}$. From Weyl's inequality, we have $\sigma_d(p_1^{-1} \tilde{U}^\top \tilde{U}) \geq \frac{1}{2}$, which implies $\sigma_d(p_1^{-1/2} \tilde{U}) \geq \frac{1}{\sqrt{2}}$. From a similar argument, $\sigma_d(p_1^{-1/2} \tilde{V}) \geq \frac{1}{\sqrt{2}}$. Let $p = p_1 \vee p_2$, from the singular value version of the Courant-Fischer minimax theorem (Theorem 7.3.8 [29]), we have

$$\begin{aligned}
\sigma_d(p^{-1} \tilde{Q}) &= \max_{S: \dim(S)=d} \min_{x \in S, x \neq 0} \frac{\|p^{-1} \tilde{U} \Sigma \tilde{V}^\top x\|_2}{\|x\|_2} \\
&= \max_{S: \dim(S)=d} \min_{x \in S, x \neq 0} \frac{\|(p^{-1/2} \tilde{U}) \Sigma (p^{-1/2} \tilde{V}^\top) x\|_2}{\|\Sigma (p^{-1/2} \tilde{V}^\top) x\|_2} \frac{\|\Sigma (p^{-1/2} \tilde{V}^\top) x\|_2}{\|p^{-1/2} \tilde{V}^\top x\|_2} \frac{\|p^{-1/2} \tilde{V}^\top x\|_2}{\|x\|_2} \\
&\geq \max_{S: \dim(S)=d} \min_{x \in S, x \neq 0} \frac{\|(p^{-1/2} \tilde{U}) \Sigma (p^{-1/2} \tilde{V}^\top) x\|_2}{\|(p^{-1/2} \tilde{U})^\dagger\|_{op} \|(p^{-1/2} \tilde{U}) \Sigma (p^{-1/2} \tilde{V}^\top) x\|_2} \\
&\quad \cdot \frac{\|\Sigma (p^{-1/2} \tilde{V}^\top) x\|_2}{\|\Sigma^{-1}\|_{op} \|\Sigma (p^{-1/2} \tilde{V}^\top) x\|_2} \frac{\|p^{-1/2} \tilde{V}^\top x\|_2}{\|x\|_2} \\
&= \sigma_d(p^{-1/2} \tilde{U}) \cdot \sigma_d(\Sigma) \max_{S: \dim(S)=d} \min_{x \in S, x \neq 0} \frac{\|p^{-1/2} \tilde{V}^\top x\|_2}{\|x\|_2} \\
&= \sigma_d(p^{-1/2} \tilde{U}) \cdot \sigma_d(\Sigma) \sigma_d(p^{-1/2} \tilde{V}^\top) \\
&\geq \sigma_d(p_1^{-1/2} \tilde{U}) \cdot \sigma_d(\Sigma) \sigma_d(p_2^{-1/2} \tilde{V}^\top) \\
&\geq \frac{1}{\sqrt{2}} \sigma_d(Q_h) \frac{1}{\sqrt{2}} \\
&= \frac{1}{2} \sigma_d(Q_h)
\end{aligned}$$

where the first inequality comes from properties of the operator norm and inverses/pseudo-inverses and the second inequality comes from replacing $p = p_1 \vee p_2$ with either p_1 or p_2 . $\square$

Using the two above lemmas, we next prove Lemma 8.

Proof of Lemma 8. Let p_1, p_2 be defined as in Lemma 21. From the previous two lemmas, it follows that with probability at least $1 - 4(|S| \wedge |A|)^{-10}$, we have $\sigma_d((p_1 \vee p_2)^{-1} \tilde{Q}) \geq \frac{1}{2} \sigma_d(Q_h)$. Next, we upper bound $\alpha = \frac{\|Q_h\|_\infty}{\sigma_d(Q_h(S^\#, A^\#))}$ assuming that Q_h is μ -incoherent with condition number κ . Let the singular value decomposition of

the rank d matrix Q_h be $Q_h = U\Sigma V^\top$. For $(s, a) \in S \times A$,

$$\begin{aligned}
|Q_h(s, a)| &= |U_s \Sigma V_a| \\
&\leq \|\Sigma\|_{op} |U_s V_a| \\
&\leq \sigma_1(Q_h) \|U_s\|_2 \|V_a\|_2 \\
&\leq \sigma_1(Q_h) \sqrt{\frac{\mu d}{|S|}} \sqrt{\frac{\mu d}{|A|}} \\
&= \frac{d\sigma_1(Q_h)\mu}{\sqrt{|S||A|}}
\end{aligned}$$

where the third inequality comes from Q_h being μ incoherent. Hence,

$$\begin{aligned}
\frac{\|Q_h\|_\infty}{\sigma_d(Q_h(S^\#, A^\#))} &\leq \frac{d\sigma_1(Q_h)\mu}{\sigma_d(Q_h(S^\#, A^\#)) \sqrt{|S||A|}} \\
&\leq \frac{d\sigma_1(Q_h)\mu}{\sigma_d(Q_h(S^\#, A^\#))(|S| \wedge |A|)} \\
&= \frac{320\sigma_1(Q_h)}{\sigma_d((p_1 \vee p_2)^{-1} Q_h(S^\#, A^\#)) \log(|S| \wedge |A|)} \\
&= \frac{640\sigma_1(Q_h)}{\sigma_d(Q_h) \log(|S| \wedge |A|)} \\
&= \frac{640\kappa}{\log(|S| \wedge |A|)}
\end{aligned}$$

where the third line comes from the definition of p_1 and p_2 and the fourth line comes from Lemma 20. Hence, $\alpha \in O(\kappa)$. Next, we upperbound the size of the anchor sets with high probability.

From the one-sided Bernstein's inequality, Proposition 28, for $C'' = \frac{25600}{3\mu d}$,

$$\begin{aligned}
\mathbb{P}(|S^\#| - \mathbb{E}[|S^\#|] \geq C'' p_1 |S|) &\leq \exp\left(-\frac{p_1^2 (C'')^2 |S|}{2(p_1 + \frac{p_1 C''}{3})}\right) \\
&\leq \exp\left(-\frac{\mu d C''}{640(1 + \frac{1}{3})} \log(|S|)\right) \\
&= |S|^{-10}.
\end{aligned}$$

With a similar argument,

$$\mathbb{P}(|A^\#| - \mathbb{E}[|A^\#|] \geq C'' p_2 |A|) \leq |A|^{-10}.$$

From our definition of p_1, p_2 , it follows that $\mathbb{E}[|S^\#|] = O(d\mu \log(|S|))$ and $\mathbb{E}[|A^\#|] = O(d\mu \log(|A|))$. A union bound on the above two events and the one in Lemma 20 asserts that

$$|S^\#| \in O(d\mu \log(|S|)), \quad |A^\#| \leq O(d\mu \log(|A|)), \quad \text{and } \alpha \in O(\kappa)$$

with probability at least $1 - 6(|S| \wedge |A|)^{-10}$.

□

A.3.4 Proof of Lemma 4

Proof of Lemma 4. Let π be an ϵ -optimal policy. We first show that $Q_H^\pi = r_2$ is a matrix with rank 2. By construction, the first two rows of Q_H^π are:

$$(Q_2^\pi)_1 = [1, 1, 1, \dots, 1], \quad (Q_2^\pi)_2 = \left[1, 1 - \left(\frac{1}{(m+1)^2} \right)^{1/2}, 1 - \left(\frac{2}{(m+1)^2} \right)^{1/2}, \dots, 1 - \left(\frac{m+1}{(m+1)^2} \right)^{1/2} \right],$$

and for any $i \in \{3, \dots, m+1\}$, the i -th row of Q_2^π is

$$(Q_2^\pi)_i = \left[1, 1 - \left(\frac{i}{(m+1)^2} \right)^{1/2}, 1 - \left(\frac{2i}{(m+1)^2} \right)^{1/2}, \dots, 1 - \left(\frac{(m+1)i}{(m+1)^2} \right)^{1/2} \right].$$

Hence, $(Q_2^\pi)_i = (1 - i^{1/2})(Q_2^\pi)_1 + i^{1/2}(Q_2^\pi)_2$, and $\text{rank}(Q_2^\pi) = 2$.

Let $h \in [H]$, to bound the rank of Q_h^π , we first note that for all $s \in S$, $|V_H^\pi(s) - 1| = \left(\frac{s\pi(s)}{(m+1)^2} \right)^{1/2} \leq \epsilon$ since π is ϵ -optimal. It follows that $\pi(s) < \frac{\epsilon^2(m+1)^2}{s}$, and if $\frac{\epsilon^2(m+1)^2}{s} < 1$, $\pi(s)$ must equal 0, which is the optimal action. Hence, there are at most $s \leq \epsilon^2(m+1)^2$ number of states in which π can deviate from the optimal policy. The value function of an ϵ -optimal policy is

$$V_H^\pi(s) = \begin{cases} 1, & \text{if } s = 0, \\ 1, & \text{if } s > \epsilon^2(m+1)^2, \\ 1 - \left(\frac{s\pi(s)}{(m+1)^2} \right)^{1/2}, & \text{otherwise.} \end{cases}$$

Since π is ϵ -optimal, we have $1 - V_{h+1}^\pi(s) \leq \epsilon$ for all $s \in S$. Due to the construction of the dynamics, if one starts at state s at time step h , one will be at either state s (choosing action $\pi(s) = s$ at each time step) or state 0 (taking any other sequence of actions). Thus, $V_{h+1}^\pi(s) = V_H^\pi(s)$ or $V_{h+1}^\pi(s) = 1$ depending on the sequence of action. It follows that $V_{h+1}^\pi(s) \geq V_H^\pi(s)$. It follows that if $s = 0$ or $s > \epsilon^2(m+1)^2$, $V_{h+1}^\pi(s) = 1$. Otherwise, $V_{h+1}^\pi(s) \geq \left(\frac{s\pi(s)}{(m+1)^2}\right)^{1/2}$ for $s \leq \lfloor \epsilon^2(m+1)^2 \rfloor$.

We next compute the Q function at the time step h to show that we can upper-bound the rank of Q_h by the number of states that π_h deviates from the optimal policy. Specifically, for each $s \leq \lfloor \epsilon^2(m+1)^2 \rfloor$, let $\pi_h(s) = s$ for $h \in [H]$. It follows that

$$\begin{aligned} Q_h^\pi(s, a) &= r_h(s, a) + \sum_{s'=0}^m P_h(s'|s, a) V_{h+1}^\pi(s') \\ &= 0 + P_h(0|s, a) V_{h+1}^\pi(0) + \sum_{s'=\lfloor \epsilon^2(m+1)^2 \rfloor+1}^m P_h(s'|s, a) V_{h+1}^\pi(s') + \sum_{s'=1}^{\lfloor \epsilon^2(m+1)^2 \rfloor} P_h(s'|s, a) V_{h+1}^\pi(s') \\ &= P_h(0|s, a) + \sum_{s'=\lfloor \epsilon^2(m+1)^2 \rfloor+1}^m P_h(s'|s, a) + \sum_{s'=1}^{\lfloor \epsilon^2(m+1)^2 \rfloor} P_h(s'|s, a) \left(1 - \left(\frac{s\pi(s)}{(m+1)^2}\right)^{1/2}\right). \end{aligned}$$

In matrix form, it follows that

$$\begin{aligned} Q_h^\pi &\geq \begin{bmatrix} \mathbf{0} & 1 & 1 & \dots & 1 \\ 1 & \mathbf{0} & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & \dots & \mathbf{0} & 1 \\ 1 & 1 & \dots & 1 & \mathbf{0} \end{bmatrix} + \sum_{s'=\lfloor \epsilon^2(m+1)^2 \rfloor+1}^m E^{s'} + \sum_{s'=1}^{\lfloor \epsilon^2(m+1)^2 \rfloor} E^{s'} \left(1 - \left(\frac{s\pi(s)}{(m+1)^2}\right)^{1/2}\right) \\ &= J_{m \times m} - \sum_{s'=1}^{\lfloor \epsilon^2(m+1)^2 \rfloor} E^{s'} \left(\frac{s\pi(s)}{(m+1)^2}\right)^{1/2}. \end{aligned}$$

Thus, at most $\lfloor \epsilon^2(m+1)^2 \rfloor$ -rows of Q_h^π are different from the all-ones row. It follows that $\text{rank}(Q_h^\pi) \leq 1 + \lfloor \epsilon^2(m+1)^2 \rfloor$, and each state $s \leq \lfloor \epsilon^2(m+1)^2 \rfloor$ that π_h performs optimally at tightens the above upperbound on the rank by one.

Since the bound holds for arbitrary $h \in [H - 1]$, it follows that it holds for all $h \in [H - 1]$. $\square$

A.3.5 Proof of Lemma 10 (Entrywise Bounds for Matrix Estimation)

Lemma 10 provides bounds for the entrywise error amplification of the matrix estimation method as a function of on k and α , assuming that $S^\#$ and $A^\#$ are (k, α) -anchor states and actions for matrix Q_h .

Proof of Lemma 10. Let $S^\#$ and $A^\#$ be (k, α) -anchor states and actions for matrix Q_h . For all $(s, a) \in \Omega^\# = S^\# \times A^\#$, assume that $\hat{Q}_h(s, a)$ satisfies $|\hat{Q}_h(s, a) - Q_h(s, a)| \leq \eta^\#$, and for all $(s, a) \in \Omega \setminus \Omega^\#$, assume that $\hat{Q}_h(s, a)$ satisfies $|\hat{Q}_h(s, a) - Q_h(s, a)| \leq \eta$. We follow the same argument as the proof of Proposition 13 in [55] except we upperbound equations (22) and (23) with $\|Q_h\|_\infty$ instead of $V_{\max}$. Following the steps in [55], i.e., using the triangle inequality and from the definition of the operator norm, for all $(s, a) \in S \times A$, since $S^\#$ and $A^\#$ are (k, α) -anchor states and actions,

$$\begin{aligned} |\bar{Q}_h(s, a) - Q_h(s, a)| &\leq \sqrt{2} \left\| [\hat{Q}_h(S^\#, A^\#)]^\dagger \right\|_{op} \left\| \hat{Q}_h(S^\#, a) \hat{Q}_h(s, A^\#) - Q_h(S^\#, a) Q_h(s, A^\#) \right\|_F \\ &\quad + \left\| [\hat{Q}_h(S^\#, A^\#)]^\dagger - [Q_h(S^\#, A^\#)]^\dagger \right\|_{op} \left\| Q_h(S^\#, a) Q_h(s, A^\#) \right\|_F. \end{aligned}$$

Following the steps in the proof of Proposition 13, we upperbound the first operator norm term with Weyl's inequality and our assumption on ϵ and the second

operator norm term with a classic result from perturbing pseudoinverses,

$$\begin{aligned} \left\| [\hat{Q}_h(S^\#, A^\#)]^\dagger \right\|_{op} &\leq \frac{2}{\sigma_d(Q_h(S^\#, A^\#))} \\ \left\| [\hat{Q}_h(S^\#, A^\#)]^\dagger - [Q_h(S^\#, A^\#)]^\dagger \right\|_{op} &\leq 2(1 + \sqrt{5}) \frac{\eta^\# k}{\sigma_d(Q_h(S^\#, A^\#))^2}. \end{aligned}$$

Since for all $s, s' \in S$ and $a, a' \in A$,

$$\begin{aligned} |\hat{Q}_h(s', a) \hat{Q}_h(s, a') - Q_h(s', a) Q_h(s, a')| &\leq |(Q_h(s', a) + \eta)(Q_h(s, a') + \eta) - Q_h(s', a) Q_h(s, a')| \\ &\leq \eta |Q_h(s', a)| + \eta |Q_h(s, a')| + \eta^2 \\ &\leq 2\eta \|Q_h\|_\infty + \eta^2, \end{aligned}$$

then, $\left\| \hat{Q}_h(S^\#, a) \hat{Q}_h(s, A^\#) - Q_h(S^\#, a) Q_h(s, A^\#) \right\|_F \leq (2\eta \|Q_h\|_\infty + \eta^2)k$. Because $|Q_h(s', a) Q_h(s, a')| \leq \|Q_h\|_\infty^2$ for all $s, s' \in S$ and $a, a' \in A$, clearly $\left\| Q_h(S^\#, a) Q_h(s, A^\#) \right\|_F \leq \|Q_h\|_\infty^2 k$. Using these inequalities gives that for all $(s, a) \in S \times A$,

$$|\bar{Q}_h(s, a) - Q_h(s, a)| \leq (6\sqrt{2}\alpha k\eta + 2(1 + \sqrt{5})\alpha^2 k^2 \eta^\#) \in O(\alpha k\eta + \alpha^2 k^2 \eta^\#) \quad (\text{A.4})$$

since $\eta \leq \|Q_h\|_\infty$. □

A.3.6 Inductive Arguments for Theorems 5, 6, and 7

We next present the missing proofs of our sample complexity bounds in Section 2.7. Recall that for ease of notation,

$$N_{s,a,h} := \begin{cases} N_h^\# & \text{if } (s, a) \in \Omega_h^\# = S_h^\# \times A_h^\# \\ N_h & \text{otherwise.} \end{cases}$$

Proof of Theorem 5. Assume that Q_h^* is rank d and has suboptimality gap $\Delta_{\min}$ (Assumptions 1 and 2), and $S_h^\#, A_h^\#$ are (k, α) -anchor states and actions for Q_h^* for all $h \in [H]$. Let $N_{H-t} = \frac{2(t+1)^2(c')^2 k^2 \alpha^2 \log(2H|S||A|/\delta)}{\Delta_{\min}^2}$, $N_{H-t}^\# = \alpha^2 k^2 N_{H-t}$, where c' satisfies the

inequality in Lemma 10, for all $h \in [H]$. We prove the correctness of LR-MCPI with high probability with induction on t that the learned policy $\hat{\pi}_{H-t}$ is an optimal policy with probability at least $1 - \delta(t+1)/H$.

The base case occurs at step $t = 0$ in which case our estimates, $\hat{Q}_H(s, a) = \frac{1}{N(s, a, H)} \sum_{i=1}^{N(s, a, H)} r_H^i(s, a)$ over Ω_H , are only averages of realizations $r_H^i \sim R_H(s, a)$. Since $R_H(s, a)$ has bounded support for all $(s, a) \in S \times A$, from Hoeffding's inequality (Theorem 27) with our choice of $N(s, a, H)$,

$$\begin{aligned} |\hat{Q}_H(s, a) - Q_H^*(s, a)| &\leq \frac{\Delta_{\min}}{2c'k\alpha} & \forall (s, a) \in \Omega_H \\ |\hat{Q}_H(s, a) - Q_H^*(s, a)| &\leq \frac{\Delta_{\min}}{2c'k^2\alpha^2} & \forall (s, a) \in \Omega_H^\# \end{aligned}$$

with probability at least $1 - \delta/H$ because $|\Omega_h| \leq |S||A|$. Step 2 of LR-MCPI gives

$$|\bar{Q}_H(s, a) - Q_H^*(s, a)| \leq \frac{\Delta_{\min}}{2}$$

for all $(s, a) \in S \times A$ from Lemma 10. From Step 3 of LR-MCPI, the identified policy is $\hat{\pi}_H(s) = \operatorname{argmax}_{a \in A} \bar{Q}_H(s, a)$. Assume for sake of contradiction that there exists an $s \in S$ such that $Q_H^*(s, \hat{\pi}_H(s)) < Q_H^*(s, \pi_H^*(s))$. Let $\hat{\pi}_H(s) = a, \pi_H^*(s) = a^*$. Hence,

$$\begin{aligned} Q_H^*(s, a^*) - Q_H^*(s, a) &= Q_H^*(s, a^*) - \bar{Q}_H(s, a) + \bar{Q}_H(s, a) - Q_H^*(s, a) \\ &\leq Q_H^*(s, a^*) - \bar{Q}_H(s, a^*) + \frac{\Delta_{\min}}{2} \\ &\leq \Delta_{\min} \end{aligned}$$

where the first inequality comes from how $\hat{\pi}_H(s)$ is defined and the matrix estimation step. Hence, we reach a contradiction since $Q_H^*(s, a^*) - Q_H^*(s, a)$ is less than the suboptimality gap. Thus, $\hat{\pi}_H(s)$ is an optimal policy. Hence, the base case holds.

Next, let $x \in \{0, \dots, H-1\}$. Assume that the inductive hypothesis, the policy

$\hat{\pi}_{H-x}$ found in Step 4 of LR-MCPI is an optimal policy with probability at least $1 - \delta(x+1)/H$, holds.

Following Step 1 of LR-MCPI, we have $\hat{Q}_{H-x-1}(s, a) = \hat{r}_{H-x-1}^{\text{cum}}(s, a)$, which is an unbiased estimate of $Q_{H-x-1}^*(s, a)$ and also bounded. Hence, from Hoeffding's inequality (Theorem 27), with the choice of $N_{H-x-1} = \frac{2(x+2)^2(c')^2k^2\alpha^2 \log(2H|S||A|/\delta)}{\Delta_{\min}^2}$, $N_{H-x-1}^\# = \alpha^2k^2N_{H-x-1}$, it follows that

$$\begin{aligned} |\hat{Q}_{H-x-1}(s, a) - Q_{H-x-1}^*(s, a)| &\leq \frac{\Delta_{\min}}{2c'k\alpha} & \forall (s, a) \in \Omega_{H-x-1} \\ |\hat{Q}_{H-x-1}(s, a) - Q_{H-x-1}^*(s, a)| &\leq \frac{\Delta_{\min}}{2c'k^2\alpha^2} & \forall (s, a) \in \Omega_{H-x-1}^\# \end{aligned}$$

with probability $1 - \frac{\delta}{H|S||A|}$. Step 2 of LR-MCPI gives

$$|\bar{Q}_{H-x-1} - Q_{H-x-1}^*|_\infty \leq \frac{\Delta_{\min}}{2}$$

from Lemma 10. From a union bound, it follows that $\hat{\pi}_{H-x}$ is an optimal policy and the above event occur with probability at least $1 - \delta(x+2)/H$. From Step 3 of LR-MCPI, the identified policy is $\hat{\pi}_{H-x-1}(s) = \operatorname{argmax}_{a \in A} \bar{Q}_{H-x-1}(s, a)$. Assume for sake of contradiction that there exists an $s \in S$ such that $Q_{H-x-1}^*(s, \hat{\pi}_{H-x-1}(s)) < Q_{H-x-1}^*(s, \pi_{H-x-1}^*(s))$. Let $\hat{\pi}_{H-x-1}(s) = a, \pi_{H-x-1}^*(s) = a^*$. Hence,

$$\begin{aligned} &Q_{H-x-1}^*(s, a^*) - Q_{H-x-1}^*(s, a) \\ &= Q_{H-x-1}^*(s, a^*) - \bar{Q}_{H-x-1}(s, a) + \bar{Q}_{H-x-1}(s, a) - Q_{H-x-1}^*(s, a) \\ &\leq Q_{H-x-1}^*(s, a^*) - \bar{Q}_{H-x-1}(s, a^*) + \frac{\Delta_{\min}}{2} \\ &\leq \Delta_{\min} \end{aligned}$$

where the first inequality comes from how $\hat{\pi}_{H-x-1}(s)$ is defined and the matrix estimation step. Hence, we reach a contradiction since $Q_{H-x-1}^*(s, a^*) - Q_{H-x-1}^*(s, a)$ is less than the suboptimality gap. Thus, $\hat{\pi}_{H-x-1}(s)$ is an optimal policy, and the inductive step holds for $x+1$. It follows from mathematical induction that the learned policy $\hat{\pi}$ is an optimal policy with probability at least $1 - \delta$.

Next, we bound the number of required samples. The number of samples used is

$$\sum_{t=0}^{H-1} (k(|A| + |S|))N_{H-t}(t+1) + k^2 N_{H-t}^\#(t+1)$$

where the $t+1$ comes from the length of the rollout. With our choice of N_{H-t} , it follows that

$$\begin{aligned} & \sum_{t=0}^{H-1} (k(|A| + |S|))N_{H-t}(t+1) \\ &= \sum_{t=0}^{H-1} (k(|A| + |S|)) \frac{2(t+1)^3(c')^2 k^2 \alpha_{H-t}^2 \log(2H|S||A|/\delta)}{\Delta_{\min}^2} + \frac{2(t+1)^3(c')^2 k^4 \alpha_{H-t}^4 \log(2H|S||A|/\delta)}{\Delta_{\min}^2} \\ &\leq \left(\frac{2c'^2 k^3 \alpha^2 (|S| + |A|) \log(2H|S||A|/\delta)}{\Delta_{\min}^2} + \frac{k^6 \alpha^4 c'^2 \log(2H|S||A|/\delta)}{\Delta_{\min}^2} \right) \sum_{t=0}^{H-1} (t+1)^3 \\ &\in \tilde{O} \left(\frac{k^3 \alpha^2 (|S| + |A|) H^4}{\Delta_{\min}^2} + \frac{k^6 \alpha^4 H^4}{\Delta_{\min}^2} \right). \end{aligned}$$

□

Proof of Theorem 6. This proof follows the same steps as the previous one. Assume that for all ϵ -optimal policies π , Q_h^π is rank d (Assumption 3), and $S_h^\#, A_h^\#$ are (k, α) -anchor states and actions for $Q_h^{\hat{\pi}}$, where $\hat{\pi}$ is the learned policy from Low Rank Monte Carlo Policy Iteration for all $h \in [H]$. Let $N_{H-t} = \frac{2(t+1)^2(c')^2 k^2 \alpha^2 H^2 \log(2H|S||A|/\delta)}{\epsilon^2}$, $N_{H-t}^\# = \alpha^2 k^2 N_{H-t}$, where c' satisfies the inequality in Lemma 10, for all $h \in [H]$. We prove the correctness of LR-MCPI with high probability with induction on t that the learned policy $\hat{\pi}_{H-t}$ is $\epsilon(t+1)/H$ -optimal policy with probability at least $1 - \delta(t+1)/H$.

The base case occurs at step $t = 0$ in which case our estimates, $\hat{Q}_H(s, a) = \frac{1}{N_{s,a,H}} \sum_{i=1}^{N_{s,a,H}} r_H^i(s, a)$ over Ω_H , are only averages of realizations $r_H^i \sim R_H(s, a)$. Since $R_H(s, a)$ has bounded support for all $(s, a) \in S \times A$, from Hoeffding's inequality

(Theorem 27) with our choice of $N_{s,a,H}$,

$$\begin{aligned} |\hat{Q}_H(s, a) - Q_H^*(s, a)| &\leq \frac{\epsilon}{2c'k\alpha H} & \forall (s, a) \in \Omega_h \\ |\hat{Q}_H(s, a) - Q_H^*(s, a)| &\leq \frac{\epsilon}{2c'k^2\alpha^2 H} & \forall (s, a) \in \Omega_h^\# \end{aligned}$$

with probability at least $1 - \delta/H$ because $|\Omega_h| \leq |S||A|$. Step 2 of LR-MCPI gives

$$|\bar{Q}_H(s, a) - Q_H^*(s, a)| \leq \frac{\epsilon}{2H}$$

for all $(s, a) \in S \times A$ from Lemma 10. Assume for sake of contradiction that there exists an $s \in S$ such that $Q_H^*(s, \hat{\pi}_H(s)) < Q_H^*(s, \pi_H^*(s)) - \epsilon/H$. Let $\hat{\pi}_H(s) = a, \pi_H^*(s) = a^*$. Hence,

$$\begin{aligned} Q_H^*(s, a^*) - Q_H^*(s, a) &= Q_H^*(s, a^*) - \bar{Q}_H(s, a) + \bar{Q}_H(s, a) - Q_H^*(s, a) \\ &\leq Q_H^*(s, a^*) - \bar{Q}_H(s, a^*) + \frac{\epsilon}{2H} \\ &\leq \frac{\epsilon}{H} \end{aligned}$$

where the first inequality comes from how $\hat{\pi}_H(s)$ is defined and the matrix estimation step. Hence, we reach a contradiction since $Q_H^*(s, a^*) - Q_H^*(s, a)$ is less ϵ/H . Thus, $\bar{Q}_H$ and $\hat{\pi}_H$ are both ϵ/H -optimal, and the base case holds.

Next, let $x \in \{0, \dots, H-1\}$. Assume that the inductive hypothesis, the policy $\hat{\pi}_{H-x}$ and action-value function estimate $\bar{Q}_{H-x}$ found in Step 3 of LR-MCPI are $\epsilon(x+1)/H$ -optimal with probability at least $1 - \delta(x+1)/H$, holds.

Following Step 1 from LR-MCPI, we have $\hat{Q}_{H-x-1}(s, a) = \hat{r}_{H-x-1}^{\text{cum}}(s, a)$, which is bounded and an unbiased estimate of $Q^{\hat{\pi}}(s, a)$ for $\hat{\pi} = \{\hat{\pi}_h\}_{H-x \leq h \leq H}$, which is an ϵ -optimal policy. Hence, from Hoeffding's inequality (Theorem 27), with the choice of $N_{H-x-1} = \frac{2(x+2)^2(c')^2 H^2 \alpha^2 k^2 \log(2H|S||A|/\delta)}{\epsilon^2}$, $N_{H-x-1}^\# = \alpha^2 k^2 N_{H-x-1}$, it follows that

$$\begin{aligned} |\hat{Q}_{H-x-1}(s, a) - Q_{H-x-1}^{\hat{\pi}}(s, a)| &\leq \frac{\epsilon}{2c'\alpha k H} & \forall (s, a) \in \Omega_{H-x-1} \\ |\hat{Q}_{H-x-1}(s, a) - Q_{H-x-1}^{\hat{\pi}}(s, a)| &\leq \frac{\epsilon}{2c'\alpha^2 k^2 H} & \forall (s, a) \in \Omega_{H-x-1}^\# \end{aligned}$$

with probability $1 - \frac{\delta}{H|S||A|}$. Step 2 of LR-MCPI gives

$$\|\bar{Q}_{H-x-1} - Q_{H-x-1}^{\hat{\pi}}\|_{\infty} \leq \frac{\epsilon}{2H}$$

from Lemma 10. The union bound asserts that the above error guarantee and $\hat{\pi}_{H-x}$ and $\bar{Q}_{H-x}^{\hat{\pi}}$ are $(x+1)\epsilon/H$ holds with probability at least $1 - \delta(x+2)/H$. From step 3 of LR-MCPI, the identified policy is $\hat{\pi}_{H-x-1}(s) = \operatorname{argmax}_{a \in A} \bar{Q}_{H-x-1}(s, a)$. For all $(s, a) \in S \times A$,

$$\begin{aligned} |\bar{Q}_{H-x-1}(s, a) - Q_{H-x-1}^*(s, a)| &\leq |\bar{Q}_{H-x-1}(s, a) - Q_{H-x-1}^{\hat{\pi}}(s, a)| \\ &\quad + |Q_{H-x-1}^{\hat{\pi}}(s, a) - Q_{H-x-1}^*(s, a)| \\ &\leq \frac{\epsilon}{2H} + \left| \mathbb{E}_{s' \sim P_{H-x-1}(\cdot|s, a)} [V_{H-x}^{\hat{\pi}}(s') - V_{H-x}^*(s')] \right| \\ &\leq \frac{\epsilon}{2H} + \left| \mathbb{E}_{s' \sim P_{H-x-1}(\cdot|s, a)} [(x+1)\epsilon/H] \right| \\ &= \frac{(2x+3)\epsilon}{2H}. \end{aligned}$$

Thus, $\bar{Q}_{H-x-1}$ is $\frac{\epsilon(x+2)}{H}$ -optimal. It follows from the construction of $\hat{\pi}_{H-x-1}(s)$ that

$$\bar{Q}_{H-x-1}(s, \hat{\pi}_{H-x-1}(s)) \geq \bar{Q}_{H-x-1}(s, a'),$$

where $a' = \operatorname{argmax}_a Q_{H-x-1}^*(s, a)$. Hence, for all $s \in S$,

$$\begin{aligned} |V_{H-x-1}^*(s) - V_{H-x-1}^{\hat{\pi}}(s)| &\leq |Q_{H-x-1}^*(s, a') - \bar{Q}_{H-x-1}(s, \hat{\pi}_{H-x-1}(s))| \\ &\quad + |\bar{Q}_{H-x-1}(s, \hat{\pi}_{H-x-1}(s)) - Q_{H-x-1}^{\hat{\pi}}(s, \hat{\pi}_{H-x-1}(s))| \\ &\leq \frac{(2x+3)\epsilon}{2H} + \frac{\epsilon}{2H} \\ &= \frac{(x+2)\epsilon}{H}. \end{aligned}$$

Thus, $\hat{\pi}_{H-x-1}(s)$ and $\bar{Q}_{H-x-1}$ are $(x+2)\epsilon/H$ -optimal, and the inductive step holds for $x+1$. It follows from mathematical induction that the learned policy $\hat{\pi}$ and action-value function are ϵ -optimal with probability at least $1 - \delta$.

Next, we bound the number of required samples. The number of samples used is

$$\sum_{t=0}^{H-1} (k(|A| + |S|))N_{H-t}(t+1) + k^2 N_{H-t}^{\#}(t+1)$$

where the $t+1$ comes from the length of the rollout. With our choice of N_{H-t} , it follows that

$$\begin{aligned} & \sum_{t=0}^{H-1} (k(|A| + |S|))N_{H-t}(t+1) \\ &= \sum_{t=0}^{H-1} (k(|A| + |S|)) \frac{2(t+1)^3(c')^2 k^2 \alpha^2 H^2 \log(2H|S||A|/\delta)}{\epsilon^2} + \frac{2(t+1)^3(c')^2 k^6 \alpha^4 H^2 \log(2H|S||A|/\delta)}{\epsilon^2} \\ &\in \tilde{O}\left(\frac{k^3 \alpha^2 (|S| + |A|) H^6}{\epsilon^2} + \frac{k^6 \alpha^4 H^6}{\epsilon^2}\right). \end{aligned}$$

□

Proof. Proof of Theorem 7 This proof follows the same steps as the previous two proofs. Assume that for any ϵ -optimal value function V_{h+1} , the matrix corresponding to $Q'_h = [r_h + [P_h V_{h+1}]]$ is rank d , and $S_h^{\#}, A_h^{\#}$ are (k, α) -anchor states and actions for $\hat{Q}'_h = [r_h + [P_h \hat{V}_{h+1}]]$, where $\hat{V}_{h+1}$ is the learned value function from Low Rank Empirical Value Iteration for all $h \in [H]$. Let $N_{H-t} = \frac{2(t+1)^2(c')^2 k^2 \alpha^2 H^2 \log(2H|S||A|/\delta)}{\epsilon^2}$, $N_{H-t}^{\#} = \alpha^2 k^2 N_{H-t}$, where c' satisfies the inequality in Lemma 10, and $Q'_h = [r_h + P_h \hat{V}_{h+1}]$ for all ϵ -optimal value functions $\hat{V}_{h+1}$ for all $h \in [H]$. We prove the correctness of LR-EVI with high probability with induction on t that

$$\|\bar{Q}_{H-t} - Q_{H-t}^*\|_{\infty} \leq \frac{\epsilon(t+1)}{H}, \quad \|\bar{Q}_{H-t} - Q_{H-t}^{\hat{\pi}}\|_{\infty} \leq \frac{\epsilon(t+1)}{H}$$

where $\bar{Q}_{H-t}$ and $\hat{\pi}_{H-t}$ are the learned Q function and policy with probability at least $1 - \delta(t+1)/H$.

The base case occurs at step $t = 0$ in which case our estimates, $\hat{Q}_H(s, a) = \frac{1}{N_{s,a,H}} \sum_{i=1}^{N_{s,a,H}} r_H^i(s, a)$ over Ω_H , are only averages of realizations $r_H^i \sim R_H(s, a)$ since

$\hat{V}_{H+1} = \vec{0}$. Since $R_H(s, a)$ has bounded support for all $(s, a) \in S \times A$, from Hoeffding's inequality (Theorem 27) with our choice of $N_{s,a,H}$,

$$\begin{aligned} |\hat{Q}_H(s, a) - Q_H^*(s, a)| &\leq \frac{\epsilon}{c'k\alpha H} & \forall (s, a) \in \Omega_h \\ |\hat{Q}_H(s, a) - Q_H^*(s, a)| &\leq \frac{\epsilon}{c'k^2\alpha^2 H} & \forall (s, a) \in \Omega_h^\# \end{aligned}$$

with probability at least $1 - \delta/H$ because $|\Omega_h| \leq |S||A|$. Step 2 of LR-EVI gives

$$|\bar{Q}_H(s, a) - Q_H^*(s, a)| \leq \frac{\epsilon}{H}$$

for all $(s, a) \in S \times A$ from Lemma 10. Since $Q_H^* = Q_H^\pi$, the base case holds.

Next, let $x \in \{0, \dots, H-1\}$. Assume that the inductive hypothesis, the action-value function estimates $\bar{Q}_{H-x}$ and learned policy $\hat{\pi}_{H-x}$ satisfy

$$\|\bar{Q}_{H-x} - Q_{H-x}^*\|_\infty \leq \frac{(x+1)\epsilon}{H}, \quad \|\bar{Q}_{H-x} - Q_{H-x}^\pi\|_\infty \leq \frac{(x+1)\epsilon}{H}$$

holds with probability at least $1 - \delta(x+1)/H$. Following Step 1 from LR-EVI, we have

$$\hat{Q}_{H-x-1}(s, a) = \hat{r}_{H-x-1}(s, a) + \mathbb{E}_{s' \sim \hat{P}_{H-x-1}(\cdot|s,a)}[\hat{V}_{H-x}(s')],$$

an unbiased estimate of $Q'_{H-x-1}(s, a) = r_{H-x-1}(s, a) + \mathbb{E}_{s' \sim P_{H-x-1}(\cdot|s,a)}[\hat{V}_{H-x}(s')]$, which is bounded. Hence, from Hoeffding's inequality (Theorem 27), with the choice of $N_{H-x-1} = \frac{(x+2)^2(c')^2k^2H^2\alpha^2\log(2H|S||A|/\delta)}{2\epsilon^2}$, $N_{H-x-1}^\# = \alpha^2k^2N_{H-x-1}$, it follows that

$$\begin{aligned} |\hat{Q}_{H-x-1}(s, a) - Q'_{H-x-1}(s, a)| &\leq \frac{\epsilon}{2c'k\alpha H} & \forall (s, a) \in \Omega_{H-x-1} \\ |\hat{Q}_{H-x-1}(s, a) - Q'_{H-x-1}(s, a)| &\leq \frac{\epsilon}{2c'k^2\alpha^2 H} & \forall (s, a) \in \Omega_{H-x-1}^\# \end{aligned}$$

with probability $1 - \frac{\delta}{H|S||A|}$. Step 2 of LR-EVI gives

$$|\bar{Q}_{H-x-1} - Q'_{H-x-1}|_\infty \leq \frac{\epsilon}{H}$$

from Lemma 10. The union bound asserts that the above error guarantee and $\bar{Q}_{H-x'}$ is close to $Q_{H-x'}^*$ and $Q_{H-x'}^\pi$ for $x \in [x]$ holds with probability at least $1 -$

$\delta(x+2)/H$. Hence, for all $(s, a) \in S \times A$,

$$\begin{aligned}
|\bar{Q}_{H-x-1}(s, a) - Q_{H-x-1}^*(s, a)| &\leq |\bar{Q}_{H-x-1}(s, a) - Q'_{H-x-1}(s, a)| + |Q'_{H-x-1}(s, a) - Q_{H-x-1}^*(s, a)| \\
&\leq \frac{\epsilon}{H} + |\mathbb{E}_{s' \sim P_{H-x-1}(\cdot|s, a)}[\max_{a \in A} \bar{Q}_{H-x}(s', a') - V_{H-x}^*(s')]| \\
&\leq \frac{\epsilon}{H} + |\mathbb{E}_{s' \sim P_{H-x-1}(\cdot|s, a)}[(x+1)\epsilon/H]| \\
&= \frac{(x+2)\epsilon}{H}
\end{aligned}$$

Thus, $\bar{Q}_{H-x-1}$ is $(x+2)\epsilon/H$ -optimal. Next, we note that

$$\begin{aligned}
|\bar{Q}_{H-x-1}(s, a) - Q_{H-x-1}^{\hat{\pi}}(s, a)| &\leq |\bar{Q}_{H-x-1}(s, a) - Q'_{H-x-1}(s, a)| + |Q'_{H-x-1}(s, a) - Q_{H-x-1}^{\hat{\pi}}(s, a)| \\
&\leq \frac{\epsilon}{H} + |\mathbb{E}_{s' \sim P_{H-x-1}(\cdot|s, a)}[\max_{a \in A} \bar{Q}_{H-x}(s', a') - V_{H-x}^{\hat{\pi}}(s')]| \\
&\leq \frac{\epsilon}{H} + |\mathbb{E}_{s' \sim P_{H-x-1}(\cdot|s, a)}[(x+1)\epsilon/H]| \\
&= \frac{(x+2)\epsilon}{H}
\end{aligned}$$

where the third inequality holds because

$$\begin{aligned}
|\max_{a \in A} \bar{Q}_{H-x}(s', a') - V_{H-x}^{\hat{\pi}}(s')| &\leq |\mathbb{E}_{a' \sim \hat{\pi}_{H-x}(s')}[\bar{Q}_{H-x}(s', a')] - \mathbb{E}_{a' \sim \hat{\pi}_{H-x}(s')}[Q_{H-x}^{\hat{\pi}}(s', a')]| \\
&\leq \|\hat{Q}_{H-x} - Q_{H-x}^{\hat{\pi}}\|_{\infty} \\
&\leq \frac{(x+1)\epsilon}{H}
\end{aligned}$$

from the induction hypothesis, and the inductive step holds for $x+1$. It follows from mathematical induction (and the triangle inequality) that the learned policy $\hat{\pi}$ and action-value function are 2ϵ and ϵ -optimal with probability at least $1 - \delta$. Scaling N_h by a factor of four results in learning an ϵ -optimal policy with probability at least $1 - \delta$ without changing the sample complexity's dependence on $|S|$, $|A|$, H , or ϵ .

Next, we bound the number of required samples. The number of samples used is

$$\sum_{t=0}^{H-1} (k(|A| + |S|))N_{H-t} + k^2 N_{H-t}^{\#}.$$

Note that there is no $(t + 1)$ term as samples are single transitions instead of rollouts. With our choice of N_{H-t} , it follows that

$$\begin{aligned} \sum_{t=0}^{H-1} (k(|A| + |S|)) N_{H-t} &= \sum_{t=0}^{H-1} k(|A| + |S|) \frac{8(t+1)^2 (c')^2 k^2 \alpha^2 H^2 \log(2H|S||A|/\delta)}{\epsilon^2} \epsilon \\ &\quad + \frac{8(t+1)^2 (c')^2 k^6 \alpha^4 H^2 \log(2H|S||A|/\delta)}{\epsilon^2} \\ &\in \tilde{O} \left(\frac{k^3 \alpha^2 (|S| + |A|) H^5}{\epsilon^2} + \frac{k^6 \alpha^4 H^5}{\epsilon^2} \right). \end{aligned}$$

□

A.3.7 Proofs for Approximately Low Rank Models

We first present the proof of Proposition 11, which shows that if the reward function and transition kernel are low rank, then for any value function estimate $\hat{V}_{h+1}$, $r_h + [P_h \hat{V}_{h+1}]$ has rank upper bounded by d .

Proof of Proposition 11. Let $\xi_R, \xi_P, r_{h,d}, [P_{h,d} \hat{V}_{h+1}]$ be defined as in Section 2.8. Then, for all $(s, a, h) \in S \times A \times [H]$,

$$\begin{aligned} |r_{h,d} + P_{h,d} \hat{V}_{h+1}(s, a) - [r_h + P_h \hat{V}_{h+1}](s, a)| \\ &\leq |[r_h - r_{h,d}](s, a)| + |[P_{h,d} - P_h] \hat{V}_{h+1}(s, a)| \\ &= \xi_R + \left| \sum_{s' \in S} \hat{V}_{h+1}(s') (P_{h,d}(s'|s, a) - P_h(s'|s, a)) \right| \\ &\leq \xi_R + (H - h) \left| \sum_{s' \in S} (P_{h,d}(s'|s, a) - P_h(s'|s, a)) \right| \\ &= \xi_R + (H - h) 2d_{\text{TV}}(P_h(\cdot|s, a), P_{h,d}(\cdot|s, a))_{\text{TV}} \\ &= \xi_R + (H - h) \xi_P \end{aligned}$$

since $\hat{V}_{h+1}(s) \in [0, H - h - 1]$.

□

We next prove that the learned policy's error is additive with respect to the approximation error.

Proof. Proof of Theorem 12 This proof follows the same steps as the proof of Theorem 7 while accounting for the approximation error. Assume that we have a (d, ξ_R, ξ_P) -approximately low-rank MDP. Let $S_h^\#, A_h^\#$ be (k, α) -anchor states and actions, c' be a constant that satisfies the inequality in Lemma 10, $N_{H-t} = \frac{2(t+1)^2(c')^2k^2\alpha^2H^2\log(2H|S||A|/\delta)}{\epsilon^2}$, $N_{H-t}^\# = \alpha^2k^2N_{H-t}$, and $Q'_h = [r_h + P_h\hat{V}_{h+1}]$ for all ϵ -optimal value functions $\hat{V}_{h+1}$ for all $h \in [H]$. We prove the correctness of LR-EVI with high probability with induction on t that

$$\|\bar{Q}_{H-t} - Q_{H-t}^*\|_\infty, \|\bar{Q}_{H-t} - Q_{H-t}^{\hat{\pi}}\|_\infty \leq (t+1)\epsilon/H + \sum_{i=0}^t (c'k^2\alpha^2 + 1)(\xi_R + i\xi_P)$$

where $\bar{Q}_{H-t}$ and $\hat{\pi}_{H-t}$ are the learned Q function and policy with probability at least $1 - \delta(t+1)/H$ for each $t \in \{0, \dots, H-1\}$.

The base case occurs at step $t = 0$ in which case our estimates, $\hat{Q}_H(s, a) = \frac{1}{N_{s,a,H}} \sum_{i=1}^{N_{s,a,H}} r_{H,i}(s, a)$ over Ω_H , are only averages of realizations $r_H^i \sim R_H(s, a)$ since $\hat{V}_{H+1} = \vec{0}$. Since $R_H(s, a)$ has bounded support for all $(s, a) \in S \times A$, from Hoeffding's inequality (Theorem 27) with our choice of $N_{s,a,H}$

$$\begin{aligned} |\hat{Q}_H(s, a) - Q_H^*(s, a)| &\leq \frac{\epsilon}{c'k\alpha H} & \forall (s, a) \in \Omega_h \\ |\hat{Q}_H(s, a) - Q_H^*(s, a)| &\leq \frac{\epsilon}{c'k^2\alpha^2 H} & \forall (s, a) \in \Omega_h^\# \end{aligned}$$

with probability at least $1 - \delta/H$ because $|\Omega_h| \leq |S||A|$. Under the event above, it follows that $|\hat{Q}_H(s, a) - Q_{H,d}^*(s, a)| \leq \frac{\epsilon}{c'_H\alpha^2k^2H} + \xi_R$ or $|\hat{Q}_H(s, a) - Q_{H,d}^*(s, a)| \leq \frac{\epsilon}{c'_H\alpha kH} + \xi_R$ for all $(s, a) \in \Omega_H$. Step 2 of LR-EVI gives

$$|\bar{Q}_H(s, a) - Q_{H,d}^*(s, a)| \leq \frac{\epsilon}{H} + Ck^2\alpha^2\xi_R$$

for all $(s, a) \in S \times A$ from Lemma 10 for some positive constant C . By definition of the approximation error,

$$|\bar{Q}_H(s, a) - Q_H^*(s, a)| \leq \frac{\epsilon}{H} + (Ck^2\alpha^2 + 1)\xi_R \quad \forall (s, a) \in S \times A.$$

Since $Q_H^* = Q_H^{\hat{\pi}}$, the base case holds.

Next, let $x \in \{0, \dots, H-1\}$. Assume that the inductive hypothesis, the action-value function estimates $\bar{Q}_{H-x}$ and learned policy $\hat{\pi}_{H-x}$ satisfy

$$\|\bar{Q}_{H-x} - Q_{H-x}^*\|_\infty \leq \frac{(x+1)\epsilon}{H}, \quad \|\bar{Q}_{H-x} - Q_{H-x}^{\hat{\pi}}\|_\infty \leq \frac{(x+1)\epsilon}{H} + \sum_{i=0}^x (Ck^2\alpha^2 + 1)(\xi_R + i\xi_P)$$

holds with probability at least $1 - \delta(x+1)/H$. At step $x+1$, following Step 1 from LR-EVI, we have

$$\hat{Q}_{H-x-1}(s, a) = \hat{r}_{H-x-1}(s, a) + \mathbb{E}_{s' \sim \hat{P}_{H-x-1}(\cdot|s, a)}[\hat{V}_{H-x}(s')],$$

an unbiased estimate of $Q'_{H-x-1}(s, a) = r_{H-x-1}(s, a) + \mathbb{E}_{s' \sim P_{H-x-1}(\cdot|s, a)}[\hat{V}_{H-x}(s')]$. Furthermore, $\hat{Q}_{H-x-1}(s, a) \in [0, x+2]$ is a bounded random variable because of bounded rewards. Hence, from Hoeffding's inequality (Theorem 27), with the choice of $N_{H-x-1} = \frac{(x+2)^2(c')^2k^2\alpha^2H^2\log(2H|S||A|/\delta)}{2\epsilon^2}$, $N_{H-x-1}^\# = \alpha^2k^2N_{H-x-1}$, it follows that

$$\begin{aligned} |\hat{Q}_{H-x-1}(s, a) - Q'_{H-x-1}(s, a)| &\leq \frac{\epsilon}{2c'k\alpha H} & \forall (s, a) \in \Omega_{H-x-1} \\ |\hat{Q}_{H-x-1}(s, a) - Q'_{H-x-1}(s, a)| &\leq \frac{\epsilon}{2c'k^2\alpha^2H} & \forall (s, a) \in \Omega_{H-x-1}^\# \end{aligned}$$

with probability $1 - \frac{\delta}{H|S||A|}$. Under the event above, it follows that $|\hat{Q}_{H-x-1}(s, a) - Q'_{H-x-1}(s, a)| \leq \frac{\epsilon}{c'k^2\alpha^2H} + \xi_R + (H-x-1)\xi_P$ for all $(s, a) \in \Omega_{H-x-1}$ where $Q'_{h,d} = r_{h,d} + [P_{h,d}\hat{V}_{h+1}]$. Step 2 of LR-EVI gives

$$|\bar{Q}_{H-x-1} - Q'_{H-x-1,d}|_\infty \leq \frac{\epsilon}{H} + Ck^2\alpha^2(\xi_R + (H-x-1)\xi_P)$$

from Lemma 10 for some positive constant C . The union bound asserts that the above error guarantee holds with probability at least $1 - \delta(x+2)/H$. Hence, for

all $(s, a) \in S \times A$,

$$\begin{aligned}
& |\bar{Q}_{H-x-1}(s, a) - Q_{H-x-1}^*(s, a)| \\
& \leq |\bar{Q}_{H-x-1}(s, a) - Q'_{H-x-1,d}(s, a)| + |Q'_{H-x-1,d}(s, a) - Q'_{H-x-1}(s, a)| \\
& \quad + |Q'_{H-x-1}(s, a) - Q_{H-x-1}^*(s, a)| \\
& \leq \frac{\epsilon}{H} + c'k^2\alpha^2 (\xi_R + (H-x-1)\xi_P) \\
& \quad + \xi_R + (H-x-1)\xi_P \\
& \quad + |\mathbb{E}_{s' \sim P_{H-x-1}(\cdot|s,a)}[\max_{a \in A} \bar{Q}_{H-x}(s', a') - V_{H-x}^*(s')]| \\
& \leq \frac{\epsilon}{H} + (1 + c'k^2\alpha^2) (\xi_R + (H-x-1)\xi_P) \\
& \quad + |\mathbb{E}_{s' \sim P_{H-x-1}(\cdot|s,a)}[(x+1)\epsilon/H + \sum_{i=0}^x (Ck^2\alpha^2 + 1) (\xi_R + i\xi_P)]| \\
& = \frac{(x+2)\epsilon}{H} + \sum_{i=0}^{x+1} (Ck^2\alpha^2 + 1) (\xi_R + i\xi_P).
\end{aligned}$$

With a similar argument, it follows that for all $(s, a) \in S \times A$,

$$|\bar{Q}_{H-x-1}(s, a) - Q_{H-x-1}^{\hat{\pi}}(s, a)| \leq \frac{(x+2)\epsilon}{H} + \sum_{i=0}^{x+1} (Ck^2\alpha^2 + 1) (\xi_R + i\xi_P).$$

Thus, the inductive step holds for $x+1$, and from mathematical induction, the lemma holds.

Choosing $t = H-1$ proves the correctness of the algorithm. Next, we bound the number of required samples. The number of samples used is the same as in the proof of Theorem 7, which implies a sample complexity of

$$\tilde{O}\left(\frac{k^3\alpha^2(|S| + |A|)H^5}{\epsilon^2} + \frac{k^6\alpha^4H^5}{\epsilon^2}\right).$$

□

A.3.8 Proofs for Continuous MDPs

We next present the proofs of the results in Section 2.9, starting with our procedure on how we obtain samples/rollouts from the discretized MDP.

Using the generative model, we simulate trajectories from M^β with the following procedure: to sample a trajectory from M^β following policy π starting at (s, a, h) , first sample a state s' from $P_h(\cdot|s, a)$. Then, we take the closest discretized state s'_β to s' , i.e., $s'_\beta = \arg \min_{s \in S} \|s - s'\|_2$, to be the observed state in the trajectory. The generative model is then used to sample from $P_{h+1}(\cdot|s'_\beta, \pi(s'_\beta))$, and we repeat until the end of the horizon to obtain a trajectory from P_h^β using the generative model on the original MDP. Lemma 22 asserts the correctness of this procedure.

Lemma 22. *Let $\tau_h^\pi(s_h, a_h)$ be a rollout obtained with the procedure detailed above starting at state-action pair (s_h, a_h) at step h with policy π , $\tau_{\beta,h}^\pi(s_h, a_h)$ be a rollout following policy π from the discretized MDP starting at state-action pair (s_h, a_h) at step h , and τ be a realization of a rollout following policy π from the discretized MDP starting at state-action pair (s_h, a_h) at step h . Then,*

$$\mathbb{P}(\tau_h(s_h, a_h) = \tau) = \mathbb{P}(\tau_h^\beta(s_h, a_h) = \tau).$$

Proof of Lemma 22. Let $\tau = (s_{h+1}, \pi(s_{h+1}), \dots, s_H, \pi(s_H))$. From the Markov Property and the procedure defined above, it follows that

$$\begin{aligned} \mathbb{P}(\tau_h(s_h, a_h) = \tau) &= \int_{\{s': \|s_{h+1} - s'\|_2 \leq \beta\}} P(s'|s_h, a_h) ds' \prod_{i=h+1}^{H-1} \int_{\{s': \|s_{i+1} - s'\|_2 \leq \beta\}} P(s'|s_i, \pi(s_i)) ds' \\ &= P_h^\beta(s_{h+1}|s_h, a_h) \prod_{i=h+1}^{H-1} P_h^\beta(s_{i+1}|s_i, \pi(s_i)) \\ &= \mathbb{P}(\tau_h^\beta(s_h, a_h) = \tau). \end{aligned}$$

□

We next present the proof of Lemma 13, which allows us to use Q_h^β to estimate Q_h^* .

Proof of Lemma 13. We prove this lemma via induction. For $h = H$, by construction of the β -nets, for any $s \in S, a \in A$ and $s' \in S^\beta, a' \in A^\beta$ such that $\|s - s'\|_2 \leq \beta, \|a - a'\|_2 \leq \beta$,

$$|Q_H^*(s, a) - Q_H^\beta(s', a')| \leq 2L\beta$$

because Q_H^* is L -Lipschitz. For all $s \in S$ and $s' \in S^\beta$ such that $\|s - s'\|_2 \leq \beta$, let $a_{\max} = \operatorname{argmax}_{a \in A} Q_H^*(s, a)$ and $d_A(a)$ be the function that maps the action a to the closest action in A^β , which is at most β away. It follows that

$$\begin{aligned} |V_H^*(s) - V_H^\beta(s')| &= |Q_H^*(s, a_{\max}) - \max_{a^* \in A^\beta} Q_H^\beta(s', a^*)| \\ &\leq |Q_H^*(s, a_{\max}) - Q_H^\beta(s', d(a_{\max}))| \\ &\leq 2L\beta \end{aligned}$$

where the first inequality comes from $Q_H^*(s, a_{\max})$ being an upperbound of $Q_H^\beta(s', a')$ and the max operator, and the second inequality comes from Q_H^* being Lipschitz. Next, assume that for any $s \in S$ and $s' \in S^\beta$ such that $\|s - s'\|_2 \leq \beta$, $|V_{H-t+1}^*(s) - V_{H-t+1}^\beta(s')| \leq 2(H - t + 1)L\beta$. Let $d_S(s)$ be the function that maps the state s to the closest state in S^β , which is at most β away. For any $s \in S, a \in A$ and

$s' \in S^\beta, a' \in A^\beta$ such that $\|s - s'\|_2 \leq \beta, \|a - a'\|_2 \leq \beta$,

$$\begin{aligned}
& |Q_{H-t}^*(s, a) - Q_{H-t}^\beta(s', a')| \\
& \leq |Q_{H-t}^*(s, a) - Q_{H-t}^*(s', a')| + |Q_{H-t}^*(s', a') - Q_{H-t}^\beta(s', a')| \\
& \leq 2L\beta + |\mathbb{E}_{s^* \sim P(\cdot|s', a')} [V_{H-t+1}^*(s^*)] - \mathbb{E}_{s^* \sim P^\beta(\cdot|s', a')} [V_{H-t+1}^\beta(s^*)]| \\
& = 2L\beta + \left| \int_{s^* \in S} P_{H+1}(s^*|s', a') V_{H-t+1}^*(s^*) ds^* - \sum_{s^* \in S^\beta} P_{H+1}^\beta(s^*|s', a') V_{H-t+1}^\beta(s^*) \right| \\
& = 2L\beta + \left| \int_{s^* \in S} P_{H+1}(s^*|s', a') V_{H-t+1}^*(s^*) ds^* \right. \\
& \quad \left. - \sum_{s^* \in S^\beta} \int_{\{s'' \in S : \|s^* - s''\|_2 \leq \beta\}} V_{H-t+1}^\beta(s^*) P_h(s''|s', a') ds'' \right| \\
& = 2L\beta + \left| \int_{s^* \in S} P_{H+1}(s^*|s', a') V_{H-t+1}^*(s^*) ds^* - \int_{s^* \in S} P_{H+1}(s^*|s', a') V_{H-t+1}^\beta(s^*) ds^* \right| \\
& \leq 2L\beta + |\mathbb{E}_{s^* \sim P(\cdot|s', a')} 2(H-t+1)L\beta| \\
& = 2(H-t)L\beta
\end{aligned}$$

where the fourth line comes from the definition of P_h^β . For all $s \in S$ and $s' \in S^\beta$ such that $\|s - s'\|_2 \leq \beta$, let $a_{\max} = \operatorname{argmax}_{a \in A} Q_{H-t}^*(s, a)$. It follows that

$$\begin{aligned}
|V_{H-t}^*(s) - V_{H-t}^\beta(s')| &= |Q_{H-t}^*(s, a_{\max}) - Q_{H-t}^\beta(s', a'_{\max})| \\
&\leq |Q_{H-t}^*(s, a_{\max}) - Q_{H-t}^\beta(s', d_A(a_{\max}))| \\
&\leq 2(H-t)L\beta
\end{aligned}$$

Thus, from induction, for any $s \in S, a \in A$ and $s' \in S^\beta, a' \in A^\beta$ such that $\|s - s'\|_2 \leq \beta, \|a - a'\|_2 \leq \beta$, for all $h \in [H]$,

$$|Q_h^*(s, a) - Q_h^\beta(s', a')| \leq 2L(H-h+1)\beta, \quad |V_h^*(s, a) - V_h^\beta(s', a')| \leq 2L(H-h+1)\beta.$$

□

To prove the desired sample complexity bounds, we first state a lemma on covering numbers that upperbounds the number of points required for our β -nets.

Lemma 23 (Theorem 14.2 from [68]). *Let $\Theta \subset \mathbb{R}^n$. Then,*

$$N(\Theta, \epsilon) \leq \left(\frac{3}{\epsilon}\right)^n \frac{\text{Vol}(\Theta)}{\text{Vol}(B)}$$

where $N(\Theta, \epsilon)$ is the covering number of Θ , and B is the unit norm ball in $\mathbb{R}^n$.

We next present the sample complexity bound of LR-MCPI when M^β satisfies Assumption 3 and its proof.

Theorem 24. *Let $Q_{h,\beta}^\pi = [Q_h^\pi(s, a)]_{(s,a) \in S^\beta \times A^\beta}$, the action-value function of policy π at step h on only the discretized state-action pairs. After discretizing the continuous MDP, let Assumption 3 hold on M^β . Furthermore, assume that $S_h^\#, A_h^\#$ are (k, α) -anchor states and actions for $Q_h^{\pi,\beta}$ for all $h \in [H]$. Let $\bar{Q}_h$ be the action-value function estimates that Low Rank Monte Carlo Policy Iteration for Step 1 return for all $h \in [H]$ when run on M^β . For any $s \in S, a \in A$ and $s' \in S^\beta, a' \in A^\beta$ such that $\|s - s'\|_2 \leq \beta, \|a - a'\|_2 \leq \beta$ and $h \in [H]$,*

$$|\bar{Q}_h(s', a') - Q_h^*(s, a)| \leq \epsilon$$

with probability at least $1 - \delta$ when $\beta = \frac{\epsilon}{4LH}$, $N_{H-t} = \frac{8(t+1)^2(c')^2 H^2 k^2 \alpha^2 \log(2H|S||A|/\delta)}{\epsilon^2}$, $N_{H-t}^\# = \alpha^2 k^2 N_{H-t}$, and c' satisfies the inequality in Lemma 10 for all $t \in \{0, \dots, H-1\}$. Furthermore, at most $\tilde{O}\left(\frac{k^3 \alpha^2 H^{n+6}}{\epsilon^{n+2} \text{Vol}(B)}\right)$ number of samples are required with the same probability where B is the unit norm ball in $\mathbb{R}^n$.

Proof of Theorem 24. After discretizing the continuous MDP to get M^β for $\beta = \frac{\epsilon}{4LH}$, we note that $|S^\beta|, |A^\beta| \in O(\frac{H^n}{\epsilon^n \text{Vol}(B)})$ from Lemma 23. Since the required assumptions for Theorem 6 hold on M^β , it follows that each $\bar{Q}_h$ is $\epsilon(H-h+1)/H$ -optimal for all $h \in [H]$ on M^β when running LR-MCPI with $N_{H-t} = \frac{8(t+1)^2(c')^2 H^2 k^2 \alpha^2 \log(2H|S||A|/\delta)}{\epsilon^2}$, $N_{H-t}^\# = \alpha^2 k^2 N_{H-t}$ using at most $\tilde{O}\left(\frac{k^3 \alpha^2 H^{n+6}}{\epsilon^{n+2} \text{Vol}(B)}\right)$ samples with probability at least $1 - \delta$. Since $\beta = \frac{\epsilon}{4LH}$, from Lemma 13, for any $s \in S, a \in A$ and

$s' \in S^\beta, a' \in A^\beta$ such that $\|s - s'\|_2 \leq \beta, \|a - a'\|_2 \leq \beta$ and for all $t \in \{0, \dots, H-1\}$,

$$\begin{aligned} |\bar{Q}_{H-t}(s', a') - Q_{H-t}^*(s, a)| &\leq |\bar{Q}_{H-t}(s', a') - Q_{H-t}^\beta(s', a')| + |Q_{H-t}^\beta(s', a') - Q_{H-t}^*(s, a)| \\ &\leq \frac{\epsilon}{2} + 2L(t+1)\beta \\ &\leq \epsilon. \end{aligned}$$

Hence, an ϵ -optimal Q function on the continuous space is $\bar{Q}_h^c(s, a) = \bar{Q}_h(s', a')$, where (s', a') is the discretized state-action pair closest to (s, a) . $\square$

Proof of Theorem 14. After discretizing the continuous MDP to get M^β for $\beta = \frac{\epsilon}{4LH}$, we note that $|S^\beta|, |A^\beta| \in O(\frac{H^n}{\epsilon^n \text{Vol}(B)})$ from Lemma 23. Since Assumption 4 holds on M^β , from Theorem 7, it follows that each $\bar{Q}_h$ is $\epsilon/2$ -optimal for all $h \in [H]$ on M^β when running LR-EVI with $N_{H-t} = \frac{4(t+1)^2(c')^2 k^2 \alpha^2 H^2 \log(2H|S||A|/\delta)}{\epsilon^2}$, $N_{H-t}^\# = \frac{4(t+1)^2(c')^2 k^4 \alpha^4 H^2 \log(2H|S||A|/\delta)}{\epsilon^2}$ using at most $\tilde{O}\left(\frac{k^3 \alpha^2 H^{n+5}}{\epsilon^{n+2} \text{Vol}(B)}\right)$ samples with probability at least $1 - \delta$. Since $\beta = \frac{\epsilon}{4LH}$, from Lemma 13, for any $s \in S, a \in A$ and $s' \in S^\beta, a' \in A^\beta$ such that $\|s - s'\|_2 \leq \beta, \|a - a'\|_2 \leq \beta$ and for all $t \in \{0, \dots, H-1\}$,

$$\begin{aligned} |\bar{Q}_{H-t}(s', a') - Q_{H-t}^*(s, a)| &\leq |\bar{Q}_{H-t}(s', a') - Q_{H-t}^\beta(s', a')| + |Q_{H-t}^\beta(s', a') - Q_{H-t}^*(s, a)| \\ &\leq \frac{\epsilon}{2} + 2L(t+1)\beta \\ &\leq \epsilon. \end{aligned}$$

Hence, an ϵ -optimal Q function on the continuous space is $\bar{Q}_h^c(s, a) = \bar{Q}_h(s', a')$, where (s', a') is the discretized state-action pair closest to (s, a) . $\square$

A.3.9 Proofs for Infinite-Horizon Discounted MDPs

In this section, we present the omitted proofs from Appendix 2.10. We first prove that for any estimate of the value function, $r + P\hat{V}_t$ has rank that is at most d .

Proof of Proposition 15. Let MDP $M = (S, A, P, R, \gamma)$ satisfy Assumption 7. For the Tucker rank $(|S|, |S|, d)$ case, it follows that

$$\begin{aligned} r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[\hat{V}(s')] &= \sum_{i=1}^d W(s, i) V(a, i) + \gamma \sum_{s' \in S} \sum_{i=1}^d U(s', s, i) V(a, i) \hat{V}(s') \\ &= \sum_{i=1}^d V(a, i) \left(W(s, i) + \gamma \sum_{s' \in S} U(s', s, i) \hat{V}(s') \right) \end{aligned}$$

Thus, $r + \gamma[P\hat{V}]$ has rank upper bounded by d , and the Tucker rank $(|S|, d, |A|)$ case follows the same steps. $\square$

To prove the correctness of LR-EVI for the infinite-horizon setting, we first show that the error of the Q -function decreases in each iteration, Lemma 17.

Proof of Lemma 17. Let $Q'_{t+1} = r + \gamma P \bar{V}_t$ and $t \in [T - 1]$. From proposition 15, Q'_{t+1} has rank at most d for all $t \in [T - 1]$. Following step 1 from LR-EVI, $\hat{Q}_{t+1}(s, a) = \frac{1}{N_{t+1}} \sum_{i=1}^{N_{t+1}} R(s, a) + \gamma \bar{V}_t(s'_i)$ for all $(s, a) \in \Omega_{t+1}$. Hence, $\hat{Q}_{t+1}(s, a)$ is an unbiased estimate of $Q'_{t+1}(s, a)$ for all $(s, a) \in \Omega_t$. Furthermore, because of bounded rewards, $\hat{Q}_{t+1}(s, a) \in [0, \frac{1}{1-\gamma}]$ is a bounded random variable. With our choice of $N_{t+1} = \frac{2(c')^2 k^2 \alpha^2 \log(2T|S||A|/\delta)}{(1-\gamma)^4 B_t^2}$, $N_{t+1}^\# = N_{t+1} \alpha^2, k^2$, it follows from Hoeffding's inequality that for all $(s, a) \in \Omega_{t+1}$,

$$\begin{aligned} |\hat{Q}_{t+1}(s, a) - Q'_{t+1}(s, a)| &\leq \frac{(1-\gamma)B_t}{2c'\alpha k} \quad \forall (s, a) \in \Omega_{t+1} \\ |\hat{Q}_{t+1}(s, a) - Q'_{t+1}(s, a)| &\leq \frac{(1-\gamma)B_t}{2c'\alpha^2 k^2} \quad \forall (s, a) \in \Omega_{t+1}^\# \end{aligned}$$

with probability at least $1 - \frac{\delta}{T|S||A|}$. Step 2 of LR-EVI gives that for all $(s, a) \in S \times A$

$$|\bar{Q}_{t+1}(s, a) - Q'_{t+1}(s, a)| \leq \frac{(1-\gamma)B_t}{2}$$

from Lemma 10. Hence, for all $(s, a) \in S \times A$,

$$\begin{aligned}
|\bar{Q}_{t+1}(s, a) - Q^*(s, a)| &\leq |\bar{Q}_{t+1}(s, a) - Q'_{t+1}(s, a)| + |Q'_{t+1}(s, a) - Q^*(s, a)| \\
&\leq \frac{(1 - \gamma)B_t}{2} + |\gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[\bar{V}_t(s') - V^*(s')]| \\
&\leq \frac{(1 - \gamma)B_t}{2} + \gamma B_t \\
&= \frac{(1 + \gamma)B_t}{2}.
\end{aligned}$$

From step 4 of LR-EVI, the estimate of the value function is defined as $\bar{V}_{t+1}(s) = \max_{a \in A} \bar{Q}_{t+1}(s, a)$ for all $s \in S$. It follows that $|\bar{V}_{t+1}(s) - V^*(s)| \leq \frac{(1 + \gamma)B_t}{2}$. $\square$

Proof of Theorem 16.] Since the value function estimate is initialized as the zero vector, $|\bar{V}_0 - V^*|_\infty \leq \frac{1}{1 - \gamma} = B_0$. We prove the correctness of this algorithm by repeatedly applying Lemma 17 T times. From the union bound, the $\bar{Q}_T$ that the algorithm returns satisfies

$$|\bar{Q}_T(s, a) - Q^*(s, a)| \leq \left(\frac{1 + \gamma}{2}\right)^T \left(\frac{1}{1 - \gamma}\right)$$

with probability at least $1 - \delta$. With $T = \frac{\ln(\epsilon(1 - \gamma))}{\ln(\frac{1 + \gamma}{2})}$, it follows that $(\frac{1 + \gamma}{2})^T (\frac{1}{1 - \gamma}) = \epsilon$, so

$$|\bar{Q}_T(s, a) - Q^*(s, a)| \leq \epsilon$$

with probability at least $1 - \delta$. Note that since B_t is strictly decreasing with respect to t , it follows that $N_t = \frac{2(c')^2 \alpha^2 k^2 \log(2T/\delta)}{(1 - \gamma)^4 B_{t-1}^2}$, $N_t^\# = \alpha^2 k^2$ are strictly increasing with respect to t . Furthermore, since $B_{T-1} > \epsilon$, $N_t \in \tilde{O}\left(\frac{\alpha^2 k^2}{(1 - \gamma)^4 \epsilon^2}\right)$ for all $t \in [T]$. It follows that the sample complexity of the algorithm is

$$\tilde{O}\left(\frac{\alpha^2 k^3 (|S| + |A|)}{\epsilon^2 (1 - \gamma)^{-4}} + \frac{\alpha^4 k^6}{\epsilon^2 (1 - \gamma)^{-4}}\right).$$

$\square$

A.3.10 Proofs for LR-EVI with Matrix Estimation using Nuclear Norm Regularization

In this subsection, we present the omitted proofs from Section 2.11. We first prove a lemma that gives us the matrix estimation guarantee in our desired form.

Lemma 25. *Assume that for any ϵ -optimal value function $\hat{V}_{h+1}$, the matrix corresponding to $[r_h + [P_h \hat{V}_{h+1}]]$ is rank d , μ -incoherent, and has condition number bounded by κ . Then, for*

$$p_h = \frac{\mu^3 d^2 \kappa^2 H^4 C_{cvx}^2 \log(n)}{\epsilon^2 n},$$

where C_{cvx} is defined as in Theorem 18 with probability $1 - O(n^{-3})$, we have

$$\|\hat{Q}_h - r_h + [P_h \hat{V}_{h+1}]\|_\infty \leq \frac{\epsilon}{H}.$$

Proof of Lemma 25. Since $Q_h(s, a)$ is bounded by $H - h$, the estimates in Step 2 of LR-EVI-cvx are bounded random variables. Hence, they are unbiased with sub-Gaussian parameter $H - h$ [66]. Let $Q' = r_h + [P_h \hat{V}_{h+1}]$. From Theorem 18, with probability $1 - O(n^{-3})$,

$$\|\bar{Q}_h - Q'_h\|_\infty \leq \frac{C_{cvx}(H - h)}{\sigma_r(Q'_h)} \sqrt{\frac{\mu n \log n}{p_h}} \|Q'\|_\infty.$$

Let Q' have singular value decomposition $U\Sigma V^T$. Then, for $(s, a) \in S \times A$,

$$\begin{aligned} |Q'_h(s, a)| &= |e_s^T U \Sigma V^T e_a| \\ &\leq \|U(s)\|_2 \|\Sigma\|_{op} \|V(a)\|_2 \\ &\leq \frac{\mu d}{n} \sigma_1(Q'_h) \\ &\leq \frac{\mu d \kappa}{n} \sigma_d(Q'_h) \end{aligned}$$

where the second inequality comes from incoherence and the last inequality comes from bounded condition number. Plugging this inequality into the application of Theorem 18 gives

$$\|\bar{Q}_h - Q'_h\|_\infty \leq \mu dk C_{\text{cvx}}(H - h) \sqrt{\frac{\mu \log n}{p_h n}}.$$

From our choice of p_h , we get the desired result. $\square$

Next, we prove a helper lemma that follows the same steps as the helper lemmas needed to prove Theorems 5, 6, and 7. Similar lemmas can be proved in the suboptimality gap or all ϵ -optimal π have low-rank Q^π setting.

Lemma 26. *Let ϵ , p_{H-t} , and λ be defined as in Theorem 19. Then, the learned policy and action-value function estimate satisfy*

$$\|\bar{Q}_{H-t} - Q_{H-t}^*\|_\infty \leq \frac{\epsilon(t+1)}{H}, \quad \|\bar{Q}_{H-t} - Q_{H-t}^{\hat{\pi}}\|_\infty \leq \frac{\epsilon(t+1)}{H}$$

with probability at least $1 - O((t+1)n^{-3})$ for all $t \in \{0, \dots, H_1\}$.

Proof of Lemma 26. We prove this with induction on t . At step $t = 0$, it follows that from Lemma 25 with probability $1 - O(n^{-3})$,

$$\|\bar{Q}_H - Q_H^*\|_\infty \leq \frac{\epsilon}{H}.$$

Since $Q_H^* = Q_H^{\hat{\pi}}$, the base case holds.

Let $x \in [H - 1]$. Assume that the inductive hypothesis,

$$\|\bar{Q}_{H-x} - Q_{H-x}^*\|_\infty \leq \frac{\epsilon(x+1)}{H}, \quad \|\bar{Q}_{H-x} - Q_{H-x}^{\hat{\pi}}\|_\infty \leq \frac{\epsilon(x+1)}{H}$$

with probability at least $1 - O((x+1)n^{-3})$, holds. Following the steps of LR-EVI with the convex program based matrix estimation method, it follows that with probability $1 - O(n^{-3})$,

$$\|\bar{Q}_{H-s-1} - Q'_{H-s-1}\|_\infty \leq \frac{\epsilon}{H}$$

where $Q'_{H-x-1} = r_{H-x-1} + P_{H-x-1} \hat{V}_{H-x}$. The union bound asserts that the above error guarantee holds with probability at least $1 - O((x+2)n^{-3})$. Hence, for all $(s, a) \in S \times A$,

$$\begin{aligned}
|\bar{Q}_{H-x-1}(s, a) - Q^*_{H-x-1}(s, a)| &\leq |\bar{Q}_{H-x-1}(s, a) - Q'_{H-x-1}(s, a)| + |Q'_{H-x-1}(s, a) - Q^*_{H-x-1}(s, a)| \\
&\leq \frac{\epsilon}{H} + |E_{s' \sim P_{H-x-1}(\cdot|s, a)}[\hat{V}_{H-x}(s') - V^*_{H-x}(s')]| \\
&\leq \frac{\epsilon}{H} + |E_{s' \sim P_{H-h}(\cdot|s, a)}[(x+1)\epsilon/H]| \\
&= \frac{(x+2)\epsilon}{H}.
\end{aligned}$$

Following the same steps,

$$\begin{aligned}
|\bar{Q}_{H-x-1}(s, a) - \hat{Q}^{\hat{\pi}}_{H-x-1}(s, a)| &\leq |\bar{Q}_{H-x-1}(s, a) - Q'_{H-x-1}(s, a)| + |Q'_{H-x-1}(s, a) - \hat{Q}^{\hat{\pi}}_{H-x-1}(s, a)| \\
&\leq \frac{\epsilon}{H} + |E_{s' \sim P_{H-x-1}(\cdot|s, a)}[\hat{V}_{H-x}(s') - V^{\hat{\pi}}_{H-x}(s')]| \\
&\leq \frac{\epsilon}{H} + |E_{s' \sim P_{H-h}(\cdot|s, a)}[(x+1)\epsilon/H]| \\
&= \frac{(x+2)\epsilon}{H}.
\end{aligned}$$

Hence, from mathematical induction, the lemma holds. $\square$

We now present the proof of the main result of this section. Similarly, the same steps can be used to prove similar results in our other low-rank settings.

Proof of Theorem 19. We prove the correctness of the algorithm by applying Lemma 26 at time step 1, which occurs with probability at least $1 - O(Hn^{-3})$. Next, the number of samples used is $\sum_{t=0}^{H-1} |\Omega_{H-t}|$. By definition of our sampling procedure, $k = |\Omega_h| \sim \text{Bin}(n^2, p_h)$. Hence, from the one-sided Bernstein's inequality

ity, Proposition 28, for $h \in [H]$ and $C'' = \frac{\sqrt{8}}{C_{\text{vxx}} \sqrt{3}}$, it follows that

$$\begin{aligned} \mathbb{P}(|\Omega_h| - \mathbb{E}[|\Omega_h|] \geq C'' p_h n^2) &\leq \exp\left(-\frac{p_h^2 (C'')^2 n^2}{2(p_h + \frac{p_h C''}{3})}\right) \\ &\leq \exp\left(-\frac{3p_h C'' n^2}{8}\right) \\ &\leq \exp\left(-\mu^3 d^2 \kappa^2 H^4 n \log(n)/\epsilon^2\right). \end{aligned}$$

Since $\mathbb{E}[|\Omega_h|] = n^2 p_h = C_{\text{vxx}} \mu^3 d^2 \kappa^2 H^4 n \log(n)/\epsilon^2$, from the union bound, it follows that $|\Omega_h| \in O(H^4 n \log(n)/\epsilon^2)$ for all $h \in [H]$ with probability at least $1 - \exp\left(-\mu^3 d^2 \kappa^2 H^4 n \log(n)/\epsilon^2\right)$. Hence, the sample complexity is upper bounded by

$$\sum_{t=0}^{H-1} |\Omega_{H-t}| \in \tilde{O}\left(\frac{\mu^3 H^5 n}{\epsilon^2}\right)$$

with probability at least $1 - O(Hn^{-3}) - \exp\left(-\mu^3 d^2 \kappa^2 H^4 n \log(n)/\epsilon^2\right)$.

□

A.4 Additional Theorems for Reference

We present the following lemmas, propositions, and theorems for the readers' convenience.

Theorem 27 (Hoeffding's Inequality [66]). *Let $X_1, \dots, X_n$ be independent, and X_i have mean μ_i and sub-Gaussian parameter σ_i . Then, for all $t \geq 0$, we have*

$$\mathbb{P}\left[\sum_{i=1}^n (X_i - \mu_i) \geq t\right] \leq \exp\left(-\frac{t^2}{2 \sum_{i=1}^n \sigma_i^2}\right).$$

Proposition 28 (Proposition 2.14 (One-sided Bernstein's Inequality) [66]). *Given n independent random variables such that $X_i \leq b$ almost surely, we have*

$$\mathbb{P}\left(\sum_{i=1}^n (X_i - \mathbb{E}[X_i]) \geq cn\right) \leq \exp\left(-\frac{nc^2}{2(\frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i^2] + \frac{bc}{3})}\right).$$

Theorem 29 (Matrix Bernstein [62]). *Let $X^{(1)}, \dots, X^{(n)} \in \mathbb{R}^{d_1 \times d_2}$ be independent zero-mean matrices satisfying*

$$\|X^{(i)}\|_{op} \leq b, \quad a.s.$$

$$\max\{\|\sum_{i=1}^n \mathbb{E}[X^{(i)\top} X^{(i)}]\|_{op}, \|\sum_{i=1}^n \mathbb{E}[X^{(i)} X^{(i)\top}]\|_{op}\} \leq n\sigma^2.$$

Then

$$\mathbb{P}\left(\left\|\sum_{i=1}^n X^{(i)}\right\|_{op} \geq t\right) \leq (d_1 + d_2) \exp\left(-\frac{t^2}{2(n\sigma^2 + \frac{bt}{3})}\right).$$

Theorem 30 (Singular Value Courant-Fischer Minimax Theorem (Theorem 7.3.8 [29])). *Let $A \in \mathbb{R}^{m \times n}$, and $q = \min(m, n)$, let $\sigma_1(A), \sigma_2(A), \dots, \sigma_q(A)$ be the ordered singular values of A , and let $k \in [q]$. Then,*

$$\sigma_k(A) = \min_{S: \dim(S)=m-k+1} \max_{x: 0 \neq x \in S} \frac{\|Ax\|_2}{\|x\|_2}$$

and

$$\sigma_k(A) = \max_{S: \dim(S)=k} \min_{x: 0 \neq x \in S} \frac{\|Ax\|_2}{\|x\|_2}.$$

APPENDIX B
APPENDICES FOR CHAPTER 3

B.1 Notation Table

Symbol	Definition
$\mathcal{S}, S$	State space, Cardinality of $\mathcal{S}$
$\mathcal{A}, A$	Action space, Cardinality of $\mathcal{A}$
H, K	Finite Horizon, Number of Episodes ($T = KH$)
M	Number of Source MDPs
$r_{m,h}, P_{m,h}$	Reward Function, Transition Kernel for Source MDP m at step h
r_h, P_h	Reward Function, Transition Kernel for the Target MDP at step h
$Q_{m,h}^\pi, Q_h^\pi$	Action-value Function at Step h Following π for Source MDP m and the Target MDP
$Q_{m,h}^*, Q_h^*$	Optimal Action-value Function at Step h for Source MDP m and the Target MDP
$V_{m,h}^\pi, V_h^\pi$	Value Function at Step h Following π for Source MDP m and the Target MDP
$V_{m,h}^*, V_h^*$	Optimal Value Function at Step h for Source MDP m and the Target MDP
d	Rank of the Low-dimensional Mode of $P_{m,h}$
d'	Rank of the Low-dimensional Mode of P_h
α	Transfer-ability Coefficient
$G_{m,h}(a), G_h(a)$	True Latent Representation of Each Action in the Source and Target MDPs in the (S, S, d) Tucker Rank Setting
$\tilde{G}_h$	Concatenation of $G_{m,h}$ for $m \in [M]$
$\hat{G}_h$	Latent Representation Estimate Computed in the Source Phase
LR-EVI($\cdot$)	Low Rank Empirical Value Iteration [54]
Λ_h^s	Gram Matrix for State s at Step h
w_h^s	Regularized Least Squares Solution for State s at Step h
$T_{k,h}^s$	Episodes before Episode k , in which one was at State s at Step h
$\mu(X)$	Incoherence of Matrix X
$\kappa(X)$	Condition Number of Matrix X
$\sigma_i(X)$	i -th Singular Value of X
$\ X\ _{op}$	Operator Norm of Matrix X
$\ X\ _\infty$	Maximum of the Absolute Value of Each Entry of X

Table B.1: Notation Table

B.2 Table of Regret Bounds with Known Latent Representations

	Tucker Rank	Regret Bound with Known Latent Representation
Theorem 14	(S, S, d)	$\tilde{O}(\sqrt{d^3 H^3 S T})$
Theorem 6	(S, d, A)	$\tilde{O}(\sqrt{d^3 H^3 A T})$
Theorem 3.1 [37]	(d, S, A)	$\tilde{O}(\sqrt{d^3 H^3 T})$
Theorem 5.1 [28], Theorem 5.6 [79]	(d, S, A)	$\tilde{\Theta}(\sqrt{d^2 H^2 T})$

Table B.2: Theoretical guarantees of our algorithms alongside results from literature in the different Tucker rank settings. To the best of our knowledge, there are no algorithms tailored to the (d, d, d) Tucker rank setting. Instead, one should use a linear MDP algorithm.

Since each Tucker rank setting captures tabular MDPs, we can construct lower bounds with a reduction from the bound in [34]; in the (S, d, A) Tucker rank setting, the regret is at least $\tilde{\Omega}(\sqrt{d A H^2 T})$, and in the (S, S, d) setting, the regret is at least $\tilde{\Omega}(\sqrt{d A H^2 T})$. As mentioned, our dependence on d and H are sub optimal because we modify LSVI-UCB [37], which has a sub-optimal dependence on d and H . Modifying the algorithm from [28] to achieve the optimal dependence on d and H in the (S, S, d) and (S, d, A) Tucker rank settings is an interesting problem for future research.

B.3 Thresholding Procedure in the (S, S, d) Tucker Rank Setting

When the feature representations of each source MDP lies in orthogonal d -dimensional subspaces, we cannot discard any unique dimension of the M subspaces as we cannot tell which of the d -dimensions match the ones in the target

MDP without interacting with it. However, when the feature mappings from the source MDPs lie in intersecting subspaces, i.e.,

$$\text{rank}(\tilde{G}_h) = \text{rank}\left(\sqrt{\frac{A}{d\mu}} \begin{bmatrix} G_{1,h} & \dots & G_{M,h} \end{bmatrix}\right) = d''$$

for $d' \leq d'' \leq dM$ (recall the target MDP has Tucker rank (S, S, d')), we use the following procedure: after computing $\tilde{G}_h$ via concatenation of $\hat{G}_{m,h}$ for all $m \in [M]$, we perform a singular value decomposition of $\tilde{G}_h$ and threshold the singular values to keep only d'' dimensions. With this procedure, our regret bound now depends on d'' instead of dM at the cost of an increased source sample complexity. We only present the theorem and proof in the (S, S, d) Tucker rank setting, but our results and methods extend to the other Tucker rank settings.

Theorem 3. *Suppose Assumptions 8, 9, and 10 hold, and assume the source MDP latent factors span a d'' -dimensional space. Let $\delta \in (0, 1)$. Furthermore, assume that, for any $\epsilon \in (0, \sqrt{SH/T})$, $Q_{m,h} = r_{m,h} + P_{m,h}V_{m,h+1}$ has rank d and is μ -incoherent with condition number κ for all ϵ -optimal value functions V_{h+1} . Let $\lambda = 1$ and $\beta_{k,h}^s$ be a function of $d, H, |T_{k,h}^s|, |S|, M, T$. Then, for $T \geq \frac{S}{\alpha^2}$, using at most $\tilde{O}(d^7\mu^5\kappa^4(A+A^2)M^3H^4\alpha^2T/(Sd''))$ samples in the source problems, our algorithm has regret $\tilde{O}(\sqrt{(d''H)^3ST})$ with probability at least $1 - \delta$.*

The proof of Theorem 3 follows the proof of Theorem 2, except we require a larger source sample complexity to guarantee with high probability that the $d'' + 1$ -th singular value of the concatenated feature representation is sufficiently small. With this procedure, we ensure that our feature representation removes the unneeded dimensions. See Appendix B.9.1 for proof of the above theorem.

B.3.1 Thresholding Procedure without Knowledge of d''

While the above procedure requires knowledge of the dimension of the subspace spanned by the latent features of the source MDPs, one can also threshold small singular values of the concatenated feature mapping as the misspecification error will still be small. The specific procedure is as follows: after performing the singular value decomposition on the concatenation of $\hat{G}_{m,h}$, we first find the smallest value of $t \in [dM]$ that satisfies

$$\sigma_{t+1} \leq \sqrt{\frac{tHSd\mu}{\alpha^2 T(dM-t)M^2 A}},$$

and threshold $\sigma_{t+1}, \dots, \sigma_{dM}$. Let

$$\mathcal{T} = \left\{ t \in [dM] \mid \sigma_{t+1} \leq \sqrt{\frac{tHSd\mu}{\alpha^2 T(dM-t)M^2 A}} \right\}$$

be the set of t that satisfies the inequality on the singular values. Then, with the specified procedure, our algorithm admits a target regret bound that depends on t instead of dM . Note that t is minimally 1 and at most dM .

Theorem 4. *Suppose Assumptions 8, 9, and 10 hold. Let $\delta \in (0, 1)$ and $t = \min_{t' \in \mathcal{T}} t'$. Furthermore, assume that, for any $\epsilon \in (0, \sqrt{SH/T})$, $Q_{m,h} = r_{m,h} + P_{m,h}V_{m,h+1}$ has rank d and is μ -incoherent with condition number κ for all ϵ -optimal value functions V_{h+1} . Let $\lambda = 1$ and $\beta_{k,h}^s$ be a function of $d, H, |T_{k,h}^s|, |S|, M, T$. Then, for $T \geq \frac{S}{\alpha^2}$, using at most $\tilde{O}(d^5 \mu^5 \kappa^4 (1 + A/S) M^3 H^4 \alpha^2 T)$ samples in the source problems, our algorithm has regret $\tilde{O}(\sqrt{(tH)^3 ST})$ with probability at least $1 - \delta$.*

With an increase in a factor of dM to the source sample complexity, we can potentially improve the regret bound of the target by decreasing the dimension of the feature representation from dM to t . See Appendix B.9.1 for the proof of the above theorem.

B.4 Properties of the Transfer-ability Coefficient α

In this section, we prove properties of α and provide easy and hard instances.

We first bound the values of α .

Lemma 1. *Let α be defined according to Definition 5. Then, we have that $\frac{1}{dM} \leq \alpha < \infty$.*

Proof. Since Assumption 12 holds and that the latent factors are normalized, it follows that for all $i \in [d']$ and any latent factor matrix,

$$\begin{aligned} \|F_h(:, i)\|_2 &= \left\| \sum_{j \in [d], m \in [M]} B_h(i, j, m) F_{m,h}(:, j) \right\|_2 \\ 1 &\leq \sum_{j \in [d], m \in [M]} \|B_h(i, j, m) F_{m,h}(:, j)\|_2 \\ &\leq \sum_{j \in [d], m \in [M]} |B_h(i, j, m)| \|F_{m,h}(:, j)\|_2 \\ &= \sum_{j \in [d], m \in [M]} |B_h(i, j, m)|. \end{aligned}$$

Given that $1 \leq \sum_{j \in [d], m \in [M]} |B_h(i, j, m)|$, it follows that the minimum of $\max_{j \in [d], m \in [M]} |B_h(i, j, m)|$ is attained when $|B_h(i, j, m)| = \frac{1}{dM}$. Thus, $\alpha \geq \frac{1}{dM}$.

Let $\gamma \in (0, 1)$. Consider the construction where the target latent factor is $F_T = [1, 0]$ and the source latent factors from source MDPs one and two are

$$F_1 = [1 - \gamma, \gamma \sqrt{2/\gamma - 1}], \quad F_2 = [1 - \gamma, -\gamma \sqrt{2/\gamma - 1}].$$

Clearly, the latent factors are orthonormal. Then, it follows that the only linear combination of F_1 and F_2 that results in F is

$$F_T = cF_1 + cF_2,$$

which implies that $c = \frac{1}{2-2\gamma}$. Thus, $\alpha = \frac{1}{2-2\gamma}$, and it follows that we can make α arbitrarily large because for any positive constant C , we can choose γ so that $\alpha > C$. $\square$

As the proof of Lemma 1 uses a construction in which α can be arbitrarily large, we next provide simple constructions where α is small. In the rank one case where all source latent factors equal the target latent factor, α attains its minimum of $\alpha = \frac{1}{M}$. Next, we consider the construction where

$$F_T = \left[\frac{1}{\sqrt{d}}, \dots, \frac{1}{\sqrt{d}} \right],$$

and there are M source MDPs with the standard basis vectors in $\mathbb{R}^d$ as the latent factors equally distributed among the M rank one source MDPs. Thus,

$$F_T = \sum_{m \in M} \frac{\sqrt{d}}{M} F_m,$$

and $\alpha = \sqrt{d}/M$.

We next present a transfer RL instance with large α . Consider the following construction with the goal of trying to transfer the state latent factors. Without loss of generality, let the size of the state space and action space be $4n$ for some $n \in \mathbb{N}_+$. Furthermore, we refer to the i -th quarter of the states as s_i for $i \in [4]$ and the i -th half of the actions as a_i for $i \in [2]$ because our construction is a block MDP with four latent states and two latent actions. Let $H = 2$. In each MDP, the transition kernel does not vary with h while the reward function is time dependent. Assume that the transition kernels have Tucker rank (S, d, A) (Assumption 11). For ease of notation, we refer to the i -th row as any state in s_i and the i -th column as any state in a_i . The target MDP's latent factors are

$$F_h = \begin{bmatrix} \sqrt{1/(2n)} & 0 \\ \sqrt{1/(2n)} & 0 \\ 0 & \sqrt{1/(2n)} \\ 0 & \sqrt{1/(2n)} \end{bmatrix}, \quad W_1 = \begin{bmatrix} \sqrt{1/(2n)} & \sqrt{1/(8n)} \\ \sqrt{1/(2n)} & \sqrt{1/(8n)} \end{bmatrix}, \quad W_2 = \begin{bmatrix} \sqrt{1/(8n)} & \sqrt{1/(2n)} \\ \sqrt{1/(8n)} & \sqrt{1/(2n)} \end{bmatrix}$$

for all $h \in [H]$, and

$$U_h(s_1|a_1) = U_h(s_2|a_1) = [3\sqrt{2}/(8\sqrt{n}), \sqrt{2}/(8\sqrt{n})]$$

$$U_h(s_1|a_2) = U_h(s_2|a_2) = [\sqrt{2}/(8\sqrt{n}), 3\sqrt{2}/(8\sqrt{n})]$$

$$U_h(s_3|a_1) = U_h(s_4|a_1) = [3\sqrt{2}/(8\sqrt{n}), \sqrt{2}/(8\sqrt{n})]$$

$$U_h(s_3|a_2) = U_h(s_4|a_2) = [\sqrt{2}/(8\sqrt{n}), 3\sqrt{2}/(8\sqrt{n})].$$

Thus, the reward functions and transition kernels for the target MDP are

$$r_1 = \begin{bmatrix} \frac{1}{2n} & \frac{1}{2n} \\ \frac{1}{2n} & \frac{1}{2n} \\ \frac{1}{4n} & \frac{1}{4n} \\ \frac{1}{4n} & \frac{1}{4n} \end{bmatrix} \quad r_2 = \begin{bmatrix} \frac{1}{4n} & \frac{1}{4n} \\ \frac{1}{4n} & \frac{1}{4n} \\ \frac{1}{2n} & \frac{1}{2n} \\ \frac{1}{2n} & \frac{1}{2n} \end{bmatrix}$$

and the transition kernels are

$$P_h(s_1|\cdot, \cdot) = P_h(s_2|\cdot, \cdot) = \begin{bmatrix} \frac{3}{8n} & \frac{1}{8n} \\ \frac{3}{8n} & \frac{1}{8n} \\ \frac{1}{8n} & \frac{3}{8n} \\ \frac{1}{8n} & \frac{3}{8n} \end{bmatrix} \quad P_h(s_3|\cdot, \cdot) = P_h(s_4|\cdot, \cdot) = \begin{bmatrix} \frac{1}{8n} & \frac{3}{8n} \\ \frac{1}{8n} & \frac{3}{8n} \\ \frac{3}{8n} & \frac{1}{8n} \\ \frac{3}{8n} & \frac{1}{8n} \end{bmatrix}$$

for all $h \in [H]$. Using the Bellman equations, it follows that the optimal Q function for the above MDP is

$$Q_1^* = \begin{bmatrix} \frac{13}{16n} & \frac{15}{16n} \\ \frac{13}{16n} & \frac{15}{16n} \\ \frac{11}{16n} & \frac{9}{16n} \\ \frac{11}{16n} & \frac{9}{16n} \end{bmatrix}, \quad Q_2^* = \begin{bmatrix} \frac{1}{4n} & \frac{1}{4n} \\ \frac{1}{4n} & \frac{1}{4n} \\ \frac{1}{2n} & \frac{1}{2n} \\ \frac{1}{2n} & \frac{1}{2n} \end{bmatrix}.$$

and both Q^* s have F_h as orthonormal latent factors. Next, we present the or-

thonormal latent factors for the first source MDP

$$F_{1,h} = \begin{bmatrix} 1/\sqrt{2n} & 1/\sqrt{4n} \\ -1/\sqrt{2n} & 1/\sqrt{4n} \\ 0 & 1/\sqrt{4n} \\ 0 & 1/\sqrt{4n} \end{bmatrix}, W_{1,1} = \begin{bmatrix} 3/\sqrt{4n} & 3/\sqrt{4n} \\ 1/\sqrt{4n} & 1/\sqrt{4n} \end{bmatrix}, W_{1,2} = \begin{bmatrix} 1/\sqrt{4n} & 1/\sqrt{4n} \\ 2/\sqrt{4n} & 2/\sqrt{4n} \end{bmatrix}.$$

It follows that the reward functions are

$$r_{1,1} = \begin{bmatrix} (3 + 3\sqrt{2})/(4n) & (1 + \sqrt{2})/(4n) \\ (3 - 3\sqrt{2})/(4n) & (1 - \sqrt{2})/(4n) \\ 3/(4n) & 1/(4n) \\ 3/(4n) & 1/(4n) \end{bmatrix}, r_{1,2} = \begin{bmatrix} (1 + \sqrt{2})/(4n) & (1 + \sqrt{2})/(2n) \\ (1 - \sqrt{2})/(4n) & (1 - \sqrt{2})/(2n) \\ 1/(4n) & 1/(2n) \\ 1/(4n) & 1/(2n) \end{bmatrix}.$$

with $P_{1,h}(\cdot|\cdot, \cdot) = 1/(4n)$. Since $P_{1,h}$ is the uniform transition kernel (which has the second column of $F_{1,h}$ as its latent factor), it follows that $PV_{1,2}^*(\cdot, \cdot)$ is a rank one matrix with every entry being the same positive constant. Thus, $Q_{1,1}^* = r_{1,1} + P_{1,1}V_{1,2}^*$ and $Q_{1,2}^* = r_{1,2}$ are rank two matrices with $F_{1,h}$ as latent factors.

The second source MDP has the following latent factors

$$F_{2,h} = \begin{bmatrix} \frac{1}{c\sqrt{2n}} + \frac{1}{c\beta\sqrt{4n}} & 1/\sqrt{4n} \\ -\frac{1}{c\sqrt{2n}} + \frac{1}{c\beta\sqrt{4n}} & 1/\sqrt{4n} \\ -1/(c\beta\sqrt{4n}) & 1/\sqrt{4n} \\ -1/(c\beta\sqrt{4n}) & 1/\sqrt{4n} \end{bmatrix}, W_{2,1} = \begin{bmatrix} 1/\sqrt{2n} & 0 \\ 0 & 1/\sqrt{2n} \end{bmatrix}, W_{2,2} = \begin{bmatrix} 0 & 1/\sqrt{2n} \\ 1/\sqrt{2n} & 0 \end{bmatrix}$$

where $c = \sqrt{1 + 1/\beta^2}$ for all $h \in [H]$ with $P_{2,h}(\cdot|\cdot, \cdot) = 1/(4n)$. Since $P_{2,h}$ is the uniform transition kernel (which has the second column of $F_{2,h}$ as its latent factor), it follows that $PV_{2,2}^*(\cdot, \cdot)$ is a rank one matrix with every entry being the same positive constant. Thus, $Q_{2,1}^* = r_{2,1} + P_{2,1}V_{2,2}^*$ and $Q_{2,2}^* = r_{2,2}$ are rank two matrices with $F_{2,h}$ as latent factors. Now, we prove that $\alpha \in \Omega(\beta)$.

Corollary 31. *Consider the above construction that satisfies Assumptions 11 and 12.*

Then, α defined in Definition 5 satisfies $\alpha \in \Omega(\beta)$.

Proof. Consider the above construction with $\beta > 0$. First, we show that when expressing the first column of F_h as a linear combination of the columns of $F_{1,h}$ and $F_{2,h}$, one coefficient is $\Omega(\beta)$.

We first note that one must use the first column in $F_{2,h}$ or else the system of linear equations is inconsistent. Thus, we solve

$$a \begin{bmatrix} 1/\sqrt{2n} \\ -1/\sqrt{2n} \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 1/\sqrt{4n} \\ 1/\sqrt{4n} \\ 1/\sqrt{4n} \\ 1/\sqrt{4n} \end{bmatrix} + d \begin{bmatrix} \frac{1}{c\sqrt{2n}} + \frac{1}{c\beta\sqrt{4n}} \\ -\frac{1}{c\sqrt{2n}} + \frac{1}{c\beta\sqrt{4n}} \\ -1/(c\beta\sqrt{4n}) \\ -1/(c\beta\sqrt{4n}) \end{bmatrix} = \begin{bmatrix} 1/\sqrt{22n} \\ 1/\sqrt{n} \\ 0 \\ 0 \end{bmatrix}$$

It follows that $d = bc\beta$. Thus, $b = 1/\sqrt{2}$, and $d \in \Omega(\beta)$ because $c \geq 1$.

To show that $\alpha \in \Omega(\beta)$, we assume for sake of contradiction that there exists some set of basis vectors $F'_{1,h}$ and $F_{2,h'}$ that span the subspace defined by $F_{1,h}$ and $F_{2,h}$, respectively, such that we can express the any rotation of the first column of F_h as a linear combination of the columns of $F'_{1,h}$ and $F_{2,h'}$ using coefficients a, b, c, d such that $|a|, |b|, |c|, |d| \in o(\beta)$. Since $F_h, F_{1,h}$, and $F_{2,h}$ are orthonormal latent factors, it follows that there exist rotation matrices R, R_1, R_2 with $\|R\|_\infty, \|R_1\|_\infty, \|R_2\|_\infty \in O(1)$ such that $F'_{1,h} = R_1 F_{1,h}$ and $F_{2,h'} = R_2 F_{2,h}$. It follows that R^{-1} exists and has entries with constant magnitude. By construction we have

$$F_h(:, 0) = aR^{-1}R_1F_{1,h}(:, 0) + bR^{-1}R_1F_{1,h}(:, 1) + cR^{-1}R_2F_{2,h}(:, 0) + dR^{-1}R_2F_{2,h}(:, 1).$$

Since the entries of the rotation matrices and their inverse are bounded by a constant, the above equation states that we can represent the first column of F_h as a linear combination of the columns of $F_{1,h}$ and $F_{2,h}$ using only constant coefficients. Thus, we reach a contradiction because we've shown that one entry

of $dR^{-1}R_2 \in \Omega(\beta)$. Therefore, $\alpha \in \Omega(\beta)$. $\square$

B.5 (S, d, A) Tucker Rank Setting

In this section, we present the assumptions, algorithm, and results for the (S, d, A) Tucker rank setting. In this setting, we assume the following structure on the source and target MDPs.

Assumption 11. *In each of the M source MDPs, the reward functions have rank d , and the transition kernels have Tucker rank (S, d, A) . Let the target MDP's reward functions have rank d' and transition kernels have Tucker rank (S, d', A) where $d' \leq dM$. Thus, there exists $S \times d \times A$ tensors $U_{m,h}$, $S \times d' \times A$ tensors U_h , $S \times d$ μ -incoherent matrices $F_{m,h}$, $S \times d'$ μ -incoherent matrices F_h with orthonormal columns, and $A \times d$ matrices $W_{m,h}$, $A \times d'$ matrices W_h such that*

$$P_{m,h}(s'|s, a) = \sum_{i \in [d]} F_{m,h}(s, i) U_{m,h}(s', a, i), \quad r_{m,h}(s, a) = \sum_{i \in [d]} F_{m,h}(s, i) W_{m,h}(a, i)$$

and

$$P_h(s'|s, a) = \sum_{i \in [d']} F_h(s, i) U_h(s', a, i), \quad r_h(s, a) = \sum_{i \in [d']} F_h(s, i) W_h(a, i)$$

where $\|\sum_{s' \in S} g(s') U_{m,h}(s', :, a)\|_2, \|\sum_{s' \in S} g(s') U_h(s', :, a)\|_2, \|W_h(a)\|_2, \|W_{m,h}(a)\|_2 \leq \sqrt{S/(d\mu)}$ for all $s \in S, a \in \mathcal{A}, h \in [H]$, and $m \in [M]$ for any function $g : S \rightarrow [0, 1]$.

Similarly, to allow transfer of a latent representation between the source and target MDPs, we assume that the subspace spanned by the set of all source latent factors contains the space spanned by the set of target latent factors.

Assumption 12. *Suppose Assumption 11 holds. The target MDP latent factors F_h and source MDP latent factors $F_{m,h}$ satisfy for all $h \in [H]$, $\text{Span}(\{F_h(:, i)\}_{i \in [d']}) \subseteq \text{Span}(\{F_{m,h}(:, i)\}_{i \in [d], m \in [M]})$.*

Before defining our transfer-ability coefficient, we first introduce the following notation. We define the set $\mathcal{B}_h(F_h, \{F_{h,m}\}_{m \in M})$ to contain all coefficients of the linear combinations, i.e., $B_h \in \mathbb{R}^{d', d, M}$ and $B_h \in \mathcal{B}_h(F_h, \{F_{h,m}\}_{m \in M})$ if for all $i \in [d'], h \in [H]$, $F_h(\cdot, i) = \sum_{j \in [d], m \in [M]} B_h(i, j, m) F_{m,h}(\cdot, j)$. Furthermore, as the source MDPs latent factors of $\mathcal{Q}_{m,h}^*$ are not unique, we define α to measure the difficulty given the best set of latent factors from the source MDPs. Thus, α precisely measures the challenge involved in transferring the latent representation.

Definition 5 (Transfer-ability Coefficient). *Given a transfer RL problem that satisfies Assumptions 11, 12, and 10, let $\mathcal{F}$ be the set of all $\mathbb{R}^{S \times d}$ matrices with orthonormal columns that span the column space of $\mathcal{Q}_{m,h}^*$ for all $m \in [M], h \in [H]$. Then, we define α as*

$$\alpha := \max_{h \in [H]} \min_{F \in \mathcal{F}} \min_{B \in \mathcal{B}_h(F_h, F)} \max_{i \in [d'], j \in [d], m \in [M]} |B(i, j, m)|.$$

We now prove our lower bound in this Tucker rank setting.

B.5.1 Information Theoretic Lower Bound

To formalize the importance of α in transfer learning, we prove a lower bound that shows that a dependence on α in the sample complexity of the source phase is necessary to benefit from transfer learning.

Theorem 5. *There exist two transfer RL instances such that (i) they satisfy Assumptions 11 and 12, (ii) they cannot be distinguished without observing $\Omega(\alpha^2)$ samples in the source phase, and (3) they have target state latent features that are orthogonal to each other.*

Proof. Suppose all source and target MDPs (S, A, P, H, r) share the same state

space, action space, and horizon $H = 1$ and assume that $\mathcal{S} = \mathcal{A} = [2n]$ for some $n \in \mathbb{N}_+$. For ease of notation, we let s_1 and a_1 refer to any state and action in $[n]$, respectively, and s_2 and a_2 refer to any state and action in $\{n + 1, \dots, 2n\}$, respectively. We will present the latent factors and optimal Q functions as block vectors and matrices with s_1, s_2, a_1, a_2 as the blocks containing n entries. The initial state distribution in the target MDP is uniform over $\mathcal{S}$. We now present two transfer RL problems with similar Q functions (with rows s_1, s_2 and columns a_1, a_2) that satisfy Assumptions 11 and 12 but have orthogonal target state latent factors. For ease of notation, the superscript i of $Q_{m,h}^{*,i}, Q_h^{*,i}$ denotes transfer RL problem. Then, the optimal Q functions for transfer RL problem one are

$$Q_{1,1}^{*,1} = n \begin{bmatrix} \sqrt{1/(2n)} \\ \sqrt{1/(2n)} \end{bmatrix} \begin{bmatrix} \sqrt{1/n} & 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 0 \end{bmatrix}, \quad Q_{2,1}^{*,1} = n \begin{bmatrix} \sqrt{1/(2n)} \\ \sqrt{1/(2n)} \end{bmatrix} \begin{bmatrix} 0 & \sqrt{1/n} \end{bmatrix} = \begin{bmatrix} 0 & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} \end{bmatrix},$$

$$Q_1^{*,1} = n \begin{bmatrix} \sqrt{1/(2n)} \\ \sqrt{1/(2n)} \end{bmatrix} \begin{bmatrix} \sqrt{\frac{1}{n}} & -\sqrt{\frac{1}{n}} \end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 \\ 1/2 & -1/2 \end{bmatrix},$$

and the optimal Q functions for transfer RL problem two are

$$Q_{1,1}^{*,2} = n \begin{bmatrix} \sqrt{1/(2n)} \\ \sqrt{1/(2n)} \end{bmatrix} \begin{bmatrix} \sqrt{1/n} & 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 0 \end{bmatrix} \quad Q_{2,1}^{*,2} = nF' \begin{bmatrix} 0 & \sqrt{1/n} \end{bmatrix} = \begin{bmatrix} 0 & (1 + 1/\alpha)/(c\sqrt{2}) \\ 0 & (1 - 1/\alpha)/(c\sqrt{2}) \end{bmatrix},$$

$$Q_1^{*,2} = n \begin{bmatrix} \sqrt{1/(2n)} \\ -\sqrt{1/(2n)} \end{bmatrix} \begin{bmatrix} \sqrt{\frac{1}{n}} & -\sqrt{\frac{1}{n}} \end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix},$$

where

$$F' = \begin{bmatrix} (\sqrt{1/(2n)} + \sqrt{1/(2n\alpha^2)})/\sqrt{1 + 1/\alpha^2} & (\sqrt{1/(2n)} - \sqrt{1/(2n\alpha^2)})/\sqrt{1 + 1/\alpha^2} \end{bmatrix}$$

and $c \in (1, 2dM)$ because $\alpha \geq 1/(dM)$ from Lemma 1 for $\alpha > 0$. The incoherence μ , rank d , and condition number κ of the above Q functions are $O(1)$. Furthermore, the above construction satisfies Assumption 11 with Tucker rank $(S, 1, A)$ and Assumption 12 because the source and target MDPs share the same state latent

factor $[\sqrt{1/(2n)}, \sqrt{1/(2n)}]$ in the first transfer RL problem while in the second transfer RL problem,

$$\begin{bmatrix} \sqrt{1/(2n)} & -\sqrt{1/(2n)} \end{bmatrix} = -\alpha \begin{bmatrix} \sqrt{1/(2n)} & \sqrt{1/(2n)} \end{bmatrix} + \alpha c F'^\top.$$

Furthermore, note that the target latent factors,

$$F^1 = \begin{bmatrix} \sqrt{1/(2n)} \\ \sqrt{1/(2n)} \end{bmatrix} \text{ and } F^2 = \begin{bmatrix} \sqrt{1/(2n)} \\ -\sqrt{1/(2n)} \end{bmatrix}$$

are orthogonal to each other.

The learner is given either transfer RL problem one or two with equal probability with a generative model in the source problem. When interacting with the source MDPs, the learner specifies a state-action pair (s, a) and source MDP m and observes a realization of a shifted and scaled Bernoulli random variable X . The distribution of X is that $X = 1$ with probability $(Q_{m,1}^{*,i}(s, a) + 1)/2$ and $X = -1$ with probability $(1 - Q_{m,1}^{*,i}(s, a))/2$. Furthermore, the learner is given the knowledge that the state latent features lie in the function class $\mathcal{F} = \{F^1, F^2\}$ and must use the knowledge obtained from interacting with the source MDP to choose one to use in the target MDP, which is no harder than receiving no information about the function class.

To distinguish between the two transfer RL problems, one needs to differentiate between $Q_{2,1}^{*,1}$ and $Q_{2,1}^{*,2}$. By construction, the magnitude of the largest entrywise difference between $Q_{2,1}^{*,1}$ and $Q_{2,1}^{*,2}$ is lower bounded by $\Omega(1/\alpha)$.

If the learner observes Z samples in the source MDPs, where $Z \leq O(\alpha^2)$ for some constant $C > 0$, then the probability of correctly identifying F is upper bounded by 0.76 (Lemma 5.1 with $\delta = 0.24$ [8]). Thus, if the learner does not observe $\Omega(\alpha^2)$ samples in the source phase, then the learner returns the incorrect

feature mapping that is orthogonal to the true feature mapping with probability at least 0.24. $\square$

Theorem 5 states that unless one incurs a sample complexity of $\Omega(\alpha^2)$ in the source phase, the latent representation F learned in the source phase is useless in the target phase as one would incur linear regret using F .

Our algorithm in this setting is essentially the same as the one used in the (S, S, d) Tucker rank setting except we construct our latent factors using the other set of singular vectors and compute Gram matrices and coefficients for each action instead of each state.

Algorithm 3 Source Phase

Require: $\{N_h\}_{h \in [H]}$

- 1: **for** $m \in [M]$ **do**
 - 2: Run LR-EVI($\{N_h\}_{h \in [H]}$) on source MDP m to obtain $\bar{Q}_{m,h}$.
 - 3: Compute the singular value decomposition of $\bar{Q}_{m,h} = \hat{F}_{m,h} \hat{\Sigma}_{m,h} \hat{G}_{m,h}^\top$.
 - 4: **end for**
 - 5: Compute feature mapping $\hat{F} = \{\hat{F}_h\}_{h \in [H]}$ with $\hat{F}_h = \left\{ \sqrt{\frac{S}{d\mu}} \hat{F}_{m,h}(:, i) | i \in [d], m \in [M] \right\}$.
-

We first present the theoretical guarantees of LSVI-UCB-(S, d, A) and defer their proofs to Appendix B.9.2.

Before proving our main positive result in the (S, d, A) Tucker rank setting, we first present the definitions and theorems of LSVI-UCB-(S, d, A) and defer their proofs to Appendix B.9.2. If one is given the true latent low-rank representation of each state, then LSVI-UCB-(S, d, A) admits the following regret bound.

Theorem 6. *Assume that Assumption 11 holds and the learner is given the true latent state representation $F = \{F_h\}_{h \in [H]}$. Then, there exists a constant $c > 0$ such that for any*

Algorithm 4 Target Phase: LSVI-UCB-(S, d, A)

Require: $\lambda, \beta_{k,h}^a, \hat{F}$

- 1: **for** $k \in [K]$ **do**
- 2: Receive initial state s_1^k .
- 3: **for** $h = H, \dots, 1$ **do**
- 4: **for** $a \in \mathcal{A}$ **do**
- 5: Set $\Lambda_h^a \leftarrow \sum_{t \in T_{k-1,h}^a} \hat{F}_h(s_h^t) \hat{F}_h(s_h^t)^\top + \lambda \mathbf{I}$.
- 6: /* Compute Gram matrix Λ_h^a */
- 7: Set $w_h^a \leftarrow (\Lambda_h^a)^{-1} \sum_{t \in T_{k-1,h}^a} \hat{F}_h(s_h^t) [r_h(s_h^t, a) + \max_{a' \in \mathcal{A}} Q_{h+1}(s_{h+1}^t, a')]$.
- 8: /* Estimate w_h^a via regularized least squares */
- 9: **end for**
- 10: Set $Q_h(\cdot, \cdot) \leftarrow \min \left(H, \langle w_h, \hat{F}_h(\cdot) \rangle + \beta_{k,h}^a \sqrt{\hat{F}_h(\cdot)^\top (\Lambda_h^a)^{-1} \hat{F}_h(\cdot)} \right)$.
- 11: /* Estimate w_h^s via regularized least squares */
- 12: **end for**
- 13: **for** $h \in [H]$ **do**
- 14: Take action $a_h^k \leftarrow \max_{a \in \mathcal{A}} Q_h(s_h^k, a)$, and observe s_{h+1}^k .
- 15: **end for**
- 16: **end for**

$\delta \in (0, 1)$, if we set $\lambda = 1, \beta_{k,h}^a = cdH \sqrt{\iota}, \iota = \log(2dT A/\delta)$, then with probability at least $1 - \delta$, the total regret of LSVI-UCB-TR is $\tilde{O}(\sqrt{d^3 A H^3 T})$.

Theorem 6 states that LSVI-UCB-TR completely removes the regret bound's dependence on the size of the state space by utilizing the latent state representation F . To measure the algorithm's robustness with respect to misspecification error of the low-rank representation, we first define ξ -approximate (S, d, A) Tucker rank MDPs (similarly to Assumption B from [37]).

Assumption 13 (ξ -approximate (S, d, A) Tucker rank MDP). *Let $\xi \in [0, 1]$. Then, an MDP $(S, \mathcal{A}, P, r, H)$ is a ξ -approximate (S, d, A) Tucker rank MDP with given feature map F if there exist H unknown A -by- d matrices $W = \{W_h\}_{h \in [H]}$ and S -by- d -by- A tensors $U = \{U_h\}_{h \in [H]}$ such that*

$$|r_h(s, a) - F_h(s)^\top W_h(a)| \leq \xi,$$

$$\left| \sum_{s' \in \mathcal{S}} (P_h(s'|s, a) - F_h(s)^\top U_h(s', a)) \right| \leq \xi$$

for all $h \in [H]$ where $\|W_h\|_2, \|\sum_{s' \in \mathcal{S}} g(s') U_h(s', a)\|_2 \leq \sqrt{S/(d\mu)}$ for any function $g : \mathcal{S} \rightarrow [0, 1]$.

The following result shows that similar to LSVI-UCB, our algorithm is robust to the misspecified setting.

Theorem 7. *Suppose that the learner is given a feature mapping F' that satisfies Assumption 13. Then, using $F = \sqrt{S/(d\mu)}F'$ as the feature mapping for any $\delta \in (0, 1)$ and $\lambda = 1, \beta_{k,h}^a = O(Hd\sqrt{\iota} + \xi\sqrt{dT_{k,h}^a|H})$ for $\iota = \log(2dT A/\delta)$, the regret of LSVI-UCB-TR is $\tilde{O}(\sqrt{d^3 H^3 A T} + \xi d H T)$ with probability at least $1 - \delta$.*

The above theorem states that the misspecification error adds a term that is linear in T to the regret bound of LSVI-UCB-(S, d, A). Thus, if one's latent state representation is close enough, i.e., ξ is so small that there exists some positive constant c such that $\xi d H T \leq c \sqrt{d^3 H^3 A T}$, then LSVI-UCB-(S, d, A) admits a regret bound that is independent of the state space, which is the motivation for our main result. We now present our main theorem.

Theorem 8. *Suppose Assumptions 11, 12, and 10 hold, and set $\delta \in (0, 1)$. Furthermore, assume that, for any $\epsilon \in (0, \sqrt{AH/T})$, $Q_{m,h} = r_{m,h} + P_{m,h} V_{m,h+1}$ has rank d and is μ -incoherent with condition number κ for all ϵ -optimal value functions V_{h+1} . Also, let $\lambda = 1$ and $\beta_{k,h}^a$ be a function of $d, H, |T_{k,h}^a|, |A|, M, T$. Then, for $T \geq \frac{A}{\alpha^2}$, using at most $\tilde{O}(d^4 \mu^5 \kappa^4 (S/A + 1) M^2 H^4 \alpha^2 T)$ samples in the source problems, our algorithm has regret at most $\tilde{O}(\sqrt{(dMH)^3 AT})$ with probability at least $1 - \delta$.*

Theorem 8 states that using transfer learning improves the performance on the target problem by removing the regret bound's dependence on the state

space. Thus, with enough samples from the source MDPs one can recover a regret bound in the target problem that matches the best regret bound for any algorithm in the (S, d, A) Tucker rank setting with **known** latent feature representation F concerning A and T . We now prove Theorem 8.

Proof. Suppose the assumption stated in Theorem 8 hold. Then, from [54], running LR-EVI with a sample complexity of $C' d^4 \mu^5 \kappa^4 (S+A) M H^4 \alpha^2 T \log(2S A d M H / \delta) / A$ on each source MDP results in $\bar{Q}_{m,h}$ functions with singular value decomposition $\bar{Q}_{m,h} = \hat{F}_{m,h} \hat{\Sigma}_{m,h} \hat{G}_{m,h}$ that are $\sqrt{HA} / (16\alpha\mu\kappa\sqrt{dMT})$ -optimal with probability at least $1 - \delta/2$.

Since our estimates are $\sqrt{HA} / (16\alpha\sqrt{dMT})$ -optimal, it follows that $\bar{\gamma} < 1/2$ in the singular vector perturbation bound, Corollary 33. Since $\|Q_{m,h}^*\|_\infty \geq C$, from Corollary 33, it follows that there exists rotation matrices $R_{m,h}$ such that

$$\|(\hat{F}_{m,h} R_{m,h} - F_{m,h})(s)\|_2 \leq \frac{d \sqrt{H\mu A}}{\alpha \sqrt{SMT}}$$

for all $s \in \mathcal{S}$ where the optimal Q function have singular value decomposition $Q_{m,h}^* = F_{m,h} \Sigma_{m,h} G_{m,h}^\top$.

Let $\tilde{F}$ be the $S \times dM$ matrix from joining all the source state latent factor matrices, and $\hat{\tilde{F}}_h$ be the $S \times dM$ matrix from joining all $\hat{F}_{m,h}$. Let R_h be the $d \times dM$ matrix that joins $R_{m,h}$ for all $m \in [M]$. Then, from Assumption 12 and Definition

5, it follows that for all $(s', s, a) \in \mathcal{S} \times \mathcal{S} \times \mathcal{A}$, that

$$\begin{aligned}
r_h(s, a) &= \sum_{i \in [d]} F_h(s, i) W_h(a, i) \\
&= \sum_{i \in [d]} \left(\sum_{j \in [d], m \in [M]} B(i, j, m) F_{h,m}(s, j) \right) W_h(a, i) \\
&= \sum_{i, j \in [d], m \in [M]} B(i, j, m) F_{m,h}(s, j) W_h(a, i) \\
&= \tilde{F}_h(s) W'_h(a)^\top
\end{aligned}$$

and $P(s' | \cdot, \cdot) = \tilde{F}_h U'_h(s')^\top$ (with the same argument) for some $A \times dM$ matrices $W'_h, U'_h(s')$. Note that W'_h and $U'_h(s')$ are matrices that join $W_{m,h}$ and $U_{m,h}(s')$, respectively, with entries that at most α times larger than the original entry.

Therefore, it follows for all $(s', s, a, h) \in \mathcal{S} \times \mathcal{S} \times \mathcal{A} \times [H]$,

$$\begin{aligned}
|r_h(s, a) - \hat{\tilde{F}}_h(s) R_h W'_h(a)^\top| &= |\tilde{F}_h(s) W'_h(a)^\top - \hat{\tilde{F}}_h(s) R_h W'_h(a)^\top| \\
&= |(\tilde{F}_h(s) - \hat{\tilde{F}}_h(s) R_h) W'_h(a)| \\
&\leq \alpha \left| \sum_{m \in M} (F_{m,h}(s) - \hat{F}_{m,h}(s) R_{m,h}) W_{m,h}(a) \right| \\
&\leq \alpha \sum_{m \in M} \|F_{m,h}(s) - \hat{F}_{m,h}(s) R_{m,h}\|_2 \|W_{m,h}(a)\|_2 \\
&\leq \alpha \sqrt{\frac{S}{d\mu}} \sum_{m \in M} \|F_{m,h}(s) - \hat{F}_{m,h}(s) R_{m,h}\|_2 \\
&\leq \sqrt{\frac{dHMA}{T}}
\end{aligned}$$

from the Cauchy-Schwarz inequality and our singular vector perturbation

bound. From the same logic,

$$\begin{aligned}
\left| \sum_{s' \in \mathcal{S}} (P_h(s'|s, a) - F_h(s)^\top U_h(s', a)) \right| &= |\tilde{F}_h(s) U'_h(s', a) - \hat{\tilde{F}}_h(s) R_h U'_h(s', a)^\top| \\
&= |(\tilde{F}_h(s) - \hat{\tilde{F}}_h(s) R_h) \sum_{s' \in \mathcal{S}} U'_h(s', a)^\top| \\
&\leq \alpha \left| \sum_{m \in M} (F_{m,h}(s) - \hat{F}_{m,h}(s) R_{m,h}) \sum_{s' \in \mathcal{S}} U_{m,h}(s', a) \right| \\
&\leq \alpha \sum_{m \in M} \|F_{m,h}(s) - \hat{F}_{m,h}(s) R_{m,h}\|_2 \left\| \sum_{s' \in \mathcal{S}} U_{m,h}(s', a)^\top \right\|_2 \\
&\leq \sqrt{\frac{dHMA}{T}}.
\end{aligned}$$

Therefore, $\hat{\tilde{F}}_h(s)$ satisfies Assumption 13 with $\xi = \sqrt{\frac{dMHA}{T}}$. Then it follows that running LSVI-UCB-TR using $\hat{\tilde{F}}_h(s)$ with $\delta' = \delta/2$ during the target phase of the algorithm admits a regret bound of

$$Regret(T) \leq C''(\sqrt{d^3 M^3 H^3 A T} + dMH \sqrt{\frac{dMHA}{T}}) \in \tilde{O}(\sqrt{d^3 M^3 H^3 A T})$$

with probability at least $1 - \delta$ from a final union bound. Furthermore, the sample complexity in the source phase is

$$\tilde{O}\left(\frac{d^4 \mu^5 \kappa^4 (S + A) M^2 H^4 \alpha^2 T}{A}\right),$$

which proves our result. $\square$

B.5.2 Simulator to Simulator Transfer RL

In this section, we consider the same transfer reinforcement learning setting with the low Tucker rank assumptions except the learner is given access to the generative model in both the source and target problems. In this setting, we

isolate the statistical difficulty of the transfer RL problem by removing the challenge of exploration. Thus, instead of regret, the goal in this setting is to learn an ϵ -optimal Q function in the target phase using as few samples in the source and target phases as possible. The main benefit of having access to a generative model in the target phase is that the learner can choose the best states to observe covariates from when performing the regression step at each time step. In this setting, our algorithm still uses LR-EVI in the source phase (Algorithm 3 and adapts the algorithm from [39] for the finite horizon setting in the target phase. Learning Q^* over the support of an approximately optimal design ρ on our latent factors $\hat{F}$ allows us to construct a good estimate of Q^* over all state-action pairs. The optimal design problem that we solve in this algorithm is

$$G(\rho) = \sum_{s \in S} \rho(s) \hat{F}(s) \hat{F}(s)^\top, \quad g(\rho) = \max_{s \in S} \|\hat{F}\|_{G(\rho)^{-1}}$$

where ρ is a probability distribution over S . From Theorems 4.3 and 4.4 [39], one can compute a ρ such that $g(\rho) \leq 2d$, with the core set of states, or anchor states, $S^\# = \{s \in S \mid \rho(s) > 0\}$ having size at most $4d \log(\log(d)) + 16$ in a polynomial number of computations. This ensures that both the error amplification from using the low-rank structure and the number of states we need to observe Q^* is not too large. Our algorithm is tailored to the (S, d, A) Tucker rank setting but can easily be modified for the other Tucker rank settings.

Algorithm 5 Target Phase

Require: $\{N_h^t\}_{h \in [H]}, \hat{F}$

- 1: **for** $h = H, \dots, 1$ **do**
- 2: Compute $\mathcal{S}_h^\#, \rho_h$ according to the above optimal design problem with $\hat{F}_h$.
- 3: **for** $(s, a) \in \mathcal{S}_h^\# \times \mathcal{A}$ **do**
- 4: Collect N_h^t samples of the reward function and one-step transition to estimate $\hat{Q}_h(s, a)$ with

$$\hat{Q}_h(s, a) = \hat{r}_h(s, a) + \mathbb{E}_{s' \sim \hat{P}_h(\cdot|s, a)}[\hat{V}_{h+1}(s')],$$

where $\hat{r}_h(s, a)$ denotes the empirical average of the N_h^t samples of the reward function, and $\hat{P}_h(\cdot|s, a)$ denotes the empirical distribution over the N_h^t samples.

- 5: **end for**
- 6: Use the linear structure to estimate $\bar{Q}$ over all state-action pairs with

$$\hat{\theta}_{a,h} = G(\rho)^{-1} \sum_{s \in \mathcal{S}_h^\#} \rho(s) \hat{F}_h(s) \hat{Q}_h(s, a), \quad \bar{Q}_h(s, a) = \hat{F}_h(s) \theta_{a,h}.$$

- 7: Compute the value function and policy using the Q function,

$$\hat{V}_h(s) = \max_{a \in \mathcal{A}} \bar{Q}_h(s, a), \quad \hat{\pi}_h(s) = \arg \max_{a \in \mathcal{A}} \bar{Q}_h(s, a).$$

- 8: **end for**
-

In the source phase, our algorithm learns estimates of $Q_{m,h}^*$ and then computes the singular value decomposition to construct approximate feature mappings using the singular subspaces with respect to the state. In step (a) of the target phase, one first approximately solves the optimal design problem to compute the anchor states. After obtaining estimates of the Q function on the anchor states, one uses the linear structure in step (c) to estimate the full Q function over all states with the least squares solution. The above algorithm admits the following sample complexities for learning an ϵ -optimal Q function in the target phase.

Theorem 9. *Suppose Assumptions 11, 12, and 10 hold, and set $\delta \in (0, 1)$. Furthermore,*

assume that for any $\epsilon > 0$, $Q_{m,h} = r_{m,h} + P_{m,h}V_{m,h+1}$ has rank d and is μ -incoherent with condition number κ for all ϵ -optimal value functions V_{h+1} . Then, for $\epsilon \leq \alpha d^2$, using at most

$$\tilde{O}\left(\frac{d^7 \mu^3 \kappa^4 (S + A) H^9 \alpha^2 M^3}{\epsilon^2}\right)$$

samples in the source problems, Algorithm 5 learns an ϵ -optimal policy in the target MDP using at most

$$\tilde{O}\left(\frac{d^2 M^2 H^5 A}{\epsilon^2}\right)$$

samples with probability at least $1 - \delta$.

Theorem 9 states that with enough samples in the source phase, one can remove the dependence on the size of the state space in the target phase; in tabular finite-horizon MDPs with (S, d, A) Tucker rank transition kernels, one needs to observe at least $\tilde{\Omega}(d(S + A)H^3/\epsilon^2)$ samples from a generative model to learn an ϵ -optimal policy. Furthermore, the source sample complexity's dependence on S, A , and H are reasonable while the dependence on α is optimal due to our lower bound. We can easily construct similar algorithms that admit the corresponding theoretical guarantees in the other Tucker rank settings. We now prove Theorem 9.

Proof. In the source phase, using

$$\tilde{O}\left(\frac{d^6 \mu^3 \kappa^4 (S + A) H^9 \alpha^2 M^4}{\epsilon^2}\right)$$

samples guarantees that we learn $\epsilon/(128H^2\alpha M^{3/2}\kappa d^2)$ -optimal $\hat{Q}_{m,h} = \hat{F}_{m,h}\hat{\Sigma}_{m,h}\hat{G}_{m,h}^\top$ functions for each source MDP with probability at least $1 - \delta/2$. Since $\alpha d^2 \geq \epsilon$ by assumption, we have $\bar{\gamma} < 1/2$ in the singular vector perturbation bound, Corollary 33. Since $\|Q_{m,h}^*\|_\infty \geq C$, it follows from Corollary 33 that there exists

rotation matrices $R_{m,h}$ such that

$$\|F_{m,h} - \hat{F}_{m,h}R\|_{2,\infty} \leq \frac{\epsilon \sqrt{\mu}}{8H^2\alpha M^{3/2} \sqrt{S}}$$

holds for all $m \in [M], h \in [H]$ with probability at least $1 - \delta/2$. Let $\tilde{F}$ be the $S \times dM$ matrix from joining all the source state latent factor matrices, and $\hat{\tilde{F}}_h$ be the $S \times dM$ matrix from joining all $\tilde{F}_{m,h}$. Let R_h be the $d \times dM$ matrix that joins $R_{m,h}$ for all $m \in [M]$. Then, from Assumption 12 and Definition 5, it follows that for all $(s', s, a) \in \mathcal{S} \times \mathcal{S} \times \mathcal{A}$, that

$$\begin{aligned} r_h(s, a) &= \sum_{i \in [d]} F_h(s, i) W_h(a, i) \\ &= \sum_{i \in [d]} \left(\sum_{j \in [d], m \in [M]} B(i, j, m) F_{h,m}(s, j) \right) W_h(a, i) \\ &= \sum_{i, j \in [d], m \in [M]} B(i, j, m) F_{m,h}(s, j) W_h(a, i) \\ &= \tilde{F}_h(s) W'_h(a)^\top \end{aligned}$$

and $P(s'|\cdot, \cdot) = \tilde{F}_h U'_h(s')^\top$ (with the same argument) for some $A \times dM$ matrices $W'_h, U'_h(s')$. Note that W'_h and $U'_h(s')$ are matrices that join $W_{m,h}$ and $U_{m,h}(s')$, respectively, with entries that at most α times larger than the original entry. Therefore, it follows for all $(s', s, a, h) \in \mathcal{S} \times \mathcal{S} \times \mathcal{A} \times [H]$,

$$\begin{aligned} |r_h(s, a) - \hat{\tilde{F}}_h(s) R_h W'_h(a)^\top| &= |\tilde{F}_h(s) W'_h(a)^\top - \hat{\tilde{F}}_h(s) R_h W'_h(a)^\top| \\ &= |(\tilde{F}_h(s) - \hat{\tilde{F}}_h(s) R_h) W'_h(a)| \\ &\leq \alpha \sum_{m \in M} (F_{m,h}(s) - \hat{F}_{m,h}(s) R_{m,h}) W_{m,h}(a) \\ &\leq \alpha \sum_{m \in M} \|F_{m,h}(s) - \hat{F}_{m,h}(s) R_{m,h}\|_2 \|W_{m,h}(a)\|_2 \\ &\leq \alpha \sqrt{\frac{S}{d\mu}} \sum_{m \in M} \|F_{m,h}(s) - \hat{F}_{m,h}(s) R_{m,h}\|_2 \\ &\leq \frac{\epsilon}{8 \sqrt{dM} H^2} \end{aligned}$$

from the Cauchy-Schwarz inequality and our singular vector perturbation bound. From the same logic,

$$\begin{aligned}
|PV(s, a) - [\hat{F}_h R_h U'_h V](s, a)| &= \left| \sum_{s' \in S} P_h(s'|s, a) V(s') - \hat{F}_h(s) R_h U'_h(s', a)^\top V(s') \right| \\
&= \left| \sum_{s' \in S} \tilde{F}_h(s) U'_h(s', a)^\top V(s') - \hat{F}_h(s) R_h U'_h(s', a)^\top V(s') \right| \\
&\leq H |(\tilde{F}_h(s) - \hat{F}_h(s) R) \sum_{s' \in S} U'_h(s', a)| \\
&\leq H \alpha \sum_{m \in M} \|F_{m,h}(s) - \hat{F}_{m,h}(s) R\|_2 \sum_{s' \in S} U_{m,h}(s', a) \\
&\leq \frac{H \alpha \sqrt{S}}{\sqrt{d\mu}} \sum_{m \in M} \|F_{m,h}(s) - \hat{F}_{m,h}(s) R\|_2 \\
&\leq \frac{\epsilon}{8 \sqrt{dMH}}.
\end{aligned}$$

From the above result, it follows that for any value function V , the misspecification error of using the approximate feature mapping of the corresponding Q function $Q' = r + PV$ satisfies

$$\begin{aligned}
|Q'_h(s, a) - Q'_{h,d}(s, a)| &\leq |r_h(s, a) - \hat{F}(s) R W(a)^\top| + |PV(s, a) - [\hat{F}_h R_h U'_h V](s, a)| \\
&\leq \frac{\epsilon}{8 \sqrt{dMH}^2} + \frac{\epsilon}{8 \sqrt{dMH}} \leq \frac{\epsilon}{4H \sqrt{dM}}.
\end{aligned}$$

To prove our main theorem, we first present the following helper lemma and defer the proof to later in this section.

Lemma 2. *Suppose the setting of Theorem 9 holds and that the misspecification error of the bellman update for any value function using an approximate feature mapping satisfies*

$$|Q'_h(s, a) - Q'_{h,d}(s, a)| \leq \frac{\epsilon}{4H \sqrt{dM}}.$$

Then, the learned Q function from our algorithm at time step h satisfies

$$\|\bar{Q}_h - Q_h^*\|_\infty, \|\bar{Q}_h - \hat{Q}_h\|_\infty \leq \frac{\epsilon(H - h + 1)}{H}$$

for $h \in [H]$ for $N_h = H^4 d M C^2 \log(SAH/\delta)/\epsilon^2$ with probability at least $1 - (H - h + 1)\delta/(2H)$.

From a union bound and applying Lemma 2, it follows that Linear Q -Learning learns an ϵ -optimal policy with probability at least $1 - \delta$. Therefore, the sample complexity of the algorithm is

$$\sum_{h \in H} N_h |S_h^\#| A \leq \tilde{O}\left(\frac{d^2 M^2 H^5 A}{\epsilon^2}\right).$$

It follows from the triangle inequality that the learned policy is 2ϵ -optimal by replacing 2ϵ with ϵ' .

□

We now prove Lemma 2.

Proof. We proceed via induction. At time step H (the base case), it follows from Hoeffding's inequality for $N_H = H^4 d M C^2 \log(SAH/\delta)/\epsilon^2$ for some constant $C > 0$,

$$|\hat{Q}_H(s, a) - Q_H^*(s, a)| \leq \frac{\epsilon}{4\sqrt{dMH}}$$

for $(s, a) \in S_h^\# \times \mathcal{A}$. From Proposition 4.5 from [39] and our assumption on the misspecification error of the approximate feature mapping, it follows that

$$|\bar{Q}_H(s, a) - Q_H^*(s, a)| \leq \frac{\epsilon}{H}.$$

Since $Q_H^\# = Q_H^*$, it follows that the base case holds.

Next, assume that

$$\|\bar{Q}_{h+1} - Q_{h+1}^*\|_\infty, \|\bar{Q}_{h+1} - Q_{h+1}^\pi\|_\infty \leq \frac{\epsilon(H-h)}{H}$$

holds with probability at least $1 - (H-h)\delta/(2H)$. We define Q'_h as $Q'_h = r_h + P_h \hat{V}_{h+1}$. Let $Q'_{h,d} = F_h(W_h + \sum_{s' \in S} \hat{V}_{h+1}(s') U_h(s'))$. Then, it follows that from Hoeffding's inequality for $N_h = H^4 C^2 dM \log(SAH/\delta)/\epsilon^2$ for some constant $C > 0$, for all $(s, a) \in S_h^\# \times \mathcal{A}$, in step (b) of our algorithm,

$$|\hat{Q}_h(s, a) - Q'_h(s, a)| \leq \frac{\epsilon}{4H \sqrt{dM}}.$$

From our assumption on the misspecification error of the approximate feature mapping, it follows that $\|Q'_h - Q'_{h,d}\|_\infty \leq \frac{\epsilon}{4H \sqrt{dM}}$. Then, from Proposition 4.5 [39], we have

$$|\bar{Q}_h(s, a) - Q'_{h,d}(s, a)| \leq \frac{\epsilon}{2H} + \frac{\epsilon}{2H} = \frac{\epsilon}{H}$$

for all $(s, a) \in S \times \mathcal{A}$. Since

$$|Q_h^*(s, a) - Q'_h(s, a)| \leq \left| \sum_{s' \in S} (V_{h+1}^*(s') - \hat{V}_{h+1}(s')) P_h(s'|s, a) \right| \leq \frac{\epsilon(H-h)}{H}$$

and

$$|Q_h^\pi(s, a) - Q'_h(s, a)| \leq \left| \sum_{s' \in S} (V_{h+1}^*(s') - \hat{V}_{h+1}(s')) P_h(s'|s, a) \right| \leq \frac{\epsilon(H-h)}{H},$$

holds from the inductive hypothesis for all $(s, a) \in S \times \mathcal{A}$, it follows that

$$\begin{aligned} |\bar{Q}_h(s, a) - Q_h^*(s, a)| &\leq |\bar{Q}_h(s, a) - Q'_h(s, a)| + |Q'_h(s, a) - Q_h^*(s, a)| \\ &\leq \frac{\epsilon}{H} + \frac{\epsilon(H-h)}{H} = \frac{\epsilon(H-h+1)}{H}, \end{aligned}$$

and similarly,

$$|Q_h^\pi(s, a) - \bar{Q}_h(s, a)| \leq \frac{\epsilon(H-h+1)}{H}.$$

Thus, the inductive hypothesis holds as a union bound asserts that the above result holds with probability at least $1 - (H-h+1)\delta/(2H)$. $\square$

B.6 (d, S, A) Tucker Rank Setting

In the low-rank MDP transfer RL setting, [6] provide an algorithm that admits a source sample complexity of $\tilde{O}(A\alpha^3MT \log(\Phi))$ and target regret of $\tilde{O}(\bar{\alpha}H^2d^{3/2}\sqrt{T})$ with high probability in the transfer RL setting with low-rank MDPs. We first present the transfer learning assumptions in [6] needed to prove a lower bound in their setting. Low rank MDPs assume the following structure with shared latent representation ϕ ,

$$P_{m,h}(s'|s, a) = \mu_{m,h}(s')^\top \phi_h(s, a),$$

$$P_h(s'|s, a) = \mu_h(s')^\top \phi_h(s, a),$$

where $\|\phi_h(s, a)\|_2 \leq 1$ and $\|\int \mu(s)g(s)dx\|_2 \leq \sqrt{d}$ for any function $g : S \rightarrow [0, 1]$ [4] for all $m \in [M], h \in [H]$. To allow transfer learning to occur, [6] assume the following:

Assumption 14 (Assumption 2.2 [6]). *For any $h \in [H]$ and $s' \in S$, there exists $\alpha_{m,h}(s') \in \mathbb{R}$ such that $\mu_h(s') = \sum_{m \in [M]} \alpha_{m,h}(s') \mu_{m,h}(s')$,*

and define the task-relatedness coefficient as $\alpha = \max_{m \in [M], h \in [H], s \in S} \alpha_{m,h}(s)$.

They leave determining the optimal dependence on α as an interesting open question. Similarly to our lower bound in Theorem 1, we prove that one must incur a source sample complexity of $\Omega(\alpha^2)$ to benefit from using transfer learning. Note that in this version of transfer RL, there is no reward function in the source tasks.

Theorem 10. *There exist two transfer RL instances such that (i) they satisfy Assumptions 15 and 14, (ii) they cannot be distinguished without observing $\Omega(\alpha^2)$ samples in the source phase, and (3) they have target state-action latent features that are orthogonal to each other.*

Proof. We consider two low-rank transfer RL problems, in which the source MDPs have similar transition kernels but differing ϕ . All source and target MDPs $(\mathcal{S}, \mathcal{A}, P, H, r)$ share the same state space, action space, and horizon $H = 2$ with $M = 2$. The learner is given access to a generative model in the source problems. To introduce the MDPs (without loss of generality assume that $|\mathcal{S}|, |\mathcal{A}|$ are even), we first define the feature representations: the feature mapping ϕ_i for transfer RL problem i is

$$\phi_1(s_1, a_1) = \phi_1(s_2, a_2) = [1, 0]^\top, \phi_1(s_1, a_2) = \phi_2(s_2, a_1) = [0, 1]^\top,$$

$$\phi_2(s_1, a_1) = \phi_2(s_2, a_2) = [0, 1]^\top, \phi_2(s_1, a_2) = \phi_2(s_2, a_1) = [1, 0]^\top$$

where s_1 refers to states $1, \dots, |\mathcal{S}|/2$, s_2 refers to states $|\mathcal{S}|/2 + 1, \dots, |\mathcal{S}|$, and similarly for actions. We will present the transition kernels matrices with s_1, s_2, a_1, a_2 as the blocks containing $|\mathcal{S}|/2$ or $|\mathcal{A}|/2$ entries. Clearly, the two feature representations are orthogonal to each other. Next, the transition kernels $P_{m,1}^i$ for transfer

RL problem i and source MDP m are

$$\begin{aligned}
P_{1,1}^1(s_1|\cdot, \cdot) &= [1/2, 1/2 - 1/\alpha]\phi_1(\cdot, \cdot) \\
&= \begin{bmatrix} 1/2 & 1/2 - 1/\alpha \\ 1/2 - 1/\alpha & 1/2 \end{bmatrix} \\
P_{1,1}^1(s_2|\cdot, \cdot) &= [1/2, 1/2 + 1/\alpha]\phi_1(\cdot, \cdot) \\
&= \begin{bmatrix} 1/2 & 1/2 + 1/\alpha \\ 1/2 + 1/\alpha & 1/2 \end{bmatrix} \\
P_{1,1}^2(s_1|\cdot, \cdot) &= [1/2, 1/2 - 1/\alpha]\phi_2(\cdot, \cdot) \\
&= \begin{bmatrix} 1/2 - 1/\alpha & 1/2 \\ 1/2 & 1/2 - 1/\alpha \end{bmatrix} \\
P_{1,1}^2(s_2|\cdot, \cdot) &= [1/2, 1/2 + 1/\alpha]\phi_2(\cdot, \cdot) \\
&= \begin{bmatrix} 1/2 + 1/\alpha & 1/2 \\ 1/2 & 1/2 + 1/\alpha \end{bmatrix} \\
P_{2,1}^1(s_1|\cdot, \cdot) &= P_{2,1}^1(s_2|\cdot, \cdot) = P_{2,1}^2(s_2|\cdot, \cdot) = P_{2,1}^2(s_1|\cdot, \cdot) \\
&= [1/2, 1/2]\phi_1(\cdot, \cdot) = [1/2, 1/2]\phi_2(\cdot, \cdot) \\
&= \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} \\
P_T^i(\cdot|\cdot, \cdot) &= \mu_T(\cdot)\phi_i(\cdot, \cdot),
\end{aligned}$$

where $\mu_T(\cdot) = [0, 1]$ in both transfer RL problems. It follows that

$$\mu_T(s_1) = \alpha(\mu_1^2(s_1) - \mu_1^1(s_1)) = \alpha(\mu_2^2(s_1) - \mu_2^1(s_1))$$

and similar results hold for s_2 , so α satisfies the definition of the task-relatedness coefficient. It follows that in both settings, $\alpha_{\max} = \alpha$. The target phase reward

functions are defined as

$$r_1^1(\cdot, \cdot) = r_2^1(\cdot, \cdot) = [1, 0]\phi_1(\cdot, \cdot)$$

$$r_1^2(\cdot, \cdot) = r_2^2(\cdot, \cdot) = [1, 0]\phi_2(\cdot, \cdot),$$

and the initial state distribution is uniform across all states. By construction, in transfer RL problem 1, the only state action pairs that receive reward are $(s_1, a_1), (s_2, a_2)$. In contrast, in transfer RL problem 2, the only state action pairs that receive reward are $(s_1, a_1), (s_2, a_1)$.

The learner is given either transfer RL problem one or two with equal probability with a generative model in the source problem. When interacting with the source MDPs, the learner specifies a state-action pair (s, a) and source MDP m and observes a transition from $P_{1,m}^i$. Furthermore, the learner is given the knowledge that the state latent features lie in the function class $\Phi = \{\phi_1, \phi_2\}$ and must use the knowledge obtained from interacting with the source MDP to choose one to use in the target MDP, which is no harder than receiving no information about the function class.

To distinguish between the two transfer RL problems, one needs to differentiate between $P_{1,1}^1$ and $P_{1,1}^2$. By construction, the magnitude of the largest entry-wise difference between $P_{1,1}^1$ and $P_{1,1}^2$ is lower bounded by $\Omega(1/\alpha^2)$. If the learner observes Z transitions in the source MDPs, where $Z \leq O(\alpha^2)$ for some constant $C > 0$, then the probability of correctly identifying ϕ is upper bounded by 0.76 [8]. Thus, if the learner does not observe $\Omega(\alpha^2)$ samples in the source phase, then the learner returns the incorrect feature mapping that is orthogonal to the true feature mapping with probability at least 0.24. $\square$

Theorem 10 states that one must incur dependence on α in the source sample

complexity to benefit from transfer learning as using a feature mapping that is orthogonal to the true one results in regret linear in T .

As the low rank MDP setting is analogous to our (d, S, A) Tucker rank setting, we next present our assumptions in this setting,

Assumption 15. *In each of the M source MDPs, the reward functions have rank d , and the transition kernels have Tucker rank (d, S, A) . Let the target MDP's reward functions have rank d' and transition kernels have Tucker rank (d', S, A) where $d' \leq dM$. Thus, there exists $S \times d$ tensors $U_{m,h}$, $S \times d'$ tensors U_h , $SA \times d$ matrices with orthonormal columns $\phi_{m,h}$, $SA \times d'$ matrices with orthonormal columns ϕ_h , and d -dimensional vectors $W_{m,h}$, and d' -dimensional vectors W_h such that*

$$P_{m,h}(s'|s, a) = \sum_{i \in [d]} \phi_{m,h}(s, a, i) U_{m,h}(s', i),$$

$$r_{m,h}(s, a) = \sum_{i \in [d]} \phi_{m,h}(s, a, i) W_{m,h}(i)$$

and

$$P_h(s'|s, a) = \sum_{i \in [d']} \phi_h(s, a, i) U_h(s', i),$$

$$r_h(s, a) = \sum_{i \in [d']} \phi_h(s, a, i) W_h(i)$$

where $\|\sum_{s' \in \mathcal{S}} g(s') U_{m,h}(s')\|_2, \|\sum_{s' \in \mathcal{S}} g(s') U_h(s')\|_2 \leq \sqrt{d}, \|W_h\|_2$, and $\|W_{m,h}(s)\|_2 \leq \sqrt{S/\mu}$ for all $s \in \mathcal{S}$, $a \in \mathcal{A}$, $h \in [H]$, and $m \in [M]$ for any function $g : \mathcal{S} \rightarrow [0, 1]$.

While [6] assume that the target latent representation is a linear combination of the source latent representation, we allow for a more general setting to enable transfer learning. Specifically, we assume that the space spanned by the target latent representations is a subset of the space spanned by the source latent representations.

Assumption 16. Suppose Assumption 15 holds. For latent factors ϕ_h from the target MDP and $\phi_{m,h}$ for the source MDPs, let $\phi_h = \{\phi_h, \phi_{h,m}\}_{m \in [M]}$. Then, there exists a non-empty set $\mathcal{B}(\phi_h)$ such that the elements in $\mathcal{B}(\phi_h)$ are $B_h \in \mathbb{R}^{d', d, M}$ such that

$$\phi_h(s, i) = \sum_{j \in [d], m \in [M]} B_h(i, j, m) \phi_{m,h}(s, j)$$

for all $i \in [d']$, $s \in S$, $h \in [H]$.

As in the previous Tucker rank settings, we present the transfer-ability coefficient to quantify the difficulty transfer learning.

Definition 6. Given a transfer RL problem that satisfies Assumptions 15 and 16. Let Φ be the set of all $\mathbb{R}^{S \times A \times d}$ latent factor matrices with orthonormal columns that span the column space of $P_{m,h}$ for all $m \in [M]$, $h \in [H]$. Then, we define α as

$$\alpha := \min_{\phi \in \Phi} \min_{B \in \mathcal{B}(\phi)} \max_{i \in [d'], j \in [d], m \in [M]} |B(i, j, m)|.$$

[6] provide an alternate definition of α , which also measures the difficulty of transfer RL, and leave determining the optimal dependence on α as an interesting open question. Similarly to our lower bound in Theorem 1, we prove that one must incur a source sample complexity of $\Omega(\alpha^2)$ to benefit from using transfer learning. Note that in this Tucker rank setting, there is no reward function in the source tasks.

Theorem 11. There exist two transfer RL instances such that (i) they satisfy Assumptions 15 and 16, (ii) they cannot be distinguished without observing $\Omega(\alpha^2)$ samples in the source phase, and (3) they have target state-action latent features that are orthogonal to each other.

Proof. Consider the construction where all source and target MDPs $(S, \mathcal{A}, P, 1, r)$ share the same state space, action space, and horizon with $H = 1$. As the hori-

zon is one, we use the construction used to prove Theorem 5. We consider two transfer RL problems i, i' , in which the source MDPs have similar Q_1^* , but with orthogonal target ϕ_T . For ease of notation, we let a_1 refer to any action in $[n]$, and a_2 refer to any action in $\{n+1, \dots, 2n\}$, respectively. The initial state distribution in the target MDP is uniform over $\mathcal{S}$. We now present two transfer RL problems with similar Q functions that satisfy Assumptions 11 and 12 but have orthogonal target state latent factors. For ease of notation, the superscript i of $Q_{m,h}^{*,i}, Q_h^{*,i}$ denotes transfer RL problem. Then, the optimal Q functions (with columns representing a_1 and a_2 and the row being any state) for transfer RL problem one are

$$\begin{aligned} Q_{1,1}^{*,1} &= W_{1,h}^1 \phi_{1,1}^1 \begin{bmatrix} \sqrt{n} \\ \sqrt{n} \end{bmatrix} \begin{bmatrix} \sqrt{1/(2n)} & \sqrt{1/(2n)} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}, \\ Q_{2,1}^{*,1} &= W_{2,h}^1 \phi_{1,1}^1 \begin{bmatrix} \sqrt{n} \\ \sqrt{n} \end{bmatrix} \begin{bmatrix} \sqrt{1/(2n)} & \sqrt{1/(2n)} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}, \\ Q_1^{*,1} &= W_h \phi_1^1 = \begin{bmatrix} \sqrt{n} \\ \sqrt{n} \end{bmatrix} \begin{bmatrix} \sqrt{1/(2n)} & \sqrt{1/(2n)} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \end{aligned}$$

and for transfer RL problem two are

$$\begin{aligned} Q_{1,1}^{*,2} &= W_{1,h}^2 \phi_{1,1}^2 \begin{bmatrix} \sqrt{n} \\ \sqrt{n} \end{bmatrix} \begin{bmatrix} \sqrt{1/(2n)} & \sqrt{1/(2n)} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}, \\ Q_{2,1}^{*,2} &= W_{2,h}^2 \phi_{2,1}^2 \begin{bmatrix} \sqrt{n} \\ \sqrt{n} \end{bmatrix} G' = \begin{bmatrix} (1+1/\alpha)/(c\sqrt{2}) & (1-1/\alpha)/(c\sqrt{2}) \end{bmatrix}, \\ Q_2^{*,1} &= W_h \phi_1^2 = \begin{bmatrix} \sqrt{n} \\ \sqrt{n} \end{bmatrix} \begin{bmatrix} \sqrt{1/(2n)} & -\sqrt{1/(2n)} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \end{aligned}$$

and $c = \sqrt{1+1/\alpha^2}$. Note that $c \in (1, 2dM)$ because $\alpha \geq 1/(dM)$ from Lemma 1 for $\alpha > 0$. The incoherence μ , rank d , and condition number κ of the above Q

functions are $O(1)$. Furthermore, the above construction satisfies Assumption 15 with Tucker rank $(1, S, A)$ and Assumption 16 because the source and target MDPs share the same feature representation $\phi_1^1 = \phi_{1,1}^1$ in the first transfer RL problem while in the second transfer RL problem,

$$\phi_1^2 = -\alpha\phi_{2,1}^2 + \alpha\phi_{2,1}^2.$$

Note that ϕ_1^1 and ϕ_2^2 are orthogonal to each other by construction.

The learner is given either transfer RL problem one or two with equal probability with a generative model in the source problem. When interacting with the source MDPs, the learner specifies a state-action pair (s, a) and source MDP m and observes a realization of a shifted and scaled Bernoulli random variable X . The distribution of X is that $X = 1$ with probability $(Q_{m,1}^{*,i}(s, a) + 1)/2$ and $X = -1$ with probability $(1 - Q_{m,1}^{*,i}(s, a))/2$. Furthermore, the learner is given the knowledge that the state latent features lie in the function class $\Phi = \{\phi_1, \phi_2\}$ and must use the knowledge obtained from interacting with the source MDP to choose one to use in the target MDP, which is no harder than receiving no information about the function class.

To distinguish between the two transfer RL problems, one needs to differentiate between $Q_{2,1}^{*,1}$ and $Q_{2,1}^{*,2}$. By construction, the magnitude of the largest entrywise difference between $Q_{2,1}^{*,1}$ and $Q_{2,1}^{*,2}$ is lower bounded by $\Omega(1/\alpha)$.

If the learner observes Z samples in the source MDPs, where $Z \leq O(\alpha^2)$ for some constant $C > 0$, then the probability of correctly identifying G is upper bounded by 0.76 (Lemma 5.1 with $\delta = 0.24$ [8]). Thus, if the learner does not observe $\Omega(\alpha^2)$ samples in the source phase, then the learner returns the incorrect feature mapping that is orthogonal to the true feature mapping with probability at least 0.24. $\square$

Theorem 11 states that one must incur dependence on α in the source sample complexity to benefit from transfer learning. With this lower bound, we've shown in all three Tucker rank settings that α determines the effectiveness of transfer learning under our transfer learning assumptions. We now present our algorithm that admits an efficient source sample complexity and target regret bound.

Algorithm 6 Source Phase

Require: N

- 1: **for** $(s, a) \in \mathcal{S} \times \mathcal{A}, h \in [H], m \in [M]$ **do**
- 2: Collect N transitions from source MDP m at time step h at state-action pair (s, a) .
- 3: Use the samples to estimate the transition kernel with

$$\hat{P}_{m,h}(s'|s, a) = \frac{1}{N} \sum_{i \in [N]} \mathbf{1}_{X_{i,s,a}=s'},$$

where $X_{i,s,a} \sim P_{m,h}(\cdot|s, a)$.

- 4: **end for**
- 5: Compute the feature mapping $\hat{\phi} = \{\hat{\phi}_h\}_{h \in [H]}$ with

$$\hat{\phi}_h = \left\{ \frac{\sqrt{S}}{d\mu} \hat{F}_{m,h}(\cdot, i) \hat{G}_{n,h}(\cdot, j) | i, j \in [d], m, n \in [M] \right\}.$$

Algorithm 7 Target Phase: LSVI-UCB

Require: $\lambda, \beta_{k,h}, \hat{\phi}$

- 1: Run LSVI-UCB with $\lambda, \beta_{k,h}$, and $\hat{\phi}$.
-

In contrast to our approaches in the other Tucker rank settings, one learns the latent representation in the source MDPs through the transition kernels. One potential issue arises when all entries of $P(\cdot|\cdot, a)$ are small; for example, each entry of the uniform transition kernel is $1/S$, which is not lower bounded by a constant. Thus, we let $D = \min_{m \in [M], h \in [H], a \in \mathcal{A}} \|P_{m,h}(\cdot|\cdot, a)\|_\infty$, and the dependence on D is unsurprising as one needs to ensure the noise is sufficiently small to learn

the latent representations from $P_{m,h}$. We now present and prove the main result in this Tucker rank setting.

Theorem 12. *Suppose Assumptions 11, 12, and 10 hold in the (d, S, A) Tucker Rank setting, and set $\delta \in (0, 1)$. Furthermore, assume that for any $\epsilon \in (0, \sqrt{H/T})$, $P_{m,h}(\cdot|\cdot, a)$ has rank d and is μ -incoherent with condition number κ for all $a \in A$. Let $D = \min_{m \in [M], h \in [H], a \in \mathcal{A}} \|P_{m,h}(\cdot|\cdot, a)\|_\infty$. Then, for $T \geq H/\alpha^2$, using at most*

$$\tilde{O}\left(\frac{\alpha^2 d \mu^2 \kappa^2 M^2 T S A}{D^2}\right),$$

samples in the source problems, our algorithm has regret at most

$$\tilde{O}\left(\sqrt{(dMH)^3 T}\right)$$

with probability at least $1 - \delta$ for $\lambda = 1$ and $\beta_{h,k}$, which is a function of d, H, T, M .

Proof. Suppose the setting of Theorem 12 holds. Recall that we let $D = \min_{m \in [M], h \in [H], a \in \mathcal{A}} \|P_{m,h}(\cdot|\cdot, a)\|_\infty$. Then, observing N transitions from each state-action pairs ensures that for each $(s', s, a) \in \mathcal{S} \times \mathcal{S} \times A$ and for each source MDP at each time step,

$$|P_{m,h}(s'|s, a) - \hat{P}_{m,h}(s'|s, a)| \leq \frac{\sqrt{H}}{\alpha \mu \kappa D \sqrt{dMT}}$$

with probability at least $1 - \delta/(2S^2AMH)$ where $\hat{P}_{m,h}(s'|s, a) = \frac{1}{N} \sum_{i \in [N]} \mathbf{1}_{X_i(s,a)=s'}$ and $X_i \sim P(\cdot|s, a)$ for

$$N = \frac{\alpha^2 d \mu^2 \kappa^2 MT \log(4S^2AMH/\delta)}{2HD^2}.$$

Let $\hat{P}_{m,h}(\cdot|\cdot, a) = \hat{F}_{m,h}^a \hat{\Sigma}_{m,h} \hat{G}_{m,h}^\top$ be the singular value decomposition of our estimate where $\hat{F}$ corresponds to the state one transitions from. Since $T \geq \alpha^2/H$, it follows that $\bar{\gamma} \leq 1/2$. Thus, 33 states that there exists rotation matrices $R_{m,h,a}$ such that

$$\|(\hat{F}_{m,h}^a R_{m,h,a} - F_{m,h}^a)(s)\|_2 \leq \frac{d \sqrt{H\mu}}{\alpha \sqrt{MTS}},$$

for all $s \in \mathcal{S}$ where the true transition kernel $P_{m,h}(\cdot|\cdot, a)$ has singular value decomposition $P_{m,h}(\cdot|\cdot, a) = F_{m,h}^a \Sigma_{m,h} G_{m,h}^\top$. We can re-express our terms in the more common Low rank MDP setting with $\phi_{m,h}(s, a) = F_{m,h}^a(s)$ and $\mu_{m,h}(s') = \Sigma_{m,h} G_{m,h}^\top$.

Let $\tilde{F}$ be the $SA \times dM$ matrix from joining all the source state latent factor matrices from $P(\cdot|\cdot, a)$ for each $a \in A$, and $\hat{\tilde{F}}_h$ be the $SA \times dM$ matrix from joining all $\tilde{F}_{m,h}$. Let R_h be the $d \times dM$ matrix that joins $R_{m,h}$ for all $m \in [M]$. Then, from Assumption 12 and Definition 5, it follows that for all $(s', s, a) \in \mathcal{S} \times \mathcal{S} \times A$, that

$$\begin{aligned} r_h(s, a) &= \sum_{i \in [d]} F_h^a(s, i) W_h(i) \\ &= \sum_{i \in [d]} \left(\sum_{j \in [d], m \in [M]} B(i, j, m) F_{h,m}^a(s, j) \right) W_h(i) \\ &= \sum_{i, j \in [d], m \in [M]} B(i, j, m) F_{m,h}^a(s, j) W_h(i) \\ &= \tilde{F}_h^a(s) W_h'^\top \end{aligned}$$

and $P(s'|s, a) = \tilde{F}_h^a(s) U_h'(s')^\top$ (with the same argument) for some dM -dimension vector $W_h', U_h'(s')$. Note that W_h' and $U_h'(s')$ are vectors that join $W_{m,h}$ and $U_{m,h}(s')$, respectively, with entries that at most α times larger than the original entry.

Therefore, it follows for all $(s', s, a, h) \in \mathcal{S} \times \mathcal{S} \times \mathcal{A} \times [H]$,

$$\begin{aligned} |r_h(s, a) - \hat{\tilde{F}}_h^a(s) R_h W_h'^\top| &= |\tilde{F}_h^a(s) W_h'^\top - \hat{\tilde{F}}_h^a(s) R_h W_h'^\top| \\ &= |(\tilde{F}_h^a(s) - \hat{\tilde{F}}_h^a(s) R_h) W_h'^\top| \\ &\leq \alpha \left| \sum_{m \in M} (F_{m,h}^a(s) - \hat{F}_{m,h}^a(s) R_{m,h}) W_{m,h} \right| \\ &\leq \alpha \sum_{m \in M} \|F_{m,h}^a(s) - \hat{F}_{m,h}^a(s) R_{m,h}\|_2 \|W_{m,h}\|_2 \\ &\leq \alpha \sqrt{\frac{S}{d\mu}} \sum_{m \in M} \|F_{m,h}^a(s) - \hat{F}_{m,h}^a(s) R_{m,h}\|_2 \\ &\leq \sqrt{\frac{dHM}{T}} \end{aligned}$$

from the Cauchy-Schwarz inequality and our singular vector perturbation bound. From the same logic,

$$\begin{aligned}
\left| \sum_{s' \in \mathcal{S}} (P_h(s'|s, a) - \hat{\tilde{F}}_h^a(s) R_h U'_h(s'))^\top \right| &\leq \left| \sum_{s' \in \mathcal{S}} P_h(s'|s, a) - \hat{\tilde{F}}_h^a(s) R_h U'_h(s')^\top \right| \\
&= \left| \sum_{s' \in \mathcal{S}} \tilde{F}_h^a(s) U'_h(s') - \hat{\tilde{F}}_h^a(s) R_h U'_h(s')^\top \right| \\
&\leq \alpha \left| \sum_{m \in M} (F_{m,h}^a(s) - \hat{F}_{m,h}^a(s) R_{m,h}) \sum_{s' \in \mathcal{S}} U_{m,h}(s') \right| \\
&\leq \alpha \sum_{m \in M} \|F_{m,h}^a(s) - \hat{F}_{m,h}^a(s) R_{m,h}\|_2 \left\| \sum_{s' \in \mathcal{S}} U_{m,h}(s')^\top \right\|_2 \\
&\leq \sqrt{\frac{dHM}{T}}.
\end{aligned}$$

Therefore, $\hat{\tilde{F}}_h^a(s)$ satisfies Assumption B in [37] with $\xi = \sqrt{\frac{dMH}{T}}^1$. Then it follows that running LSVI-UCB using $\hat{\tilde{F}}_h^a(s)$ with $\delta' = \delta/2$ during the target phase of the algorithm admits a regret bound of

$$\text{Regret}(T) \leq C''(\sqrt{d^3 M^3 H^3 T} + dMH \sqrt{\frac{dMH}{T}}) \in \tilde{O}(\sqrt{d^3 M^3 H^3 T})$$

with probability at least $1 - \delta$ from a final union bound. Furthermore, the sample complexity in the source phase is

$$\tilde{O}\left(\frac{\alpha^2 d \mu^2 \kappa^2 M^2 T S A}{D^2}\right),$$

which proves our result. $\square$

B.7 (d, d, d) Tucker Rank Setting

In this section, we present the assumptions, algorithm, and proof for Theorem 13. In the (d, d, d) Tucker rank setting, we assume the following low-rank struc-

¹While the misspecified assumption in [37] uses the total variation distance, they refer to it as $|\sum_{s' \in \mathcal{S}} P(s'|s, a) - \hat{P}(s'|s, a)|$ in the second to last inequality in the proof of Lemma C.1 and the second to last equation on page 23

ture on the MDPs.

Assumption 17. *In each of the M source MDPs, the reward functions have rank d , and the transition kernels have Tucker rank (d, d, d) . The target MDP's reward functions have rank d' and transition kernels have Tucker rank (d, d', d') where $d' \leq dM$. Thus, there exists $d \times d \times d$ tensors $U_{m,h}$, $d \times d' \times d'$ tensors U_h , $S \times d$ matrices $V_{m,h}$, $S \times d'$ matrices V_h , $S \times d$ matrices $F_{m,h}$, $S \times d'$ matrices F_h , $A \times d$ matrices $G_{m,h}$, and $A \times d'$ matrices G_h all with orthonormal columns, and sequences of non-increasing singular values $\{\sigma_{m,h}(i) \geq \mathbb{R}_+\}_{i \in [d]}$, $\{\sigma_h(i) \geq \mathbb{R}_+\}_{i \in [d']}$ such that*

$$P_{m,h}(s'|s, a) = \sum_{i \in [d], j \in [d'], k \in [d]} U_{m,h}(i, j, k) V_{m,h}(s', i) F_{m,h}(s, j) G_{m,h}(a, k),$$

$$r_{m,h}(s, a) = \sum_{i \in [d]} \sigma_{m,h}(i) F_{m,h}(s, i) G_{m,h}(a, i)$$

and

$$P_h(s'|s, a) = \sum_{i \in [d], j \in [d'], k \in [d']} U_h(i, j, k) V_h(s', i) F_h(s, j) G_h(a, k),$$

$$r_h(s, a) = \sum_{i \in [d']} \sigma_h(i) F_h(s, i) G_h(a, i)$$

where $\|\sum_{i \in [d], s' \in \mathcal{S}} g(s') U_{m,h}(i, :, :) V_{m,h}(s', i)\|_F, \|\sum_{s' \in \mathcal{S}, i \in [d]} g(s') U_h(i, :, :) V_h(s', i)\|_F, \sigma_{m,h}(i), \sigma_h(i) \leq \frac{\sqrt{SA}}{d\mu}$ for all $s \in \mathcal{S}$, $a \in \mathcal{A}$, $h \in [H]$, $m \in [M]$, and $i \in [d]$ for any function $g : \mathcal{S} \rightarrow [0, 1]$.

To allow transfer learning to occur, we assume that the latent factors from the source MDPs span the space spanned by the target latent factors.

Assumption 18. *For latent factors F_h from the target MDP and $F_{m,h}$ for the source MDPs, let $F = \{F_h, F_{h,m}\}_{h \in [H], m \in [M]}$. For latent factors G_h from the target MDP and $G_{m,h}$ for the source MDPs, let $G = \{G_h, G_{h,m}\}_{h \in [H], m \in [M]}$. Then, there exists a non-empty set $\mathcal{B}(F), \mathcal{C}(G)$ such that the elements in $\mathcal{B}(F)$ are $B \in \mathbb{R}^{d', d, M}$ and the elements in $\mathcal{C}(G)$ are $C \in \mathbb{R}^{d', d, M}$ such that*

$$F_h(s, i) = \sum_{j \in [d], m \in [M]} B(i, j, m) F_{m,h}(s, j)$$

and

$$G_h(a, i) = \sum_{j \in [d], m \in [M]} C(i, j, m) G_{m,h}(a, j)$$

for all $i \in [d']$, $s \in \mathcal{S}$, $a \in \mathcal{A}$.

Similarly to the (S, d, A) and (S, S, d) Tucker rank settings, we introduce the transfer-ability coefficient that measures the difficulty of applying transfer learning.

Definition 7 (Transfer-ability Coefficient). *Given a transfer RL problem that satisfies Assumptions 17, 18, and 10, let $\mathcal{F}$ be the set of all $\mathbb{R}^{S \times d}$ latent factor matrices with orthonormal columns that span the column space of $Q_{m,h}^*$ for all $m \in [M], h \in [H]$. Let $\mathcal{G}$ be the set of all $\mathbb{R}^{A \times d}$ latent factor matrices with orthonormal columns that span the row space of $Q_{m,h}^*$ for all $m \in [M], h \in [H]$. Then, we define α as*

$$\alpha := \max \left(\min_{F \in \mathcal{F}} \min_{B \in \mathcal{B}(F)} \max_{i \in [d'], j \in [d], m \in [M]} |B(i, j, m)|, \min_{G \in \mathcal{G}} \min_{C \in \mathcal{C}(F)} \max_{i \in [d'], j \in [d], m \in [M]} |C(i, j, m)| \right).$$

In contrast to the definition in the other Tucker rank settings, α bounds the largest coefficient used in the linear combination of F_h and G_h . The algorithm that we use in this setting runs LR-EVI on each of the source MDPs, constructs a feature mapping $\hat{\phi}$ using the singular vectors from the singular value decomposition of $\bar{Q}_{m,h}$, and runs LSVI-UCB with $\hat{\phi}$. Since we use both sets of singular vectors to construct $\hat{\phi}$, our feature mapping has latent dimension $d^2 M^2$.

The above algorithm admits the following result.

Theorem 13. *Suppose Assumptions 17, 18, and 10 hold in the (d, d, d) Tucker Rank setting, and set $\delta \in (0, 1)$. Furthermore, assume that for any $\epsilon \in (0, \sqrt{H/T})$, $Q_{m,h} = r_{m,h} + P_{m,h} V_{m,h+1}$ has rank d and is μ -incoherent with condition number κ for all ϵ -optimal value functions V_{h+1} . Then, assuming $T \geq H/\alpha^2$, using at most*

$$\tilde{O}\left(d^{10} \mu^5 \kappa^4 (S + A) M^2 H^4 \alpha^4 T\right)$$

Algorithm 8 Source Phase

Require: $\{N_h\}_{h \in [H]}$

- 1: **for** $m \in [M]$ **do**
- 2: Run LR-EVI($\{N_h\}_{h \in [H]}$) on source MDP m to obtain $\bar{Q}_{m,h}$.
- 3: Compute the singular value decomposition of $\bar{Q}_{m,h} = \hat{F}_{m,h} \hat{\Sigma}_{m,h} \hat{G}_{m,h}^\top$.
- 4: **end for**
- 5: Compute the feature mapping $\hat{\phi} = \{\hat{\phi}_h\}_{h \in [H]}$ with

$$\hat{\phi}_h = \left\{ \frac{\sqrt{SA}}{d\mu} \hat{F}_{m,h}(:, i) \hat{G}_{n,h}(:, j) \mid i, j \in [d], m, n \in [M] \right\}.$$

Algorithm 9 Target Phase: LSVI-UCB

Require: $\lambda, \beta_{k,h}, \hat{\phi}$

- 1: Run LSVI-UCB with $\lambda, \beta_{k,h}$, and $\hat{\phi}$.
-

samples in the source problems, our algorithm has regret at most

$$\tilde{O}\left(\sqrt{d^6 M^6 H^3 T}\right)$$

with probability at least $1 - \delta$ for $\lambda = 1$ and $\beta_{h,k}$, which is a function of d, H, T, M .

We now prove Theorem 13 using a similar argument to the one used in the proof of Theorem 8 except we show that the target MDP with our feature mapping satisfies Assumption B in [37] and apply LSVI-UCB.

Proof. Suppose the assumptions stated in Theorem 13 hold. Then, from Theorem 9[54], it follows that LR-EVI returns near-optimal Q functions $\bar{Q}_{m,h}$ functions with singular value decomposition $\bar{Q}_{m,h} = \hat{F}_{m,h} \hat{\Sigma}_{m,h} \hat{G}_{m,h}^\top$ that are $\sqrt{H}/(\kappa\alpha^2 M d^4 \mu \sqrt{T})$ -optimal with probability at least $1 - \delta/2$ using

$$\tilde{O}\left(d^{10} \mu^5 \kappa^4 (S + A) M H^4 \alpha^4 T\right)$$

samples on each source MDP. Since $T \geq H/\alpha^4$, from $\bar{\gamma} \leq \frac{1}{2}$ in Proposition 32.

Thus, 33 states that there exists rotation matrices $R_{m,h,f}, R_{m,h,g}$ such that

$$\|(\hat{F}_{m,h}R_{m,h,f} - F_{m,h})(s)\|_2 \leq \frac{\sqrt{H}}{\alpha^4 M d^2 \sqrt{TS}\mu}, \quad \|(\hat{G}_{m,h}R_{m,h,g} - G_{m,h})(s)\|_2 \leq \frac{\sqrt{H}}{\alpha^4 M d^2 \sqrt{TA}\mu}$$

for all $s \in \mathcal{S}$ where the optimal Q function have singular value decomposition

$$Q_{m,h}^* = F_{m,h}\Sigma_{m,h}G_{m,h}^\top.$$

Then, from Assumption 12 and Definition 5, it follows that for all $(s', s, a) \in \mathcal{S} \times \mathcal{S} \times \mathcal{A}$, that

$$\begin{aligned} r_h(s, a) &= \sum_{i \in [d]} \sigma_h(i) F_h(s, i) G_h(a, i) \\ &= \sum_{i \in [d]} \sigma_h(i) \left(\sum_{j \in [d], m \in [M]} B(i, j, m) F_{h,m}(s, j) \right) \left(\sum_{l \in [d], n \in [M]} C(i, l, n) G_{h,n}(s, l) \right) \\ &= \sum_{j, l \in [d], m, n \in [M]} F_{m,h}(s, j) G_{n,h}(a, l) \left(\sum_{i \in [d]} \sigma_h(i) B(i, j, m) C(i, l, n) \right) \\ &= \theta_h \phi(s, a)_h^\top \end{aligned}$$

where $\theta_h = \{\sum_{i \in [d]} \sigma_h(i) B(i, j, m) C(i, l, n)\}_{j, l \in [d], m, n \in [M]}$ and $\phi_h = \{F_{m,h}(s, j) G_{n,h}(a, l)\}_{j, l \in [d], m, n \in [M]}$

and $P(s'|\cdot, \cdot) = \mu_h(s') \phi_h^\top$ (with the same argument) for $\mu_h(s') = \{\sum_{i, j, k \in [d]} U_h(i, j, k) V_h(s', i) B(j, l, m) C(k, l, n)\}_{l, n \in [M]}$

It follows that we can express $r_h = \tilde{F}_h \tilde{\Sigma}_h \tilde{G}_h^\top$ where $\tilde{F}_h$ be the $S \times d^2 M^2$ matrix from

joining all $F_{m,h}$ dM times, $\tilde{G}_h$ be the $A \times d^2 M^2$ matrix from joining all $G_{m,h}$ dM times,

and $\tilde{\Sigma}_h$ is the diagonal matrix with entries $\theta_h(j, l, m, n)$ for all $j, l \in [d], m, n \in [M]$.

Note that the entries of $\tilde{\Sigma}_h$ and μ_h are at most $d\alpha^2$ larger than their original value.

Let $\hat{\tilde{F}}_h$ be the $S \times d^2 M^2$ matrix from joining all $\tilde{F}_{m,h} R_{m,h,f}$ dM times, and $\hat{\tilde{G}}_h$ be the $A \times d^2 M^2$ matrix from joining all $\tilde{G}_{m,h} R_{m,h,g}$ dM times. Let $\tilde{\theta}_h$ satisfy

$$\tilde{\theta}_h \hat{\phi}_h^\top = \hat{\tilde{F}}_h \tilde{\Sigma}_h \hat{\tilde{G}}_h^\top.$$

Therefore, it follows for all $(s', s, a, h) \in \mathcal{S} \times \mathcal{S} \times \mathcal{A} \times [H]$,

$$\begin{aligned}
|r_h(s, a) - \tilde{\theta}_h \hat{\phi}_h^\top(s, a)| &= |\tilde{F}_h \tilde{\Sigma}_h \tilde{G}_h^\top(s, a) - \hat{F}_h \tilde{\Sigma}_h \hat{G}_h^\top(s, a)| \\
&\leq |\tilde{F}_h \tilde{\Sigma}_h \tilde{G}_h^\top(s, a) - \hat{F}_h \tilde{\Sigma}_h \tilde{G}_h^\top(s, a)| + |\hat{F}_h \tilde{\Sigma}_h \tilde{G}_h^\top(s, a) - \hat{F}_h \tilde{\Sigma}_h \hat{G}_h^\top(s, a)| \\
&\leq dM \sum_{m \in M} \|F_{m,h}(s) - \hat{F}_{m,h}(s) R_{m,h,f}\|_2 \|\tilde{\Sigma}_h\|_{op} \max_{m \in M} \sqrt{d} \|G_{m,h}\|_\infty \\
&\quad + \sqrt{d} \max_{m \in M} \|\hat{F}_{m,h}\|_\infty \|\tilde{\Sigma}_h\|_{op} \|G_{m,h}(a) - \hat{G}_{m,h}(s) R_{m,h,g}\|_2 \\
&\leq d^{3/2} M \sum_{m \in M} \|F_{m,h}(s) - \hat{F}_{m,h}(s) R_{m,h,f}\|_2 \|\tilde{\Sigma}_h\|_{op} \max_{m \in M} \|G_{m,h}\|_\infty \\
&\quad + \max_{m \in M} (\|F_{m,h}\| + \|F_{m,h} R - \hat{F}_{m,h}\|_\infty) \|\tilde{\Sigma}_h\|_{op} \|G_{m,h}(a) - \hat{G}_{m,h}(s) R_{m,h,g}\|_2 \\
&\leq Cd^3 M \frac{\sqrt{S}}{\sqrt{\mu}} \sum_{m \in M} \|F_{m,h}(s) - \hat{F}_{m,h}(s) R_{m,h,f}\|_2 \\
&\leq C \frac{dM \sqrt{H}}{\sqrt{T}},
\end{aligned}$$

where the second inequality comes from the triangle inequality and Cauchy-Schwarz inequality, the third inequality comes from the triangle inequality, and the fourth inequality comes from the definition of incoherence, our bound on the entries of $\tilde{\Sigma}$ and the left term dominating the right term, and the final inequality is due to our singular vector perturbation bounds. Similarly,

$$\|P_h(\cdot|s, a) - \mu_h(\tilde{s}') \hat{\phi}_h(s, a)^\top\|_{TV} \leq C \frac{dM \sqrt{H}}{\sqrt{T}}$$

for some absolute constant $C > 0$ where $\mu_h(\tilde{s}')$ satisfies $\hat{F}_h \tilde{\Sigma}_h(s') \hat{G}_h^\top$ and $\tilde{\Sigma}_h(s')$ is the $d^2 M^2 \times d^2 M^2$ diagonal matrix with entries $\mu_h(s', l, k, m, n)$ with the same argument using the regularity condition on μ_h .

Therefore, $\hat{\phi}$ satisfies Assumption B in [37] with $\xi = O(dM \sqrt{H/T})$. Running LSVI-UCB with $\hat{\phi}$ admits a regret bound of

$$\text{Regret}(T) \leq C \left(\sqrt{d^6 M^6 H^3 T \iota^2} + \frac{d^3 M^3 H^{3/2} T \sqrt{\iota}}{\sqrt{T}} \right) \in \tilde{O}(d^6 M^6 H^3 T)$$

for some absolute constant $C > 0$ with probability at least $1 - \delta$ from a final union bound. The sample complexity in the source phase is

$$\tilde{O}\left(d^{10}\mu^5\kappa^4(S + A)MH^4\alpha^4T\right).$$

which completes the proof. $\square$

B.8 ℓ_∞ eigen perturbation bound with non-uniform noise: rank- r case

Suppose $A = A^* + D \in \mathbb{R}^{m \times n}$, where A^* is the true rank- r matrix and D is the noise matrix. Suppose A^* has SVD $A^* = U^* \Sigma^* V^{*\top}$, where $U^* \in^{m \times r}$, $V^* \in^{n \times r}$ are orthonormal matrices containing the left and right singular vectors, respectively, and $\Sigma^* = \text{diag}(\sigma_1^*, \dots, \sigma_r^*)$ is a diagonal matrix of the singular values. Similarly, suppose A has SVD $A = U \Sigma V^\top + U' \Sigma' V'^\top$, where $U \in^{m \times r}$, $V \in^{n \times r}$ and $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r)$ correspond to the top- r singular vectors/values, and $U' \in^{m \times (m-r)}$, $V' \in^{n \times (n-r)}$ and $\Sigma' \in^{(m-r) \times (n-r)}$ correspond to the bottom singular vectors/values.

The following proposition is a generalization of the perturbation bound in [24] to the setting with non-uniform noise.

Proposition 32. Suppose $\bar{\gamma} := \frac{\|D\|_{\text{op}}}{\sigma_r^*} < \frac{1}{2}$. There exist two orthonormal matrices $\bar{R}, \underline{R} \in^{r \times r}$ such that

$$\begin{aligned} \left\| (U - U^* \bar{R})_{i \cdot} \right\|_2 &\leq \|U_{i \cdot}^*\|_2 \cdot 8\bar{\gamma} + \|D_{i \cdot}\|_2 \cdot \frac{1}{\sigma_r^*(1 - \bar{\gamma})}, & \forall i \in [m], \\ \left\| (V - V^* \underline{R})_{j \cdot} \right\|_2 &\leq \|V_{j \cdot}^*\|_2 \cdot 8\bar{\gamma} + \|D_{j \cdot}\|_2 \cdot \frac{1}{\sigma_r^*(1 - \bar{\gamma})}, & \forall j \in [n]. \end{aligned}$$

To prove Proposition 32, we need some notations. Let $\bar{H} := U^\top U^* \in^{r \times r}$ and its SVD be $\bar{H} = \bar{A} \bar{\Lambda} \bar{B}^\top$, where $\bar{A}, \bar{B} \in^{r \times r}$ are orthonormal, and $\bar{\Lambda} = \text{diag}(\bar{\sigma}_1, \dots, \bar{\sigma}_r) \in^{r \times r}$ is a diagonal matrix. Similarly, let $\underline{H} := V^\top V^* \in^{r \times r}$ and its SVD be $\underline{H} = \underline{A} \underline{\Lambda} \underline{B}^\top$. Finally, define the orthonormal matrices $\text{sgn}(\bar{H}) := \bar{A} \bar{B}^\top$ and $\text{sgn}(\underline{H}) := \underline{A} \underline{B}^\top$. (The function $\text{sgn}(\cdot)$ is called the matrix sign function.)

We need a technical lemma, which generalizes [2, Lemma 3] to the asymmetric case.

Lemma 3 (Properties of matrix sign function). *We have $\|\bar{H}\|_{\text{op}} \leq 1$ and $\|\underline{H}\|_{\text{op}} \leq 1$.*

When $\bar{\gamma} := \frac{\|D\|_{\text{op}}}{\sigma_r^} < \frac{1}{2}$, we have*

$$\begin{aligned} \sqrt{\|\bar{H} - \text{sgn}(\bar{H})\|_{\text{op}}} &\leq \|UU^\top - U^*U^{*\top}\|_{\text{op}} \leq \frac{\max\{\|U^*D\|_{\text{op}}, \|DV^*\|_{\text{op}}\}}{(1 - \bar{\gamma})\sigma_r^*} \leq \frac{\bar{\gamma}}{1 - \bar{\gamma}}, \\ \sqrt{\|\underline{H} - \text{sgn}(\underline{H})\|_{\text{op}}} &\leq \|VV^\top - V^*V^{*\top}\|_{\text{op}} \leq \frac{\max\{\|U^*D\|_{\text{op}}, \|DV^*\|_{\text{op}}\}}{(1 - \bar{\gamma})\sigma_r^*} \leq \frac{\bar{\gamma}}{1 - \bar{\gamma}}, \end{aligned}$$

and

$$\|\Sigma \underline{H} - \bar{H} \Sigma\|_{\text{op}} \leq 2 \|D\|_{\text{op}}, \quad \|\bar{H}^\top \Sigma - \Sigma \underline{H}^\top\|_{\text{op}} \leq 2 \|D\|_{\text{op}},$$

and

$$\|\bar{H}^{-1}\|_{\text{op}} \leq \frac{(1 - \bar{\gamma})^2}{(1 - 2\bar{\gamma})}, \quad \|\underline{H}^{-1}\|_{\text{op}} \leq \frac{(1 - \bar{\gamma})^2}{(1 - 2\bar{\gamma})}.$$

Proof. We prove the bounds for $\bar{H}$. The proof for $\underline{H}$ is identical. Clearly $\|\bar{H}\|_{\text{op}} \leq \|U\|_{\text{op}} \|U^*\|_{\text{op}} = 1$.

By Weyl's inequality, we have $|\sigma_i^* - \sigma_i| \leq \|D\|_{\text{op}}, \forall i \in [m]$, hence

$$\sigma_r^* - \sigma_{r+1} \geq \sigma_r^* - \|D\|_{\text{op}} = (1 - \bar{\gamma})\sigma_r^* > \frac{1}{2}\sigma_r^* > 0$$

by assumption. On the other hand, by standard perturbation theory, $1 \geq \bar{\sigma}_1 \geq \dots \geq \bar{\sigma}_r \geq 0$ are the cosines of the principal angles $0 \leq \bar{\theta}_1 \leq \dots \leq \bar{\theta}_r \leq \pi/2$

between the column spaces of U and U^* , and $\sin \bar{\theta}_r = \|UU^\top - U^*U^{*\top}\|_{\text{op}}$. Hence $\|\text{sgn}(\bar{H}) - \bar{H}\|_{\text{op}} = 1 - \cos \bar{\theta}_r$. By Wedin's $\sin \Theta$ Theorem, we have

$$\sin \bar{\theta}_r \leq \frac{\max\{\|U^*D\|_{\text{op}}, \|DV^*\|_{\text{op}}\}}{\sigma_r^* - \sigma_{r+1}} \leq \frac{\max\{\|U^*D\|_{\text{op}}, \|DV^*\|_{\text{op}}\}}{(1 - \bar{\gamma})\sigma_r^*}.$$

Since $\cos \bar{\theta}_r \geq \cos^2 \bar{\theta}_r = 1 - \sin^2 \bar{\theta}_r$, we obtain

$$\sqrt{\|\text{sgn}(\bar{H}) - \bar{H}\|_{\text{op}}} = \sqrt{1 - \cos \bar{\theta}_r} \leq \sin \bar{\theta}_r \leq \frac{\max\{\|U^*D\|_{\text{op}}, \|DV^*\|_{\text{op}}\}}{(1 - \bar{\gamma})\sigma_r^*} \leq \frac{\|D\|_{\text{op}}}{(1 - \bar{\gamma})\sigma_r^*} = \frac{\bar{\gamma}}{1 - \bar{\gamma}}.$$

Note that

$$U^\top DV^* = U^\top (A - A^*)V^* = \Sigma V^\top V^* - U^\top U^* \Sigma^* = \Sigma \underline{H} - \bar{H} \Sigma^*, \quad \text{and}$$

Hence

$$\begin{aligned} \|\Sigma \underline{H} - \bar{H} \Sigma^*\|_{\text{op}} &\leq \|\Sigma \underline{H} - \bar{H} \Sigma^*\|_{\text{op}} + \|\bar{H}(\Sigma^* - \Sigma)\|_{\text{op}} \\ &= \|U^\top DV^*\|_{\text{op}} + \|\bar{H}(\Sigma^* - \Sigma)\|_{\text{op}} \\ &\leq \|D\|_{\text{op}} + \|\bar{H}\|_{\text{op}} \|\Sigma^* - \Sigma\|_{\text{op}} \\ &\leq 2\|D\|_{\text{op}}, \end{aligned}$$

where we use $\|\bar{H}\|_{\text{op}} \leq 1$ and $\|\Sigma^* - \Sigma\|_{\text{op}} \leq \|D\|_{\text{op}}$. Similarly,

$$U^{*\top} DV = U^{*\top} (A - A^*)V = U^{*\top} U \Sigma - \Sigma^* V^{*\top} V^\top = \bar{H}^\top \Sigma - \Sigma^* \underline{H}^\top,$$

so a similar argument as above gives $\|\bar{H}^\top \Sigma - \Sigma^* \underline{H}^\top\|_{\text{op}} \leq 2\|D\|_{\text{op}}$.

Finally, recall that $\|\bar{H} - \text{sgn}(\bar{H})\|_{\text{op}} \leq \left(\frac{\bar{\gamma}}{1 - \bar{\gamma}}\right)^2$ and note the simple identity $X^{-1} - Y^{-1} = X^{-1}(Y - X)Y^{-1}$. It follows that

$$\begin{aligned} \|\bar{H}^{-1} - \text{sgn}(\bar{H})^{-1}\|_{\text{op}} &\leq \|\bar{H}^{-1}\|_{\text{op}} \|\bar{H} - \text{sgn}(\bar{H})\|_{\text{op}} \|\text{sgn}(\bar{H})^{-1}\|_{\text{op}} \\ &= \left[\sigma_r \left(\text{sgn}(\bar{H}) + (\bar{H} - \text{sgn}(\bar{H}))\right)\right]^{-1} \|\bar{H} - \text{sgn}(\bar{H})\|_{\text{op}} \\ &\leq \frac{1}{1 - \left(\frac{\bar{\gamma}}{1 - \bar{\gamma}}\right)^2} \cdot \left(\frac{\bar{\gamma}}{1 - \bar{\gamma}}\right)^2 = \frac{\bar{\gamma}^2}{1 - 2\bar{\gamma}}. \end{aligned}$$

It follows that

$$\|H^{-1}\|_{\text{op}} \leq \|\text{sgn}(\bar{H})^{-1}\|_{\text{op}} + \|\bar{H}^{-1} - \text{sgn}(\bar{H})^{-1}\|_{\text{op}} \leq 1 + \frac{\bar{\gamma}^2}{1-2\bar{\gamma}} = \frac{(1-\bar{\gamma})^2}{1-2\bar{\gamma}}.$$

This completes the proof of Lemma 3. $\square$

We are now ready to prove Proposition 32. We have

$$\begin{aligned} U - U^* \text{sgn}(\bar{H})^\top &= AV\Sigma^{-1} - U^* \text{sgn}(\bar{H})^\top & A &= U\Sigma V^\top \\ &= (A^* + D)V\Sigma^{-1} - U^* \text{sgn}(\bar{H})^\top \\ &= U^* \Sigma^* V^{*\top} V\Sigma^{-1} - U^* \text{sgn}(\bar{H})^\top + DV\Sigma^{-1} & A^* &= U^* \Sigma^* V^{*\top} \\ &= U^* \Sigma^* \underline{H}^\top \Sigma^{-1} - U^* \text{sgn}(\bar{H})^\top + DV\Sigma^{-1}. & \underline{H} &= V^\top V^* \end{aligned}$$

But $\Sigma^* \underline{H}^\top = (\Sigma + \Sigma^* - \Sigma) \underline{H}^\top = \bar{H}^\top \Sigma + (\Sigma \underline{H}^\top - \bar{H}^\top \Sigma) + (\Sigma^* - \Sigma) \underline{H}^\top$. Hence

$$\begin{aligned} U - U^* \text{sgn}(\bar{H})^\top &= U^* \left[\bar{H}^\top \Sigma + (\Sigma \underline{H}^\top - \bar{H}^\top \Sigma) + (\Sigma^* - \Sigma) \underline{H}^\top \right] \Sigma^{-1} - U^* \text{sgn}(\bar{H})^\top + DV\Sigma^{-1} \\ &= U^* \bar{H}^\top + U^* (\Sigma \underline{H}^\top - \bar{H}^\top \Sigma) \Sigma^{-1} + U^* (\Sigma^* - \Sigma) \underline{H}^\top \Sigma^{-1} - U^* \text{sgn}(\bar{H})^\top + DV\Sigma^{-1} \\ &= U^* (\bar{H} - \text{sgn}(\bar{H}))^\top + U^* (\Sigma \underline{H}^\top - \bar{H}^\top \Sigma) \Sigma^{-1} + U^* (\Sigma^* - \Sigma) \underline{H}^\top \Sigma^{-1} + DV\Sigma^{-1}. \end{aligned}$$

For each $i \in [m]$, we can bound the i -th row of $U - U^* \text{sgn}(\bar{H})^\top$ as

$$\begin{aligned} \|(U - U^* \text{sgn}(\bar{H})^\top)_i\|_2 &\leq \|U_i^*\|_2 \left(\|\bar{H} - \text{sgn}(\bar{H})\|_{\text{op}} + \|\Sigma \underline{H}^\top - \bar{H}^\top \Sigma\|_{\text{op}} \|\Sigma^{-1}\|_{\text{op}} + \|\Sigma^* - \Sigma\|_{\text{op}} \|\underline{H}\|_{\text{op}} \|\Sigma^{-1}\|_{\text{op}} \right) \\ &\quad + \|D_i\|_2 \|V\|_{\text{op}} \|\Sigma^{-1}\|_{\text{op}} \\ &\stackrel{(i)}{\leq} \|U_i^*\|_2 \left(\left(\frac{\bar{\gamma}}{1-\bar{\gamma}} \right)^2 + 2\|D\|_{\text{op}} \cdot \frac{1}{\sigma_r^*(1-\bar{\gamma})} + \|D\|_{\text{op}} \cdot 1 \cdot \frac{1}{\sigma_r^*(1-\bar{\gamma})} \right) + \|D_i\|_2 \cdot \frac{1}{\sigma_r^*(1-\bar{\gamma})} \\ &= \|U_i^*\|_2 \left(\left(\frac{\bar{\gamma}}{1-\bar{\gamma}} \right)^2 + \frac{3\bar{\gamma}}{1-\bar{\gamma}} \right) + \|D_i\|_2 \cdot \frac{1}{\sigma_r^*(1-\bar{\gamma})} \\ &\stackrel{(ii)}{\leq} \|U_i^*\|_2 \cdot 8\bar{\gamma} + \|D_i\|_2 \cdot \frac{1}{\sigma_r^*(1-\bar{\gamma})}, \end{aligned}$$

where step (i) follows from Lemma 3 and the bound

$$\|\Sigma^{-1}\|_{\text{op}} = \frac{1}{\sigma_r} \leq \frac{1}{\sigma_r^* - \|D\|_{\text{op}}} = \frac{1}{\sigma_r^*(1-\bar{\gamma})},$$

and step (ii) holds since $\bar{\gamma} \leq \frac{1}{2}$. Setting $\bar{R} = \text{sgn}(\bar{H})^\top$ proves the first inequality in Proposition 32.

The second inequality in Proposition 32 can be established by a similar argument.

Corollary 33. *Suppose the setting of Proposition 32 holds. Furthermore, assume that A^* is μ -incoherent with condition number κ and $\|A^*\|_\infty \geq C$ for some constant $C > 0$. Then, there exist two orthonormal matrices $\bar{R}, \underline{R} \in \mathbb{R}^{r \times r}$ such that*

$$\begin{aligned} \|(U - U^* \bar{R})_{i \cdot}\|_2 &\leq \frac{16\kappa(\mu d)^{3/2}\|D\|_\infty}{C \sqrt{m}}, \quad \forall i \in [m], \\ \|(V - V^* \underline{R})_{j \cdot}\|_2 &\leq \frac{16\kappa(\mu d)^{3/2}\|D\|_\infty}{C \sqrt{n}}, \quad \forall j \in [n]. \end{aligned}$$

Proof. From the definition of the ℓ_∞ norm, there exists $(i, j) \in [m] \times [n]$ such that $\|A^*\|_\infty = |A_{i,j}|$. Then, it follows that

$$\begin{aligned} \|A^*\|_\infty &= |A_{i,j}| = |(U^* \Sigma^* V^*)_{i,j}| \\ &\leq \|\Sigma^*\|_{op} \|U_i^*\|_2 \|V_j^*\|_2 \\ &\leq \frac{\sigma_1 \mu d}{\sqrt{mn}} \\ &\leq \frac{\kappa \sigma_d \mu d}{\sqrt{mn}}. \end{aligned}$$

Furthermore, we note that $\|D\|_{op} \leq \|D\|_\infty \sqrt{mn}$. Thus, the result follows from Proposition 32 and using the definition of incoherence and the above inequalities. $\square$

B.9 Proofs for Chapter 3

B.9.1 Proofs in the (S, S, d) Tucker Rank Setting

We first prove our lower bound.

Proof of Theorem 1

Proof. Suppose all source and target MDPs (S, A, P, H, r) share the same state space, action space, and horizon $H = 1$ and assume that $\mathcal{S} = \mathcal{A} = [2n]$ for some $n \in \mathbb{N}_+$. For ease of notation, we let s_1 and a_1 refer to any state and action in $[n]$, respectively, and s_2 and a_2 refer to any state and action in $\{n + 1, \dots, 2n\}$, respectively. We will present the latent factors and optimal Q functions as block vectors and matrices with s_1, s_2, a_1, a_2 as the blocks containing n entries. The initial state distribution in the target MDP is uniform over $\mathcal{S}$. We now present two transfer RL problems with similar Q functions (with rows s_1, s_2 and columns a_1, a_2) that satisfy Assumptions 8 and 9 but have orthogonal target state latent factors. For ease of notation, the superscript i of $Q_{m,h}^{*,i}, Q_h^{*,i}$ denotes transfer RL problem for $i \in \{1, 2\}$. Then, the optimal Q functions for transfer RL problem

one are

$$\begin{aligned} Q_{1,1}^{*,1} &= n \begin{bmatrix} \sqrt{1/n} \\ 0 \end{bmatrix} \begin{bmatrix} \sqrt{1/(2n)} & \sqrt{1/(2n)} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 0 \end{bmatrix}, \\ Q_{2,1}^{*,1} &= n \begin{bmatrix} 0 \\ \sqrt{1/n} \end{bmatrix} \begin{bmatrix} \sqrt{1/(2n)} & \sqrt{1/(2n)} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}, \\ Q_1^{*,1} &= n \begin{bmatrix} \sqrt{1/(2n)} \\ -\sqrt{1/(2n)} \end{bmatrix} \begin{bmatrix} \sqrt{1/(2n)} & \sqrt{1/(2n)} \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ -1/2 & -1/2 \end{bmatrix}, \end{aligned}$$

and the optimal Q functions for transfer RL problem two are

$$\begin{aligned} Q_{1,1}^{*,2} &= n \begin{bmatrix} \sqrt{1/n} \\ 0 \end{bmatrix} \begin{bmatrix} \sqrt{1/(2n)} & \sqrt{1/(2n)} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 0 \end{bmatrix}, \\ Q_{2,1}^{*,2} &= n \begin{bmatrix} 0 \\ \sqrt{1/n} \end{bmatrix} G' = \begin{bmatrix} 0 & 0 \\ (1 + 1/\alpha)/(c\sqrt{2}) & (1 - 1/\alpha)/(c\sqrt{2}) \end{bmatrix}, \\ Q_1^{*,2} &= n \begin{bmatrix} \sqrt{1/(2n)} \\ -\sqrt{1/(2n)} \end{bmatrix} \begin{bmatrix} \sqrt{\frac{1}{(2n)}} & -\sqrt{\frac{1}{(2n)}} \end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}, \end{aligned}$$

where

$$G' = \begin{bmatrix} (\sqrt{1/(2n)} + \sqrt{1/(2n\alpha^2)})/\sqrt{1+1/\alpha^2} & (\sqrt{1/(2n)} - \sqrt{1/(2n\alpha^2)})/\sqrt{1+1/\alpha^2} \end{bmatrix},$$

and $c = \sqrt{1+1/\alpha^2}$. Note that $c \in (1, 2dM)$ because $\alpha \geq 1/(dM)$ from Lemma 1 for $\alpha > 0$. The incoherence μ , rank d , and condition number κ of the above Q functions are $O(1)$. Furthermore, the above construction satisfies Assumption 11 with Tucker rank $(S, S, 1)$ and Assumption 12 because the source and target MDPs share the same action latent factor $\begin{bmatrix} \sqrt{1/(2n)} & \sqrt{1/(2n)} \end{bmatrix}$ in the first transfer RL problem while in the second transfer RL problem,

$$\begin{bmatrix} \sqrt{1/(2n)} & -\sqrt{1/(2n)} \end{bmatrix} = -\alpha \begin{bmatrix} \sqrt{1/(2n)} & \sqrt{1/(2n)} \end{bmatrix} + \alpha c G'.$$

Note that $G^1 = \begin{bmatrix} \sqrt{1/(2n)} & \sqrt{1/(2n)} \end{bmatrix}$ and $G^2 = \begin{bmatrix} \sqrt{1/(2n)} & -\sqrt{1/(2n)} \end{bmatrix}$, so the target latent factors of the different transfer RL problems are orthogonal to each other.

The learner is given either transfer RL problem one or two with equal probability with a generative model in the source problem. When interacting with the source MDPs, the learner specifies a state-action pair (s, a) and source MDP m and observes a realization of a shifted and scaled Bernoulli random variable X . The distribution of X is that $X = 1$ with probability $(Q_{m,1}^{*,i}(s, a) + 1)/2$ and $X = -1$ with probability $(1 - Q_{m,1}^{*,i}(s, a))/2$. Furthermore, the learner is given the knowledge that the state latent features lie in the function class $\mathcal{G} = \{G^1, G^2\}$ and must use the knowledge obtained from interacting with the source MDP to choose one to use in the target MDP, which is no harder than receiving no information about the function class.

To distinguish between the two transfer RL problems, one needs to differentiate between $Q_{2,1}^{*,1}$ and $Q_{2,1}^{*,2}$. By construction, the magnitude of the largest entrywise difference between $Q_{2,1}^{*,1}$ and $Q_{2,1}^{*,2}$ is lower bounded by $\Omega(1/\alpha)$.

If the learner observes Z samples in the source MDPs, where $Z \leq O(\alpha^2)$ for some constant $C > 0$, then the probability of correctly identifying G is upper bounded by 0.76 (Lemma 5.1 with $\delta = 0.24$ [8]). Thus, if the learner does not observe $\Omega(\alpha^2)$ samples in the source phase, then the learner returns the incorrect feature mapping that is orthogonal to the true feature mapping with probability at least 0.24. $\square$

Proof of Theorem 2

We first present the theoretical guarantees of LSVI-UCB-(S, S, d) and defer their proofs to Appendix B.9.2.

Theorem 14. *Assume that Assumption 8 holds and the learner is given the true latent*

action representation $G = \{G_h\}_{h \in [H]}$. Then, there exists a constant $c > 0$ such that for any $\delta \in (0, 1)$, if we set $\lambda = 1, \beta_{k,h}^s = cdH\sqrt{\iota}, \iota = \log(2dTS/\delta)$, then with probability at least $1 - \delta$, the total regret of Algorithm 2 is $\tilde{O}(\sqrt{d^3SH^3T})$.

Theorem 14 states that LSVI-UCB-(S, S, d) completely removes the regret bound's dependence on the size of the state space by utilizing the latent action representation G . Similarly, the algorithm is robust to misspecification error.

Assumption 19 (ξ -approximate (S, S, d) Tucker rank MDP). Assume $\xi \in [0, 1]$. Then, an MDP $(S, \mathcal{A}, P, r, H)$ is a ξ -approximate (S, S, d) Tucker rank MDP with given feature map G if there exist H unknown S -by- d matrices $W = \{W_h\}_{h \in [H]}$ and S -by- S -by- d tensors $U = \{U_h\}_{h \in [H]}$ such that

$$|r_h(s, a) - W_h(s)^\top G_h(a)| \leq \xi,$$

$$\left| \sum_{s' \in S} (P_h(s'|s, a) - U_h(s', s)G_h(a)^\top) \right| \leq \xi$$

for all $h \in [H]$ where $\|W_h\|_2, \|\sum_{s' \in S} g(s')U_h(s', a)\|_2 \leq \sqrt{A/(d\mu)}$ for any function $g : S \rightarrow [0, 1]$.

Theorem 15. Under the Assumption 19, for any $\delta \in (0, 1)$, for $\lambda = 1, \beta_{k,h}^s = O(Hd\sqrt{\iota} + \xi\sqrt{kdH})$ for $\iota = \log(2dTS/\delta)$, the regret of Algorithm 2 is $\tilde{O}(\sqrt{d^3H^3ST} + \xi dHT)$ with probability at least $1 - \delta$.

We now prove Theorem 2.

Proof. Suppose the assumption stated in Theorem 2 hold. Then, from [54], running LR-EVI with a sample complexity of $C'd^4\mu^5\kappa^4(S+A)MH^4\alpha^2T \log(2SAdMH/\delta)/S$ on each source MDP results in $\bar{Q}_{m,h}$ functions with singular value decomposition $\bar{Q}_{m,h} = \hat{F}_{m,h}\hat{\Sigma}_{m,h}\hat{G}_{m,h}$ that are $\sqrt{HS}/(16\alpha\mu\kappa\sqrt{dMT})$ -optimal with probability at least $1 - \delta/2$.

Since our estimates are $\sqrt{HS}/(16\alpha\kappa\sqrt{dMT})$ -optimal, it follows that $\bar{\gamma} < 1/2$ in the singular vector perturbation bound, Corollary 33. Since $\|Q_{m,h}^*\|_\infty \geq C$, from Corollary 33, it follows that there exists rotation matrices $R_{m,h}$ such that

$$\|(\hat{G}_{m,h}R_{m,h} - G_{m,h})(a)\|_2 \leq \frac{d\sqrt{H\mu S}}{\alpha\sqrt{AMT}}$$

for all $a \in A$ where the optimal Q function have singular value decomposition

$$Q_{m,h}^* = F_{m,h}\Sigma_{m,h}G_{m,h}^\top.$$

Let $\tilde{G}$ be the $A \times dM$ matrix from joining all the source action latent factor matrices, and $\hat{\tilde{G}}_h$ be the $A \times dM$ matrix from joining all $\tilde{G}_{m,h}$. Let R_h be the $d \times dM$ matrix that joins $R_{m,h}$ for all $m \in [M]$. Then, from Assumption 9 and Definition 4, it follows that for all $(s', s, a) \in \mathcal{S} \times \mathcal{S} \times A$, that

$$\begin{aligned} r_h(s, a) &= \sum_{i \in [d]} W_h(s, i) G_h(a, i) \\ &= \sum_{i \in [d]} \left(\sum_{j \in [d], m \in [M]} B(i, j, m) G_{h,m}(a, j) \right) W_h(s, i) \\ &= \sum_{i, j \in [d], m \in [M]} B(i, j, m) G_{h,m}(a, j) W_h(s, i) \\ &= \tilde{W}'_h(s) G_h(a)^\top \end{aligned}$$

and $P(s'|\cdot, \cdot) = \tilde{G}_h U'_h(s')^\top$ (with the same argument) for some $S \times dM$ matrices $W'_h, U'_h(s')$ with entries bounded by α . Note that W'_h and $U'_h(s')$ are matrices that join $W_{m,h}$ and $U_{m,h}(s')$, respectively, with entries that at most α times larger than the original entry.

Therefore, it follows for all $(s', s, a, h) \in \mathcal{S} \times \mathcal{S} \times \mathcal{A} \times [H]$,

$$\begin{aligned}
|r_h(s, a) - \hat{\hat{G}}_h(a)R_h W'_h(s)^\top| &= |\tilde{G}_h(a)W'_h(s)^\top - \hat{\hat{G}}_h(a)R_h W'_h(s)^\top| \\
&= |(\tilde{G}_h(a) - \hat{\hat{G}}_h(a)R_h)W'_h(s)| \\
&\leq \alpha \sum_{m \in M} (G_{m,h}(a) - \hat{G}_{m,h}(a)R_{m,h})W_{m,h}(s) \\
&\leq \alpha \sum_{m \in M} \|G_{m,h}(a) - \hat{G}_{m,h}(a)R_{m,h}\|_2 \|W_{m,h}(s)\|_2 \\
&\leq \alpha \sqrt{\frac{A}{d\mu}} \sum_{m \in M} \|G_{m,h}(a) - \hat{G}_{m,h}(a)R_{m,h}\|_2 \\
&\leq \sqrt{\frac{dHMS}{T}}
\end{aligned}$$

from the Cauchy-Schwarz inequality and our singular vector perturbation bound. From the same logic,

$$\begin{aligned}
\left| \sum_{s' \in \mathcal{S}} (P_h(s'|s, a) - U_h(s', s)G_h(a)^\top) \right| &= \left| \sum_{s' \in \mathcal{S}} \tilde{G}_h(a)U'_h(s', s) - \hat{\hat{G}}_h(a)R_h U'_h(s', s)^\top \right| \\
&= |(\tilde{G}_h(a) - \hat{\hat{G}}_h(a)R_h) \sum_{s' \in \mathcal{S}} U'_h(s', s)^\top| \\
&\leq \alpha \sum_{m \in M} (G_{m,h}(a) - \hat{G}_{m,h}(a)R_{m,h}) \sum_{s' \in \mathcal{S}} U_{m,h}(s', s) \\
&\leq \alpha \sum_{m \in M} \|(G_{m,h}(a) - \hat{G}_{m,h}(a)R_{m,h})\|_2 \left\| \sum_{s' \in \mathcal{S}} U_{m,h}(s', s)^\top \right\|_2 \\
&\leq \sqrt{\frac{dHMS}{T}}.
\end{aligned}$$

Therefore, $\hat{\hat{G}}_h(a)$ satisfies Assumption 19 with $\xi = \sqrt{\frac{dMHS}{T}}$. Then it follows that running LSVI-UCB-TR using $\hat{\hat{G}}_h(a)$ with $\delta' = \delta/2$ during the target phase of the algorithm admits a regret bound of

$$\text{Regret}(T) \leq C''(\sqrt{d^3 M^3 H^3 S T} + dMHT \sqrt{\frac{dMHS}{T}}) \in \tilde{O}(\sqrt{d^3 M^3 H^3 S T})$$

with probability at least $1 - \delta$ from a final union bound. Furthermore, the sample

complexity in the source phase is

$$\tilde{O}\left(\frac{d^4\mu^5\kappa^4(S+A)M^2H^4\alpha^2T}{S}\right),$$

which proves our result. $\square$

Proof of Theorem 3

Proof. Let the assumptions of Theorem 3 hold. Then, from [54], running LR-EVI with a sample complexity of $C'd^7\mu^5\kappa^4A(S+A)M^2H^4\alpha^2T \log(2SAdMH/\delta)/(Sd'')$ on each source MDP results in $\bar{Q}_{m,h}$ functions with singular value decomposition $\bar{Q}_{m,h} = \hat{F}_{m,h}\hat{\Sigma}_{m,h}\hat{G}_{m,h}$ that are $\sqrt{Hd''S}/(16\alpha\mu M\kappa d^2\sqrt{AT})$ -optimal with probability at least $1 - \delta/2$.

Since our estimates are $\sqrt{Hd''S}/(16\alpha\mu M\kappa d^2\sqrt{AT})$ -optimal, it follows that $\bar{y} < 1/2$ in the singular vector perturbation bound, Corollary 33. Since $\|Q_{m,h}^*\|_\infty \geq C$, from Corollary 33, it follows that there exists rotation matrices $R_{m,h}$ such that

$$\|(\hat{G}_{m,h}R_{m,h} - G_{m,h})(a)\|_2 \leq \frac{\sqrt{Hd''\mu S}}{\alpha AM \sqrt{dT}}$$

for all $a \in A$ where the optimal Q function have singular value decomposition

$$Q_{m,h}^* = F_{m,h}\Sigma_{m,h}G_{m,h}^\top.$$

Let $\tilde{G}$ be the $A \times dM$ matrix from joining all the source action latent factor matrices, and $\hat{\tilde{G}}_h$ be the $A \times dM$ matrix from joining all $\bar{G}_{m,h}$. Let R_h be the $d \times dM$ matrix that joins $R_{m,h}$ for all $m \in [M]$. Then, from Assumption 9 and Definition

4, it follows that for all $(s', s, a) \in \mathcal{S} \times \mathcal{S} \times A$, that

$$\begin{aligned}
r_h(s, a) &= \sum_{i \in [d]} W_h(s, i) G_h(a, i) \\
&= \sum_{i \in [d]} \left(\sum_{j \in [d], m \in [M]} B(i, j, m) G_{h,m}(a, j) \right) W_h(s, i) \\
&= \sum_{i, j \in [d], m \in [M]} B(i, j, m) G_{m,h}(a, j) W_h(s, i) \\
&= \tilde{W}'_h(s) G_h(a)^\top
\end{aligned}$$

and $P(s'|\cdot, \cdot) = \tilde{G}_h U'_h(s')^\top$ (with the same argument) for some $S \times dM$ matrices $W'_h, U'_h(s')$ with entries bounded by α . Note that W'_h and $U'_h(s')$ are matrices that join $W_{m,h}$ and $U_{m,h}(s')$, respectively, with entries that at most α times larger than the original entry.

Let $E = \hat{\hat{G}}_h - \tilde{G}_h$, where $E = \begin{bmatrix} E_1 & \dots & E_M \end{bmatrix}$. We next bound the $d'' + 1$ smallest singular value of $\hat{\hat{G}}_h$ with our singular vector perturbation bound ($\|E\|_\infty \leq \|E\|_{2,\infty}$).

$$\sigma_{d''+1}(\hat{\hat{G}}_h) = \sigma_{d''+1}(\tilde{G}_h) + \|E\|_{op} \leq \sqrt{AdM} \|E\|_\infty \leq \frac{\sqrt{Hd''\mu S}}{\alpha \sqrt{AMT}}$$

With singular value decomposition $\hat{\hat{G}}_h = X \Sigma Y^\top$, let $G_h^{d''}$ be the thresholded feature representation estimate, i.e., $G_h^{d''} = X(:, : d'') \Sigma(:, d'', : d'') Y(:, d'')^\top$. There-

fore, it follows for all $(s', s, a, h) \in \mathcal{S} \times \mathcal{S} \times \mathcal{A} \times [H]$,

$$\begin{aligned}
& |r_h(s, a) - G_h^{d''}(a)R_h W'_h(s, : d'')^\top| \\
&= |\tilde{G}_h(a)W'_h(s)^\top - G_h^{d''}(a)R_h W'_h(s, : d'')^\top| \\
&\leq |\tilde{G}_h(a)W'_h(s)^\top - \hat{\tilde{G}}_h(a)R_h W'_h(s)^\top| + |\hat{\tilde{G}}_h(a)R_h W'_h(s)^\top - G_h^{d''}(a)R_h W'_h(s, : d'')^\top| \\
&\leq \alpha \left| \sum_{m \in M} (G_{m,h}(a) - \hat{G}_{m,h}(a)R_{m,h})W_{m,h}(s) \right| + |\hat{\tilde{G}}_h(a, d'' + 1 :)W'_h(s, d'' + 1 :)^\top| \\
&\leq \alpha \sum_{m \in M} \|G_{m,h}(a) - \hat{G}_{m,h}(a)R_{m,h}\|_2 \|W_{m,h}(s)\|_2 + \alpha \sigma_{d''+1}(\hat{\tilde{G}}_h) \sqrt{dM} \sum_{m \in [M]} \|W_{m,h}\|_2 \\
&\leq \alpha \sqrt{\frac{A}{d\mu}} \left(\sum_{m \in M} \|G_{m,h}(a) - \hat{G}_{m,h}(a)R_{m,h}\|_2 + \sigma_{d''+1}(\hat{\tilde{G}}_h) \sqrt{dM} \right) \\
&\leq \sqrt{\frac{d''HS}{T}}
\end{aligned}$$

from the Cauchy-Schwarz inequality and our singular vector perturbation bound. From the same logic,

$$\left| \sum_{s' \in \mathcal{S}} (P_h(s'|s, a) - U_h(s', s)G_h^{d''}(a)^\top) \right| \leq \sqrt{\frac{d''HS}{T}}.$$

Therefore, $G_h^{d''}(a)$ satisfies Assumption 19 with $\xi = \sqrt{\frac{d''HS}{T}}$. Then it follows that running LSVI-UCB-TR using $G_h^{d''}(a)$ with $\delta' = \delta/2$ during the target phase of the algorithm admits a regret bound of

$$\text{Regret}(T) \leq C''(\sqrt{(d'')^3 H^3 S T} + d'' H T \sqrt{\frac{d''HS}{T}}) \in \tilde{O}(\sqrt{(d'')^3 H^3 S T})$$

with probability at least $1 - \delta$ from a final union bound. Furthermore, the sample complexity in the source phase is

$$\tilde{O}\left(\frac{d^7 \mu^5 \kappa^4 (A + A^2) M^3 H^4 \alpha^2 T}{S d''}\right),$$

which proves our result. □

We now prove Theorem 4.

Proof. Suppose the assumption stated in Theorem 4 hold. Then, from [54], running LR-EVI with a sample complexity of $C'd^5\mu^5\kappa^4(S+A)M^2H^4\alpha^2T \log(2SAdMH/\delta)/S$ on each source MDP results in $\bar{Q}_{m,h}$ functions with singular value decomposition $\bar{Q}_{m,h} = \hat{F}_{m,h}\hat{\Sigma}_{m,h}\hat{G}_{m,h}$ that are $\sqrt{HS}/(16\alpha\mu\kappa\sqrt{d^2M^2T})$ -optimal with probability at least $1 - \delta/2$.

Since our estimates are $\sqrt{HS}/(16\alpha\kappa\sqrt{d^2M^2T})$ -optimal, it follows that $\bar{\gamma} < 1/2$ in the singular vector perturbation bound, Corollary 33. Since $\|Q_{m,h}^*\|_\infty \geq C$, from Corollary 33, it follows that there exists rotation matrices $R_{m,h}$ such that

$$\|(\hat{G}_{m,h}R_{m,h} - G_{m,h})(a)\|_2 \leq \frac{\sqrt{Hd\mu S}}{\alpha\sqrt{AM^2T}}$$

for all $a \in A$ where the optimal Q function have singular value decomposition $Q_{m,h}^* = F_{m,h}\Sigma_{m,h}G_{m,h}^\top$.

Let $\tilde{G}$ be the $A \times dM$ matrix from joining all the source action latent factor matrices, and $\hat{\tilde{G}}_h$ be the $A \times dM$ matrix from joining all $\tilde{G}_{m,h}$. Let R_h be the $d \times dM$ matrix that joins $R_{m,h}$ for all $m \in [M]$. Then, from Assumption 9 and Definition 4, it follows that for all $(s', s, a) \in \mathcal{S} \times \mathcal{S} \times A$, that

$$\begin{aligned} r_h(s, a) &= \sum_{i \in [d]} W_h(s, i) G_h(a, i) \\ &= \sum_{i \in [d]} \left(\sum_{j \in [d], m \in [M]} B(i, j, m) G_{h,m}(a, j) \right) W_h(s, i) \\ &= \sum_{i, j \in [d], m \in [M]} B(i, j, m) G_{m,h}(a, j) W_h(s, i) \\ &= \tilde{W}'_h(s) G_h(a)^\top \end{aligned}$$

and $P(s'|\cdot, \cdot) = \tilde{G}_h U'_h(s')^\top$ (with the same argument) for some $S \times dM$ matrices $W'_h, U'_h(s')$ with entries bounded by α . Note that W'_h and $U'_h(s')$ are matrices that

join $W_{m,h}$ and $U_{m,h}(s')$, respectively, with entries that at most α times larger than the original entry. Let $\hat{G}^t$ be the thresholded feature mapping, where t is defined as $t = \min_{t' \in \mathcal{T}} t'$ for

$$\mathcal{T} = \left\{ t \in [dM] \mid \sigma_{t+1} \leq \sqrt{\frac{tHS d\mu}{\alpha^2 T(dM - t)M^2 A}} \right\}.$$

Therefore, with the same logic as the previous proof, it follows for all $(s', s, a, h) \in \mathcal{S} \times \mathcal{S} \times \mathcal{A} \times [H]$,

$$\begin{aligned} |r_h(s, a) - \hat{G}^t(a)R_h W'_h(s)^\top| &= |\tilde{G}_h(a)W'_h(s)^\top - \hat{G}^t(a)R_h W'_h(s)^\top| \\ &\leq |(\tilde{G}_h(a) - \hat{G}_h(a)R_h)W'_h(s)^\top| + |\hat{G}_h(a)R_h W'_h(s)^\top - \hat{G}^t(a)(R_h W'_h(s))^{t,\top}| \\ &\leq \sqrt{\frac{HS}{T}} + |\hat{G}_h(a, t+1 :)(R_h W'_h(s))^{t+1 : \top}| \\ &\leq \sqrt{\frac{HS}{T}} + \sigma_{t+1} \alpha \sqrt{dM - t} \|W'_h\|_2 \\ &\leq \sqrt{\frac{HS}{T}} + \sqrt{\frac{tHS}{T}} \\ &\leq 2\sqrt{\frac{tHS}{T}} \end{aligned}$$

where the second and third inequalities comes from the same logic used in the proofs of Theorems 2 and 4, and the fourth inequality comes from the definition of t . From the same logic,

$$\left| \sum_{s' \in \mathcal{S}} (P_h(s'|s, a) - U_h(s', s)\hat{G}^t(a)^\top) \right| \leq 2\sqrt{\frac{tHS}{T}}.$$

Therefore, $\hat{G}_h(a)$ satisfies Assumption 19 with $\xi = \sqrt{\frac{tHS}{T}}$. Then it follows that running LSVI-UCB-TR using $\hat{G}_h^t(a)$ with $\delta' = \delta/2$ during the target phase of the algorithm admits a regret bound of

$$\text{Regret}(T) \leq C''(\sqrt{t^3 H^3 S T} + 2tHT\sqrt{\frac{tHS}{T}}) \in \tilde{O}(\sqrt{t^3 H^3 S T})$$

with probability at least $1 - \delta$ from a final union bound. Furthermore, the sample complexity in the source phase is

$$\tilde{O}\left(\frac{d^5 \mu^5 \kappa^4 (S + A) M^3 H^4 \alpha^2 T}{S}\right),$$

which proves our result. $\square$

B.9.2 Proofs of LSVI-UCB-(S, d, A) and LSVI-UCB-(S, S, d)

In this section, we present the proofs of Theorem 6 and 7 and the related lemmas. The proofs, theorems, and lemmas are modifications of the analogous result/proof in [37] tailored to our Tucker rank settings.

LSVI-UCB-(S, d, A)

To prove Theorem 6, we first introduce notation used throughout this section and then present auxiliary lemmas. We reiterate that the steps and analysis in this subsection are essentially the same as the proofs in [37], except we have A copies of the weight vectors and gram matrices. We let $\Lambda_{h,k}^a, w_{h,k}^a, Q_h^k, V_h^k, \pi^k$ as the parameters/estimates at episode k . Furthermore, let $V_h^k = \max_{a \in \mathcal{A}} Q_h^k(s, a)$ and π^k be the greedy policy w.r.t. Q_h^k . We let $F_h^k := F_h(s_h^k)$.

To prove the desired regret bound, we first need to prove that $\mathbb{P}V - \hat{P}V$ concentrates about 0, which differs from typical concentration inequalities for self normalized processes due to the value function not being constant. Thus, we first bound the log of the covering number of our function class for the value function to control the error of our algorithm's estimate.

Lemma 4. Let $\mathcal{V}$ denote a function class of mappings from $\mathcal{S}$ to $\mathbb{R}$ with the following parametric form

$$V(\cdot) = \min \left\{ \max_{a \in \mathcal{A}} w_a^\top F(\cdot) + \beta \sqrt{F(\cdot)^\top \Lambda_a^{-1} F(\cdot)}, H \right\}$$

where the parameters w_a, β, Λ_a satisfy $\|w_a\| \leq L$ for all $a \in \mathcal{A}$, $\beta \in [0, B]$, and the minimum eigenvalue of Λ_a is greater than λ . Assume that $\|F(s)\| \leq 1$ for all $s \in \mathcal{S}$, and let N_ϵ be the ϵ -covering number of $\mathcal{V}$ with respect to the distance $\text{dist}(V, V') = \sup_s |V(s) - V'(s)|$. Then,

$$\log(N_\epsilon) \leq d \log(1 + 4L/\epsilon) + d^2 \log(1 + 8d^{1/2} B^2 / (\lambda \epsilon^2))$$

Proof. Equivalently, we reparameterize the function class by $D_a = \beta^2 \Lambda_a^{-1}$,

$$V(\cdot) = \min \left\{ \max_{a \in \mathcal{A}} w_a^\top F(\cdot) + \sqrt{F(\cdot)^\top D_a F(\cdot)}, H \right\}$$

where $\|w_a\| \leq L$ and $\|D_a\| \leq B^2/\lambda$ for all $a \in \mathcal{A}$. For any $V_1, V_2 \in \mathcal{V}$, let them take the form in the above equation with parameters (w_a, D_a) and (w'_a, D'_a) . Since $\min(\cdot, H)$ and $\max_{a \in \mathcal{A}}$ are contraction maps,

$$\begin{aligned} \text{dist}(V_1 - V_2) &\leq \sup_{s, a} |(w_a^\top F(s) + \sqrt{F(s)^\top D_a F(s)}) - (w'_a{}^\top F(s) + \sqrt{F(s)^\top D'_a F(s)})| \\ &\leq \sup_{a, F: \|F\| \leq 1} |(w_a^\top F + \sqrt{F^\top D_a F}) - (w'_a{}^\top F + \sqrt{F^\top D'_a F})| \\ &\leq \max_{a \in \mathcal{A}} \left[\sup_{F: \|F\| \leq 1} |(w_a - w'_a)^\top F| + \sup_{F: \|F\| \leq 1} |\sqrt{F^\top (D_a - D'_a) F}| \right] \\ &\leq \max_{a \in \mathcal{A}} \left[\|w_a - w'_a\| + \sqrt{\|D_a - D'_a\|} \right] \leq \max_{a \in \mathcal{A}} \left[\|w_a - w'_a\| + \sqrt{\|D_a - D'_a\|_F} \right] \end{aligned}$$

where the holds because $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$ for any $x, y \geq 0$. Let C_w be an $\epsilon/2$ -cover of $\{w \in \mathbb{R}^d \mid \|w\| \leq L\}$ with respect to the 2-norm, and C_D be an $\epsilon^2/4$ -cover of $\{D \in \mathbb{R}^{d \times d} \mid \|D\|_F \leq d^{1/2} B^2 / \lambda\}$ with respect to the Frobenius norm. By Lemma D.5 from [37], we know that,

$$|C_w| \leq (1 + 4L/\epsilon)^d, \quad |C_D| \leq (1 + 8d^{1/2} B^2 / (\lambda \epsilon^2))^{d^2}.$$

Hence, for any V_1 , there exists a $w'_a \in C_w$ and $D'_a \in C_D$ for all $a \in \mathcal{A}$ such that V_2 parameterized by (w'_a, D'_a) satisfies $\text{dist}(V_1, V_2) \leq \epsilon$. Hence, $\mathcal{N}_\epsilon \leq |C_w||C_D|$, which gives:

$$\log(\mathcal{N}_\epsilon) \leq \log(|C_w|) + \log(|C_D|) \leq d \log(1 + 4\epsilon L/\epsilon) + d^2 \log(1 + 8d^{1/2}B^2/(\lambda\epsilon^2)).$$

□

We next bound the ℓ_2 -norm of the weight vector of any policy.

Lemma 5. *Let $\{w_h^a\}$ be the weight vector of some policy π , i.e., $Q_h^\pi(s, a) = F_h(s)^\top w_h^a$. Then,*

$$\|w_h^a\| \leq 2H \sqrt{d}.$$

Proof. Note that $Q_h^\pi(s, a) = r_h(s, a) + \sum_{s'} V_{h+1}^\pi(s') P_h(s'|s, a)$, so, using the low-rank representations of r_h, P_h , it follows that

$$\|w_h^a\|_2 = \|W_h(a, \cdot) + \sum_{s'} V^\pi(s') U_h(s', a, \cdot)\|_2 \leq \|W_h(a, \cdot)\|_2 + \|\sum_{s'} V^\pi(s') U_h(s', a, \cdot)\|_2 \leq \sqrt{d} + H \sqrt{d}.$$

Recall that since the feature mapping was scaled up ($F_h = \sqrt{\frac{S}{d\mu}} F'$) to ensure the max entries of F and W, U are on the same scale, it follows that $\|W_h(a)\|_2 \leq 1$ and $\|\sum_{s' \in \mathcal{S}} V_{h+1}^\pi(s') U_h(s', a)\|_2 \leq H$. Therefore, it follows that $\|w_h^a\|_2 \leq 2H \sqrt{d}$. □

We next bound the 2-norm of the weight vector from the algorithm.

Lemma 6. *For any k, h, a , the weight vector from the above algorithm satisfies $\|w_{h,k}^a\| \leq 2H \sqrt{dk/\lambda}$.*

Proof. For any $F_h(s) = f$, we have for any $a \in \mathcal{A}$

$$\begin{aligned}
|f^\top w_{h,k}^a| &= |f^\top (\Lambda_h^a)^{-1} \sum_{t \in T_{h,k-1}^a} F_h(s_h^t) \left[r_h(s_h^t, a) + \max_{a' \in \mathcal{A}} Q_{h+1}(s_{h+1}^t, a') \right]| \\
&\leq \left| \sum_{t \in T_{h,k-1}^a} f^\top (\Lambda_h^a)^{-1} F_h(s_h^t) \right| 2H \\
&\leq 2H \sqrt{\left(\sum_{t \in T_{h,k-1}^a} f^\top (\Lambda_h^a)^{-1} f \right) \left(\sum_{t \in T_{h,k-1}^a} F_h(s_h^t)^\top (\Lambda_h^a)^{-1} F_h(s_h^t) \right)} \\
&\leq 2H \|f\|_2 \sqrt{d |T_{h,k-1}^a| / \lambda} \leq 2H \sqrt{dk / \lambda},
\end{aligned}$$

where the second to last inequality is due to Lemma D.1 from [37] and that Rayleigh quotients are upper bounded by the largest eigenvalue of the matrix (Theorem 4.2.2 from [30]). The last inequality is due to $\|F_h(s)\|_2 \leq 1$ and $|T_{h,k-1}^a| \leq k$. $\square$

The above two lemmas are necessary in ensuring that the estimation error is small as they depend multiplicatively on the norm of the weight vectors. We next prove the main concentration lemma that controls the estimate.

Lemma 7. *Let c' be the constant in the definition of $\beta_{k,h}^a = c' dH \sqrt{t}$. Then, there exists a constant $C > 0$ that is independent of c' such that for any $\delta \in (0, 1)$, if we let the event $\mathcal{X}$ be*

$$\forall (a, k, h) \in \mathcal{A} \times [K] \times [H] : \left\| \sum_{t \in T_{h,k-1}^a} F_h(s_h^t) (V_{h+1}^t(s_{h+1}^t) - P_h V_{h+1}^t(s_h^t, a)) \right\|_{(\Lambda_{h,t}^a)^{-1}} \leq CdH \sqrt{\chi}$$

where $\chi = \log(2(c' + 1)dTA/\delta)$, then $\mathbb{P}(\mathcal{X}) \geq 1 - \delta/2$.

Proof. From Lemma 6, we have $\|w_{h,k}^a\|_2 \leq 2H \sqrt{dk/\lambda}$. By construction, the minimum eigenvalue of $\Lambda_{h,t}^a$ is lower bounded by λ . Then, from Lemma D.4 in [37] (with $\delta' = \delta/A$), Lemma 4, and taking a union bound over actions, we have that

with probability at least $1 - \delta'/2$, for all $a \in \mathcal{A}$ and $\epsilon > 0$,

$$\left\| \sum_{t \in T_{h,k-1}^a} F_h(s_h^t)(V_{h+1}^t(s_{h+1}^t) - P_h V_{h+1}^t(s_h^t, a)) \right\|_{(\Lambda_{h,t}^a)^{-1}}^2 \leq 4H^2 \left[\frac{d}{2} \log((k + \lambda)/\lambda) + \log(2A\mathcal{N}_\epsilon/\delta) \right] + 8k^2\epsilon^2/\lambda.$$

Then, from Lemma 4, it follows that with probability at least $1 - \delta'/2$, for all $a \in A$ and $\epsilon > 0$,

$$\begin{aligned} & \left\| \sum_{t \in T_{h,k-1}^a} F_h(s_h^t)(V_{h+1}^t(s_{h+1}^t) - P_h V_{h+1}^t(s_h^t, a)) \right\|_{(\Lambda_{h,t}^a)^{-1}}^2 \\ & \leq 4H^2 \left[\frac{d}{2} \log((k + \lambda)/\lambda) + \log(2A/\delta) + d \log(1 + 4L/\epsilon) + d^2 \log(1 + 8d^{1/2}\beta^2/(\lambda\epsilon^2)) \right] + 8k^2\epsilon^2/\lambda. \end{aligned}$$

Choosing $\lambda = 1, \beta_{k,h}^a = CdH\epsilon, \epsilon = dH/k$ gives that there exists some constant C' such that

$$\left\| \sum_{t \in T_{h,k-1}^a} F_h(s_h^t)(V_{h+1}^t(s_{h+1}^t) - P_h V_{h+1}^t(s_h^t, a)) \right\|_{(\Lambda_{h,t}^a)^{-1}}^2 \leq C' d^2 H^2 \log(2(c_\beta + 1)dTA/\delta).$$

□

We next recursively bound the difference between the value function maintained in the algorithm and the true value function of any policy π . We upper-bound this with their expected difference and an additional error term, which is bounded with high probability.

Lemma 8. *There exists a constant c_β such that $\beta_{k,h}^a = c_\beta dH \sqrt{\iota}$ where $\iota = \log(2dAT/p)$ and for any fixed policy π , on the event $\mathcal{X}$ defined in Lemma 7, we have for all $(s, a, h, k) \in S \times A \times [H] \times [K]$:*

$$F_h(s)^\top w_{h,k}^a - Q_h^\pi(s, a) = P_h(V_{h+1}^k - V_{h+1}^\pi)(s, a) + \Delta_h^k(s, a),$$

where $|\Delta_h^k(s, a)| \leq \beta_{k,h}^a \sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)}.$

Proof. From our low-rank assumption, it follows that

$$Q_h^\pi(s, a) = F_h(s)^\top w_h^a = (r_h + P_h V_{h+1}^\pi)(s, a),$$

which following the steps from the proof of Lemma B.4 in [37] gives

$$\begin{aligned} w_{h,k}^a - w_h^{a,\pi} &= -\lambda(\Lambda_{h,k}^a)^{-1} w_h^{a,\pi} \\ &\quad + (\Lambda_{h,k}^a)^{-1} \sum_{t \in T_{k-1,h}^a} F_h(s_h^t) (V_{h+1}^k(s_{h+1}^t) - P_h V_{h+1}^k(s_h^t, a)) \\ &\quad + (\Lambda_{h,k}^a)^{-1} \sum_{t \in T_{k-1,h}^a} F_h(s_h^t) P_h (V_{h+1}^k - V_{h+1}^\pi)(s_h^t, a) \end{aligned}$$

Now, we bound each term (q_1^a, q_2^a, q_3^a) individually, for the first term, note for all $a \in \mathcal{A}$

$$|F_h(s)^\top q_1^a| = |\lambda F_h(s)^\top (\Lambda_{h,k}^a)^{-1} w_h^{a,\pi}| \leq \sqrt{\lambda} \|w_h^{a,\pi}\| \sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)}$$

For the second term, given the event $\mathcal{X}$, we have from Cauchy Schwarz

$$|F_h(s)^\top q_2^a| \leq CdH \sqrt{\chi} \sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)}$$

where $\chi = \log(2(c_\beta + 1)dTA/p)$. For the third term,

$$\begin{aligned} F_h(s)^\top q_3^a &= F_h(s)^\top \left((\Lambda_{h,k}^a)^{-1} \sum_{t \in T_{k-1,h}^a} F_h(s_h^t) P_h (V_{h+1}^k - V_{h+1}^\pi)(s_h^t, a) \right) \\ &= F_h(s)^\top \left((\Lambda_{h,k}^a)^{-1} \sum_{t \in T_{k-1,h}^a} F_h(s_h^t) F_h(s_h^t)^\top \sum_{s' \in \mathcal{S}} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', a) \right) \\ &= F_h(s)^\top \left(\sum_{s' \in \mathcal{S}} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', a) \right) \\ &\quad - \lambda F_h(s)^\top \left((\Lambda_{h,k}^a)^{-1} \sum_{s' \in \mathcal{S}} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', a) \right) \end{aligned}$$

We note that

$$F_h(s)^\top \left(\sum_{s' \in \mathcal{S}} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', a) \right) = P_h (V_{h+1}^k - V_{h+1}^\pi)(s, a)$$

and

$$|\lambda F_h(s)^\top \left((\Lambda_{h,k}^a)^{-1} \sum_{s' \in \mathcal{S}} (V_{h+1}^k - V_{h+1}^k)(s') U_h(s', a) \right)| \leq 2H \sqrt{d\lambda} \sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)}$$

Since

$$|F_h(s)^\top w_{h,k}^a - Q_h^\pi(s, a)| = F_h(s)^\top (w_{h,k}^a - w_h^{a,\pi}) = F_h(s)^\top (q_1^a + q_2^a + q_3^a),$$

it follows that

$$|F_h(s)^\top w_{h,k}^a - Q_h^\pi(s, a) - P_h(V_{h+1}^k - V_{h+1}^\pi)(s, a)| \leq C'' dH \sqrt{\chi} \sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)}.$$

Similarly as in [37], we choose c_β that satisfies $C'' \sqrt{\log(2) + \log(c_\beta + 1)} \leq c_\beta \sqrt{\log(2)}$. $\square$

This lemma implies that by adding the appropriate bonus, Q_h^k is always an upperbound of Q_h^* with high probability, i.e., achieving optimism.

Lemma 9. *On the event $\mathcal{X}$ defined in lemma 7, we have $Q_h^k(s, a) \geq Q_h^*(s, a)$ for all $(s, a, h, k) \in \mathcal{S} \times \mathcal{A} \times [H] \times [K]$.*

Proof. We prove this with induction. The base case holds because at step $H + 1$, the value function is zero, so from Lemma 8,

$$|F_H(s)^\top w_{H,k}^a - Q_H^*(s, a)| \leq \beta_{k,H}^a \sqrt{F_H(s)^\top (\Lambda_{H,k}^a)^{-1} F_H(s)}$$

and thus,

$$Q_H^*(s, a) \leq \min \left(H, F_H(s)^\top w_{H,k}^a + \beta_{k,H}^a \sqrt{F_H(s)^\top (\Lambda_{H,k}^a)^{-1} F_H(s)} \right) = Q_H^k(s, a).$$

From the inductive hypothesis (assuming that $Q_{h+1}^k(s, a) \geq Q_{h+1}^*(s, a)$), it follows that $P_h(V_{h+1}^k - V_{h+1}^*)(s, a) \geq 0$. From Lemma 8, $F_h(s)^\top w_{h,k}^a - Q_h^*(s, a) \leq \beta_{k,h}^a \sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)}$. It follows that

$$Q_h^*(s, a) \leq \min \left(H, F_h(s)^\top w_{h,k}^a + \beta_{k,h}^a \sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)} \right) = Q_h^k(s, a).$$

$\square$

We next show that Lemma 8 transforms to the recursive formula $\delta_h^k = V_h^k(s_h^k) - V_h^{\pi_k}(s_h^k)$, which is used in proving our regret bound.

Lemma 10. *Let $\delta_h^k = V_h^k(s_h^k) - V_h^{\pi_k}(s_h^k)$ and $\xi_{h+1}^k = \mathbb{E}[\delta_{h+1}^k | s_h^k, a_h^k] - \delta_{h+1}^k$. Then, on the event $\mathcal{X}$ defined in Lemma 7, we have for any (h, k) :*

$$\delta_h^k \leq \delta_{h+1}^k + \xi_{h+1}^k + \beta_{k,h}^a \sqrt{F_h(s_h^k)^\top (\Lambda_{h,k}^{a_h^k})^{-1} F_h(s_h^k)}$$

Proof. From Lemma 8, we have for any (s, a, h, k) that

$$Q_h^k(s, a) - Q_h^{\pi_k}(s, a) \leq P_h(V_{h+1}^k - V_{h+1}^{\pi_k})(s, a) + \beta_{k,h}^a \sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)}.$$

From the definition of V^{π_k} , we have

$$\delta_h^k = Q_h^k(s_h^k, a_h^k) - Q_h^{\pi_k}(s_h^k, a_h^k).$$

□

Finally, we prove the main theorem, Theorem 6, which asserts that the regret bound of our algorithm is efficient with respect to A and T .

Proof. Suppose that the Assumptions required in Theorem 6 hold. We condition on the event $\mathcal{X}$ from Lemma 7 with $p = \delta/2$ and use the notation for δ_h^k, ξ_h^k as in Lemma 10. From Lemmas 9 and 10, we have

$$\text{Regret}(K) = \sum_{k=1}^K (V_1^*(s_1^k) - V_1^{\pi_k}(s_1^k)) \leq \sum_{k=1}^K \delta_1^k \leq \sum_{k \in [K]} \sum_{h \in [H]} \xi_h^k + \beta_{k,h}^a \sum_{k \in [K]} \sum_{h \in [H]} \sqrt{F_h(s_h^k)^\top (\Lambda_{h,k}^{a_h^k})^{-1} F_h(s_h^k)}$$

Since the observations at episode k are independent of the computed value function (this uses the trajectories from episodes 1 to $k-1$, it follows $\{\xi_h^k\}$ is a martingale difference sequence with $|\xi_h^k| \leq 2H$ for all (k, h) . Thus, from the Azuma Hoeffding inequality, we have

$$\sum_{k \in [K], h \in H} \xi_h^k \leq \sqrt{2TH^2 \log(2/p)} \leq 2H \sqrt{T\iota}$$

with probability at least $1 - \delta/2$. For the second term, we note that the minimum eigenvalue of $\Lambda_{h,k}^{a_h^k}$ is at least one by construction and $\|F_h(s)\| \leq 1$. From the Elliptical Potential Lemma (Lemma D.2 [37]), we have for all $a \in \mathcal{A}$ and $h \in [H]$,

$$\sum_{k=1}^K F_h(s_k)^\top (\Lambda_{h,k}^a)^{-1} F_h(s_k) \leq 2 \log(\det(\Lambda_{h,k+1}^a) / \det(\Lambda_{h,0}^a))$$

Furthermore, we have $\|\Lambda_{h,k+1}^a\| = \|\sum_{t \in T_{K,h}^a} F_h(s_h^t) F_h(s_h^t)^\top + \lambda I\| \leq \lambda + |T_{K,h}^a| \leq \lambda + K$. It follows that

$$\sum_{k=1}^K F_h(s_k)^\top (\Lambda_{h,k}^a)^{-1} F_h(s_k) \leq 2d \log(1 + K/\lambda) \leq 2d$$

Next, by Cauchy-Schwarz and grouping the regret by each action, it follows that

$$\begin{aligned} \sum_{k \in [K]} \sum_{h \in [H]} \sqrt{F_h(s_h^k)^\top (\Lambda_{h,k}^{a_h^k})^{-1} F_h(s_h^k)} &\leq \sum_{h \in [H]} \sqrt{K} \left[\sum_{k \in [K]} F_h(s_h^k)^\top (\Lambda_{h,k}^{a_h^k})^{-1} F_h(s_h^k) \right]^{1/2} \\ &= \sum_{h \in [H]} \sqrt{K} \left[\sum_{a \in \mathcal{A}} \sum_{t \in T_{K,h}^a} F_h(s_h^t)^\top (\Lambda_{h,t}^a)^{-1} F_h(s_h^t) \right]^{1/2} \\ &\leq \sum_{h \in [H]} \sqrt{K} \left[\sum_{a \in \mathcal{A}} 2d \right]^{1/2} \\ &\leq H \sqrt{2KA d}. \end{aligned}$$

Since $\beta_{k,h}^a = cdH \sqrt{t}$ for some constant c , from a union bound and the previous bounds, it follows that

$$\text{Regret}(K) \leq 2H \sqrt{Td} + \beta_{k,h}^a H \sqrt{2KA d} \in \tilde{O}(\sqrt{d^3 H^3 A T}).$$

with probability at least $1 - \delta$. □

To use LSVI-UCB- (S, d, A) in our transfer RL setting, we next show that it is robust to misspecification error. Similarly, we first prove the helper lemmas needed in the misspecified setting and follow the proof structure and techniques used in [37].

Lemma 11. For a ξ -approximate (S, d, A) Tucker rank MDP (Assumption 13), then for any policy π there exists corresponding weight vectors $\{w_h^a\}$ where $w_h^a = W_h(a) + \sum_{s' \in \mathcal{S}} V_{h+1}^\pi(s') U_h(s', a)$ such that for all $(s, a) \in \mathcal{S} \times \mathcal{A}$

$$|Q_h^\pi(s, a) - F_h(s) w_h^{a\top}| \leq 3H\xi.$$

Proof. Note that $Q_h^\pi(s, a) = r_h(s, a) + \sum_{s'} V_{h+1}^\pi(s') P_h(s'|s, a)$, so, using the low-rank representations of r_h, P_h , it follows that

$$\begin{aligned} |Q_h^\pi(s, a) - F_h(s) w_h^{a\top}| &\leq |r_h(s, a) - F_h(s) W_h(a)^\top| \\ &\quad + \left| \sum_{s'} V_{h+1}^\pi(s') P_h(s'|s, a) - F_h(s) \sum_{s' \in \mathcal{S}} V_{h+1}^\pi(s') U_h(s', a) \right| \\ &\leq \xi + 2H\xi \\ &\leq 3H\xi \end{aligned}$$

□

Lemma 12. Suppose Assumption 13 holds. Let $\{w_h^a\}$ be the weight vector of some policy π , i.e., $Q_h^\pi(s, a) = F_h(s)^\top w_h^a$. Then,

$$\|w_h^a\| \leq 2Hd.$$

Proof. Recall that $w_h^a = W_h(a) + \sum_{s' \in \mathcal{S}} V_{h+1}^\pi(s') U_h(s', a)$. Since the feature mapping was scaled up ($F_h = \sqrt{\frac{S}{d\mu}} F'$) to ensure the max entries of F and W, U are on the same scale, it follows that $\|W_h(a)\|_2 \leq 1$ and $\|\sum_{s' \in \mathcal{S}} V_{h+1}^\pi(s') U_h(s', a)\|_2 \leq H$. Thus, $\|w_h^a\|_2 \leq 2Hd$. □

Similarly, we bound the stochastic noise but account for the misspecification error.

Lemma 13. Suppose Assumption 13 holds. Let c'_m be the constant in the definition of $\beta_{k,h}^a = c'_m H(d\sqrt{l} + \xi\sqrt{|T_{k,h}^a|d})$. Then, there exists a constant $C > 0$ that is independent of c'_m such that for any $\delta \in (0, 1)$, if we let the event $\mathcal{X}$ be

$$\forall (a, k, h) \in \mathcal{A} \times [K] \times [H] : \left\| \sum_{t \in T_{h,k-1}^a} F_h(s_h^t) (V_{h+1}^t(s_{h+1}^t) - P_h V_{h+1}^t(s_h^t, a)) \right\|_{(\Lambda_{h,t}^a)^{-1}} \leq CdH\sqrt{\chi}$$

where $\chi = \log(2(c'_m + 1)dTA/\delta)$, then $\mathbb{P}(\mathcal{X}) \geq 1 - \delta/2$.

Proof. The proof is the same as the proof of Lemma 7 except we increase $\beta_{k,h}^a$ to account for the misspecification error. Since $\xi \leq 1$ by assumption, the modification to $\beta_{k,h}^a$ only affects C . $\square$

We next account for the noise being adversarial due to the misspecification error.

Lemma 14. Let $\{\epsilon_t\}$ be any sequence such that $|\epsilon_t| \leq B$ for any t . Then, for any $(h, k, a) \in [H] \times [K] \times \mathcal{A}$ and $F \in \mathbb{R}^d$, we have

$$|F^\top (\Lambda_{k,h}^a)^{-1} \sum_{t \in T_{k-1,h}^a} F_h(s_h^t) \epsilon_t| \leq B \sqrt{d|T_{k-1,h}^a| F^\top (\Lambda_{k,h}^a)^{-1} F}.$$

Proof. From the Cauchy-Schwarz inequality and Elliptical Potential Lemma (Lemma D.1 [37]), we have

$$\begin{aligned} |F^\top (\Lambda_{k,h}^a)^{-1} \sum_{t \in T_{k-1,h}^a} F_h(s_h^t) \epsilon_t| &\leq B \sqrt{\left(\sum_{t \in T_{k-1,h}^a} F^\top (\Lambda_{k,h}^a)^{-1} F \right) \left(\sum_{t \in T_{k-1,h}^a} F_h(s_h^t)^\top (\Lambda_{k,h}^a)^{-1} F_h(s_h^t) \right)} \\ &\leq B \sqrt{d|T_{k-1,h}^a| F^\top (\Lambda_{k,h}^a)^{-1} F}. \end{aligned}$$

$\square$

Next, we bound the error between a policies Q function and our low-rank estimate.

Lemma 15. *There exists a constant c'_m such that $\beta_{k,h}^a = c'_m H(d\sqrt{\iota} + \xi\sqrt{|T_{k,h}^a|d})$ where $\iota = \log(2dAT/p)$ and for any fixed policy π , on the event $\mathcal{X}$ defined in Lemma 13, we have for all $(s, a, h, k) \in \mathcal{S} \times \mathcal{A} \times [H] \times [K]$:*

$$F_h(s)^\top w_{h,k}^a - Q_h^\pi(s, a) = P_h(V_{h+1}^k - V_{h+1}^\pi)(s, a) + \Delta_h^k(s, a),$$

where $|\Delta_h^k(s, a)| \leq \beta_{k,h}^a \sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)} + 4H\xi$.

Proof. From Lemma 12, it follows that there exists a weight vector $w_h^a = W_h(a) + \sum_{s' \in \mathcal{S}} V_{h+1}^\pi(s') U_h(s', a)$, such that for all $(s, a) \in \mathcal{S} \times \mathcal{A}$,

$$|Q_h^\pi(s, a) - F_h(s)^\top w_h^a| \leq 2H\xi.$$

Let $\tilde{P}(\cdot|s, a) = F_h(s)^\top U_h(\cdot, a)$, so we have for any $(s, a) \in \mathcal{S} \times \mathcal{A}$, $F_h(s)^\top w_h^a = F_h(s)^\top W_h(a) + \tilde{P}V_{h+1}^\pi(s, a)$. Therefore,

$$\begin{aligned} w_{h,k}^a - w_h^{a,\pi} &= -\lambda(\Lambda_{h,k}^a)^{-1} w_h^{a,\pi} \\ &\quad + (\Lambda_{h,k}^a)^{-1} \sum_{t \in T_{k-1,h}^a} F_h(s_h^t) (V_{h+1}^k(s_{h+1}^t) - P_h V_{h+1}^k(s_h^t, a)) \\ &\quad + (\Lambda_{h,k}^a)^{-1} \sum_{t \in T_{k-1,h}^a} F_h(s_h^t) \tilde{P}_h(V_{h+1}^k - V_{h+1}^\pi)(s_h^t, a) \\ &\quad + (\Lambda_{h,k}^a)^{-1} \sum_{t \in T_{k-1,h}^a} F_h(s_h^t) (r_h(s_h^t, a) - F_h(s_h^t) W_h(a) + (P_h - \tilde{P}_h) V_{h+1}^k(s_h^t, a)) \end{aligned}$$

Now, we bound each of the four terms above $(q_1^a, q_2^a, q_3^a, q_4^a)$ individually, for the first term, note for all $a \in \mathcal{A}$

$$|F_h(s)^\top q_1^a| = |\lambda F_h(s)^\top (\Lambda_{h,k}^a)^{-1} w_h^{a,\pi}| \leq \sqrt{\lambda} \|w_h^{a,\pi}\| \sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)}$$

For the second term, given the event $\mathcal{X}$, we have from Cauchy Schwarz

$$|F_h(s)^\top q_2^a| \leq CdH \sqrt{\chi} \sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)}$$

where $\chi = \log(2(c_\beta + 1)dTA/p)$. For the third term,

$$\begin{aligned}
F_h(s)^\top q_3^a &= F_h(s)^\top \left((\Lambda_{h,k}^a)^{-1} \sum_{t \in T_{k-1,h}^a} F_h(s_h^t) \tilde{P}_h(V_{h+1}^k - V_{h+1}^\pi)(s_h^t, a) \right) \\
&= F_h(s)^\top \left((\Lambda_{h,k}^a)^{-1} \sum_{t \in T_{k-1,h}^a} F_h(s_h^t) F_h(s_h^t)^\top \sum_{s' \in \mathcal{S}} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', a) \right) \\
&= F_h(s)^\top \left(\sum_{s' \in \mathcal{S}} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', a) \right) \\
&\quad - \lambda F_h(s)^\top \left((\Lambda_{h,k}^a)^{-1} \sum_{s' \in \mathcal{S}} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', a) \right),
\end{aligned}$$

and note that

$$F_h(s)^\top \left(\sum_{s' \in \mathcal{S}} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', a) \right) = \tilde{P}_h(V_{h+1}^k - V_{h+1}^\pi)(s, a)$$

and

$$|\lambda F_h(s)^\top \left((\Lambda_{h,k}^a)^{-1} \sum_{s' \in \mathcal{S}} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', a) \right)| \leq 2H \sqrt{d\lambda} \sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)}.$$

Since $\|\tilde{P}_h(\cdot|s, a) - P_h(\cdot|s, a)\|_\infty \leq 1$ for all $(s, a) \in \mathcal{S} \times \mathcal{A}$, it follows that

$$|\tilde{P}_h(V_{h+1}^k - V_{h+1}^\pi)(s, a) - P_h(V_{h+1}^k - V_{h+1}^\pi)(s, a)| \leq |(\tilde{P}_h - P_h)(V_{h+1}^k - V_{h+1}^\pi)(s, a)| \leq 2H\xi.$$

From Lemma 14, we have $|F_h(s), q_4^a| \leq 2H\xi \sqrt{d|T_{k,h}^a| F_h(s)^\top (\Lambda_{h,k}^{-1})^{-1} F_h(s)}$. Since

$$|F_h(s)^\top w_{h,k}^a - Q_h^\pi(s, a)| = F_h(s)^\top (w_{h,k}^a - w_h^{a,\pi}) = F_h(s)^\top (q_1^a + q_2^a + q_3^a + q_4^a),$$

it follows that

$$|F_h(s)^\top w_{h,k}^a - Q_h^\pi(s, a) - P_h(V_{h+1}^k - V_{h+1}^\pi)(s, a)| \leq \sqrt{\chi}(C''d\sqrt{\chi} + 2\xi\sqrt{|T_{k,h}^a|d})H\sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)} + 4H\xi.$$

Similarly as in [37], we choose c_β that satisfies $C''\sqrt{\log(2) + \log(c_\beta + 1)} \leq c_\beta\sqrt{\log(2)}$. $\square$

Now, we prove that Q_h^k is an upperbound of Q_h^* conditioned on the event in Lemma 13.

Lemma 16. *Suppose Assumption 13 holds. On the event $\mathcal{X}$ defined in lemma 13, we have $Q_h^k(s, a) \geq Q_h^*(s, a) - 4H(H + 1 - h)\xi$ for all $(s, a, h, k) \in \mathcal{S} \times \mathcal{A} \times [H] \times [K]$.*

Proof. We prove this with induction. The base case holds because at step $H + 1$, the value function is zero, so from Lemma 15,

$$|F_H(s)^\top w_{H,k}^a - Q_H^*(s, a)| \leq \beta_{k,H}^a \sqrt{F_H(s)^\top (\Lambda_{H,k}^a)^{-1} F_H(s)} + 4H\xi$$

and thus,

$$Q_H^*(s, a) - 4H\xi \leq \min \left(H, F_H(s)^\top w_{H,k}^a + \beta_{k,H}^a \sqrt{F_H(s)^\top (\Lambda_{H,k}^a)^{-1} F_H(s)} \right) = Q_H^k(s, a).$$

From the inductive hypothesis (assuming that $Q_{h+1}^k(s, a) \geq Q_{h+1}^*(s, a) - 4H(H - h)\xi$), it follows that $P_h(V_{h+1}^k - V_{h+1}^*)(s, a) \geq -4\xi$. From Lemma 15, $F_h(s)^\top w_{h,k}^a - Q_h^*(s, a) \leq \beta_{k,h}^a \sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)} + 4H\xi$. It follows that

$$Q_h^*(s, a) - 4H(H - h + 1)\xi \leq \min \left(H, F_h(s)^\top w_{h,k}^a + \beta_{k,h}^a \sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)} \right) = Q_h^k(s, a).$$

This completes the proof. $\square$

Similarly, to the regular case, the gap in the misspecified setting has a recursive formula.

Lemma 17. *Let $\delta_h^k = V_h^k(s_h^k) - V_h^{\pi_k}(s_h^k)$ and $\xi_{h+1}^k = \mathbb{E}[\delta_{h+1}^k | s_h^k, a_h^k] - \delta_{h+1}^k$. Then, on the event $\mathcal{X}$ defined in Lemma 13, we have for any (h, k) :*

$$\delta_h^k \leq \delta_{h+1}^k + \xi_{h+1}^k + \beta_{k,h}^a \sqrt{F_h(s_h^k)^\top (\Lambda_{h,k}^{a_h^k})^{-1} F_h(s_h^k)} + 4H\xi$$

Proof. From Lemma 15, we have for any (s, a, h, k) that

$$Q_h^k(s, a) - Q_h^{\pi_k}(s, a) \leq P_h(V_{h+1}^k - V_{h+1}^{\pi_k})(s, a) + \beta_{k,h}^a \sqrt{F_h(s)^\top (\Lambda_{h,k}^a)^{-1} F_h(s)} + 4H\xi.$$

From the definition of V^{π_k} , we have

$$\delta_h^k = Q_h^k(s_h^k, a_h^k) - Q_h^{\pi_k}(s_h^k, a_h^k),$$

and substituting this into the first equation finishes the proof. $\square$

Finally, we prove the main result in the misspecified setting Theorem 7.

Proof. Suppose that the Assumptions required in Theorem 7 hold. We condition on the event $\mathcal{X}$ from Lemma 13 with $p = \delta/2$ and use the notation for δ_h^k, ξ_h^k as in Lemma 17. From Lemma 16, we have $Q_1^k(s, a) \geq Q_1^*(s, a) - 4H^2\xi$, which implies $V_1^*(s) - V_1^{\pi_k}(s) \leq \delta_1^k + 4H^2\xi$. Thus, from Lemma 15, on the event $\mathcal{X}$, it follows that

$$\begin{aligned} \text{Regret}(K) &= \sum_{k=1}^K (V_1^*(s_1^k) - V_1^{\pi_k}(s_1^k)) \leq \sum_{k=1}^K \delta_1^k + 4H^2\xi \\ &\leq \sum_{k \in [K]} \sum_{h \in [H]} \xi_h^k + \sum_{k \in [K]} \beta_{k,h}^a \sum_{h \in [H]} \sqrt{F_h(s_h^k)^\top (\Lambda_{h,k}^{a_h^k})^{-1} F_h(s_h^k)} + 4HT\xi \end{aligned}$$

since $HK = T$. Since the observations at episode k are independent of the computed value function (this uses the trajectories from episodes 1 to $k-1$, it follows $\{\xi_h^k\}$ is a martingale difference sequence with $|\xi_h^k| \leq 2H$ for all (k, h) . Thus, from the Azuma Hoeffding inequality, we have

$$\sum_{k \in [K], h \in H} \xi_h^k \leq \sqrt{2TH^2 \log(2/p)} \leq 2H \sqrt{T\iota}$$

with probability at least $1 - \delta/2$. By construction of $\beta_{k,h}^a$, we have

$$\begin{aligned} \sum_{k \in [K]} \beta_{k,h}^a \sqrt{F_h(s_h^k)^\top (\Lambda_{h,k}^{a_h^k})^{-1} F_h(s_h^k)} &\leq CH \left(\sum_{k \in [K]} d \sqrt{\iota} \sqrt{F_h(s_h^k)^\top (\Lambda_{h,k}^{a_h^k})^{-1} F_h(s_h^k)} \right. \\ &\quad \left. + \sum_{k \in [K]} \xi \sqrt{|T_{k,h}^a| d} \sqrt{F_h(s_h^k)^\top (\Lambda_{h,k}^{a_h^k})^{-1} F_h(s_h^k)} \right). \end{aligned}$$

From the Cauchy-Schwarz inequality, it follows that

$$\sum_{k \in [K]} \sqrt{F_h(s_h^k)^\top (\Lambda_{h,k}^{a_h^k})^{-1} F_h(s_h^k)} \leq (\sqrt{K}) \left(\sqrt{\sum_{k \in [K]} F_h(s_h^k)^\top (\Lambda_{h,k}^{a_h^k})^{-1} F_h(s_h^k)} \right).$$

Since the minimum eigenvalue of $\Lambda_{h,k}^{a_h^k}$ is at least one by construction and $\|F_h(s)\| \leq 1$, from the Elliptical Potential Lemma (Lemma D.2 [37]), we have for all $a \in \mathcal{A}$ and $h \in [H]$,

$$\sum_{k=1}^K F_h(s_k)^\top (\Lambda_{h,k}^a)^{-1} F_h(s_k) \leq 2 \log(\det(\Lambda_{h,k+1}^a) / \det(\Lambda_{h,0}^a))$$

Furthermore, we have $\|\Lambda_{h,k+1}^a\| = \|\sum_{t \in T_{K,h}^a} F_h(s_h^t) F_h(s_h^t)^\top + \lambda I\| \leq \lambda + |T_{K,h}^a| \leq \lambda + K$. It follows that

$$\sum_{k=1}^K F_h(s_k)^\top (\Lambda_{h,k}^a)^{-1} F_h(s_k) \leq 2d \log(1 + K/\lambda) \leq 2d\iota,$$

so grouping the episodes by actions gives

$$\left[\sum_{k \in [K]} F_h(s_h^k)^\top (\Lambda_{h,k}^{a_h^k})^{-1} F_h(s_h^k) \right]^{1/2} = \left[\sum_{a \in \mathcal{A}} \sum_{t \in T_{K,h,a}^a} F_h(s_h^t)^\top (\Lambda_{h,t}^a)^{-1} F_h(s_h^t) \right]^{1/2} \leq \sqrt{2dA\iota}.$$

For the next term, let $k_{a,i}$ refer to the episode in which action a was taken at time step h for the i th time. Then, we first re-index the summation and then use the Cauchy-Schwarz inequality to get

$$\begin{aligned} \sum_{k \in [K]} \sqrt{|T_{K,h}^a|} \sqrt{F_h(s_h^k)^\top (\Lambda_{h,k}^{a_h^k})^{-1} F_h(s_h^k)} &= \sum_{a \in \mathcal{A}} \sum_{i=1}^{|T_{K,h}^a|} \sqrt{i} \sqrt{F_{k_{a,i},h}^\top (\Lambda_{k_{a,i},h}^a)^{-1} F_{k_{a,i},h}} \\ &\leq \sum_{a \in \mathcal{A}} \left(\sum_{i=1}^{|T_{K,h}^a|} i \right)^{1/2} \left(\sum_{t \in T_{K,h}^a} F_{t,h}^\top (\Lambda_{t,h}^a)^{-1} F_{t,h} \right)^{1/2} \\ &\leq C' \sum_{a \in \mathcal{A}} |T_{K,h}^a| \left(\sum_{t \in T_{K,h}^a} F_{t,h}^\top (\Lambda_{t,h}^a)^{-1} F_{t,h} \right)^{1/2} \\ &\leq C' \sum_{a \in \mathcal{A}} |T_{K,h}^a| \sqrt{2d\iota}, \end{aligned}$$

where the last inequality comes from holds from the above elliptical potential lemma. Applying the results to the initial inequality, we have

$$\sum_{k \in [K]} \beta_{k,h}^a \sqrt{F_h(s_h^k)^\top (\Lambda_{h,k}^{a_h^k})^{-1} F_h(s_h^k)} \leq CH(\sqrt{2d^3AK\iota^2} + \xi \sqrt{d}C' \sum_{a \in \mathcal{A}} |T_{K,h}^a| \sqrt{2d\iota}) \leq C(\sqrt{2d^3AHT\iota^2} + \xi \sqrt{2d\iota}T).$$

Substituting these results into the original regret bound gives

$$\text{Regret}(K) \leq C''' (\sqrt{d^3 H^3 A T \iota^2} + d T H \xi \sqrt{\iota}).$$

for some constant $C''' > 0$ with probability at least $1 - \delta$. $\square$

LSVI-UCB-($\mathcal{S}, \mathcal{S}, d$)

We reiterate that the steps and analysis in this subsection are essentially the same as the proofs in [37], except we have S copies of the weight vectors and gram matrices. The results and proofs follow the same structure and logic as in The proof of all the theorems and lemmas follow the same steps as the proofs for the analogous result in the (S, d, A) setting except we replace $F(s)$ with $G(a)$ and estimate the Gram matrix and weight vectors for each state instead of each action.

To prove Theorem 14, we first introduce notation used throughout this section and then present auxiliary lemmas. We let $\Lambda_{h,k}^s, w_{h,k}^s, Q_h^k, V_h^k, \pi^k$ as the parameters/estimates at episode k . Furthermore, let $V_h^k = \max_{s \in \mathcal{S}} Q_h^k(s, a)$ and π^k be the greedy policy w.r.t. Q_h^k . We let $G_h^k := G_h(a_h^k)$.

We first bound the log of the covering number of our function class for the value function to control the error of our algorithm's estimate, which allows us to prove our concentration result as the value function estimate is not constant.

Lemma 18. *Let $\mathcal{V}$ denote a function class of mappings from $\mathcal{S}$ to $\mathbb{R}$ with the following parametric form for all $s \in \mathcal{S}$*

$$V(s) = \min \left\{ \max_{a \in \mathcal{A}} w_s^\top G(a) + \beta \sqrt{G(a)^\top \Lambda_s^{-1} G(a)}, H \right\}$$

where the parameters w_s, β, Λ_s satisfy $\|w_s\| \leq L$ for all $s \in \mathcal{S}$, $\beta \in [0, B]$, and the minimum eigenvalue of Λ_s is greater than λ . Assume that $\|G(a)\| \leq 1$ for all $s \in \mathcal{S}$, and let $\mathcal{N}_\epsilon$ be the ϵ -covering number of $\mathcal{V}$ with respect to the distance $\text{dist}(V, V') = \sup_s |V(s) - V'(s)|$. Then,

$$\log(\mathcal{N}_\epsilon) \leq d \log(1 + 4L/\epsilon) + d^2 \log(1 + 8d^{1/2}B^2/(\lambda\epsilon^2))$$

Proof. Equivalently, we reparameterize the function class by $D_s = \beta^2 \Lambda_s^{-1}$, for all $s \in \mathcal{S}$

$$V(s) = \min \left\{ \max_{a \in \mathcal{A}} w_s^\top G(a) + \sqrt{G(a)^\top D_s G(a)}, H \right\}$$

where $\|w_s\| \leq L$ and $\|D_s\| \leq B^2/\lambda$ for all $s \in \mathcal{S}$. For any $V_1, V_2 \in \mathcal{V}$, let them take the form in the above equation with parameters (w_s, D_s) and $(w_{s'}, D_{s'})$. Since $\min(\cdot, H)$ and $\max_{s \in \mathcal{S}}$ are contraction maps,

$$\begin{aligned} \text{dist}(V_1 - V_2) &\leq \sup_{s, a} |(w_s^\top G(a) + \sqrt{G(a)^\top D_s G(a)}) - (w_{s'}^\top G(a) + \sqrt{G(a)^\top D_{s'} G(a)})| \\ &\leq \sup_{s, G: \|G\| \leq 1} |(w_s^\top G + \sqrt{G^\top D_s G}) - (w_{s'}^\top G + \sqrt{G^\top D_{s'} G})| \\ &\leq \max_{s \in \mathcal{S}} \left[\sup_{G: \|G\| \leq 1} |(w_s - w_{s'})^\top G| + \sup_{G: \|G\| \leq 1} |\sqrt{G^\top (D_s - D_{s'}) G}| \right] \\ &\leq \max_{s \in \mathcal{S}} \left[\|w_s - w_{s'}\| + \sqrt{\|D_s - D_{s'}\|} \right] \leq \max_{s \in \mathcal{S}} \left[\|w_s - w_{s'}\| + \sqrt{\|D_s - D_{s'}\|_F} \right] \end{aligned}$$

where the third inequality holds because $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$ for any $x, y \geq 0$. Let C_w be an $\epsilon/2$ -cover of $\{w \in \mathbb{R}^d \mid \|w\| \leq L\}$ with respect to the 2-norm, and C_D be an $\epsilon^2/4$ -cover of $\{D \in \mathbb{R}^{d \times d} \mid \|D\|_F \leq d^{1/2}B^2/\lambda\}$ with respect to the Frobenius norm. By Lemma D.5 from [37], we know that,

$$|C_w| \leq (1 + 4L/\epsilon)^d, \quad |C_D| \leq (1 + 8d^{1/2}B^2/(\lambda\epsilon^2))^{d^2}.$$

Hence, for any V_1 , there exists a $w_{s'} \in C_w$ and $D_{s'} \in C_D$ for all $a \in \mathcal{A}$ such that V_2 parameterized by $(w_{s'}, D_{s'})$ satisfies $\text{dist}(V_1, V_2) \leq \epsilon$. Hence, $\mathcal{N}_\epsilon \leq |C_w||C_D|$, which

gives:

$$\log(\mathcal{N}_\epsilon) \leq \log(|C_w|) + \log(|C_D|) \leq d \log(1 + 4\epsilon L/\epsilon) + d^2 \log(1 + 8d^{1/2}B^2/(\lambda\epsilon^2)).$$

□

We next bound the ℓ_2 -norm of the weight vector of any policy.

Lemma 19. *Let $\{w_h^s\}$ be the weight vector of some policy π , i.e., $Q_h^\pi(s, a) = G_h(a)^\top w_h^s$.*

Then,

$$\|w_h^s\| \leq 2H \sqrt{d}.$$

Proof. Note that $Q_h^\pi(s, a) = r_h(s, a) + \sum_{s'} V_{h+1}^\pi(s') P_h(s'|s, a)$, so, using the low-rank representations of r_h, P_h , it follows that

$$\|w_h^s\|_2 = \|W_h(s, \cdot) + \sum_{s'} V^\pi(s') U_h(s', s, \cdot)\|_2 \leq \|W_h(s, \cdot)\|_2 + \left\| \sum_{s'} V^\pi(s') U_h(s', s, \cdot) \right\|_2.$$

Recall that since the feature mapping was scaled up ($G_h = \sqrt{\frac{A}{d\mu}} G'$) to ensure the max entries of G and W, U are on the same scale, it follows that $\|W_h(s)\|_2 \leq 1$ and $\|\sum_{s' \in \mathcal{S}} V_{h+1}^\pi(s') U_h(s', s)\|_2 \leq H$. Therefore, it follows that $\|w_h^s\|_2 \leq 2H \sqrt{d}$. □

We next bound the 2-norm of the weight vector from the algorithm.

Lemma 20. *For any k, h, s , the weight vector from the above algorithm satisfies $\|w_{h,k}^s\| \leq 2H \sqrt{dk/\lambda}$.*

Proof. For any $G_h(a) = g$, we have for all $s \in \mathcal{S}$,

$$\begin{aligned}
|g^\top w_{h,k}^s| &= |g^\top (\Lambda_h^s)^{-1} \sum_{t \in T_{k-1,h}^s} G_h(a_h^t) \left[r_h(s, a_h^t) + \max_{a' \in \mathcal{A}} Q_{h+1}(s_{h+1}^t, a') \right]| \\
&\leq \left| \sum_{t \in T_{k-1,h}^s} g^\top (\Lambda_h^s)^{-1} G_h(a_h^t) \right| 2H \\
&\leq 2H \sqrt{\left(\sum_{t \in T_{k-1,h}^s} g^\top (\Lambda_h^s)^{-1} g \right) \left(\sum_{t \in T_{k-1,h}^s} G_h(a_h^t)^\top (\Lambda_h^s)^{-1} G_h(a_h^t) \right)} \\
&\leq 2H \|g\|_2 \sqrt{d |T_{k-1,h}^s| / \lambda} \leq 2H \sqrt{dk / \lambda}
\end{aligned}$$

where the second to last inequality is due to Lemma D.1 from [37] and that Rayleigh quotients are upper bounded by the largest eigenvalue of the matrix (Theorem 4.2.2 from [30]). The last inequality is due to $\|g\|_2 \leq 1$ and $|T_{k-1,h}^s| \leq k$. $\square$

We next prove the main concentration lemma that controls the estimate.

Lemma 21. *Let c' be the constant in the definition of $\beta_{k,h}^s = c' dH \sqrt{t}$. Then, there exists a constant $C > 0$ that is independent of c' such that for any $\delta \in (0, 1)$, if we let the event $\mathcal{X}$ be*

$$\forall (s, k, h) \in \mathcal{S} \times [K] \times [H] : \left\| \sum_{t \in T_{k-1,h}^s} G_h(a_h^t) (V_{h+1}^t(s_{h+1}^t) - P_h V_{h+1}^t(s, a_h^t)) \right\|_{(\Lambda_{h,t}^s)^{-1}} \leq CdH \sqrt{\chi}$$

where $\chi = \log(2(c' + 1)dTS/\delta)$, then $\mathbb{P}(\mathcal{X}) \geq 1 - \delta/2$.

Proof. From Lemma 20, we have $\|w_{h,k}^s\|_2 \leq 2H \sqrt{dk/\lambda}$. By construction, the minimum eigenvalue of $\Lambda_{h,t}^s$ is lower bounded by λ . Then, from Lemma D.4 in [37] (with $\delta' = \delta/S$), Lemma 18, and taking a union bound over all states, we have that with probability at least $1 - \delta'/2$, for all $s \in \mathcal{S}$ and $\epsilon > 0$,

$$\left\| \sum_{t \in T_{k-1,h}^s} G_h(a_h^t) (V_{h+1}^t(s_{h+1}^t) - P_h V_{h+1}^t(s, a_h^t)) \right\|_{(\Lambda_{h,t}^s)^{-1}}^2 \leq 4H^2 \left[\frac{d}{2} \log((k + \lambda)/\lambda) + \log(2S N_\epsilon/\delta) \right] + 8k^2 \epsilon^2 / \lambda.$$

Then, from Lemma 18, it follows that with probability at least $1 - \delta'/2$, for all $a \in S$ and $\epsilon > 0$,

$$\begin{aligned} & \left\| \sum_{t \in T_{k-1,h}^s} G_h(a_h^t)(V_{h+1}^t(s_{h+1}^t) - P_h V_{h+1}^t(s, a_h^t)) \right\|_{(\Lambda_{h,t}^s)^{-1}}^2 \\ & \leq 4H^2 \left[\frac{d}{2} \log((k + \lambda)/\lambda) + \log(2S/\delta) + d \log(1 + 4L/\epsilon) + d^2 \log(1 + 8d^{1/2}\beta^2/(\lambda\epsilon^2)) \right] + 8k^2\epsilon^2/\lambda. \end{aligned}$$

Choosing $\lambda = 1, \beta_{k,h}^s = CdH\iota, \epsilon = dH/k$ gives that there exists some constant C' such that

$$\left\| \sum_{t \in T_{k-1,h}^s} G_h(a_h^t)(V_{h+1}^t(s_{h+1}^t) - P_h V_{h+1}^t(s, a_h^t)) \right\|_{(\Lambda_{h,t}^s)^{-1}}^2 \leq C' d^2 H^2 \log(2(c_\beta + 1)dTS/\delta).$$

□

We next recursively bound the difference between the value function maintained in the algorithm and the true value function of any policy π . We upper-bound this with their expected difference and an additional error term, which is bounded with high probability.

Lemma 22. *There exists a constant c_β such that $\beta_{k,h}^s = c_\beta dH \sqrt{\iota}$ where $\iota = \log(2dST/p)$ and for any fixed policy π , on the event $\mathcal{X}$ defined in Lemma 21, we have for all $(s, a, h, k) \in S \times S \times [H] \times [K]$:*

$$G_h(a)^\top w_{h,k}^s - Q_h^\pi(s, a) = P_h(V_{h+1}^k - V_{h+1}^\pi)(s, a) + \delta_h^k(s, a),$$

$$\text{where } |\delta_h^k(s, a)| \leq \beta_{k,h}^s \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)}.$$

Proof. From our low-rank assumption, it follows that

$$Q_h^\pi(s, a) = G_h(a)^\top w_h^s = (r_h + P_h V_{h+1}^\pi)(s, a),$$

which following the steps from the proof of Lemma B.4 in [37] gives

$$\begin{aligned}
w_{h,k}^s - w_h^{s,\pi} &= -\lambda(\Lambda_{h,k}^s)^{-1} w_h^{s,\pi} \\
&\quad + (\Lambda_{h,k}^s)^{-1} \sum_{t \in T_{k-1,h}^s} G_h(a_h^t)(V_{h+1}^k(s_{h+1}^t) - P_h V_{h+1}^k(s, a_h^t)) \\
&\quad + (\Lambda_{h,k}^s)^{-1} \sum_{t \in T_{k-1,h}^s} G_h(a_h^t) P_h(V_{h+1}^k - V_{h+1}^\pi)(s, a_h^t)
\end{aligned}$$

Now, we bound each term (q_1^s, q_2^s, q_3^s) individually, for the first term, note for all $s \in S, a \in \mathcal{A}$

$$|G_h(a)^\top q_1^s| = |\lambda G_h(a)^\top (\Lambda_{h,k}^s)^{-1} w_h^{s,\pi}| \leq \sqrt{\lambda} \|w_h^{s,\pi}\| \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)}$$

For the second term, given the event $\mathcal{X}$, we have from Cauchy Schwarz

$$|G_h(a)^\top q_2^s| \leq CdH \sqrt{\chi} \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)}$$

where $\chi = \log(2(c_\beta + 1)dTS/p)$. For the third term,

$$\begin{aligned}
G_h(a)^\top q_3^s &= G_h(a)^\top \left((\Lambda_{h,k}^s)^{-1} \sum_{t \in T_{k-1,h}^s} G_h(a_h^t) P_h(V_{h+1}^k - V_{h+1}^\pi)(s, a_h^t) \right) \\
&= G_h(a)^\top \left((\Lambda_{h,k}^s)^{-1} \sum_{t \in T_{k-1,h}^s} G_h(a_h^t) G_h(a_h^t)^\top \sum_{s' \in S} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', s) \right) \\
&= G_h(a)^\top \left(\sum_{s' \in S} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', s) \right) \\
&\quad - \lambda G_h(a)^\top \left((\Lambda_{h,k}^s)^{-1} \sum_{s' \in S} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', s) \right)
\end{aligned}$$

We note that

$$G_h(a)^\top \left(\sum_{s' \in S} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', s) \right) = P_h(V_{h+1}^k - V_{h+1}^\pi)(s, a)$$

and

$$|\lambda G_h(a)^\top \left((\Lambda_{h,k}^s)^{-1} \sum_{s' \in S} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', s) \right)| \leq 2H \sqrt{d\lambda} \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)}$$

Since

$$|G_h(a)^\top w_{h,k}^s - Q_h^\pi(s, a)| = G_h(a)^\top (w_{h,k}^s - w_h^{s,\pi}) = G_h(a)^\top (q_1^s + q_2^s + q_3^s),$$

it follows that

$$|G_h(a)^\top w_{h,k}^s - Q_h^\pi(s, a) - P_h(V_{h+1}^k - V_{h+1}^\pi)(s, a)| \leq C'' dH \sqrt{\chi} \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)}.$$

Similarly as in [37], we choose c_β that satisfies $C'' \sqrt{\log(2) + \log(c_\beta + 1)} \leq c_\beta \sqrt{\log(2)}$. $\square$

This lemma implies that by adding the appropriate bonus, Q_h^k is always an upperbound of Q_h^* with high probability

Lemma 23. *On the event $\mathcal{X}$ defined in lemma 21, we have $Q_h^k(s, a) \geq Q_h^*(s, a)$ for all $(s, a, h, k) \in \mathcal{S} \times \mathcal{A} \times [H] \times [K]$.*

Proof. We prove this with induction. The base case holds because at step $H + 1$, the value function is zero, so from Lemma 22,

$$|G_h(a)^\top w_{h,k}^s - Q_H^*(s, a)| \leq \beta_{k,h}^s \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)}$$

and thus,

$$Q_H^*(s, a) \leq \min \left(H, G_h(a)^\top w_{h,k}^s + \beta_{k,h}^s \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)} \right) = Q_H^k(s, a).$$

From the inductive hypothesis (assuming that $Q_{h+1}^k(s, a) \geq Q_{h+1}^*(s, a)$), it follows that $P_h(V_{h+1}^k - V_{h+1}^*)(s, a) \geq 0$. From Lemma 22, $G_h(a)^\top w_{h,k}^s - Q_h^*(s, a) \leq \beta_{k,h}^s \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)}$. It follows that

$$Q_h^*(s, a) \leq \min \left(H, G_h(a)^\top w_{h,k}^s + \beta_{k,h}^s \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)} \right) = Q_h^k(s, a).$$

$\square$

We next show that Lemma 22 transforms to the recursive formula $\delta_h^k = V_h^k(s_h^k) - V_h^{\pi_k}(s_h^k)$.

Lemma 24. *Let $\delta_h^k = V_h^k(s_h^k) - V_h^{\pi_k}(s_h^k)$ and $\xi_{h+1}^k = \mathbb{E}[\delta_{h+1}^k | s_h^k, s_h^k] - \delta_{h+1}^k$. Then, on the event $\mathcal{X}$ defined in Lemma 21, we have for any (h, k) :*

$$\delta_h^k \leq \delta_{h+1}^k + \xi_{h+1}^k + \beta_{k,h}^s \sqrt{G_h(a_h^k)^\top (\Lambda_{h,k}^{s_h^k})^{-1} G_h(a_h^k)}$$

Proof. From Lemma 22, we have for any (s, a, h, k) that

$$Q_h^k(s, a) - Q_h^{\pi_k}(s, a) \leq P_h(V_{h+1}^k - V_{h+1}^{\pi_k})(s, a) + \beta_{k,h}^s \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)}.$$

From the definition of V^{π_k} , we have

$$\delta_h^k = Q_h^k(s_h^k, a_h^k) - Q_h^{\pi_k}(s_h^k, a_h^k).$$

□

Finally, we prove the main theorem, Theorem 14.

Proof. Suppose that the Assumptions required in Theorem 14 hold. We condition on the event $\mathcal{X}$ from Lemma 21 with $p = \delta/2$ and use the notation for δ_h^k, ξ_h^k as in Lemma 24. From Lemmas 9 and 24, we have

$$\text{Regret}(K) = \sum_{k=1}^K (V_1^*(s_1^k) - V_1^{\pi_k}(s_1^k)) \leq \sum_{k=1}^K \delta_1^k \leq \sum_{k \in [K]} \sum_{h \in [H]} \xi_h^k + \beta_{k,h}^s \sum_{k \in [K]} \sum_{h \in [H]} \sqrt{G_h(a_h^k)^\top (\Lambda_{h,k}^{s_h^k})^{-1} G_h(a_h^k)}$$

Since the observations at episode k are independent of the computed value function (this uses the trajectories from episodes 1 to $k-1$, it follows $\{\xi_h^k\}$ is a martingale difference sequence with $|\xi_h^k| \leq 2H$ for all (k, h) . Thus, from the Azuma Hoeffding inequality, we have

$$\sum_{k \in [K], h \in H} \xi_h^k \leq \sqrt{2TH^2 \log(2/p)} \leq 2H \sqrt{T\iota}$$

with probability at least $1 - \delta/2$. For the second term, we note that the minimum eigenvalue of $\Lambda_{h,k}^{s_h^k}$ is at least one by construction and $\|G_h(a)\| \leq 1$. From the Elliptical Potential Lemma (Lemma D.2 [37]), we have for all $s \in \mathcal{S}$ and $h \in [H]$,

$$\sum_{k=1}^K G_h(a_h^k)^\top (\Lambda_{h,k}^s)^{-1} G_h(a_h^k) \leq 2 \log(\det(\Lambda_{h,k+1}^s) / \det(\Lambda_{h,0}^s))$$

Furthermore, we have $\|\Lambda_{h,k+1}^s\| = \|\sum_{t \in T_{K,h}^s} G_h(a_h^t) G_h(a_h^t)^\top + \lambda I\| \leq \lambda + |T_{K,h}^s| \leq \lambda + K$.

It follows that

$$\sum_{k=1}^K G_h(a_h^k)^\top (\Lambda_{h,k}^s)^{-1} G_h(a_h^k) \leq 2d \log(1 + K/\lambda) \leq 2d\iota$$

Next, by Cauchy-Schwarz and grouping the regret by each state, it follows that

$$\begin{aligned} \sum_{k \in [K]} \sum_{h \in [H]} \sqrt{G_h(a_h^k)^\top (\Lambda_{h,k}^{s_h^k})^{-1} G_h(a_h^k)} &\leq \sum_{h \in [H]} \sqrt{K} \left[\sum_{k \in [K]} G_h(a_h^k)^\top (\Lambda_{h,k}^{s_h^k})^{-1} G_h(a_h^k) \right]^{1/2} \\ &= \sum_{h \in [H]} \sqrt{K} \left[\sum_{s \in \mathcal{S}} \sum_{t \in T_{K,h}^s} G_h(a_h^t)^\top (\Lambda_{h,t}^s)^{-1} G_h(a_h^t) \right]^{1/2} \\ &\leq \sum_{h \in [H]} \sqrt{K} \left[\sum_{s \in \mathcal{S}} 2d\iota \right]^{1/2} \\ &\leq H \sqrt{2KS d\iota}. \end{aligned}$$

Since $\beta_{k,h}^s = cdH \sqrt{\iota}$ for some constant c , from a union bound and the previous bounds, it follows that

$$\text{Regret}(K) \leq 2H \sqrt{T\iota} + \beta_{k,h}^s H \sqrt{2KS d\iota} \in \tilde{O}(\sqrt{d^3 H^3 S T}).$$

with probability at least $1 - \delta$. □

Next, we prove the helper lemmas needed in the misspecified setting and follow the proof structure and techniques used in [37].

Lemma 25. *For a ξ -approximate (S, S, d) Tucker rank MDP (Assumption 19), then for any policy π there exists corresponding weight vectors $\{w_h^s\}$ where $w_h^s = \Sigma_h W_h(s) +$*

$\sum_{s' \in \mathcal{S}} V_{h+1}^\pi(s') U_h(s', s)$ such that for all $(s, a) \in \mathcal{S} \times \mathcal{A}$

$$|Q_h^\pi(s, a) - G_h(a) w_h^{s^\top}| \leq 3H\xi.$$

Proof. Note that $Q_h^\pi(s, a) = r_h(s, a) + \sum_{s'} V_{h+1}^\pi(s') P_h(s'|s, a)$, so, using the low-rank representations of r_h, P_h , it follows that

$$\begin{aligned} & |Q_h^\pi(s, a) - G_h(a) w_h^{s^\top}| \\ & \leq |r_h(s, a) - G_h(a) W_h(s)^\top| + \left| \sum_{s'} V_{h+1}^\pi(s') P_h(s'|s, a) - G_h(a) \sum_{s' \in \mathcal{S}} V_{h+1}^\pi(s') U_h(s', s) \right| \\ & \leq \xi + 2H\xi \\ & \leq 3H\xi. \end{aligned}$$

□

Lemma 26. Suppose Assumption 13 holds. Let $\{w_h^s\}$ be the weight vector of some policy π , i.e., $Q_h^\pi(s, a) = G_h(a)^\top w_h^s$. Then,

$$\|w_h^s\| \leq 2H \sqrt{d}.$$

Proof. Recall that $w_h^s = \Sigma_h W_h(s) + \sum_{s' \in \mathcal{S}} V_{h+1}^\pi(s') U_h(s', s)$. From the same argument used to prove Lemma 19, it follows that $\|w_h^s\|_2 \leq 2H \sqrt{d}$ □

Similarly, we bound the stochastic noise but account for the misspecification error.

Lemma 27. Suppose Assumption 13 holds. Let c'_m be the constant in the definition of $\beta_{k,h}^s = c'_m H(d \sqrt{\iota} + \xi \sqrt{|T_{k,h}^s|d})$. Then, there exists a constant $C > 0$ that is independent of c'_m such that for any $\delta \in (0, 1)$, if we let the event $\mathcal{X}$ be

$$\forall (a, k, h) \in \mathcal{A} \times [K] \times [H] : \left\| \sum_{t \in T_{k-1,h}^s} G_h(a_h^t) (V_{h+1}^t(s_{h+1}^t) - P_h V_{h+1}^t(s, a_h^t)) \right\|_{(\Lambda_{h,t}^s)^{-1}} \leq CdH \sqrt{\chi}$$

where $\chi = \log(2(c'_m + 1)dTS/\delta)$, then $\mathbb{P}(\mathcal{X}) \geq 1 - \delta/2$.

Proof. The proof is the same as the proof of Lemma 21 except we increase $\beta_{k,h}^s$ to account for the misspecification error. Since $\xi \leq 1$ by assumption, the modification to $\beta_{k,h}^s$ only affects C . $\square$

Next, we account for the adversarial misspecification error.

Lemma 28. *Let $\{\epsilon_t\}$ be any sequence such that $|\epsilon_t| \leq B$ for any t . Then, for any $(h, k, s) \in [H] \times [K] \times \mathcal{S}$ and $G \in \mathbb{R}^d$, we have*

$$|G^\top (\Lambda_{k,h}^s)^{-1} \sum_{t \in T_{k-1,h}^s} G_h(a_h^t) \epsilon_t| \leq B \sqrt{d |T_{k-1,h}^s| G^\top (\Lambda_{k,h}^s)^{-1} G}.$$

Proof. From the Cauchy-Schwarz inequality and Elliptical Potential Lemma (Lemma D.1 [37]), we have

$$\begin{aligned} |G^\top (\Lambda_{k,h}^s)^{-1} \sum_{t \in T_{k-1,h}^s} G_h(a_h^t) \epsilon_t| &\leq B \sqrt{\left(\sum_{t \in T_{k-1,h}^s} G^\top (\Lambda_{k,h}^s)^{-1} G \right) \left(\sum_{t \in T_{k-1,h}^s} G_h(a_h^t)^\top (\Lambda_{k,h}^s)^{-1} G_h(a_h^t) \right)} \\ &\leq B \sqrt{d |T_{k-1,h}^s| G^\top (\Lambda_{k,h}^s)^{-1} G}. \end{aligned}$$

$\square$

Next, we bound the error between a policies Q function and our low-rank estimate.

Lemma 29. *There exists a constant c'_m such that $\beta_{k,h}^s = c'_m H(d \sqrt{\iota} + \xi \sqrt{|T_{k,h}^s| d})$ where $\iota = \log(2dST/p)$ and for any fixed policy π , on the event $\mathcal{X}$ defined in Lemma 27, we have for all $(s, a, h, k) \in \mathcal{S} \times \mathcal{S} \times [H] \times [K]$:*

$$G_h(a)^\top w_{h,k}^s - Q_h^\pi(s, a) = P_h(V_{h+1}^k - V_{h+1}^\pi)(s, a) + \delta_h^k(s, a),$$

where $|\delta_h^k(s, a)| \leq \beta_{k,h}^s \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)} + 4H\xi$.

Proof. From Lemma 25, it follows that there exists a weight vector $w_h^s = W_h(a) + \sum_{s' \in \mathcal{S}} V_{h+1}^\pi(s') U_h(s', s)$, such that for all $(s, a) \in \mathcal{S} \times \mathcal{A}$,

$$|Q_h^\pi(s, a) - G_h(a)^\top w_h^s| \leq 2H\xi.$$

Let $\tilde{P}(\cdot|s, a) = G_h(a)^\top U_h(\cdot, s)$, so we have for any $(s, a) \in \mathcal{S} \times \mathcal{A}$, $G_h(a)^\top w_h^s = G_h(a)^\top W_h(a) + \tilde{P} V_{h+1}^\pi(s, a)$. Therefore,

$$\begin{aligned} w_{h,k}^s - w_h^{s,\pi} &= -\lambda(\Lambda_{h,k}^s)^{-1} w_h^{s,\pi} \\ &+ (\Lambda_{h,k}^s)^{-1} \sum_{t \in T_{k-1,h}^s} G_h(a_h^t) (V_{h+1}^k(s_{h+1}^t) - P_h V_{h+1}^k(s, a_h^t)) \\ &+ (\Lambda_{h,k}^s)^{-1} \sum_{t \in T_{k-1,h}^s} G_h(a_h^t) \tilde{P}_h(V_{h+1}^k - V_{h+1}^\pi)(s, a_h^t) \\ &+ (\Lambda_{h,k}^s)^{-1} \sum_{t \in T_{k-1,h}^s} G_h(a_h^t) (r_h(s, a_h^t) - G_h(a_h^t) W_h(a) + (P_h - \tilde{P}_h) V_{h+1}^k(s, a_h^t)) \end{aligned}$$

Now, we bound each of the four terms above $(q_1^s, q_2^s, q_3^s, q_4^s)$ individually, for the first term, note for all $a \in \mathcal{A}$

$$|G_h(a)^\top q_1^s| = |\lambda G_h(a)^\top (\Lambda_{h,k}^s)^{-1} w_h^{s,\pi}| \leq \sqrt{\lambda} \|w_h^{s,\pi}\| \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)}$$

For the second term, given the event $\mathcal{X}$, we have from Cauchy Schwarz

$$|G_h(a)^\top q_2^s| \leq CdH \sqrt{\chi} \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)}$$

where $\chi = \log(2(c_\beta + 1)dTS/p)$. For the third term,

$$\begin{aligned} G_h(a)^\top q_3^s &= G_h(a)^\top \left((\Lambda_{h,k}^s)^{-1} \sum_{t \in T_{k-1,h}^s} G_h(a_h^t) \tilde{P}_h(V_{h+1}^k - V_{h+1}^\pi)(s, a_h^t) \right) \\ &= G_h(a)^\top \left((\Lambda_{h,k}^s)^{-1} \sum_{t \in T_{k-1,h}^s} G_h(a_h^t) G_h(a_h^t)^\top \sum_{s' \in \mathcal{S}} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', s) \right) \\ &= G_h(a)^\top \left(\sum_{s' \in \mathcal{S}} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', s) \right) \\ &\quad - \lambda G_h(a)^\top \left((\Lambda_{h,k}^s)^{-1} \sum_{s' \in \mathcal{S}} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', s) \right), \end{aligned}$$

and note that

$$G_h(a)^\top \left(\sum_{s' \in \mathcal{S}} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', s) \right) = \tilde{P}_h(V_{h+1}^k - V_{h+1}^\pi)(s, a)$$

and

$$|\lambda G_h(a)^\top \left((\Lambda_{h,k}^s)^{-1} \sum_{s' \in \mathcal{S}} (V_{h+1}^k - V_{h+1}^\pi)(s') U_h(s', s) \right)| \leq 2H \sqrt{d\lambda} \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)}.$$

Since $\|\tilde{P}_h(\cdot|s, a) - P_h(\cdot|s, a)\|_\infty \leq 1$ for all $(s, a) \in \mathcal{S} \times \mathcal{A}$, it follows that

$$|\tilde{P}_h(V_{h+1}^k - V_{h+1}^\pi)(s, a) - P_h(V_{h+1}^k - V_{h+1}^\pi)(s, a)| \leq |(\tilde{P}_h - P_h)(V_{h+1}^k - V_{h+1}^\pi)(s, a)| \leq 2H\xi.$$

From Lemma 28, we have $|G_h(a), q_4^s| \leq 2H\xi \sqrt{d|T_{k,h}^s|G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)}$. Since

$$|G_h(a)^\top w_{h,k}^s - Q_h^\pi(s, a)| = G_h(a)^\top (w_{h,k}^s - w_h^{s,\pi}) = G_h(a)^\top (q_1^s + q_2^s + q_3^s + q_4^s),$$

it follows that

$$|G_h(a)^\top w_{h,k}^s - Q_h^\pi(s, a) - P_h(V_{h+1}^k - V_{h+1}^\pi)(s, a)| \leq \sqrt{\chi}(C''d\sqrt{\chi} + 2\xi\sqrt{|T_{k,h}^s|d})H\sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)} + 4H\xi.$$

Similarly as in [37], we choose c_β that satisfies $C''\sqrt{\log(2) + \log(c_\beta + 1)} \leq c_\beta\sqrt{\log(2)}$. $\square$

Now, we prove that Q_h^k is an upperbound of Q_h^* conditioned on the event in Lemma 27.

Lemma 30. *Suppose Assumption 13 holds. On the event $\mathcal{X}$ defined in lemma 27, we have $Q_h^k(s, a) \geq Q_h^*(s, a) - 4H(H + 1 - h)\xi$ for all $(s, a, h, k) \in \mathcal{S} \times \mathcal{A} \times [H] \times [K]$.*

Proof. We prove this with induction. The base case holds because at step $H + 1$, the value function is zero, so from Lemma 29,

$$|G_h(a)^\top w_{h,k}^s - Q_h^*(s, a)| \leq \beta_{k,h}^s \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)} + 4H\xi$$

and thus,

$$Q_H^*(s, a) - 4H\xi \leq \min \left(H, G_h(a)^\top w_{h,k}^s + \beta_{k,h}^s \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)} \right) = Q_H^k(s, a).$$

From the inductive hypothesis (assuming that $Q_{h+1}^k(s, a) \geq Q_{h+1}^*(s, a) - 4H(H-h)\xi$), it follows that $P_h(V_{h+1}^k - V_{h+1}^*)(s, a) \geq -4\xi$. From Lemma 29, $G_h(a)^\top w_{h,k}^s - Q_h^*(s, a) \leq \beta_{k,h}^s \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)} + 4H\xi$. It follows that

$$Q_h^*(s, a) - 4H(H-h+1)\xi \leq \min \left(H, G_h(a)^\top w_{h,k}^s + \beta_{k,h}^s \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)} \right) = Q_h^k(s, a).$$

□

Similarly, to the regular case, the gap in the misspecified setting has a recursive formula.

Lemma 31. *Let $\delta_h^k = V_h^k(s_h^k) - V_h^{\pi_k}(s_h^k)$ and $\xi_{h+1}^k = \mathbb{E}[\delta_{h+1}^k | s_h^k, a_h^k] - \delta_{h+1}^k$. Then, on the event $\mathcal{X}$ defined in Lemma 27, we have for any (h, k) :*

$$\delta_h^k \leq \delta_{h+1}^k + \xi_{h+1}^k + \beta_{k,h}^s \sqrt{G_h(a_h^k)^\top (\Lambda_{h,k}^{s_h^k})^{-1} G_h(a_h^k)} + 4H\xi$$

Proof. From Lemma 29, we have for any (s, a, h, k) that

$$Q_h^k(s, a) - Q_h^{\pi_k}(s, a) \leq P_h(V_{h+1}^k - V_{h+1}^{\pi_k})(s, a) + \beta_{k,h}^s \sqrt{G_h(a)^\top (\Lambda_{h,k}^s)^{-1} G_h(a)} + 4H\xi.$$

From the definition of V^{π_k} , we have

$$\delta_h^k = Q_h^k(s_h^k, a_h^k) - Q_h^{\pi_k}(s_h^k, a_h^k),$$

and substituting this into the first equation finishes the proof. □

Finally, we prove the main result in the misspecified setting Theorem 7.

Proof. Suppose that the Assumptions required in Theorem 15 hold. We condition on the event $\mathcal{X}$ from Lemma 27 with $p = \delta/2$ and use the notation for δ_h^k, ξ_h^k as in Lemma 31. From Lemma 16, we have $Q_1^k(s, a) \geq Q_1^*(s, a) - 4H^2\xi$, which implies $V_1^*(s) - V_1^{\pi_k}(s) \leq \delta_1^k + 4H^2\xi$. Thus, from Lemma 29, on the event $\mathcal{X}$, it follows that

$$\begin{aligned} \text{Regret}(K) &= \sum_{k=1}^K (V_1^*(s_1^k) - V_1^{\pi_k}(s_1^k)) \leq \sum_{k=1}^K \delta_1^k + 4H^2\xi \\ &\leq \sum_{k \in [K]} \sum_{h \in [H]} \xi_h^k + \sum_{k \in [K]} \beta_{k,h}^s \sum_{h \in [H]} \sqrt{G_h(a_h^k)^\top (\Lambda_{h,k}^{s_h^k})^{-1} G_h(a_h^k)} + 4HT\xi \end{aligned}$$

since $HK = T$. Since the observations at episode k are independent of the computed value function (this uses the trajectories from episodes 1 to $k-1$, it follows $\{\xi_h^k\}$ is a martingale difference sequence with $|\xi_h^k| \leq 2H$ for all (k, h) . Thus, from the Azuma Hoeffding inequality, we have

$$\sum_{k \in [K], h \in H} \xi_h^k \leq \sqrt{2TH^2 \log(2/p)} \leq 2H \sqrt{T\iota}$$

with probability at least $1 - \delta/2$. Note that

$$\begin{aligned} &\sum_{k \in [K]} \beta_{k,h}^s \sqrt{G_h(a_h^k)^\top (\Lambda_{h,k}^{s_h^k})^{-1} G_h(a_h^k)} \\ &\leq CH \left(\sum_{k \in [K]} d \sqrt{\iota} \sqrt{G_h(a_h^k)^\top (\Lambda_{h,k}^{s_h^k})^{-1} G_h(a_h^k)} + \sum_{k \in [K]} \xi \sqrt{|T_{k,h}^s|d} \sqrt{G_h(a_h^k)^\top (\Lambda_{h,k}^{s_h^k})^{-1} G_h(a_h^k)} \right). \end{aligned}$$

From the Cauchy-Schwarz inequality, it follows that

$$\sum_{k \in [K]} \sqrt{G_h(a_h^k)^\top (\Lambda_{h,k}^{s_h^k})^{-1} G_h(a_h^k)} \leq (\sqrt{K}) \left(\sqrt{\sum_{k \in [K]} G_h(a_h^k)^\top (\Lambda_{h,k}^{s_h^k})^{-1} G_h(a_h^k)} \right).$$

Let k_i refers to the episode in which state s was seen at time step h for the i th time. Then, for the other term, we first re-index the summation and then use

the Cauchy-Schwarz inequality to get

$$\begin{aligned}
\sum_{k \in [K]} \sqrt{|T_{k,h}^s|} \sqrt{G_h(a_h^k)^\top (\Lambda_{h,k}^{s_h^k})^{-1} G_h(a_h^k)} &= \sum_{s \in \mathcal{S}} \sum_{i=1}^{|T_{K,h}^s|} \sqrt{i} \sqrt{G_{k_i,h}^\top (\Lambda_{k_i,h}^a)^{-1} G_{k_i,h}} \\
&\leq \left(\sum_{i=1}^{|T_{K,h}^s|} i \right)^{1/2} \left(\sum_{t \in T_{K,h}^s} G_{k,h}^\top (\Lambda_{k,h}^{s_h^k})^{-1} G_{k,h} \right)^{1/2} \\
&\leq C' |T_{K,h}^s| \left(\sum_{t \in T_{K,h}^s} G_{k,h}^\top (\Lambda_{k,h}^{s_h^k})^{-1} G_{k,h} \right)^{1/2}
\end{aligned}$$

for some absolute constant $C' > 0$. Since the minimum eigenvalue of $\Lambda_{h,k}^{s_h^k}$ is at least one by construction and $\|G_h(s)\| \leq 1$, from the Elliptical Potential Lemma (Lemma D.2 [37]), we have for all $a \in \mathcal{A}$ and $h \in [H]$,

$$\sum_{k=1}^K G_h(s_k)^\top (\Lambda_{h,k}^a)^{-1} G_h(s_k) \leq 2 \log(\det(\Lambda_{h,k+1}^a) / \det(\Lambda_{h,0}^a))$$

Furthermore, we have $\|\Lambda_{h,k+1}^a\| = \|\sum_{t \in T_{k,h}^s} G_h(s_h^t) G_h(s_h^t)^\top + \lambda I\| \leq \lambda + |T_{k,h}^s| \leq \lambda + K$. It follows that

$$\sum_{k=1}^K G_h(s_k)^\top (\Lambda_{h,k}^a)^{-1} G_h(s_k) \leq 2d \log(1 + K/\lambda) \leq 2d\iota,$$

so grouping the episodes by actions gives

$$\left[\sum_{k \in [K]} G_h(a_h^k)^\top (\Lambda_{h,k}^{s_h^k})^{-1} G_h(a_h^k) \right]^{1/2} = \left[\sum_{s \in \mathcal{S}} \sum_{t \in T_{K,h,a}} G_h(a_h^t)^\top (\Lambda_{h,t}^a)^{-1} G_h(a_h^t) \right]^{1/2} \leq \sqrt{2dS\iota}$$

Thus,

$$\begin{aligned}
\sum_{k \in [K]} \beta_{k,h}^s \sqrt{G_h(a_h^k)^\top (\Lambda_{h,k}^{s_h^k})^{-1} G_h(a_h^k)} &\leq CH(\sqrt{2d^3 S K \iota^2} + \xi \sqrt{d} C' \sum_{s \in \mathcal{S}} |T_{k,h}^s| \sqrt{2d\iota}) \\
&\leq C(\sqrt{2d^3 S H T \iota^2} + \xi \sqrt{2d} T).
\end{aligned}$$

Substituting these results into the original regret bound gives

$$\text{Regret}(K) \leq C''' (\sqrt{d^3 H^3 S T \iota^2} + d T H \xi \sqrt{\iota}).$$

for some constant $C''' > 0$ with probability at least $1 - \delta$. $\square$

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